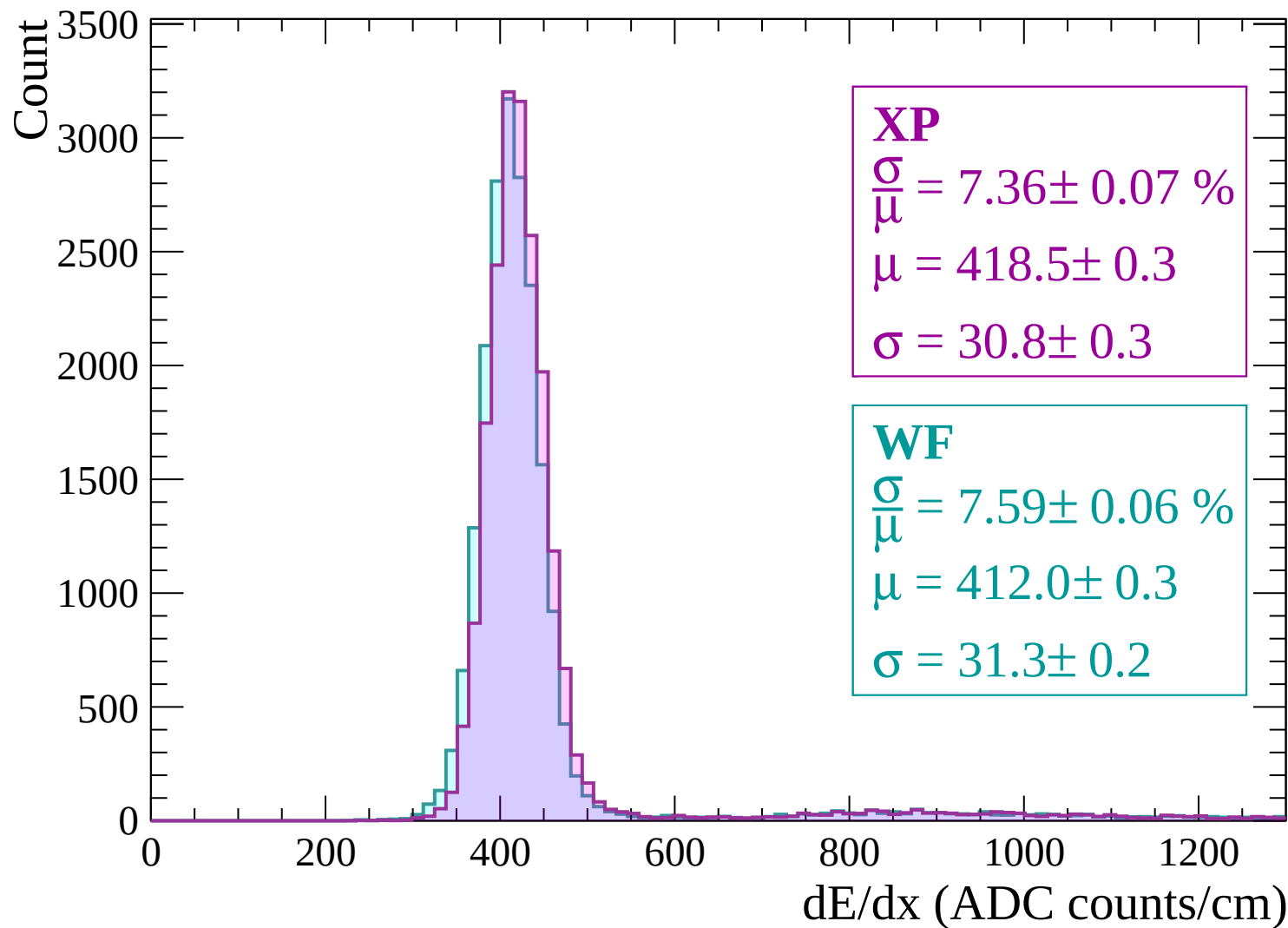
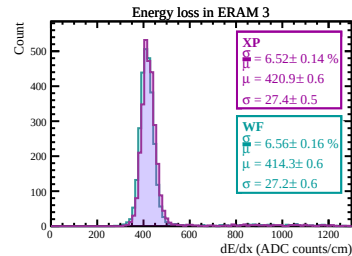
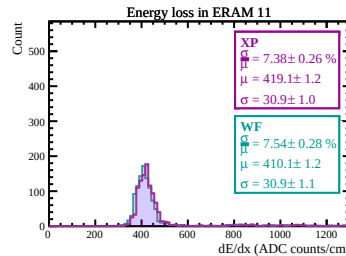
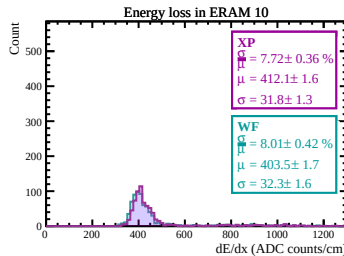
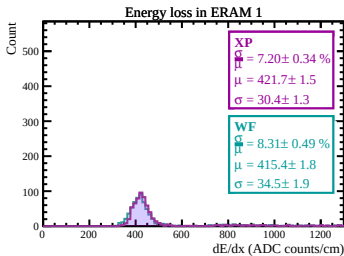
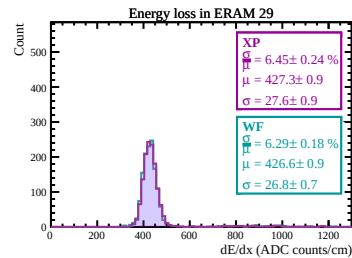
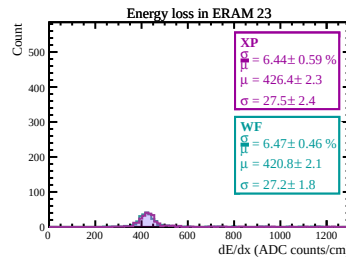
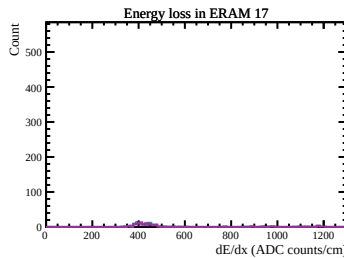
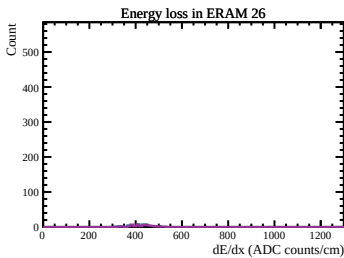
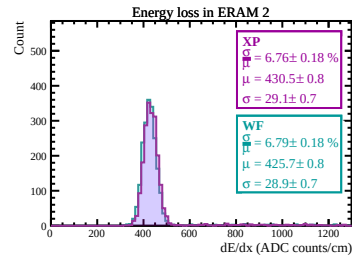
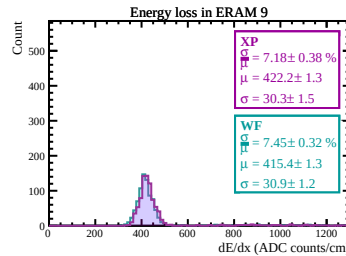
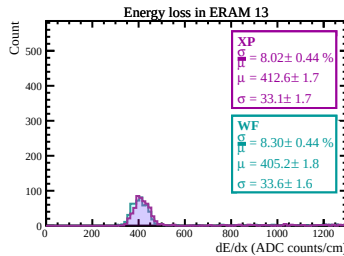
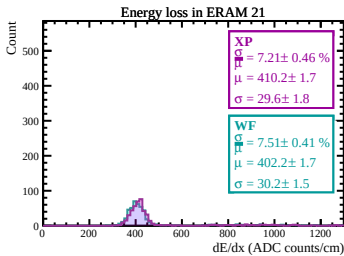
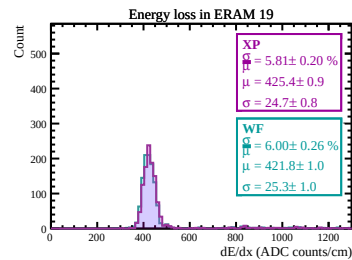
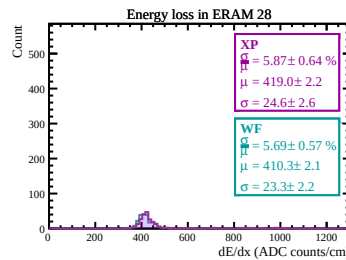
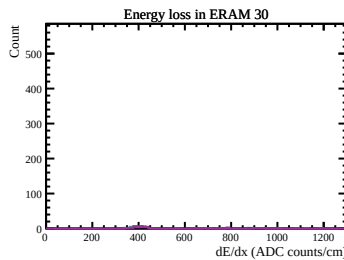
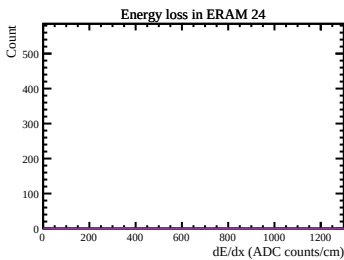
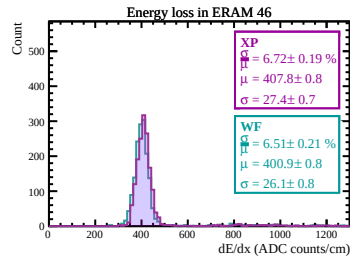
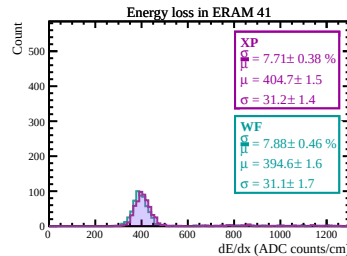
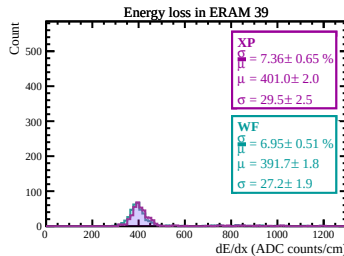
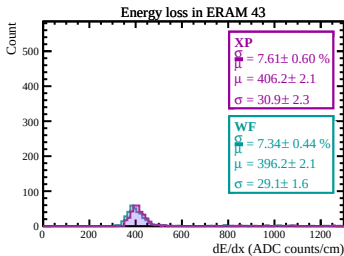
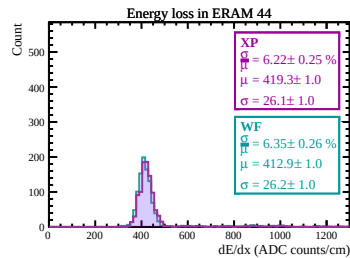
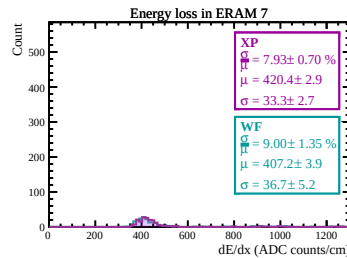
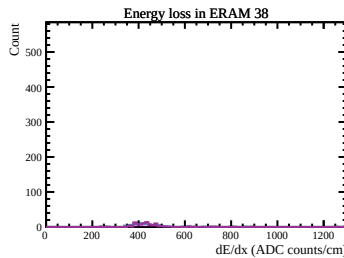
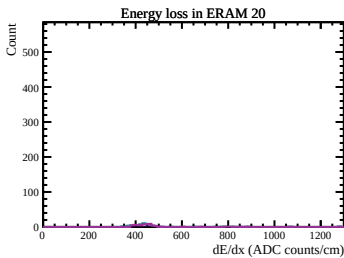
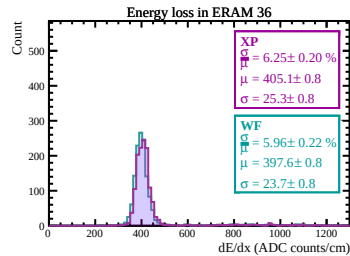
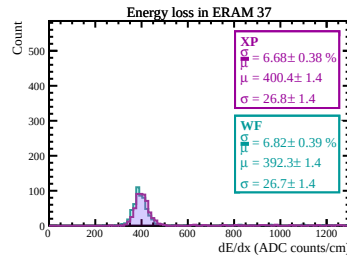
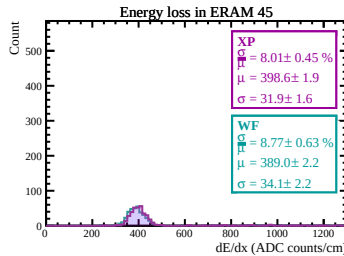
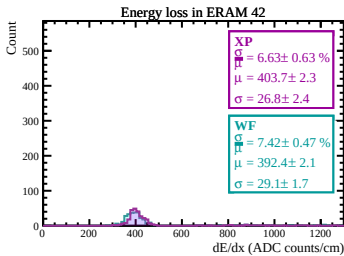
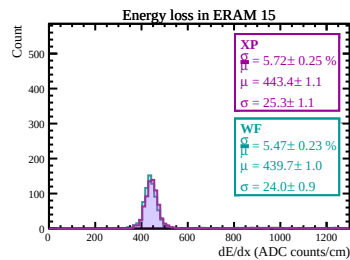
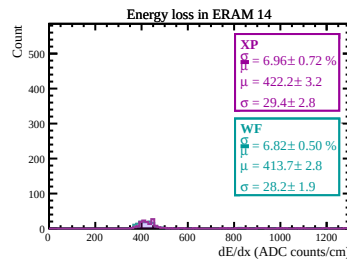
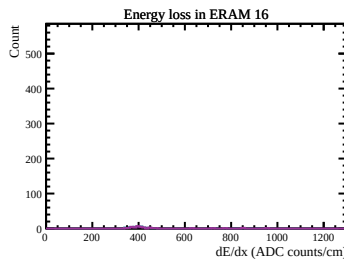
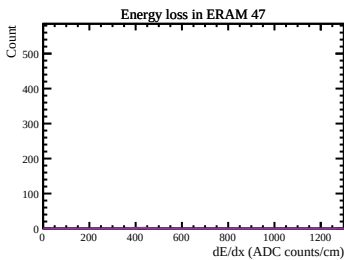
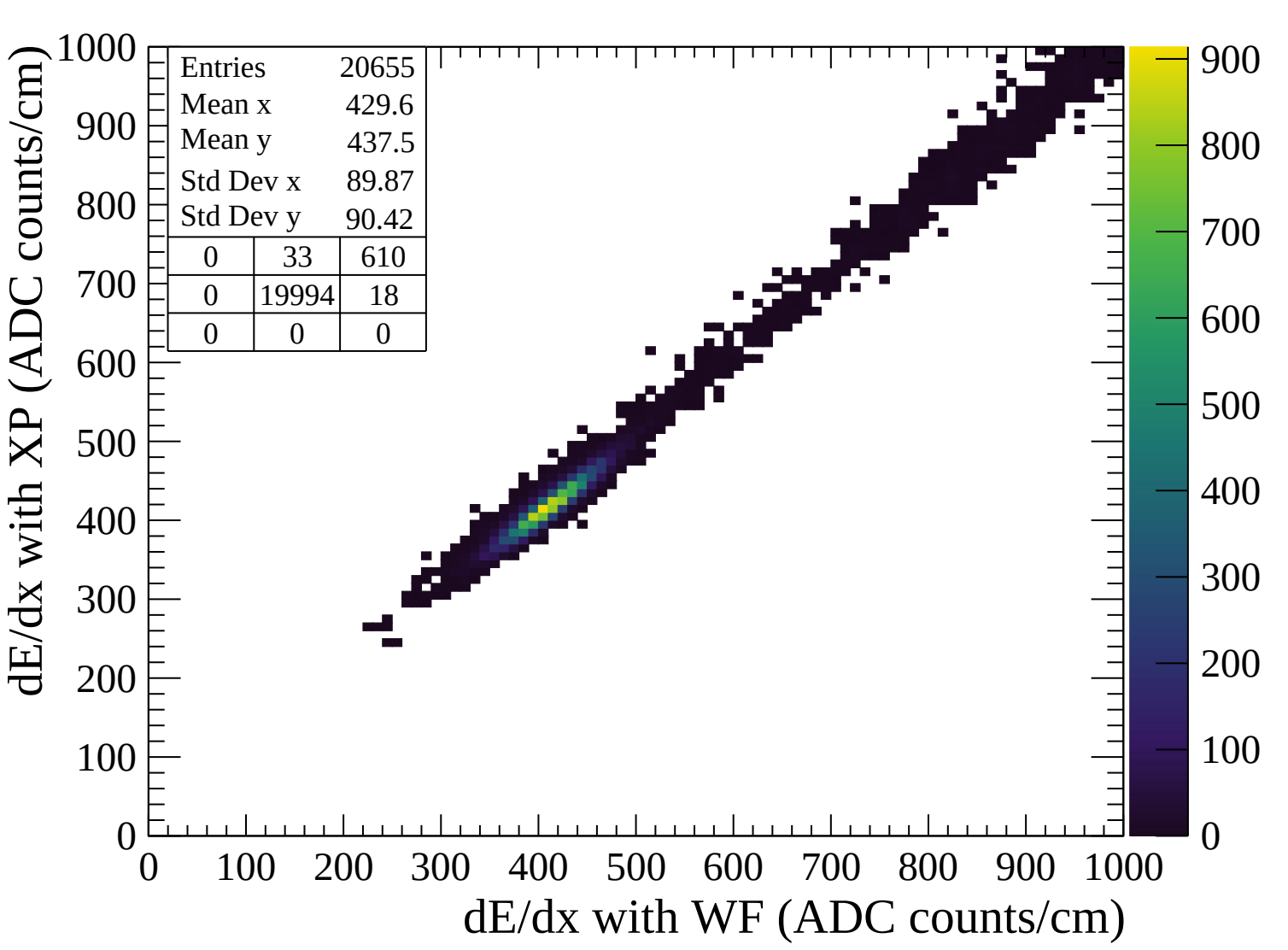
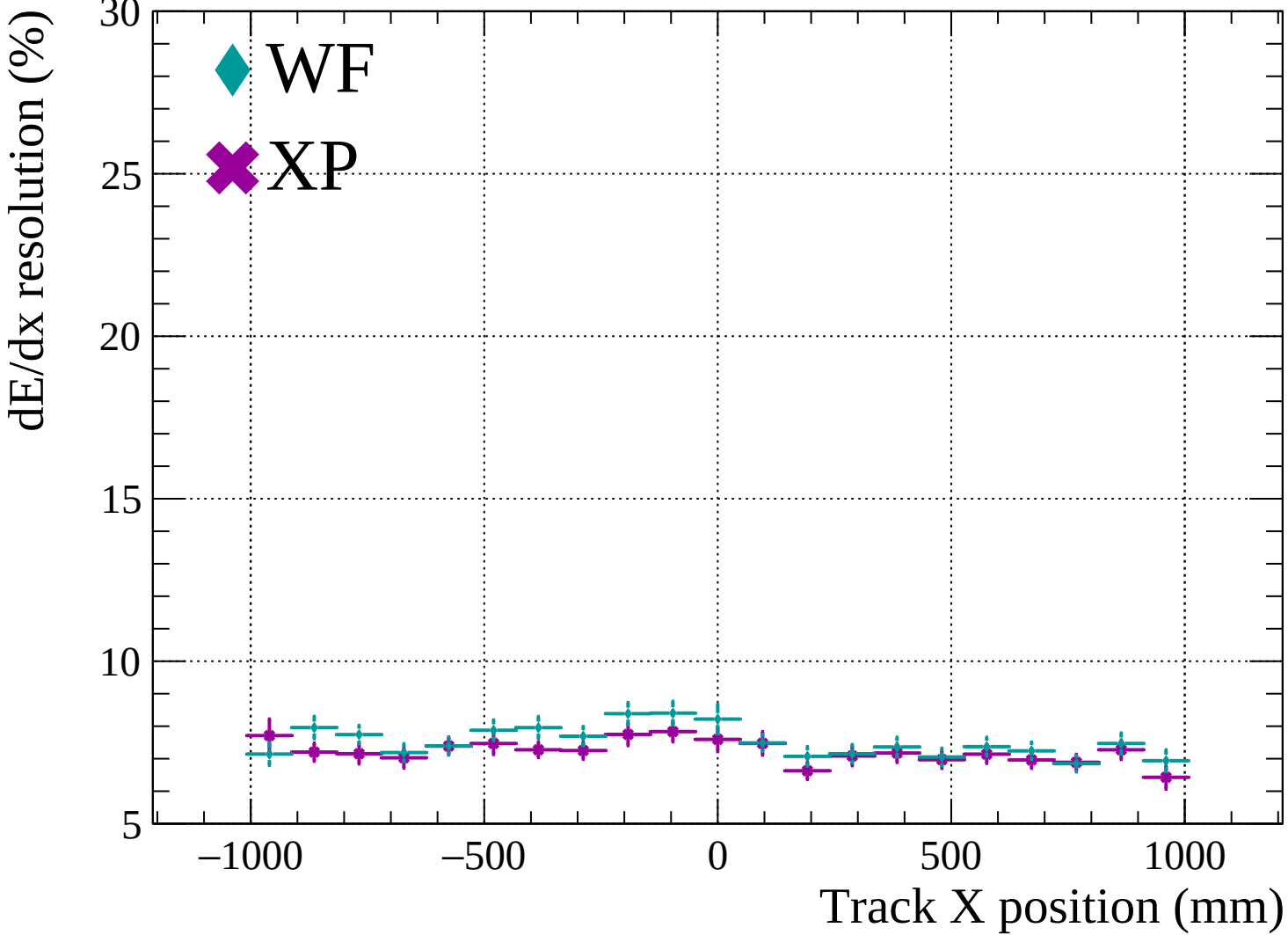


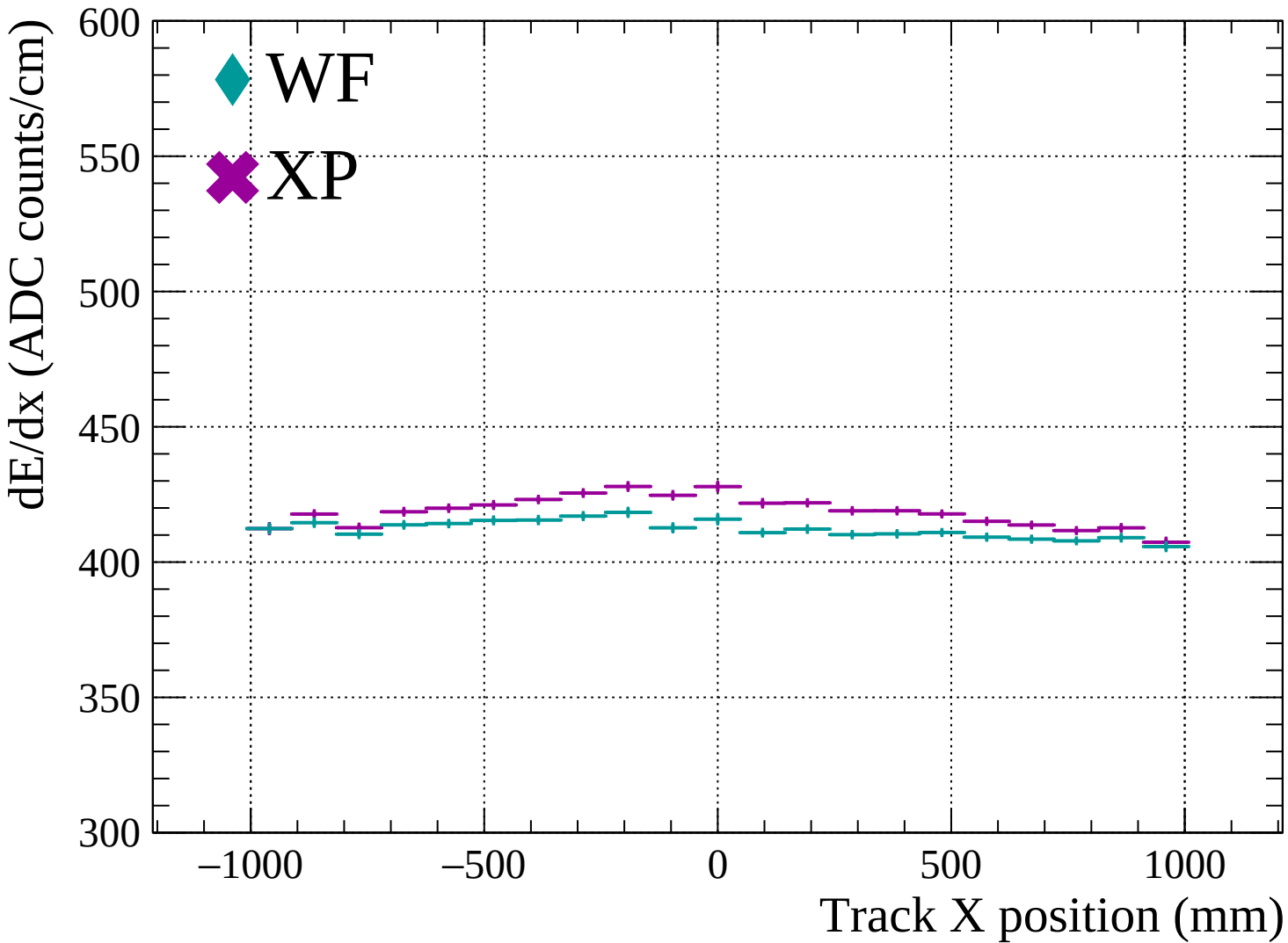


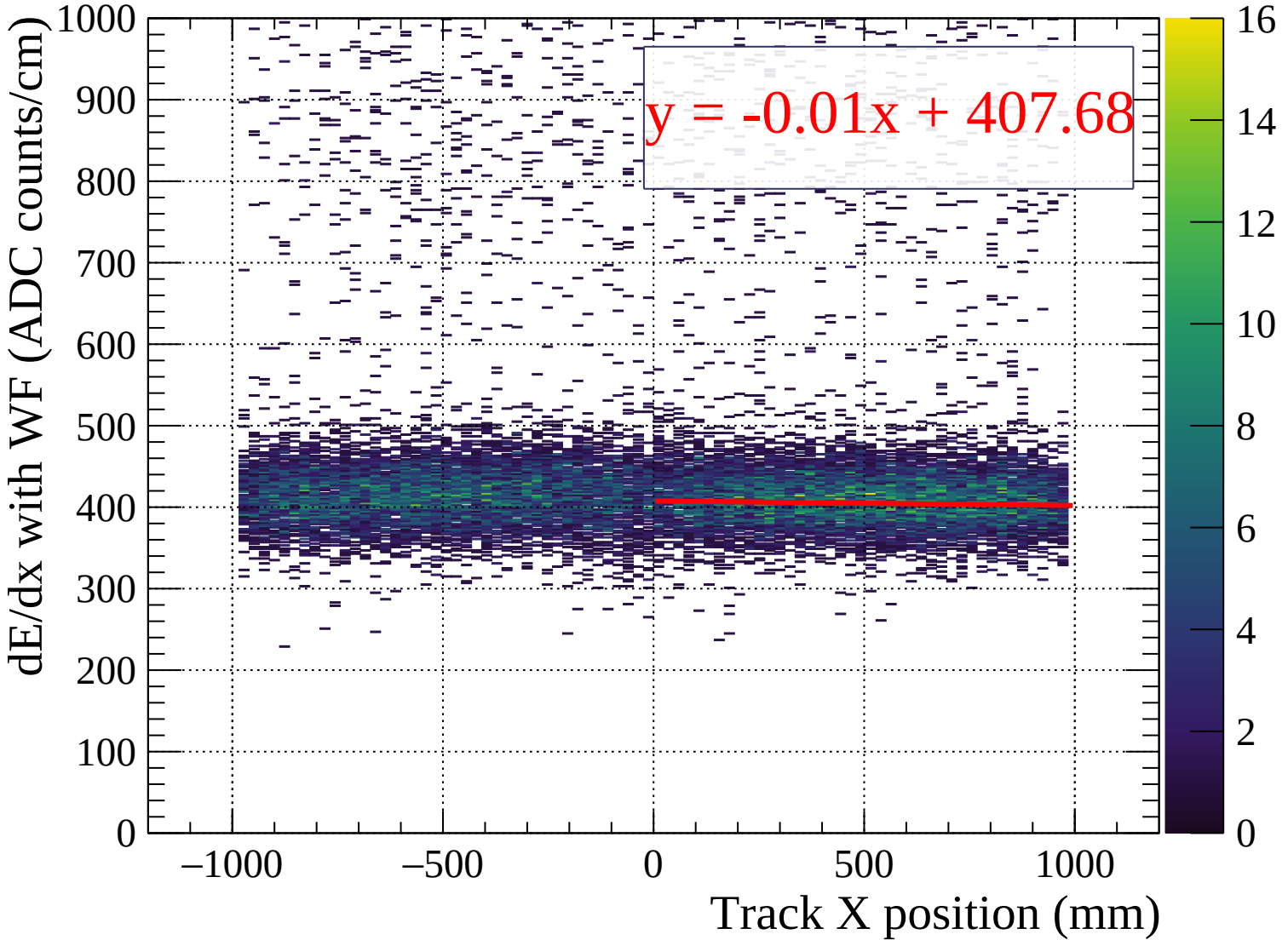


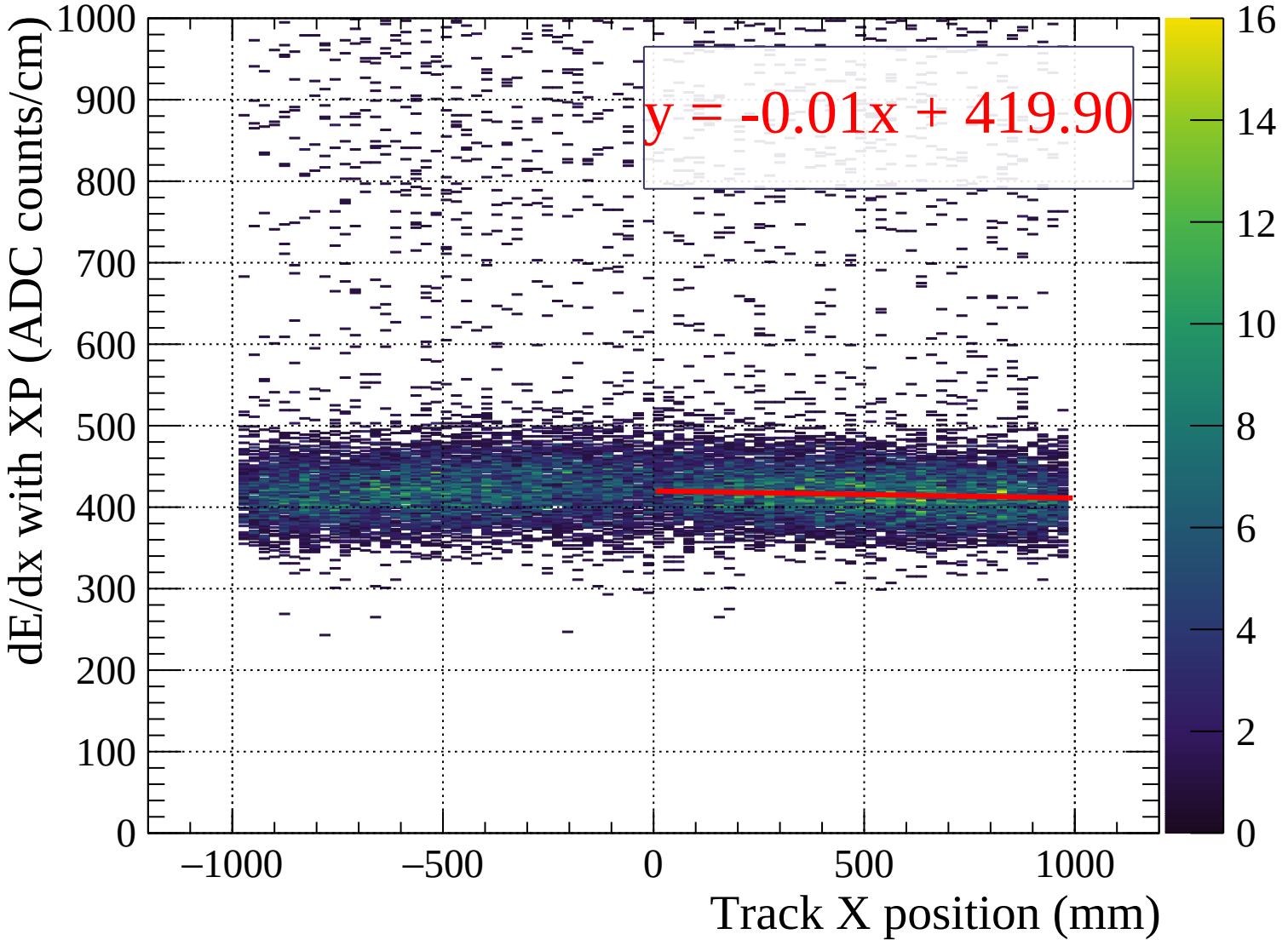


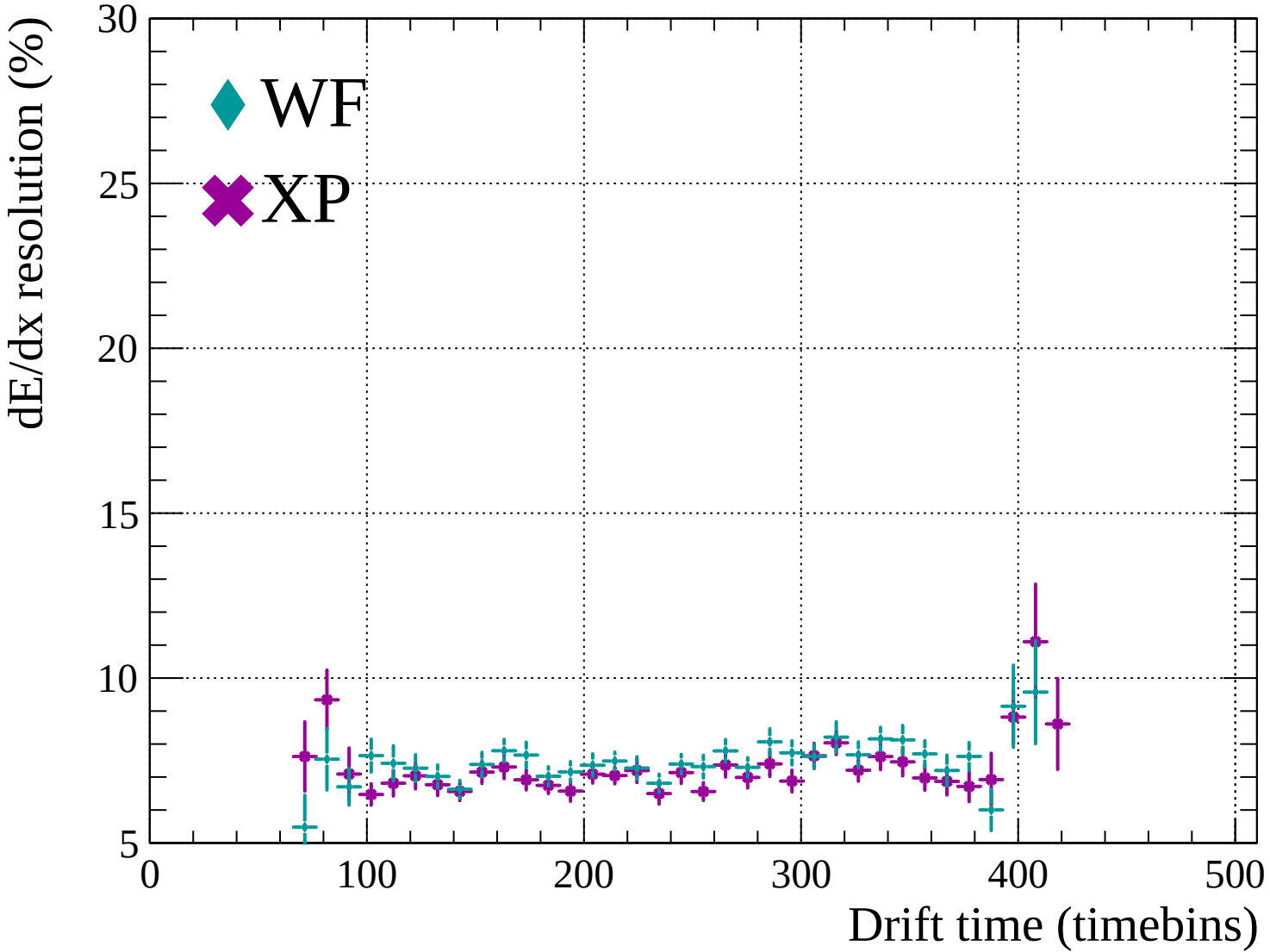


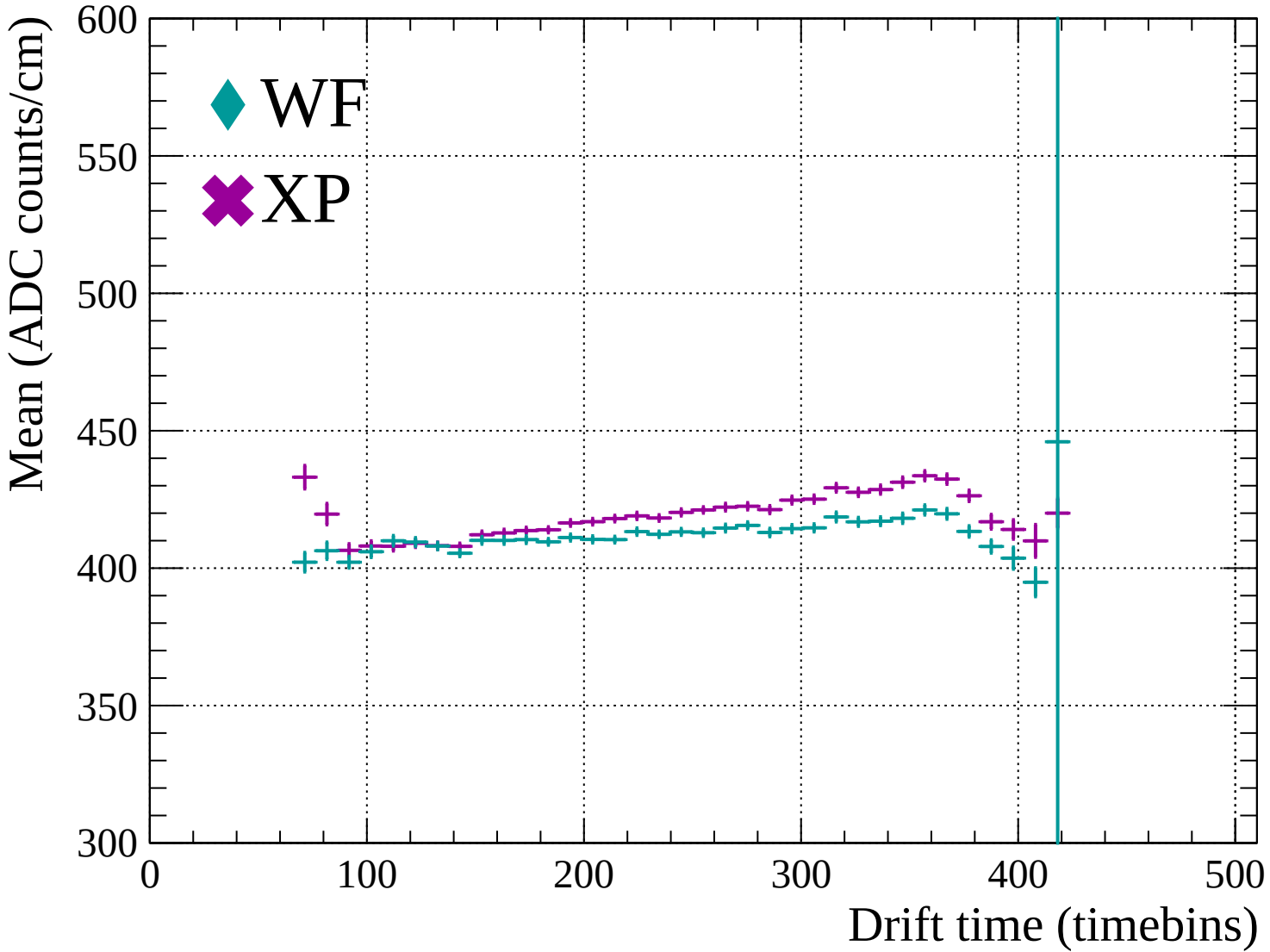


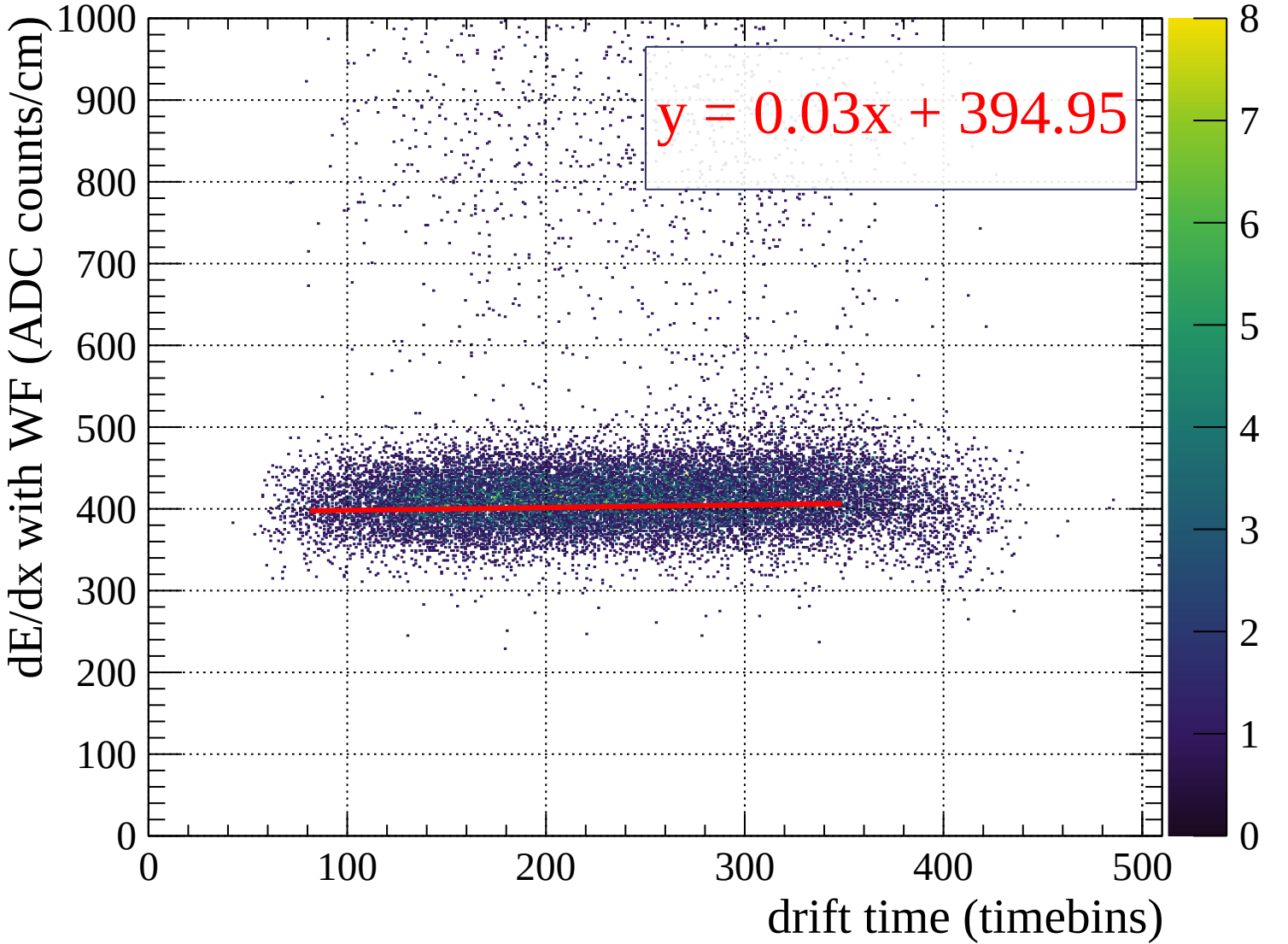


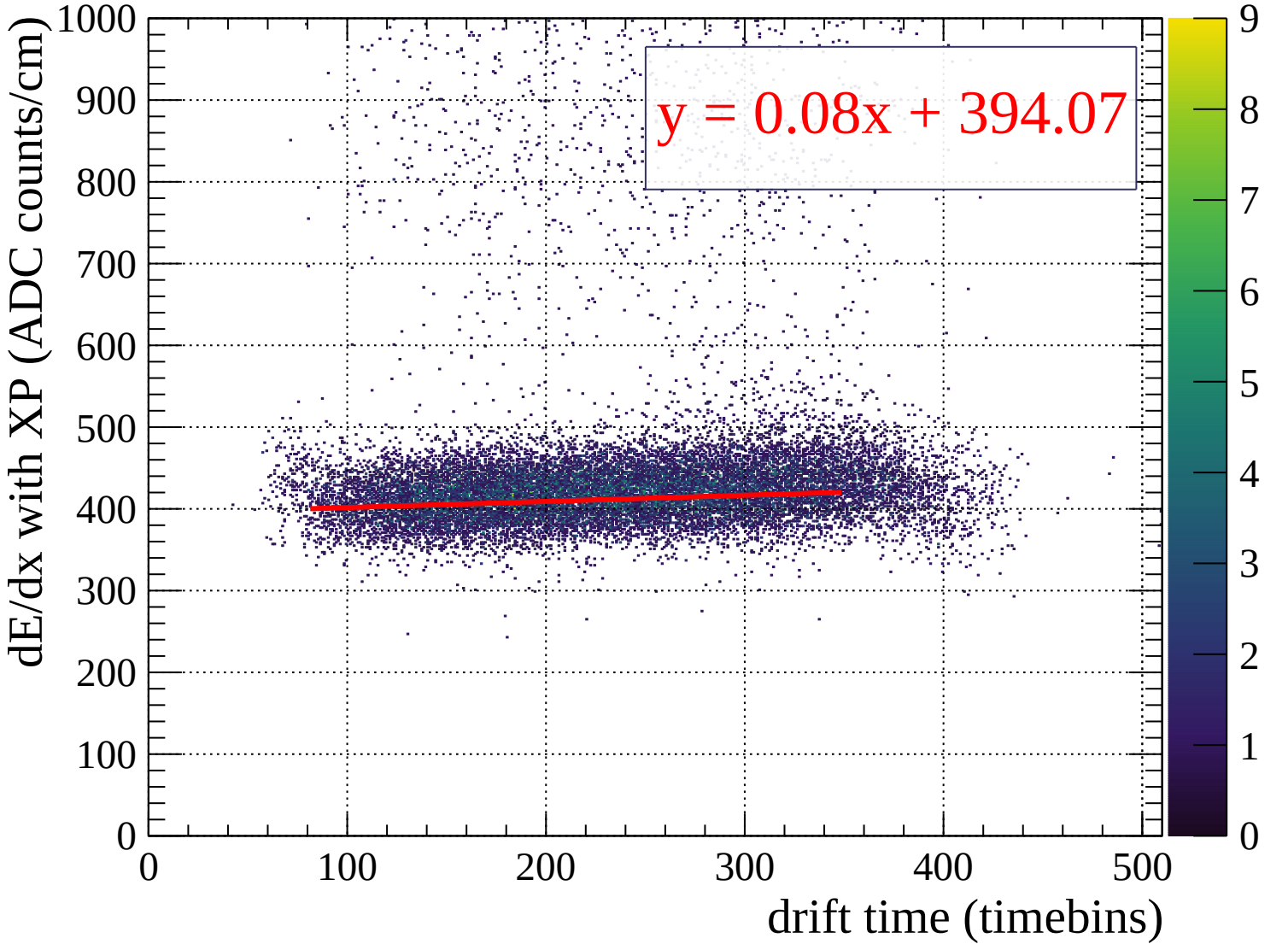


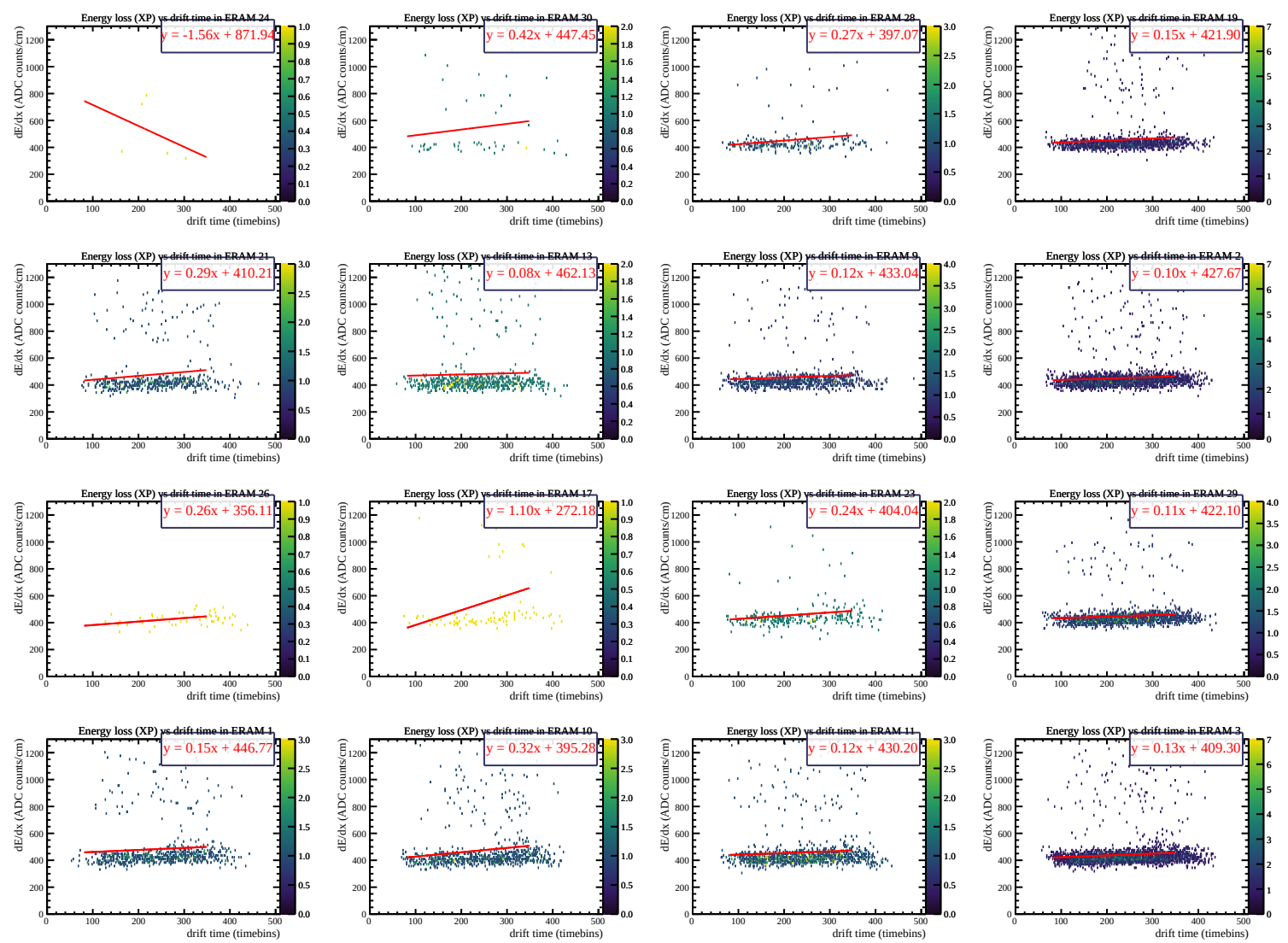


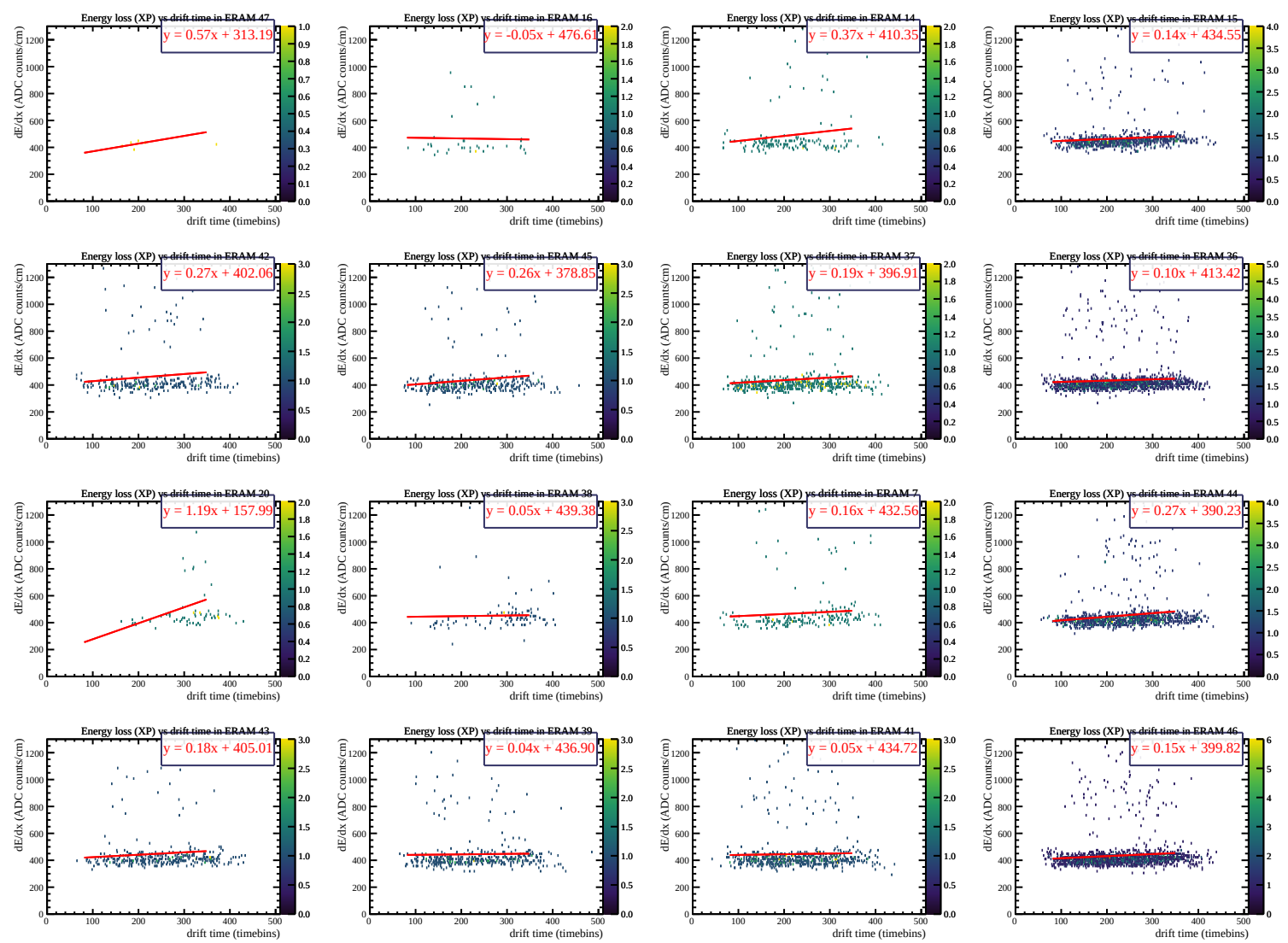


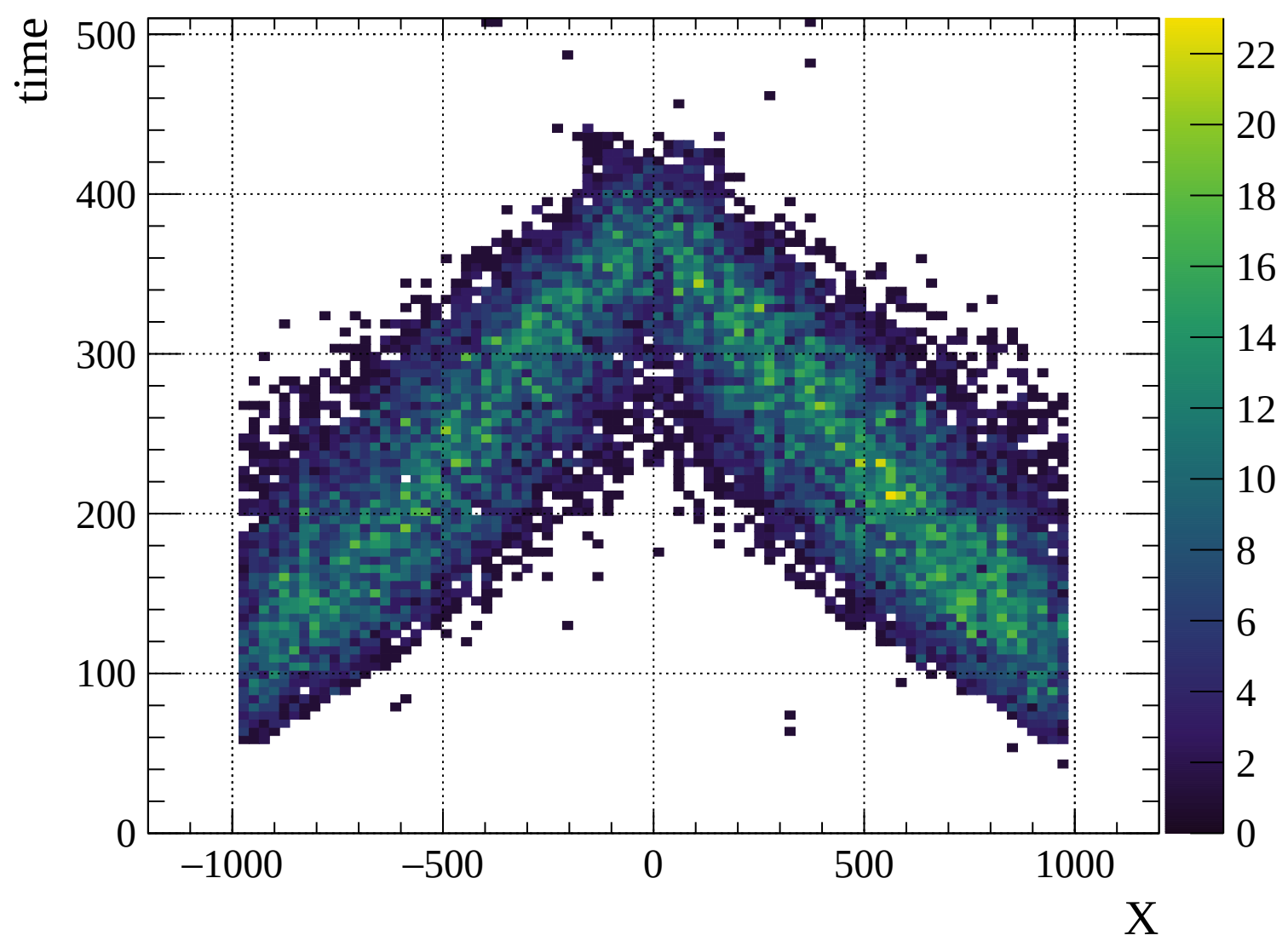


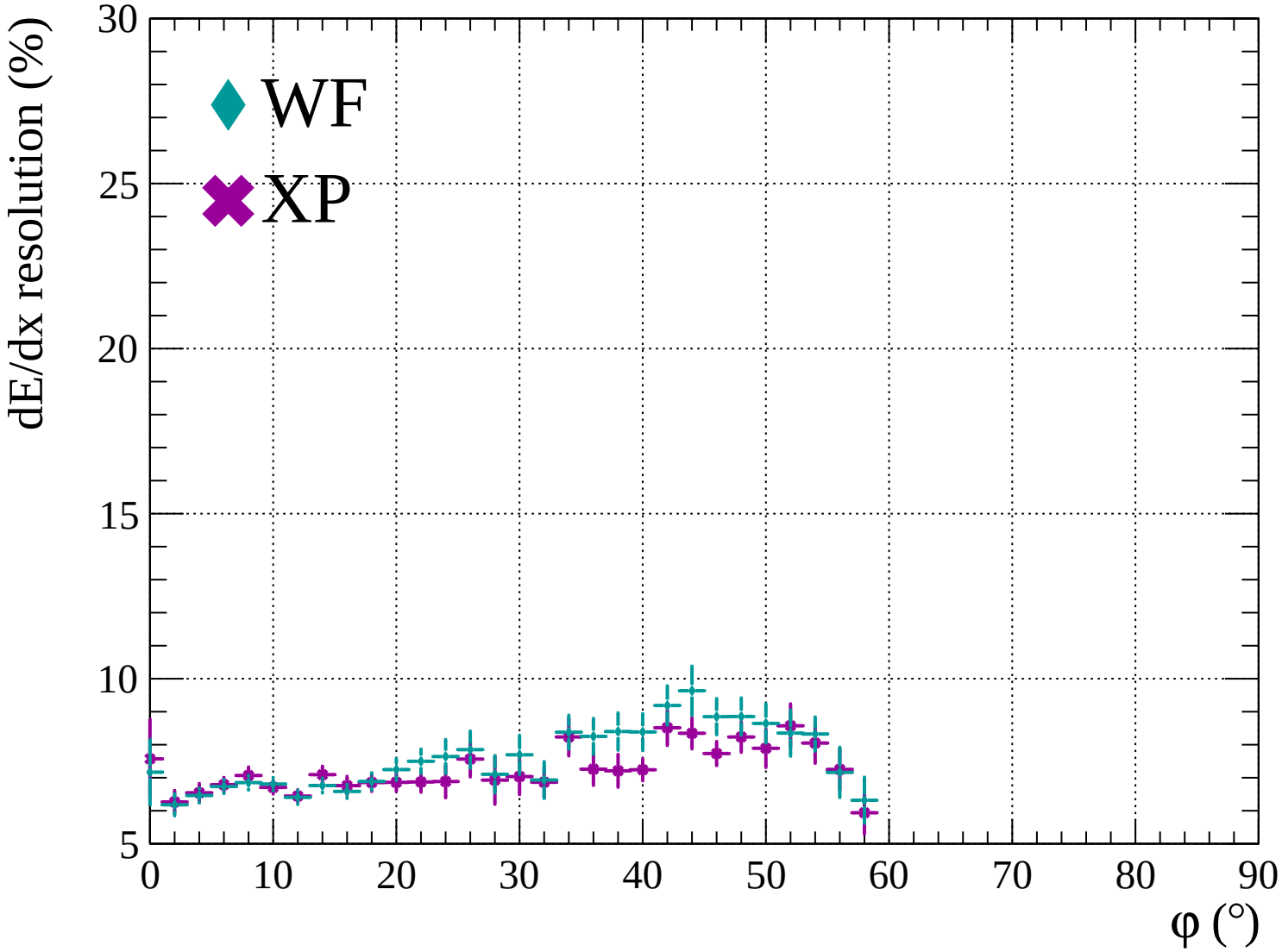


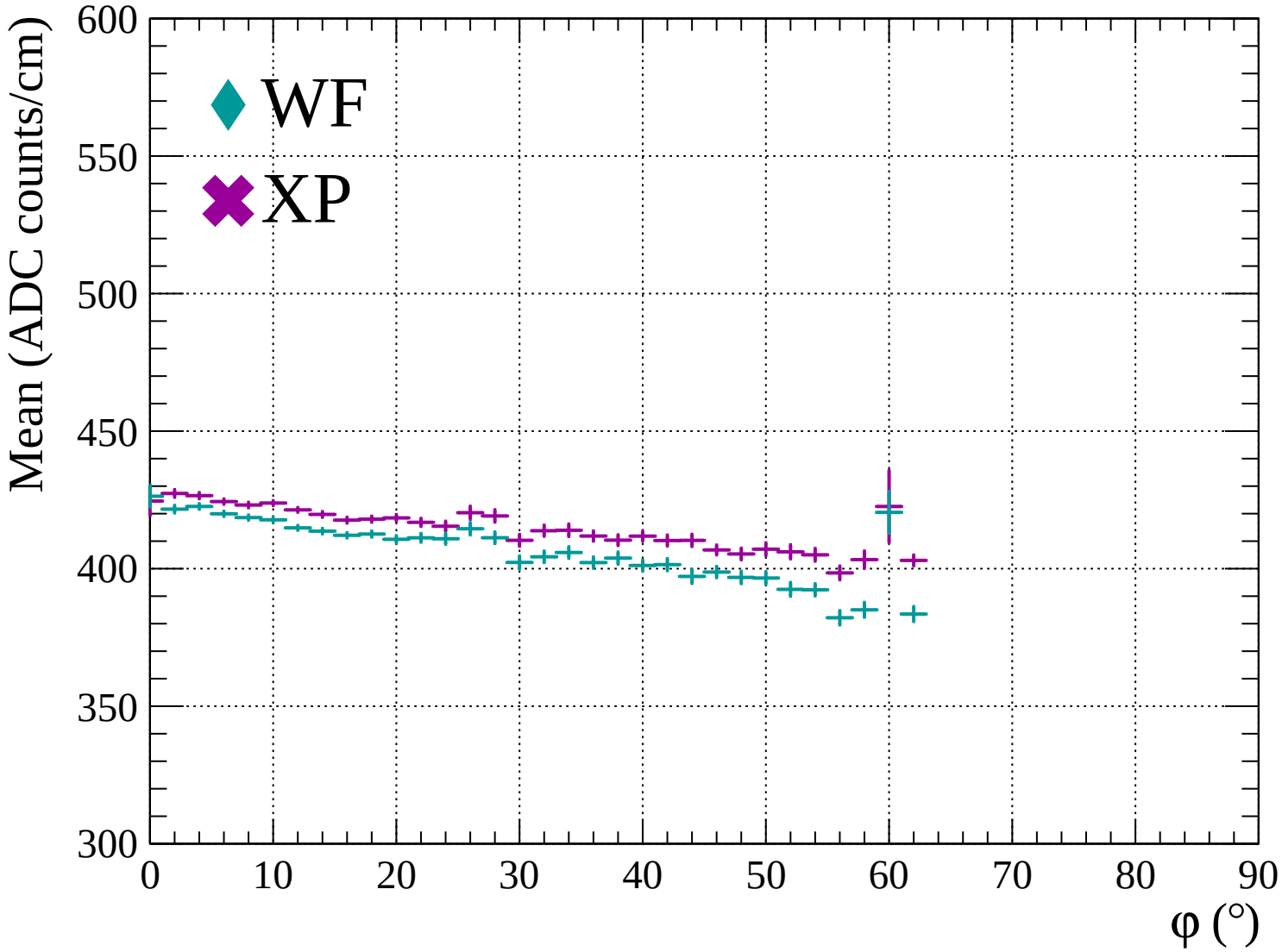


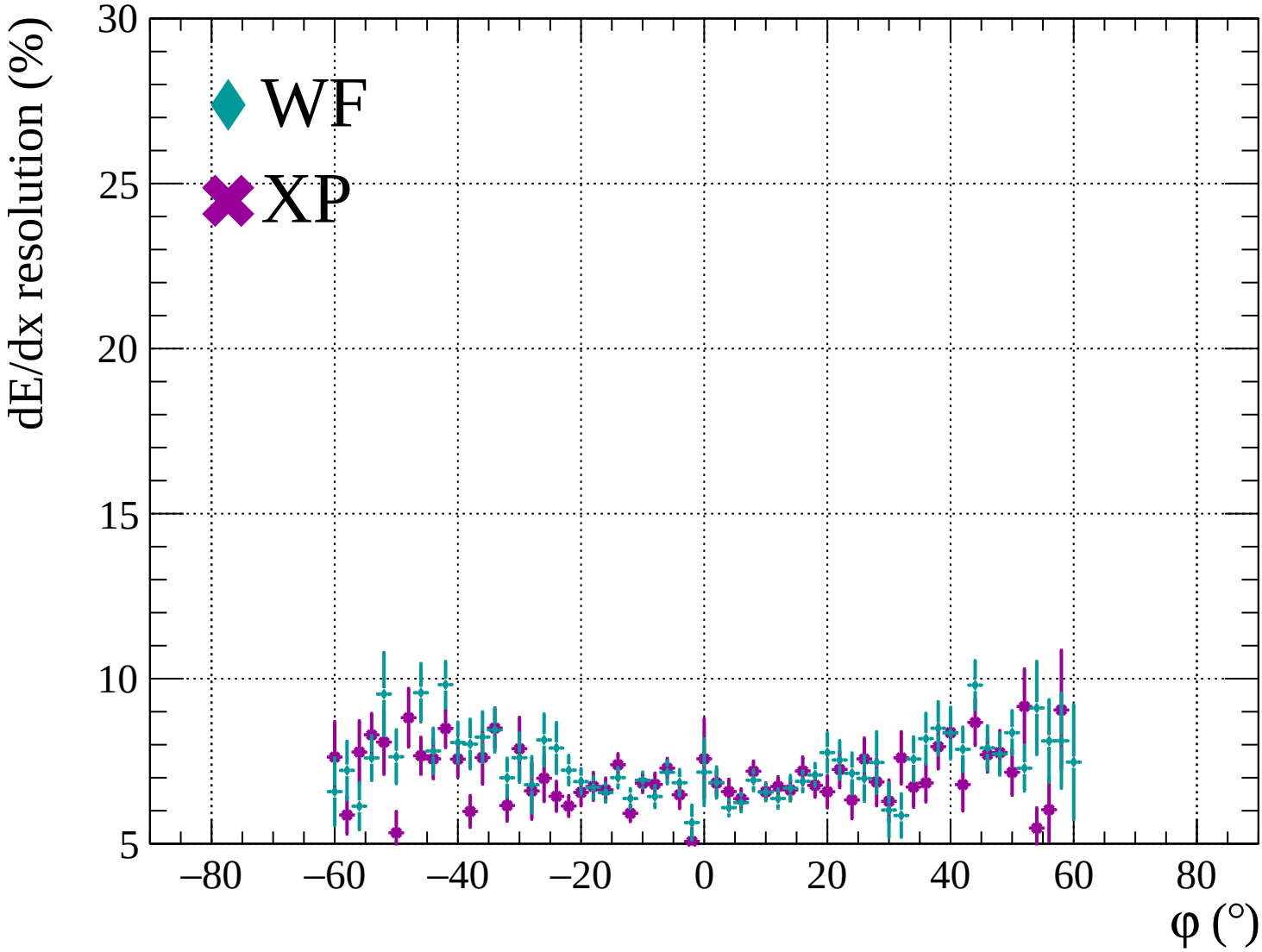


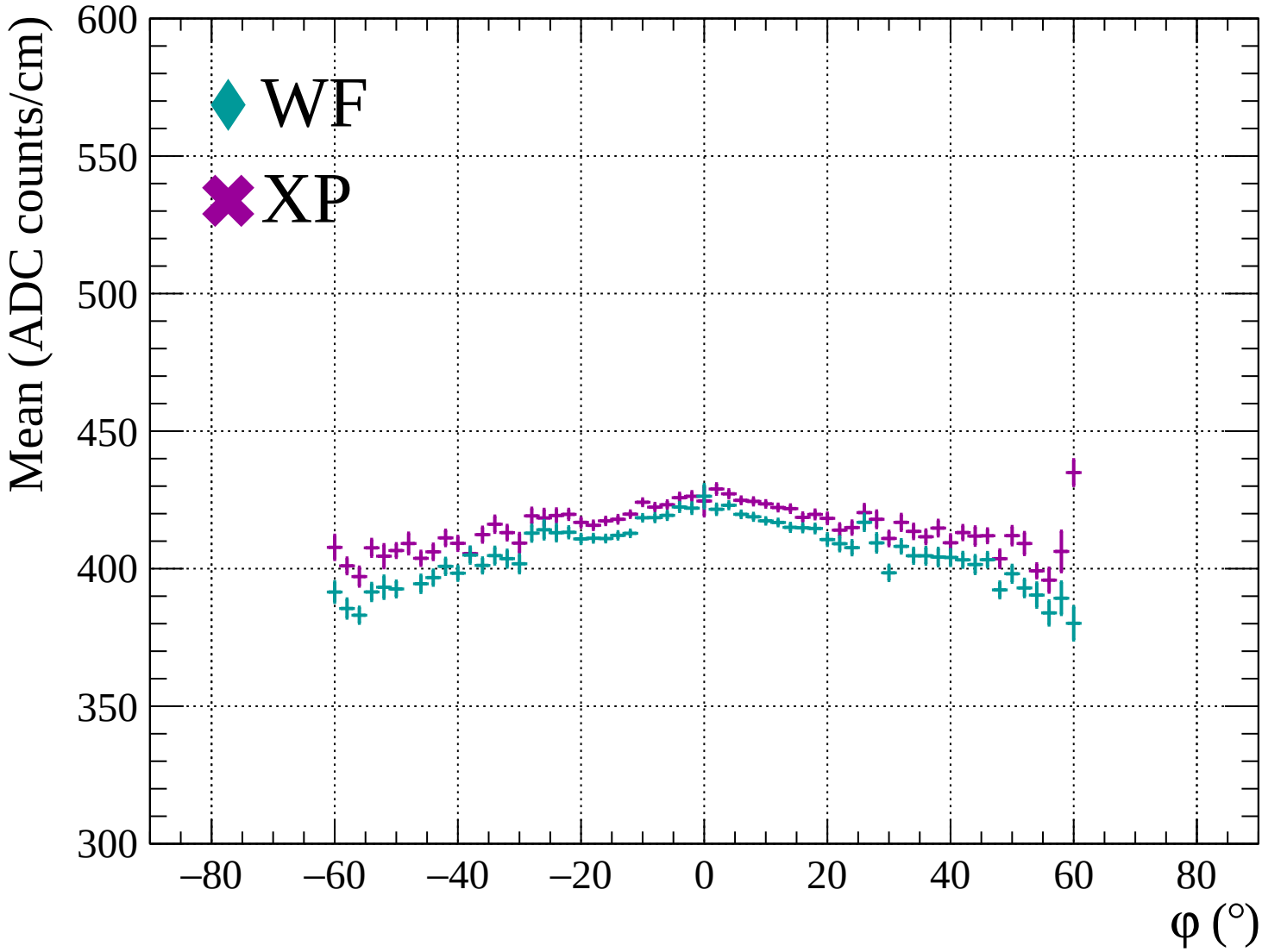


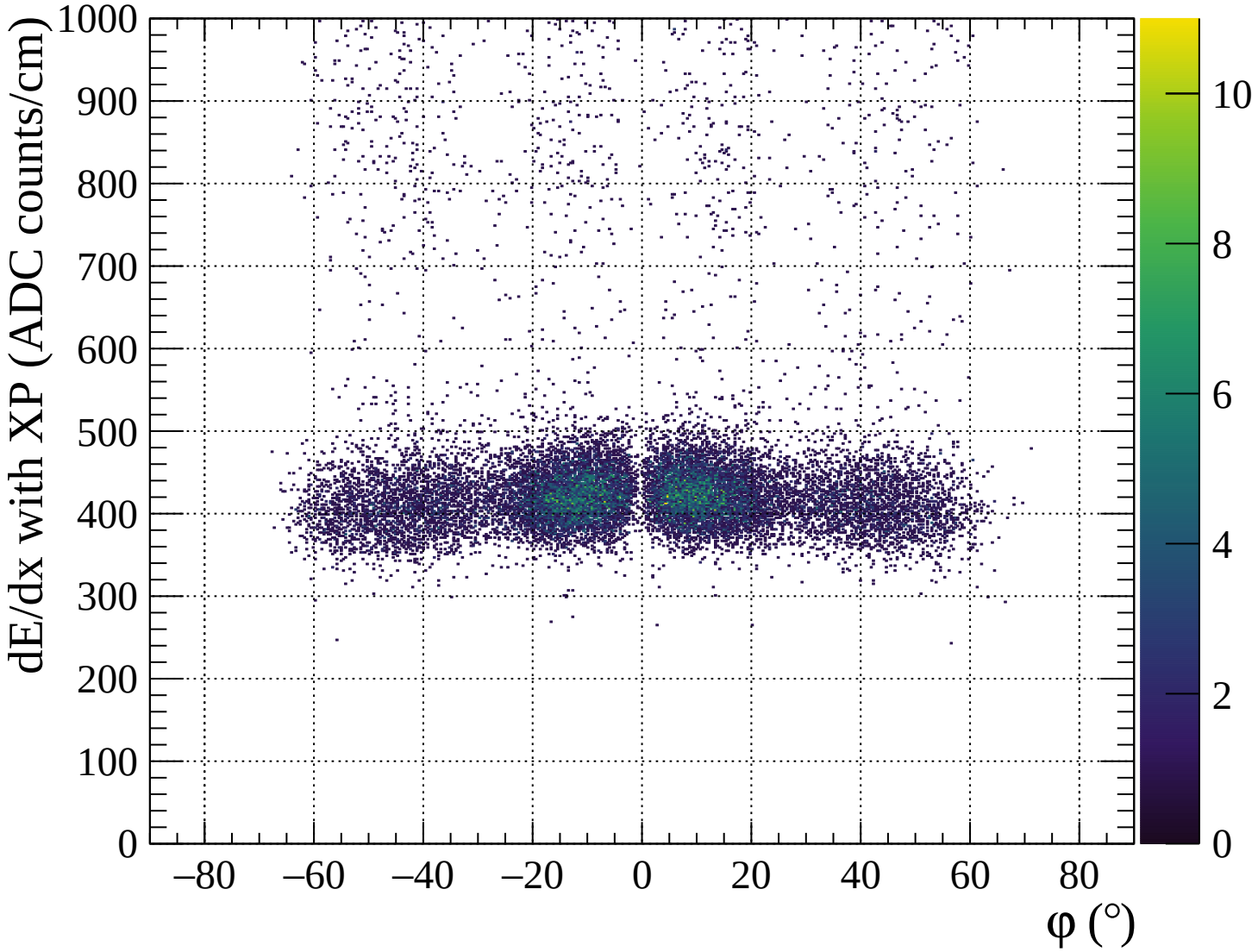


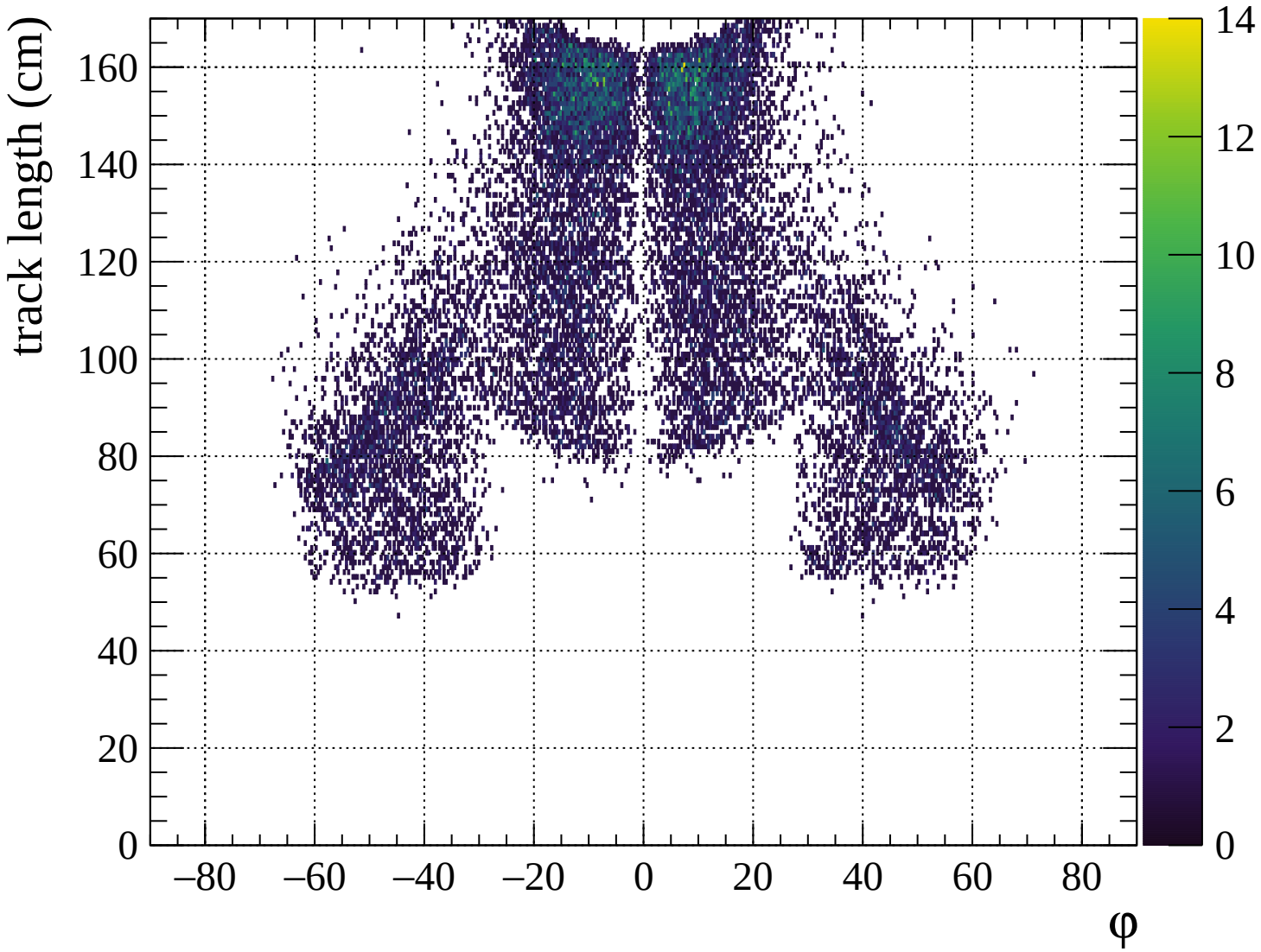


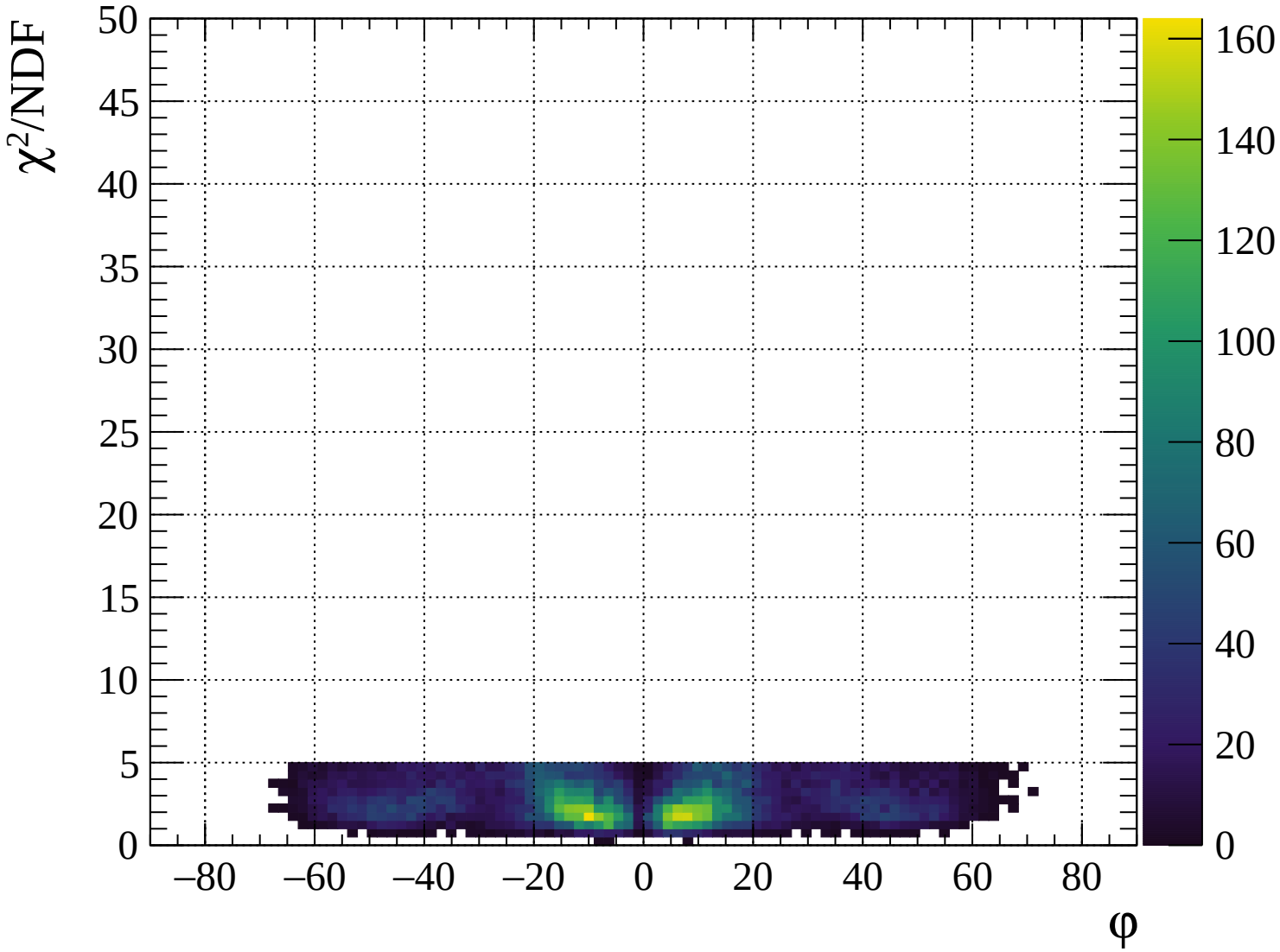


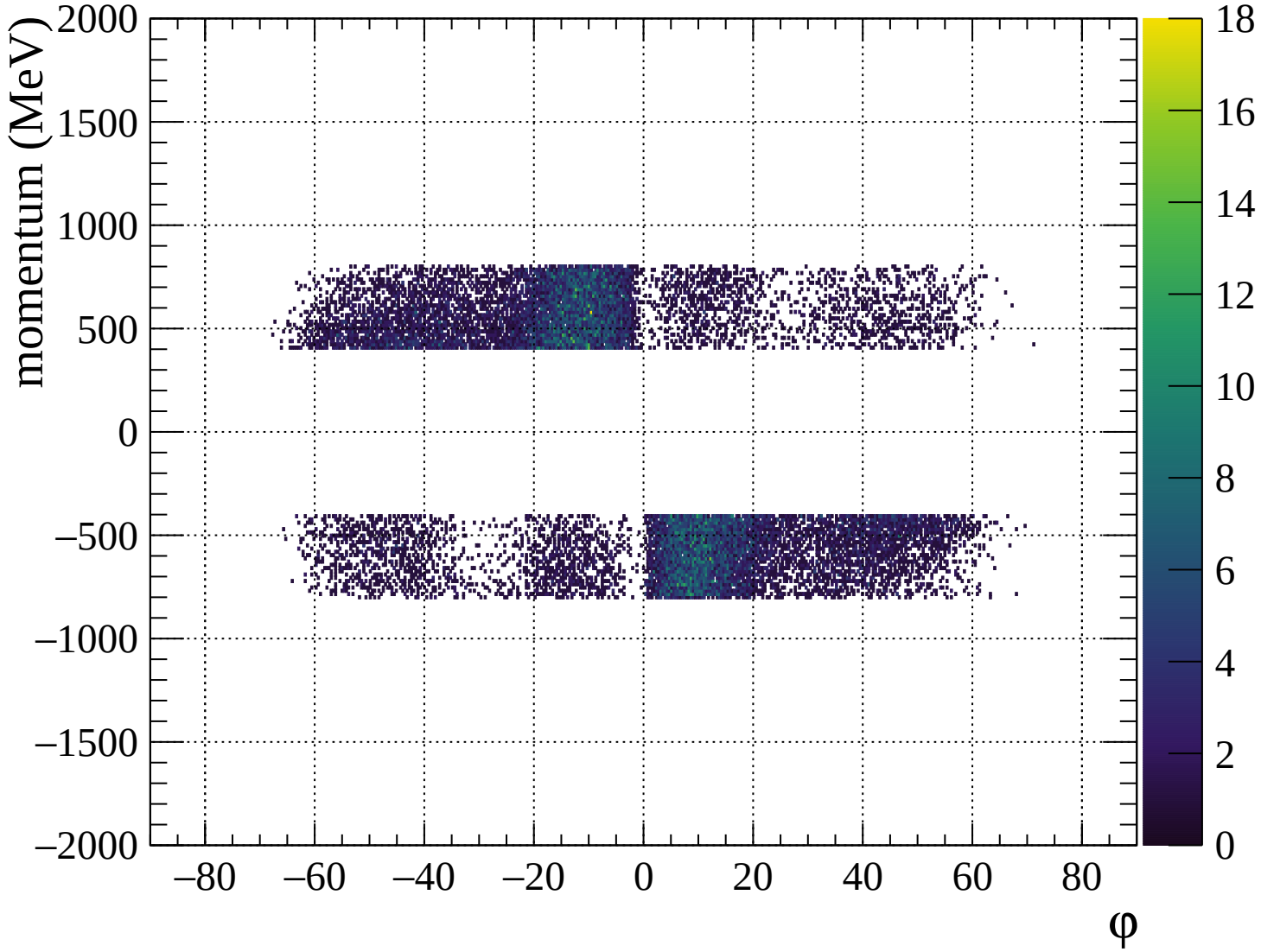


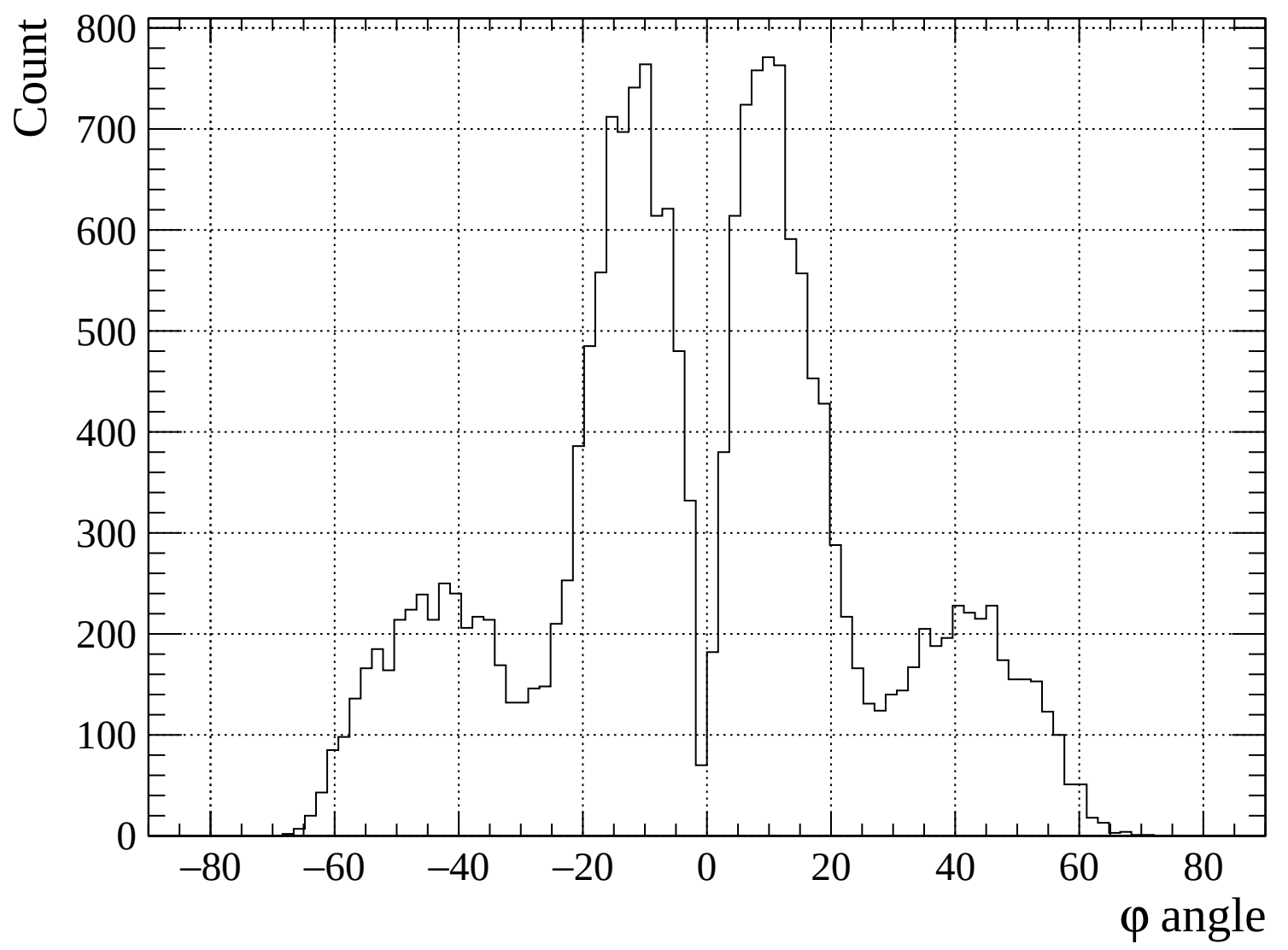


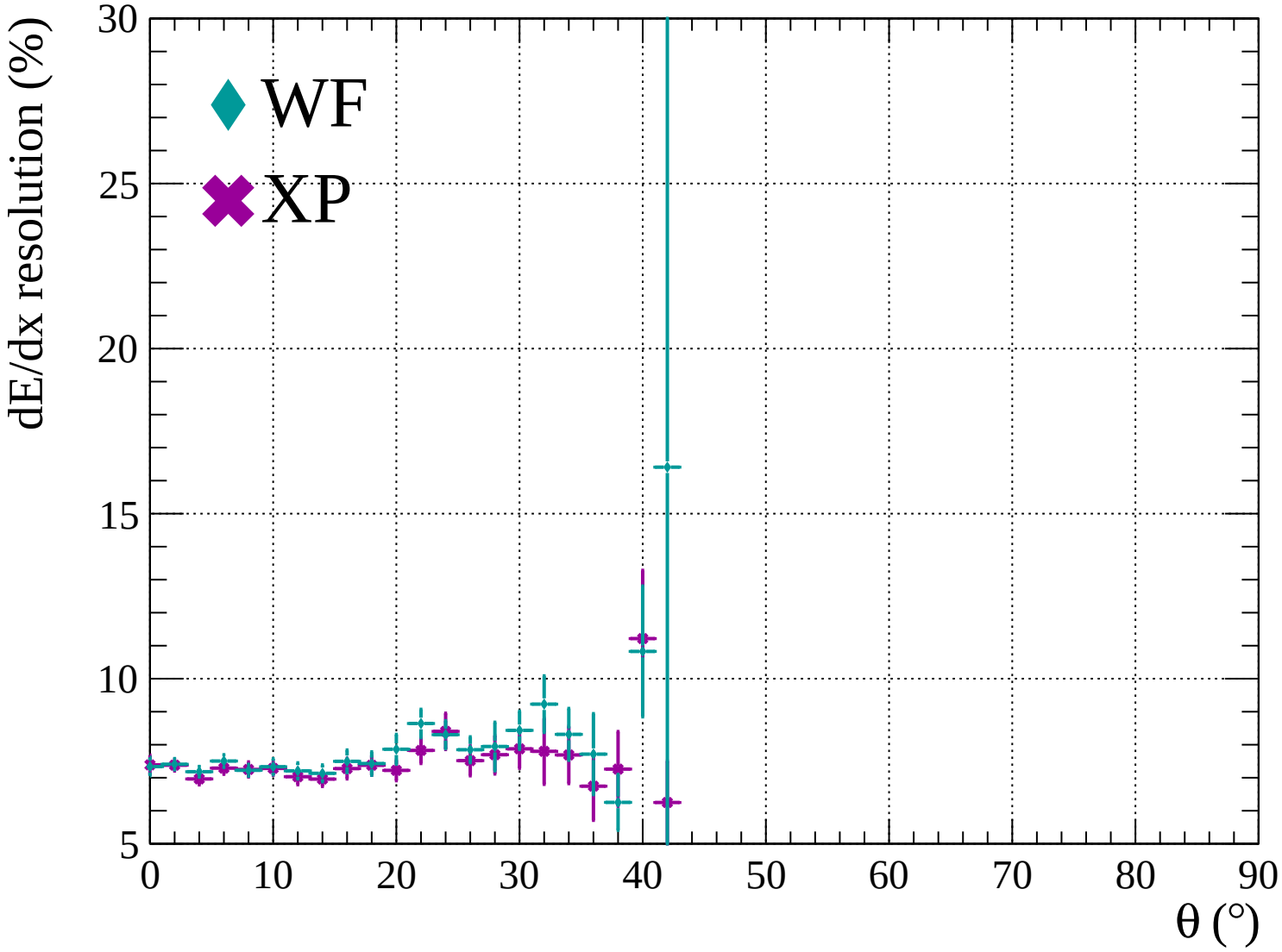


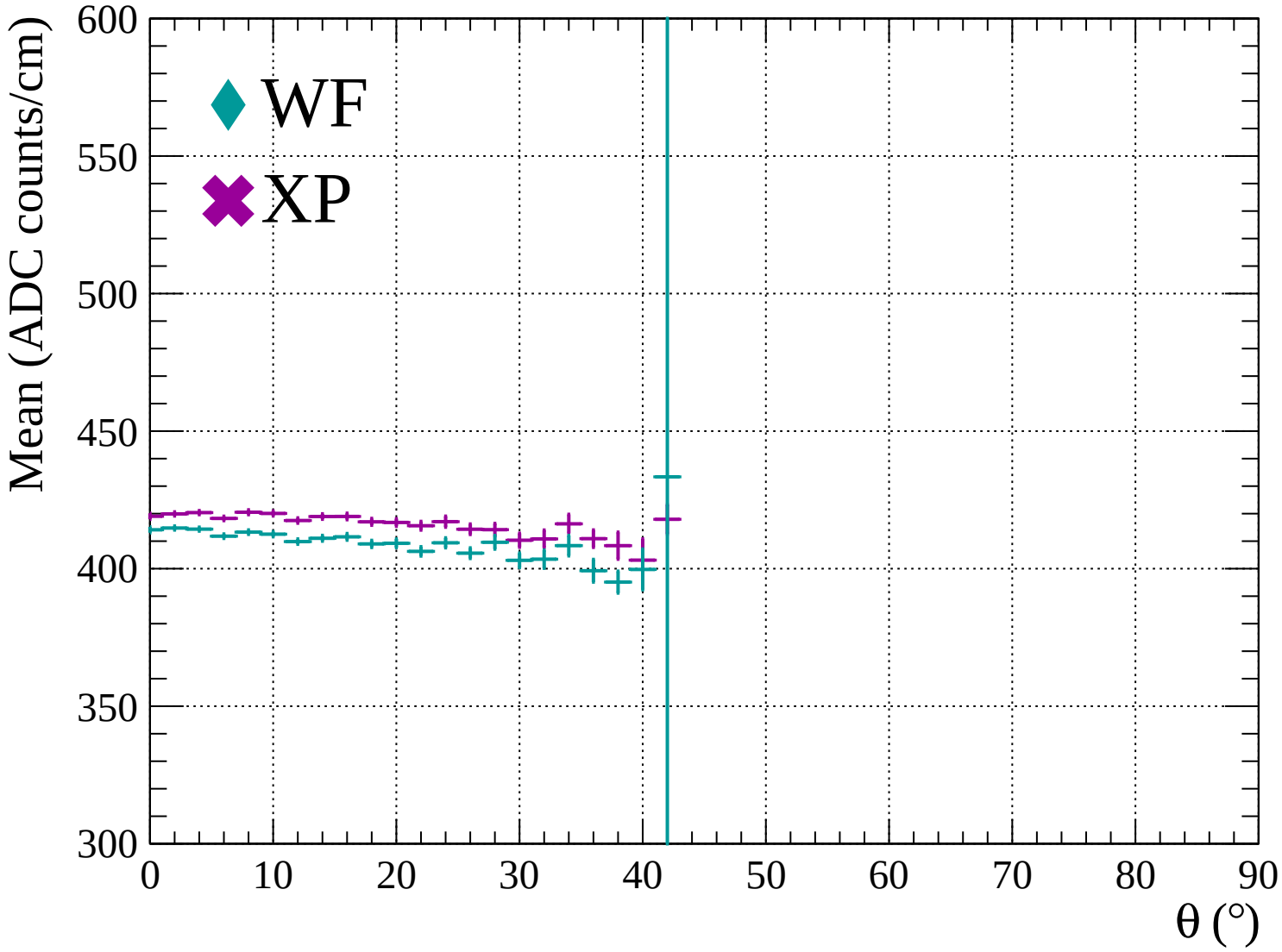


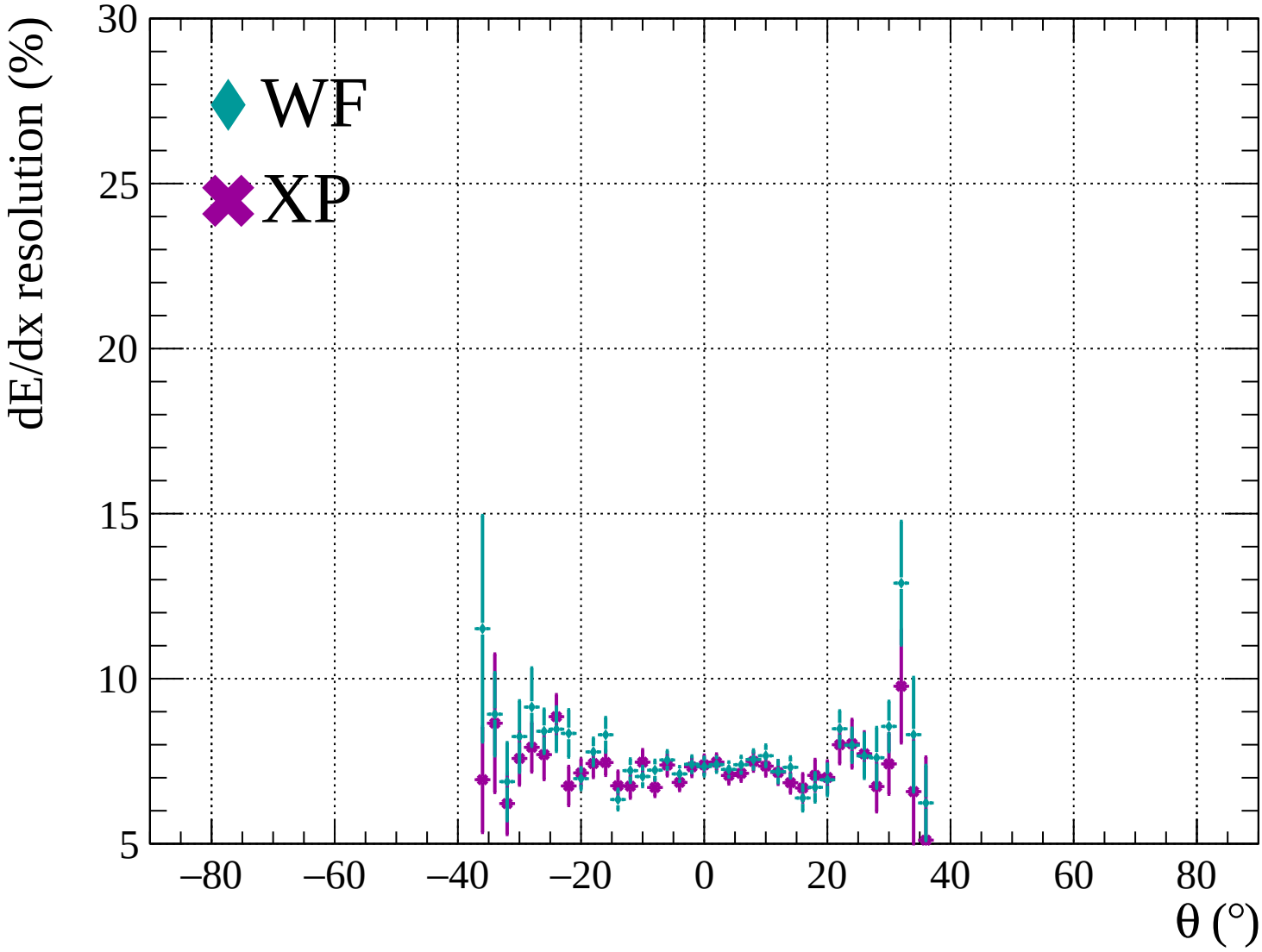


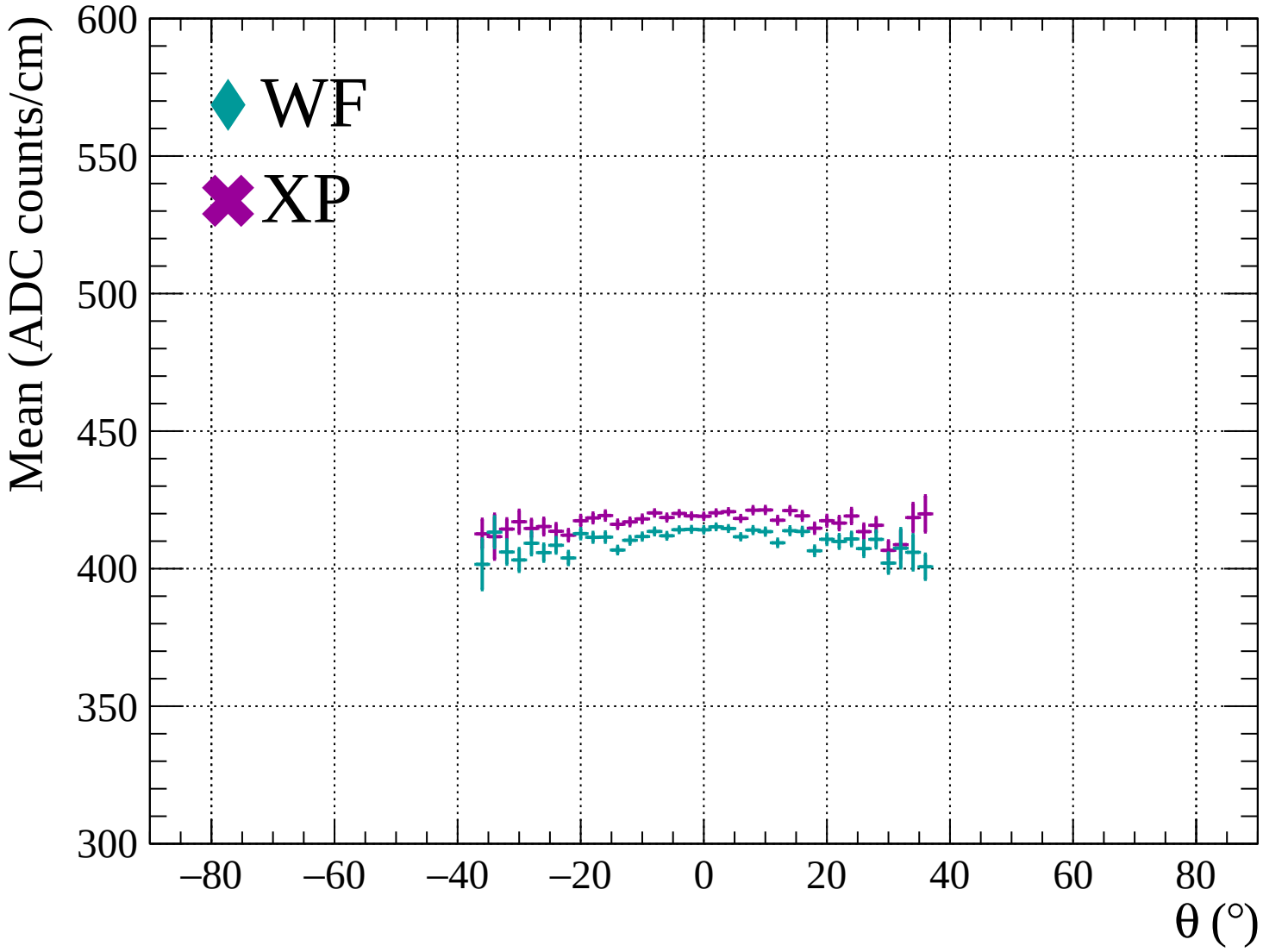


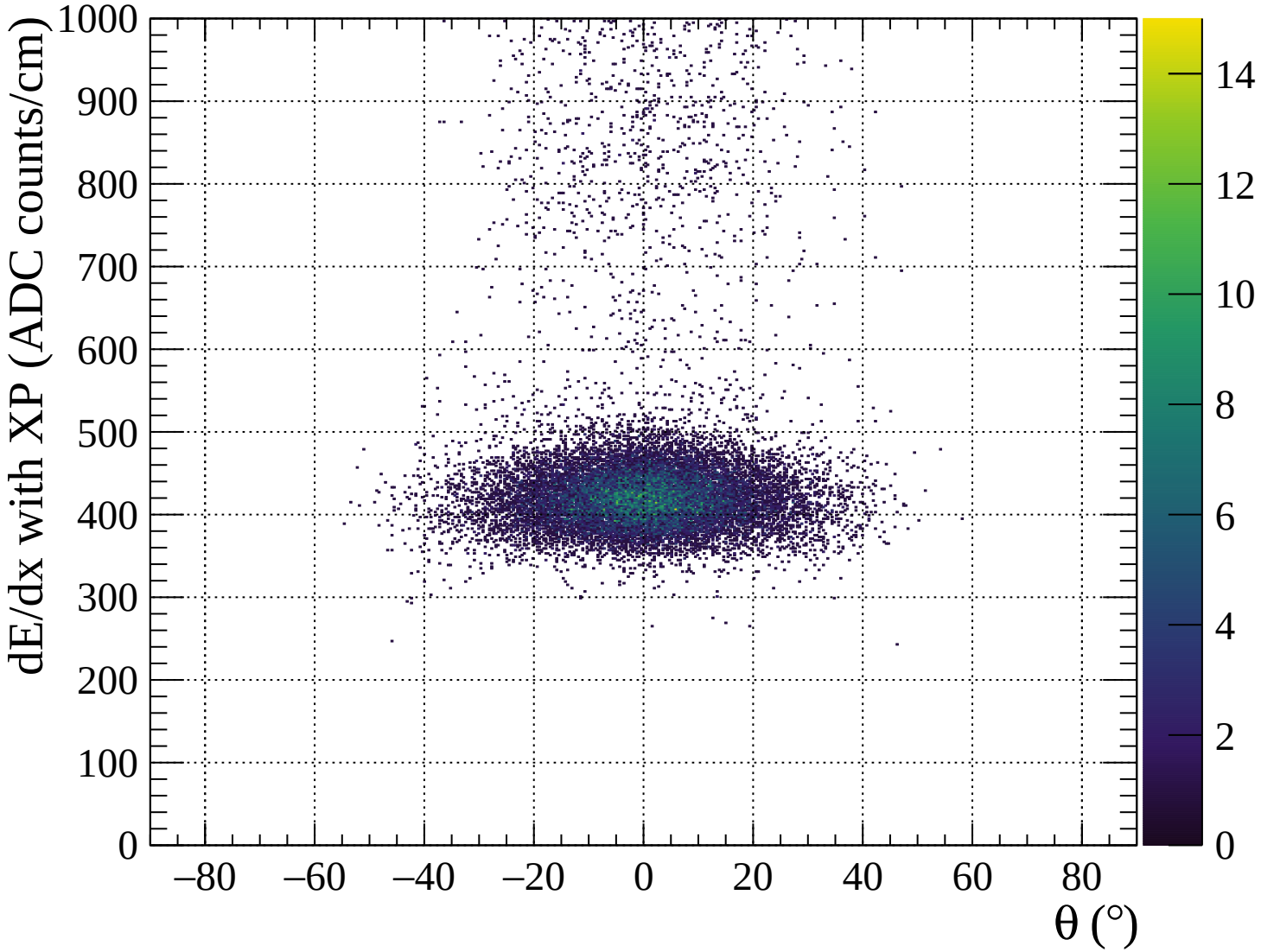


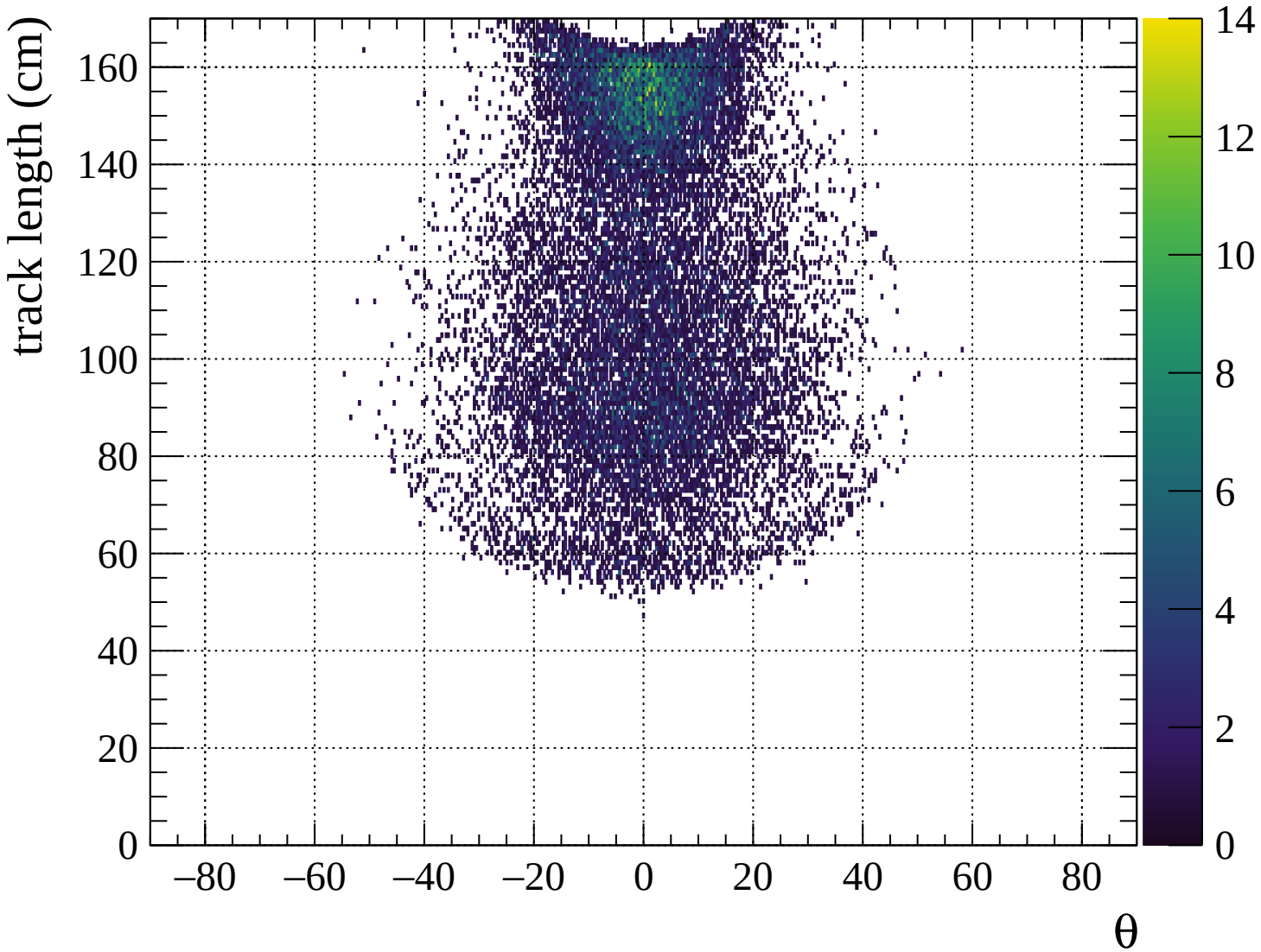


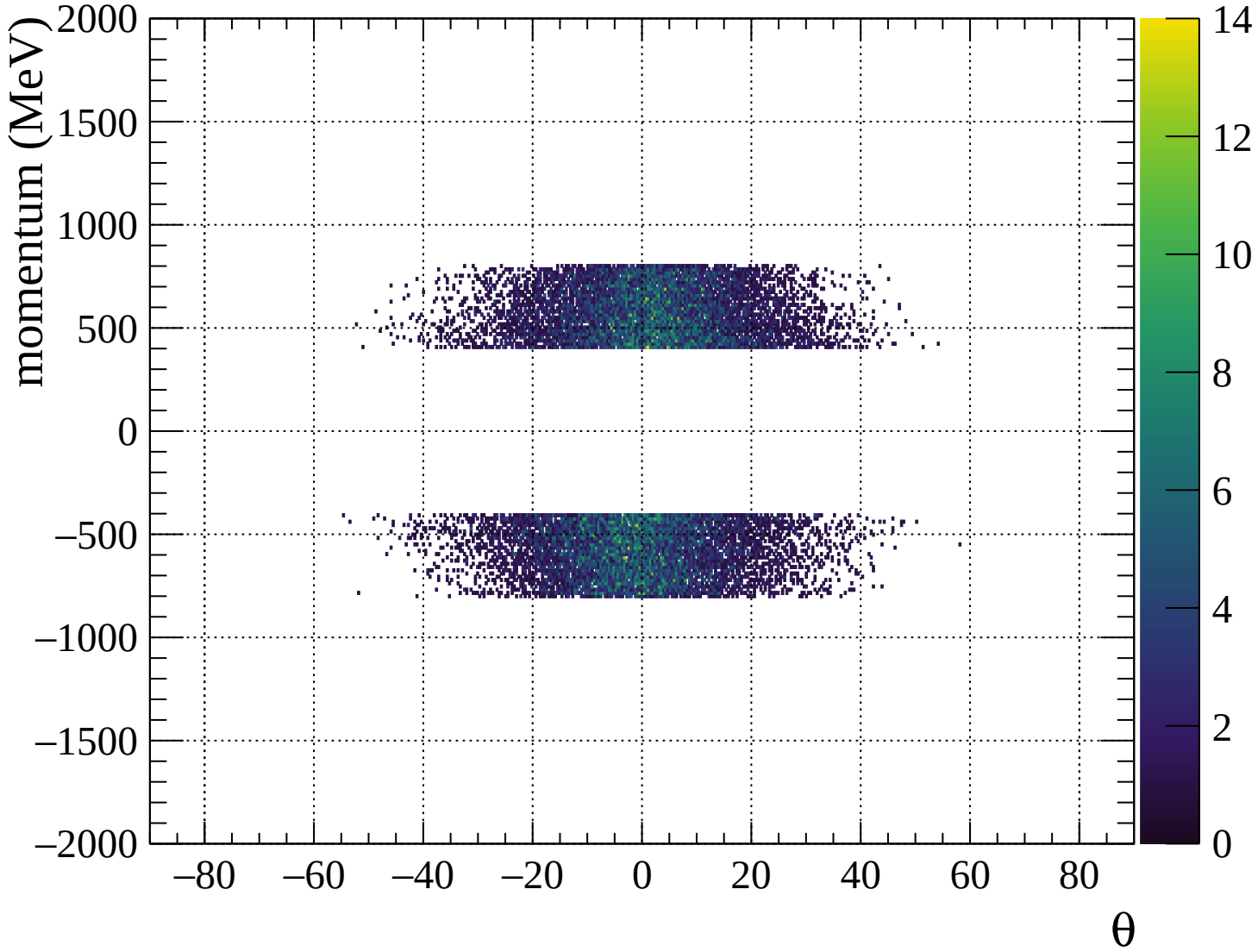


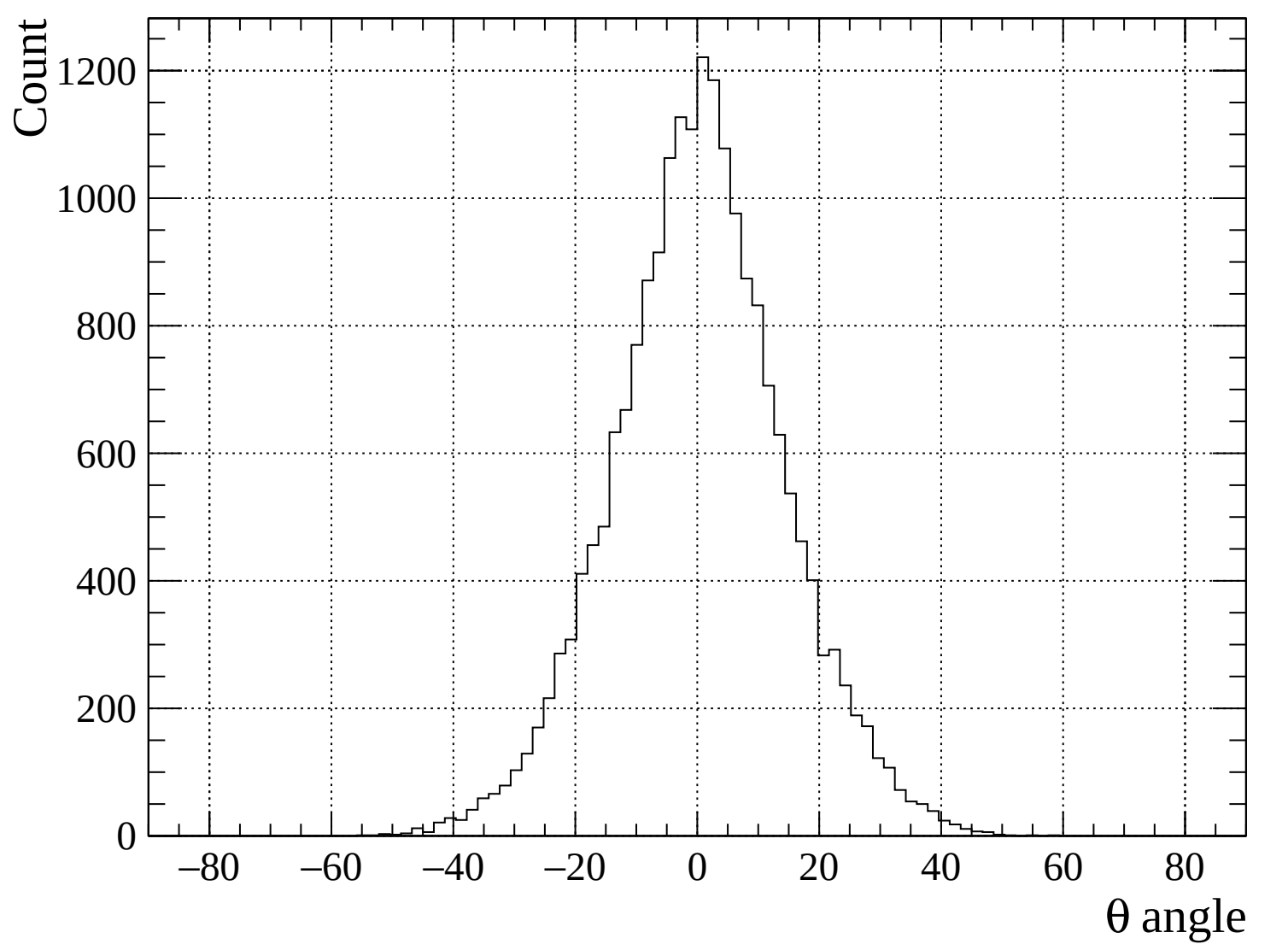


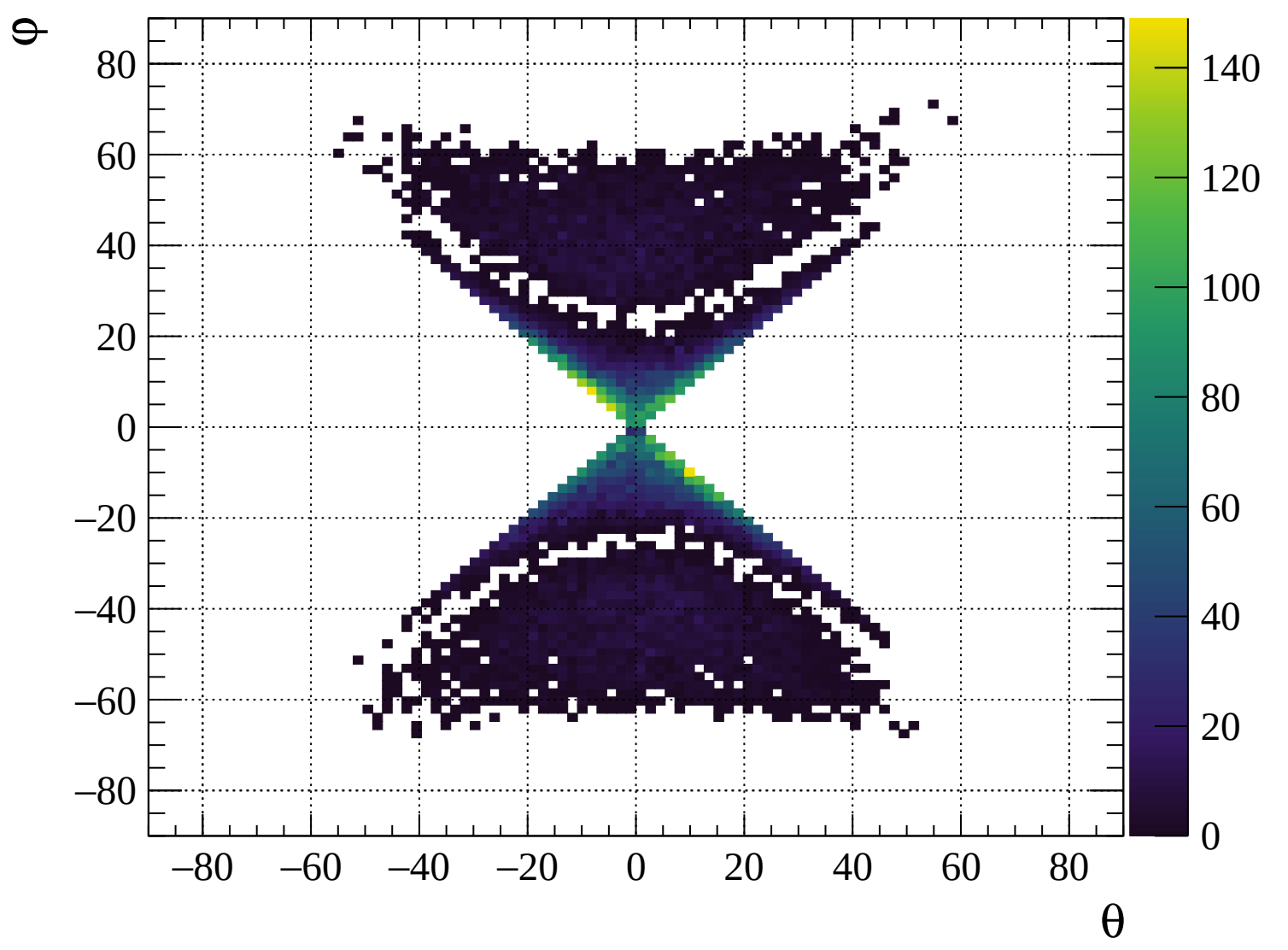


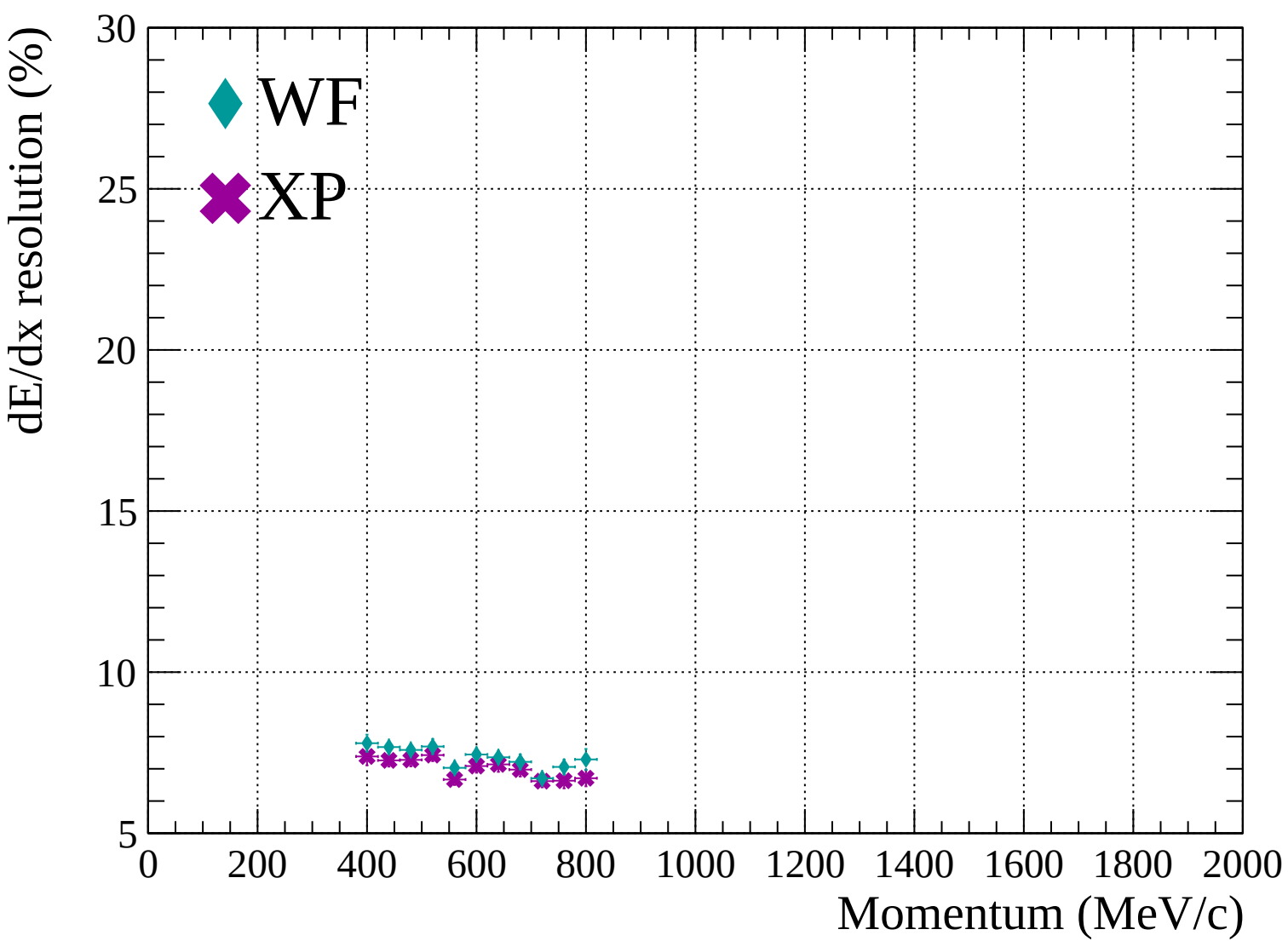


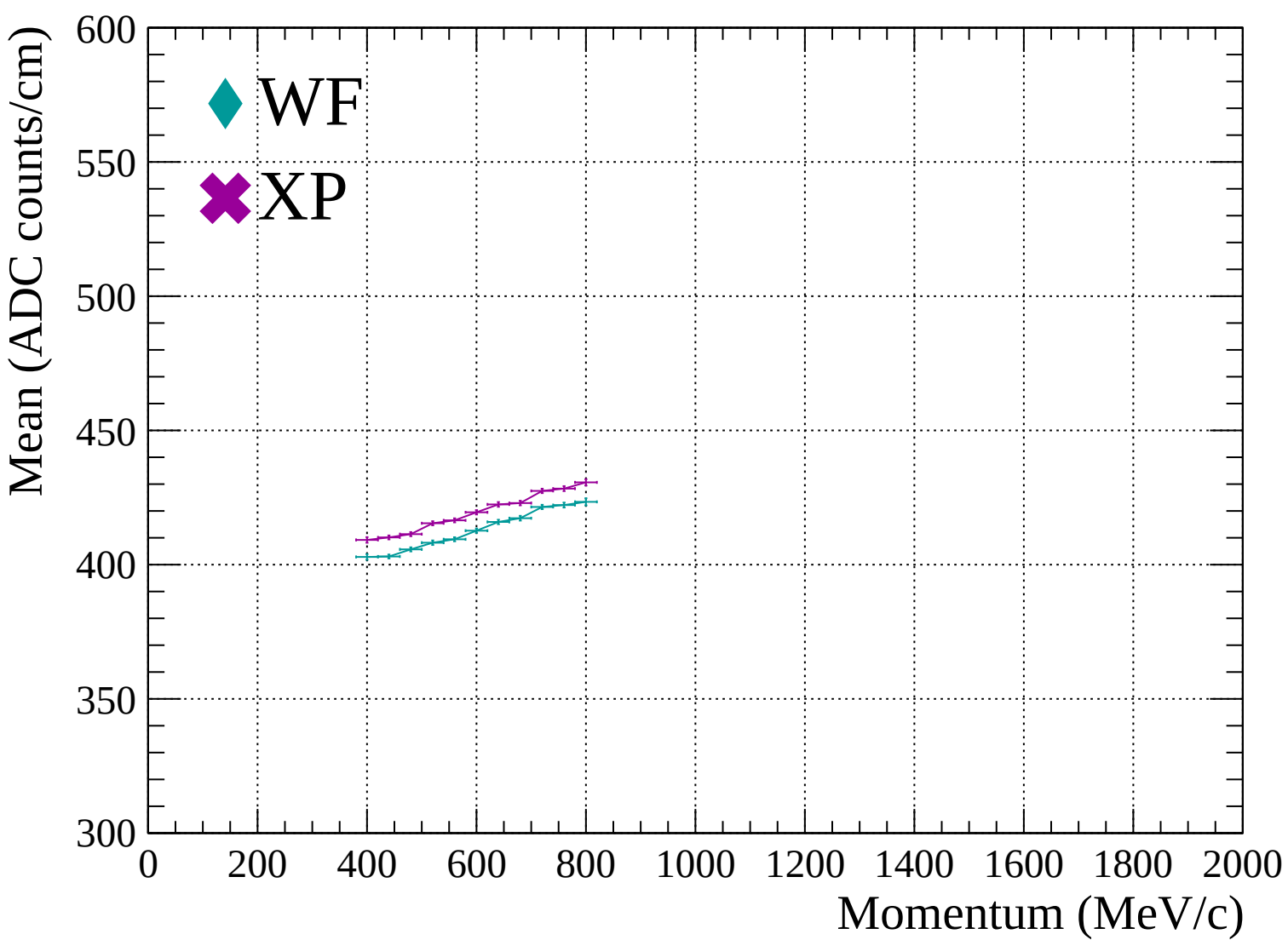


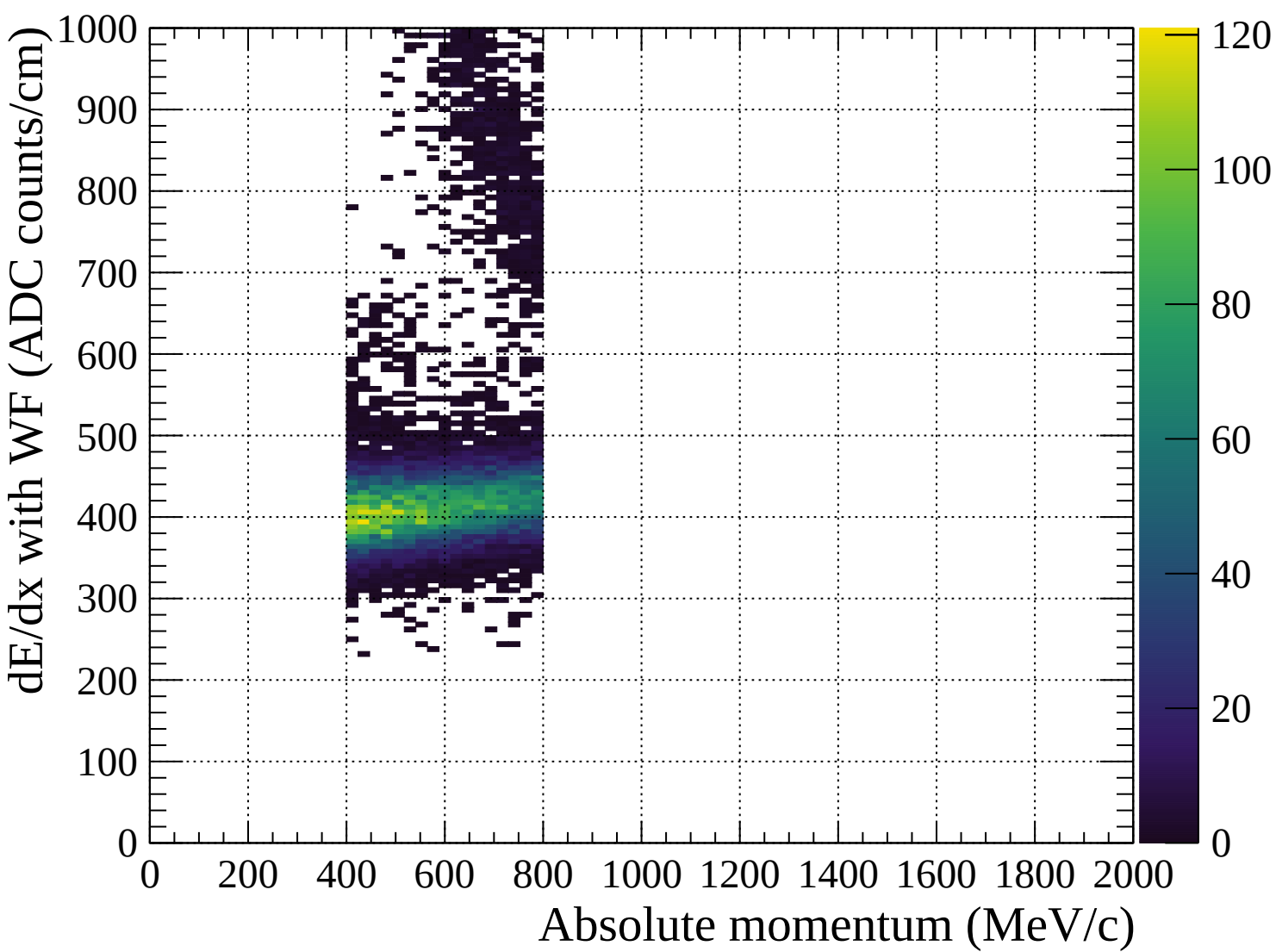


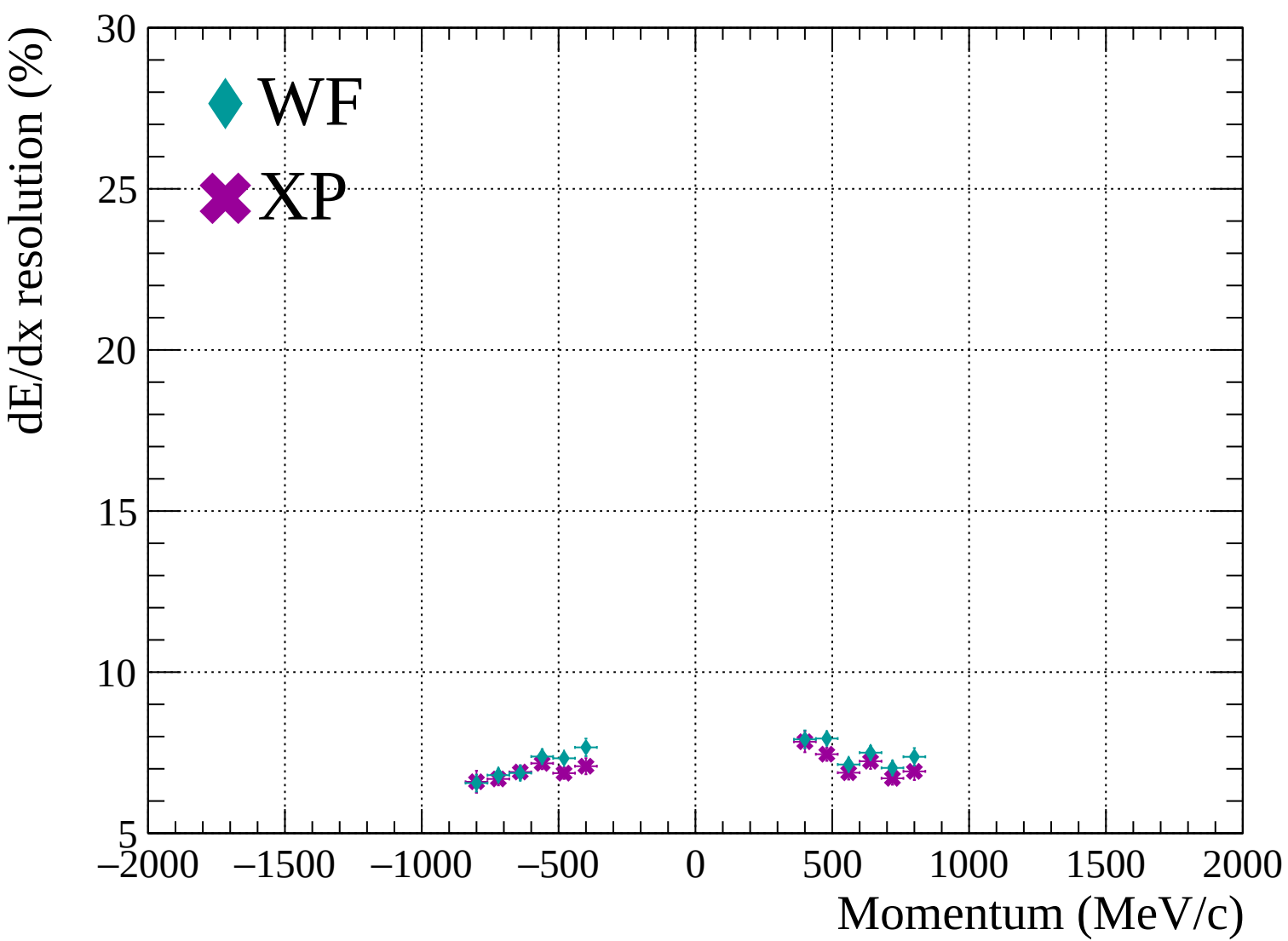


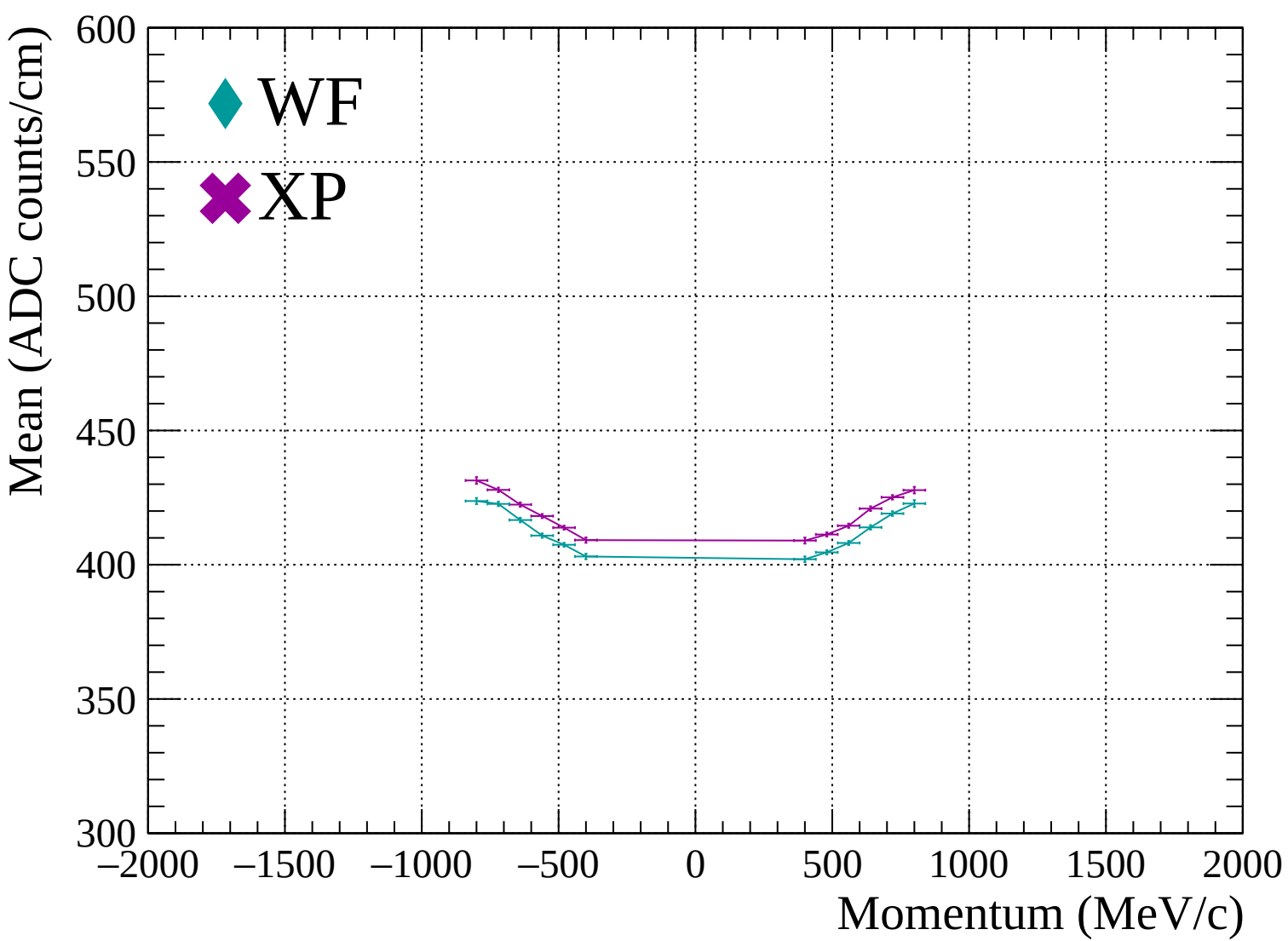


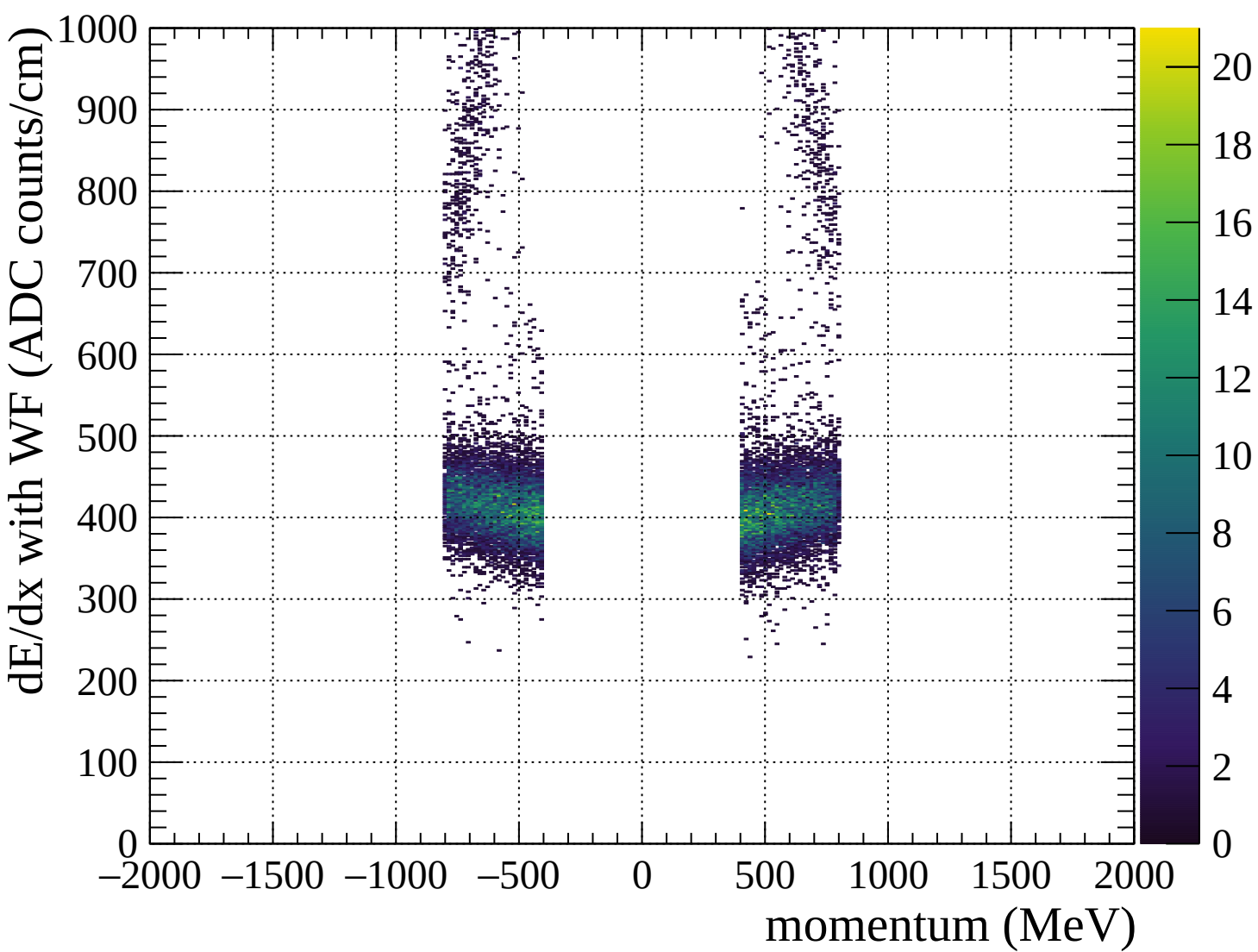


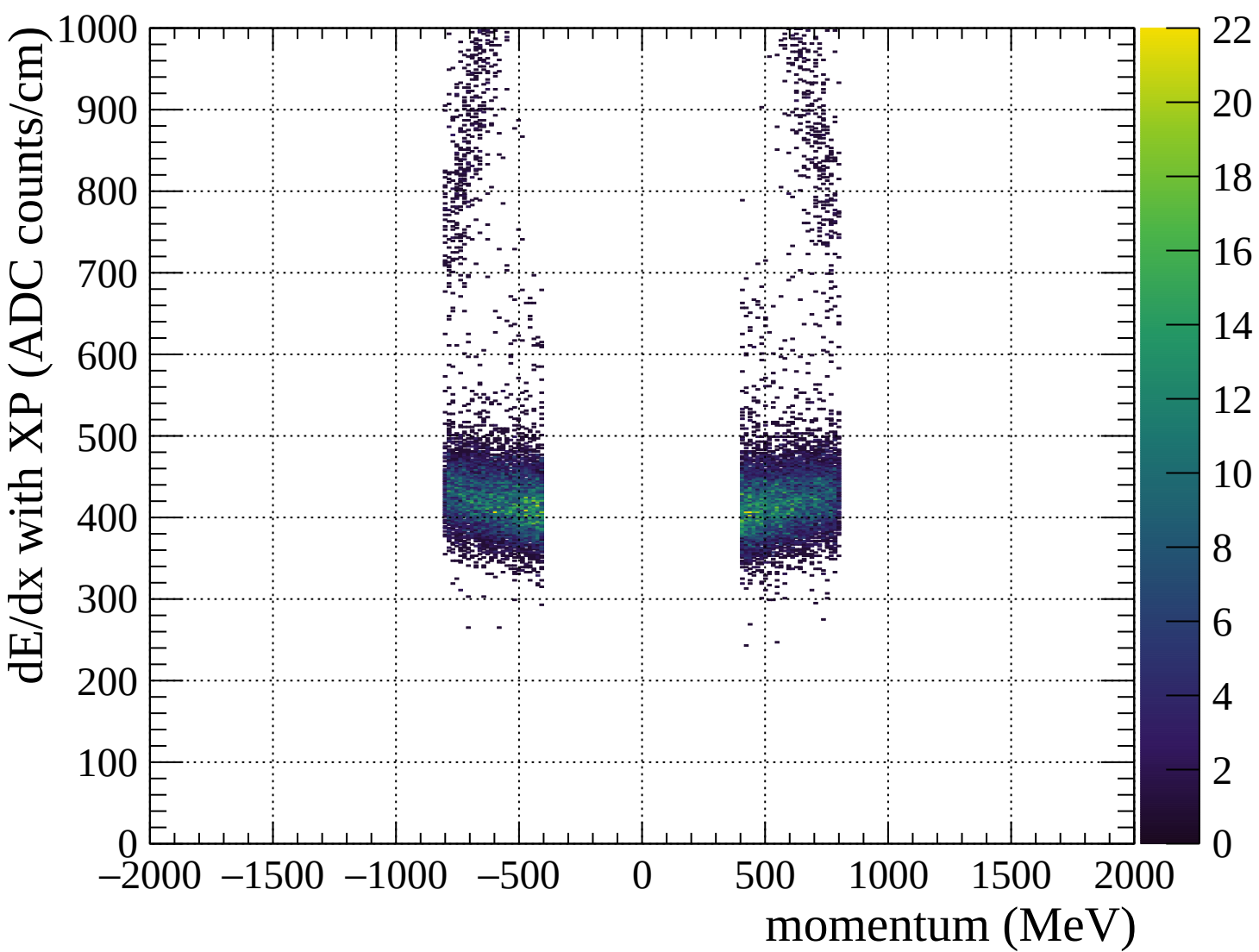


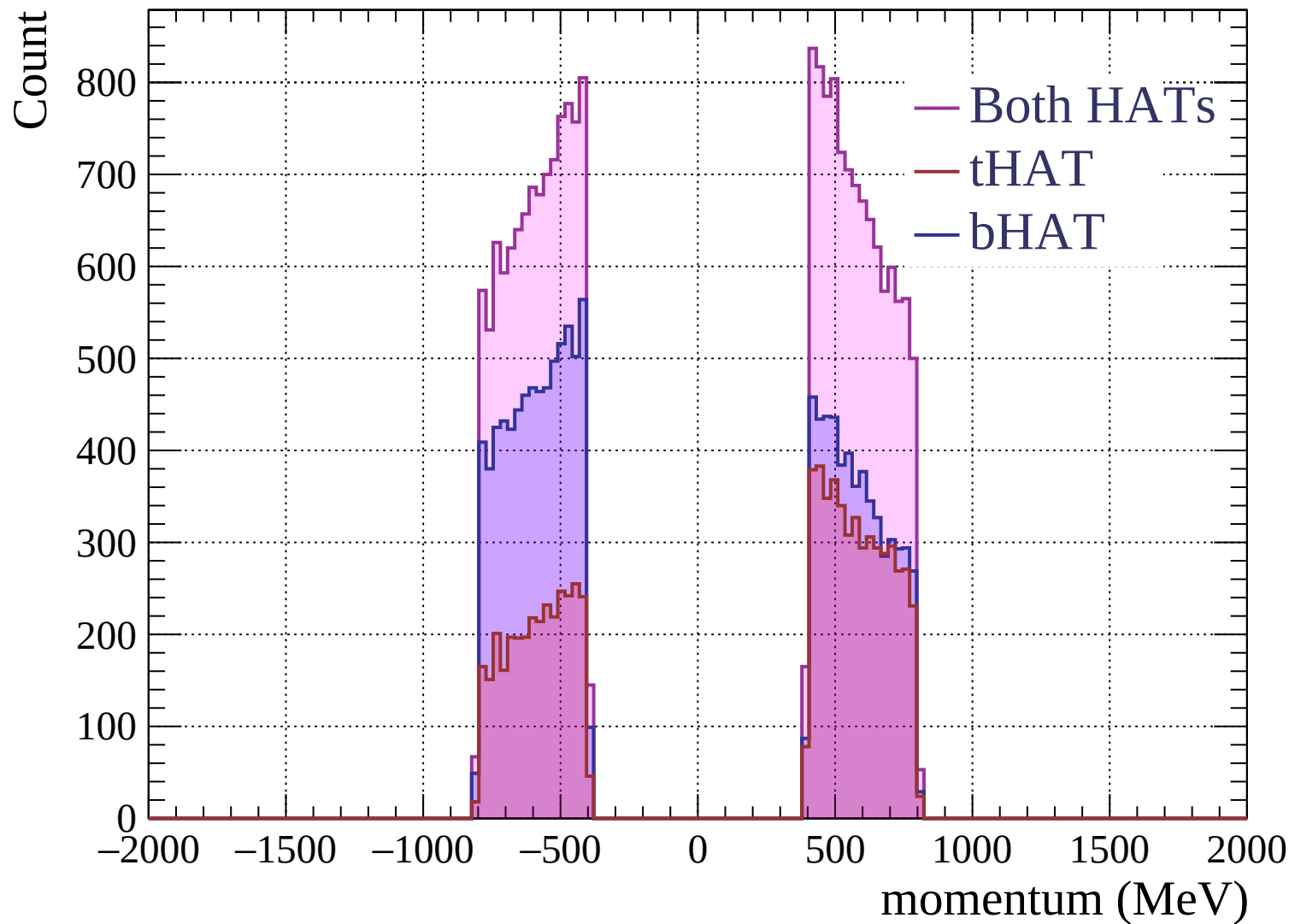


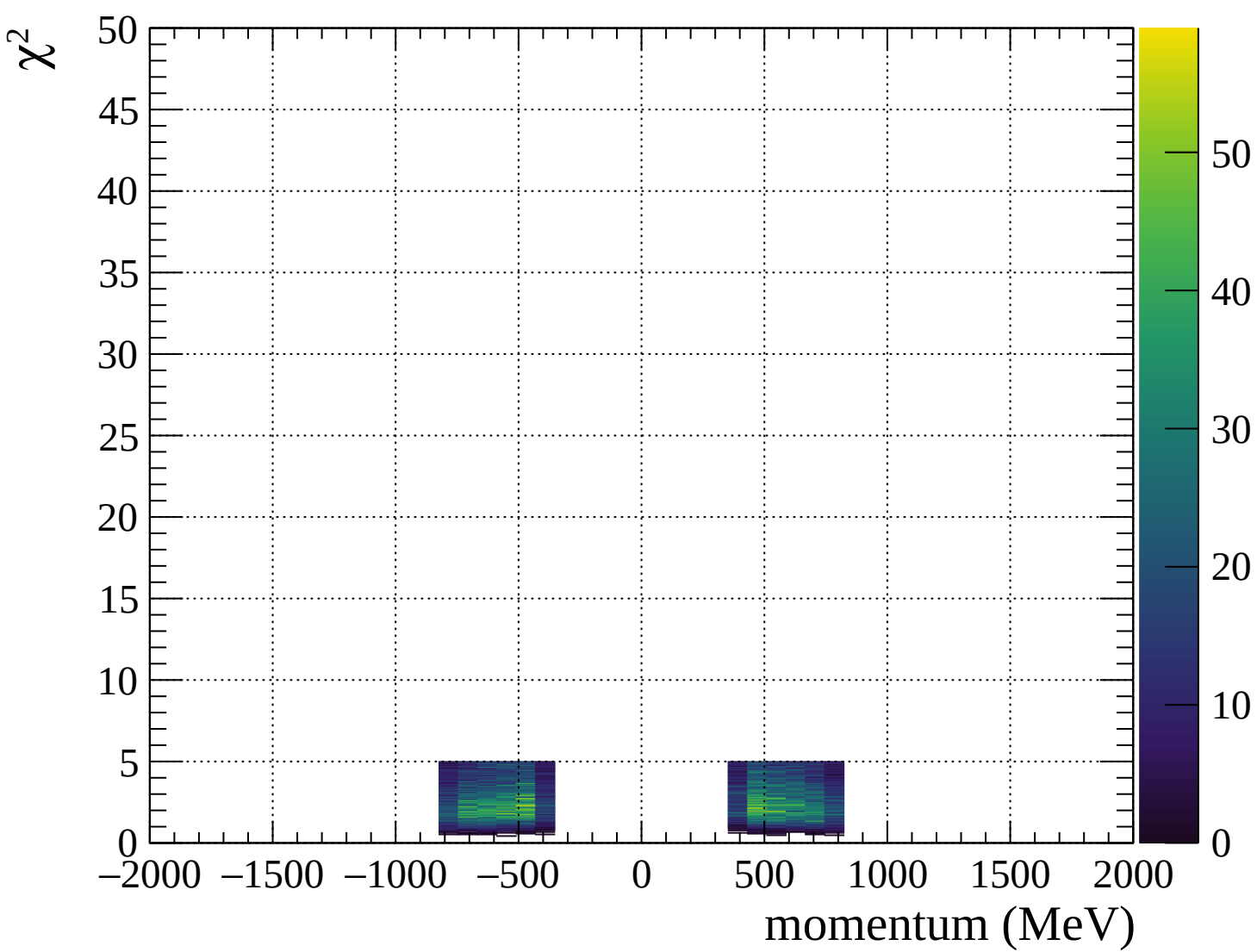




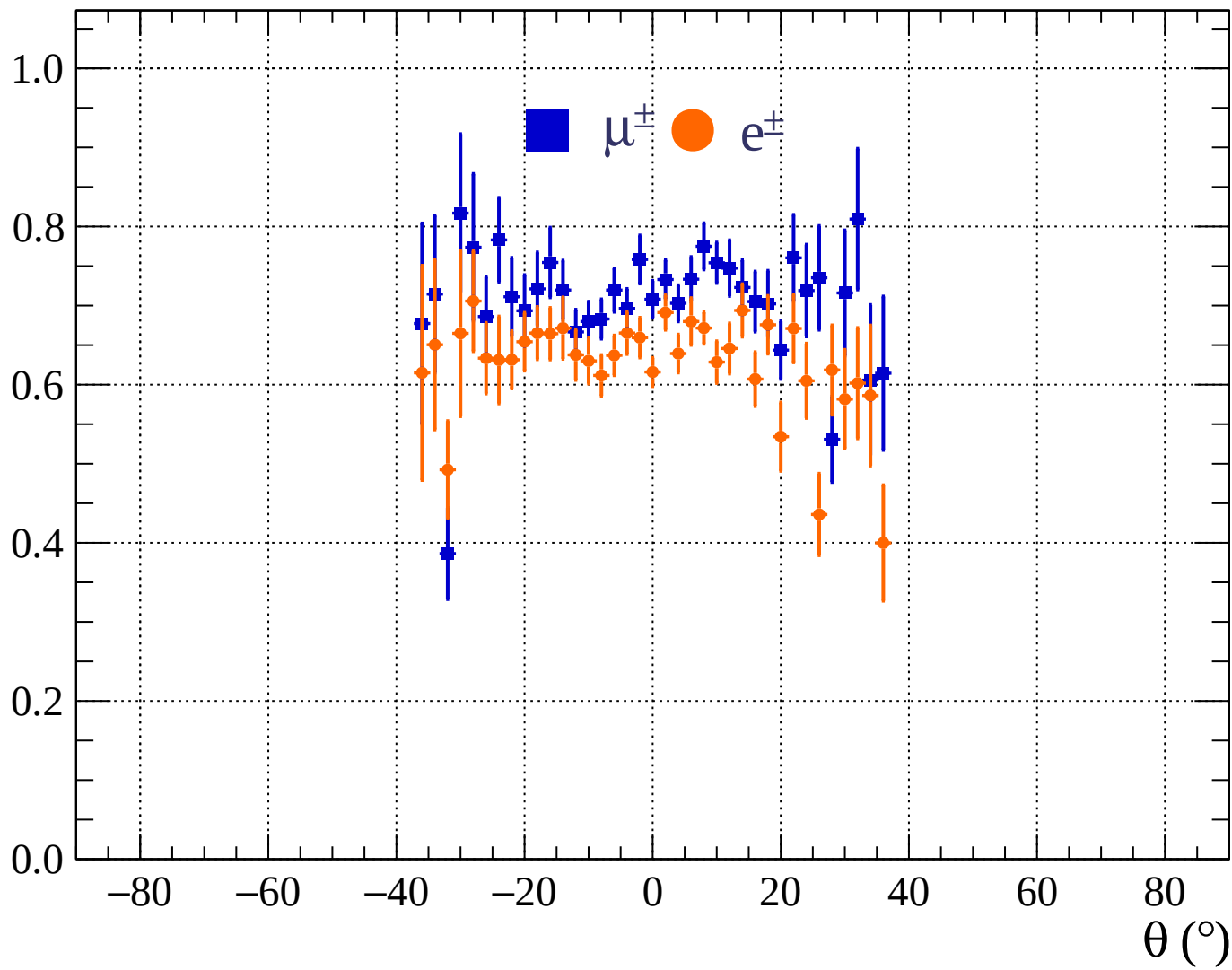


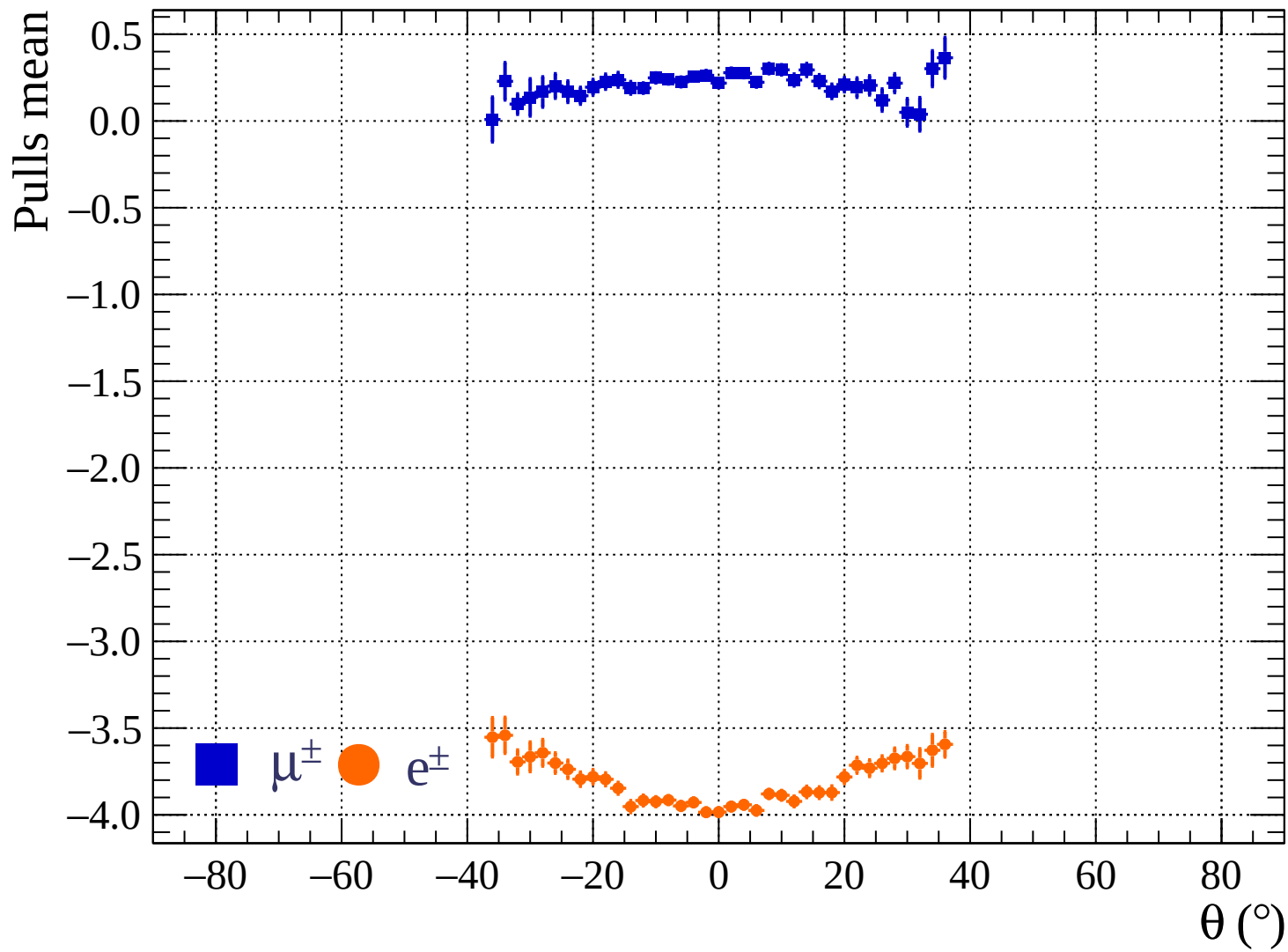


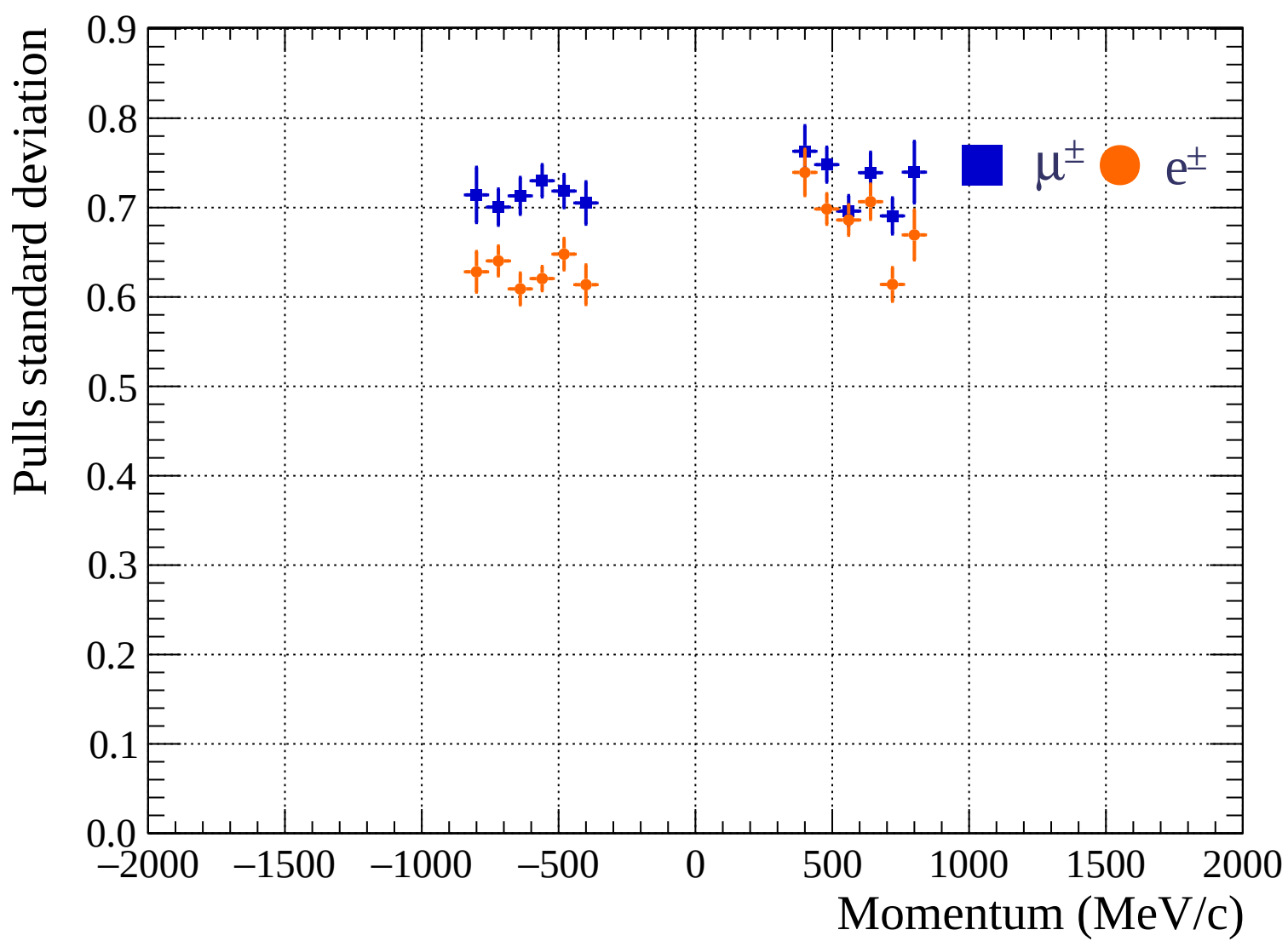


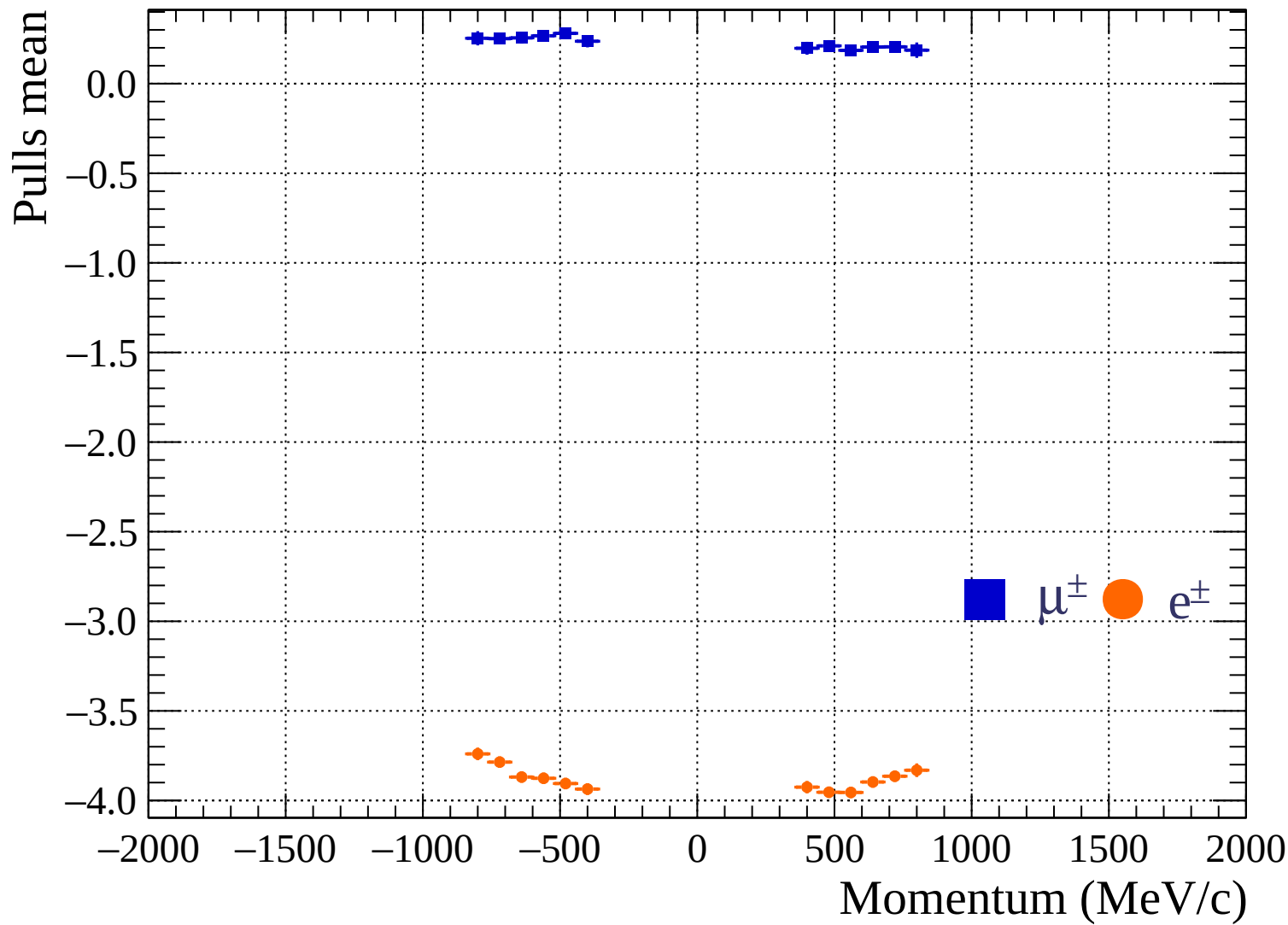


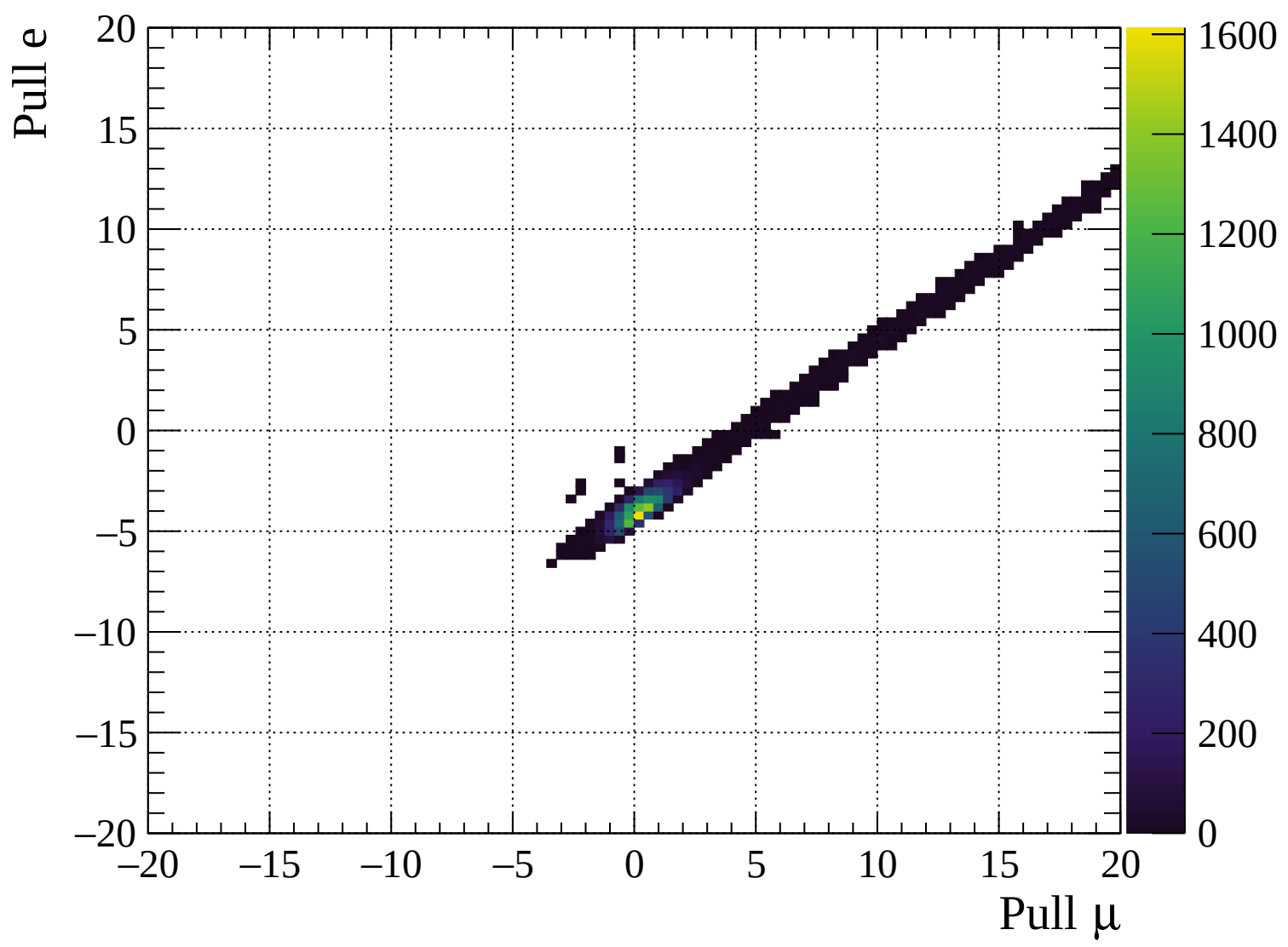
Pulls standard deviation

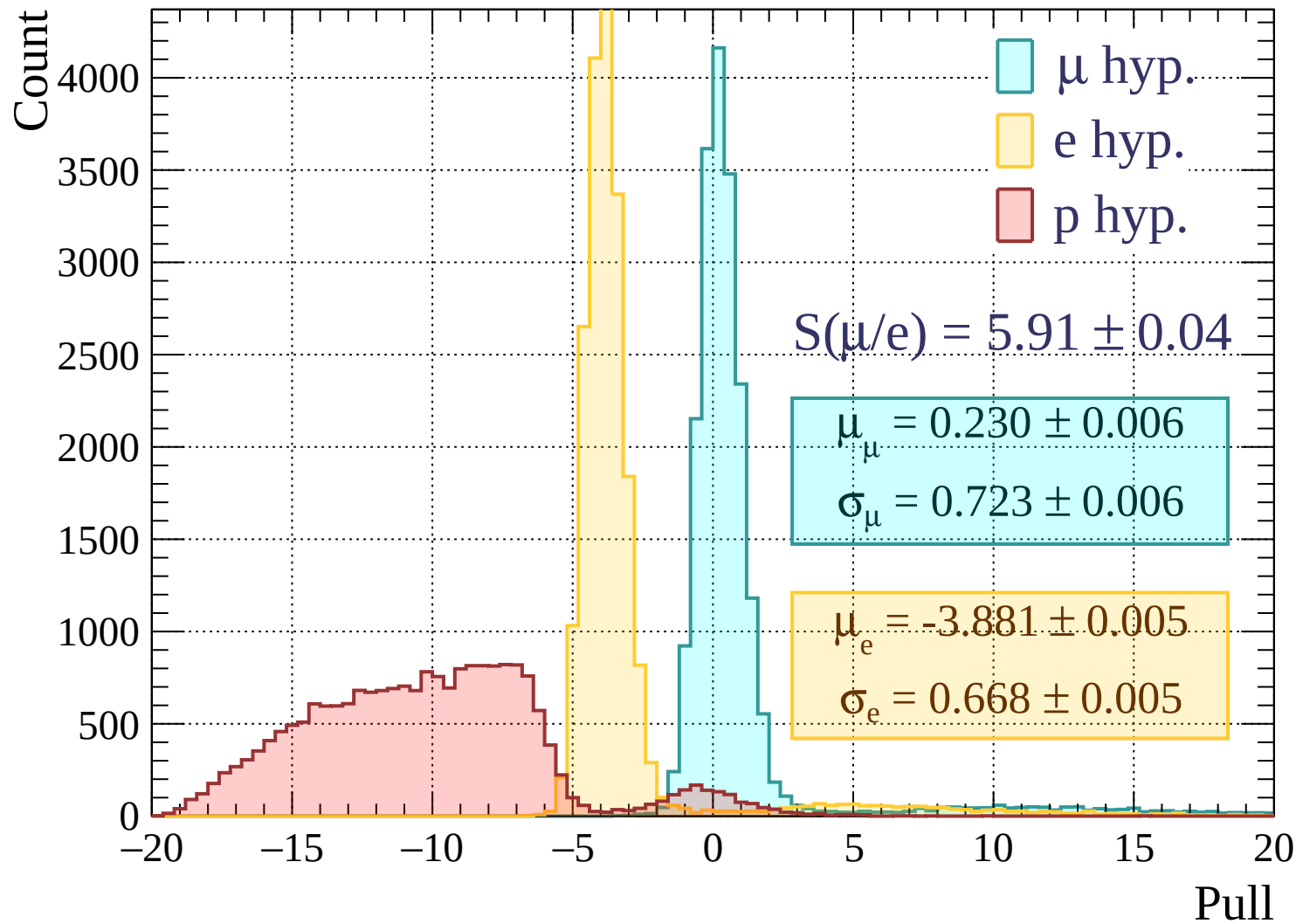


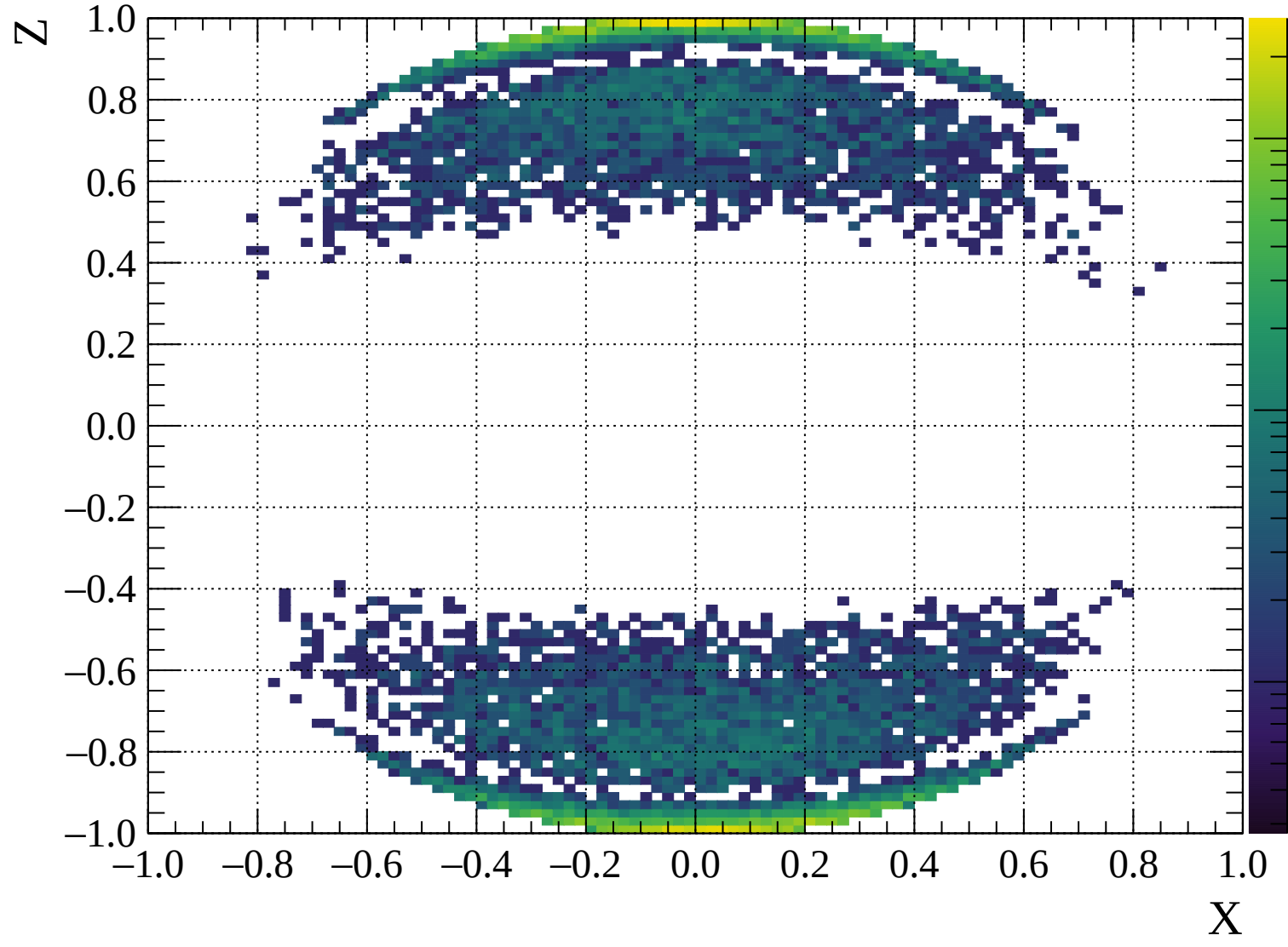


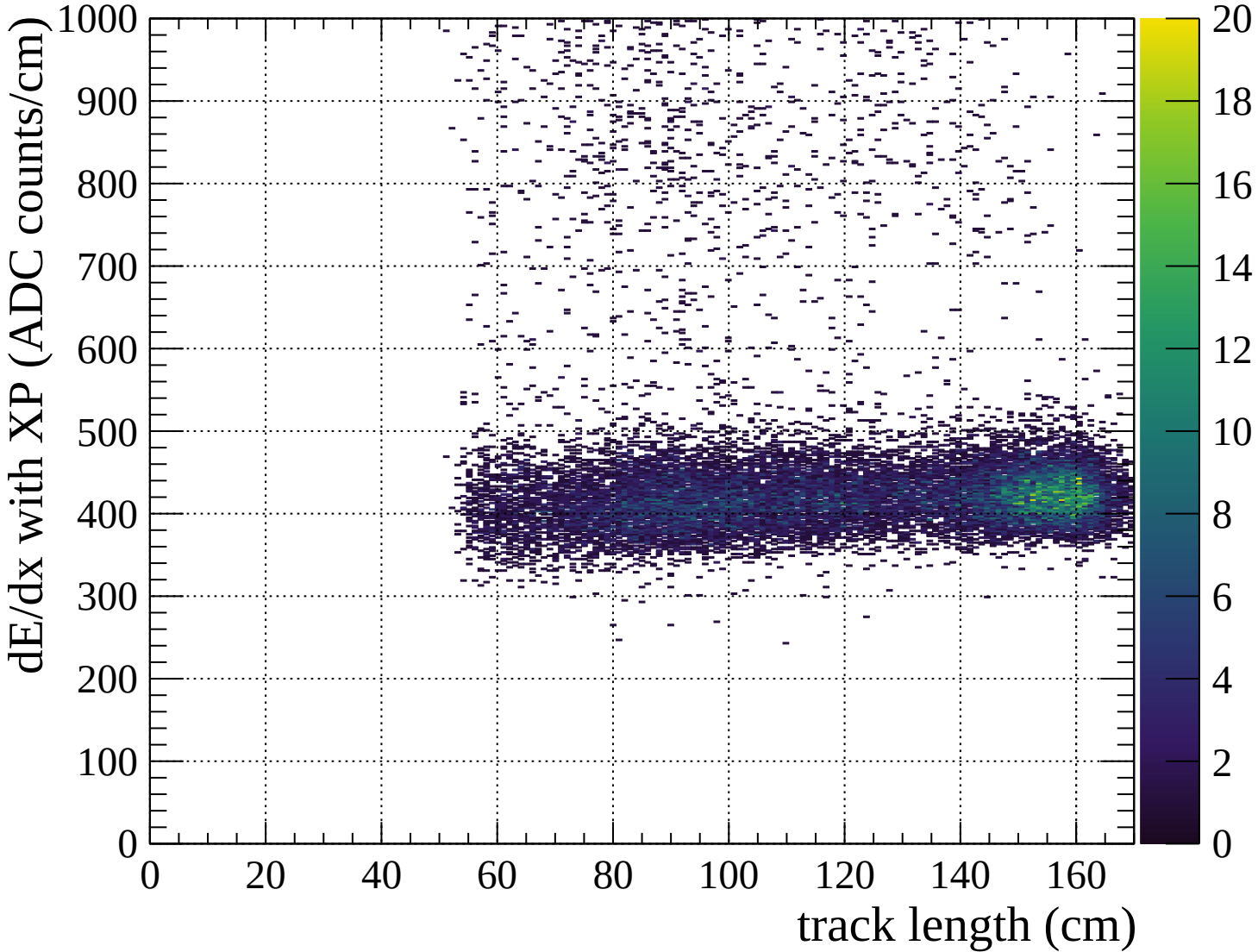


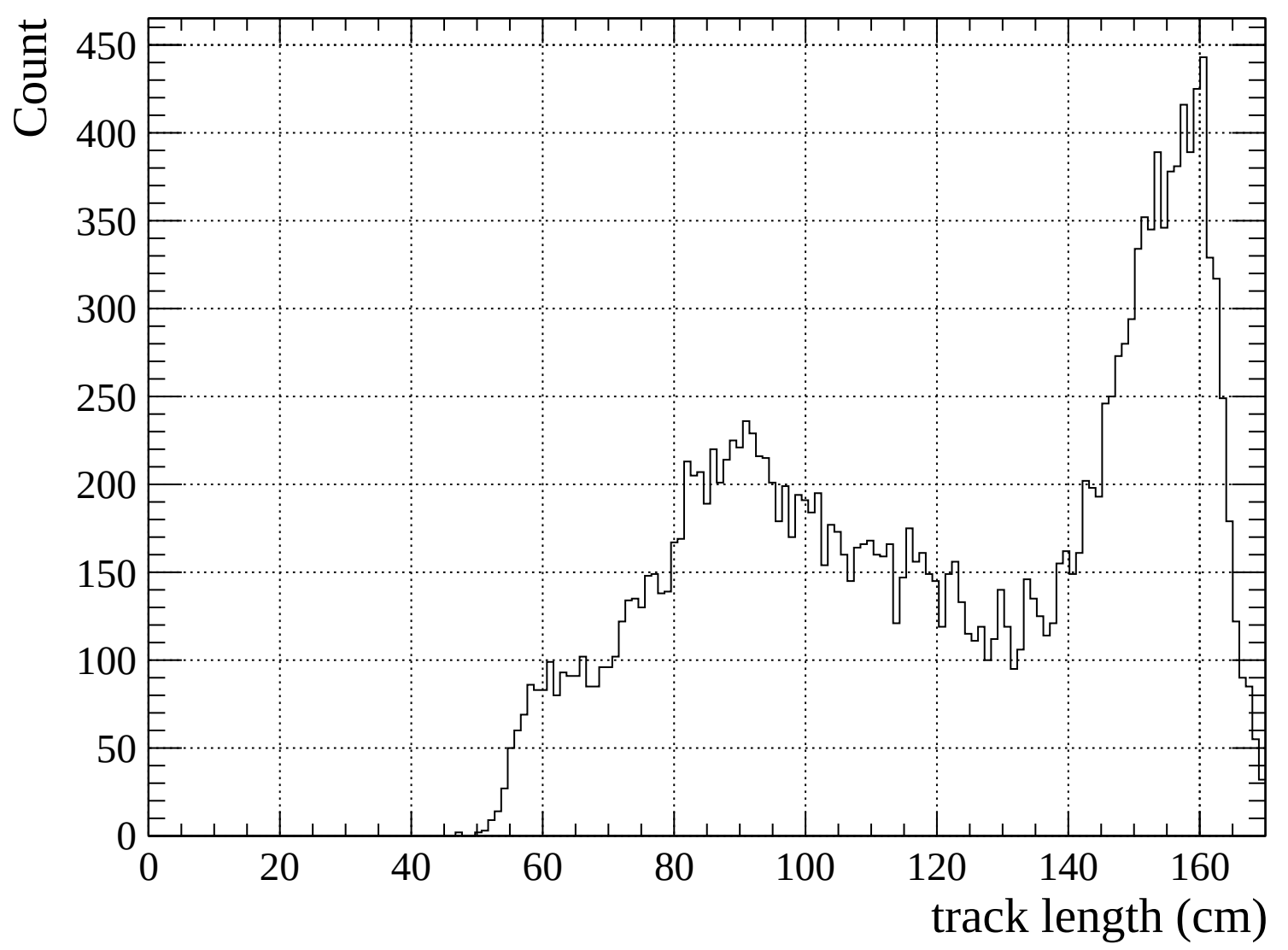


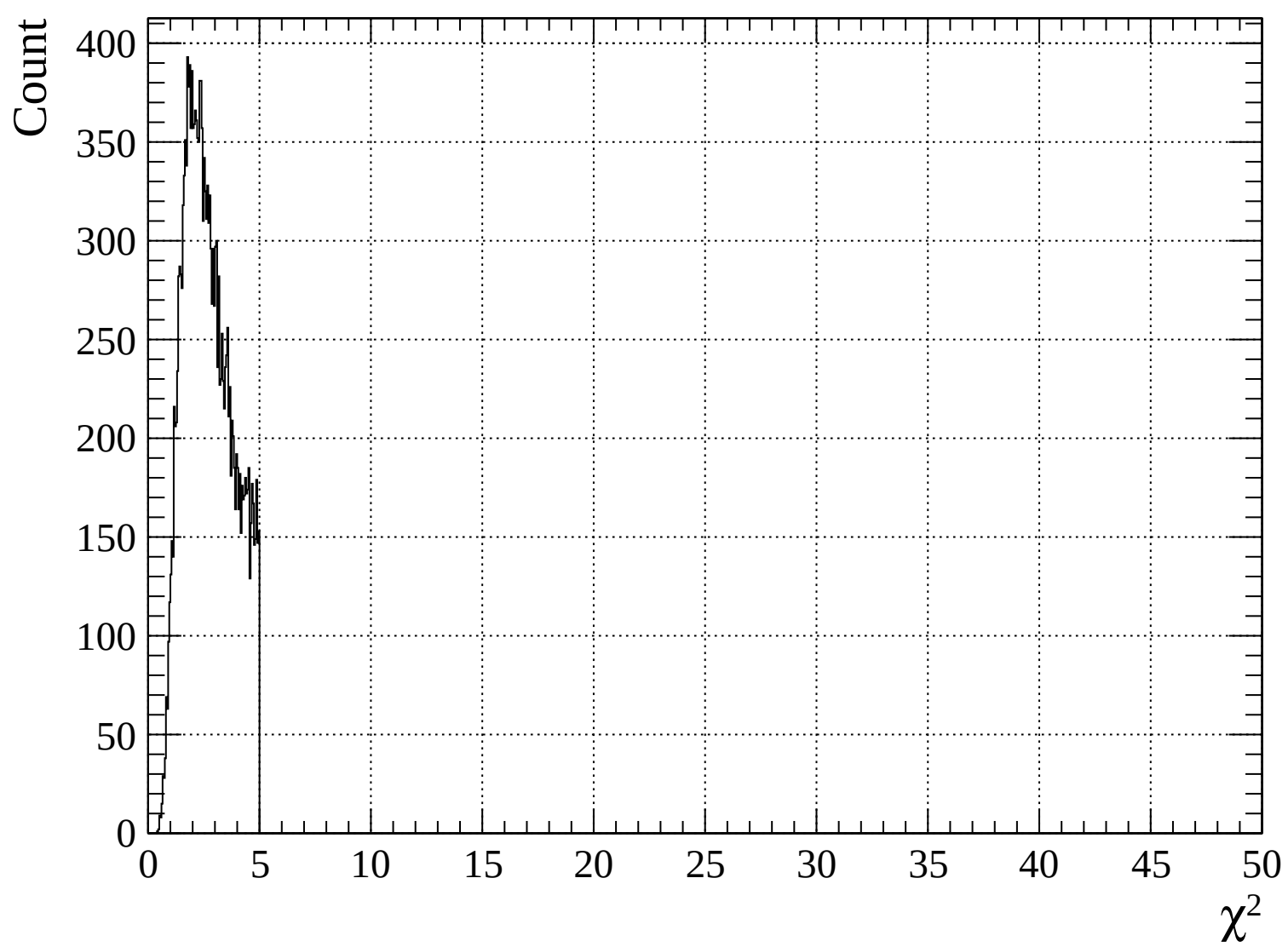


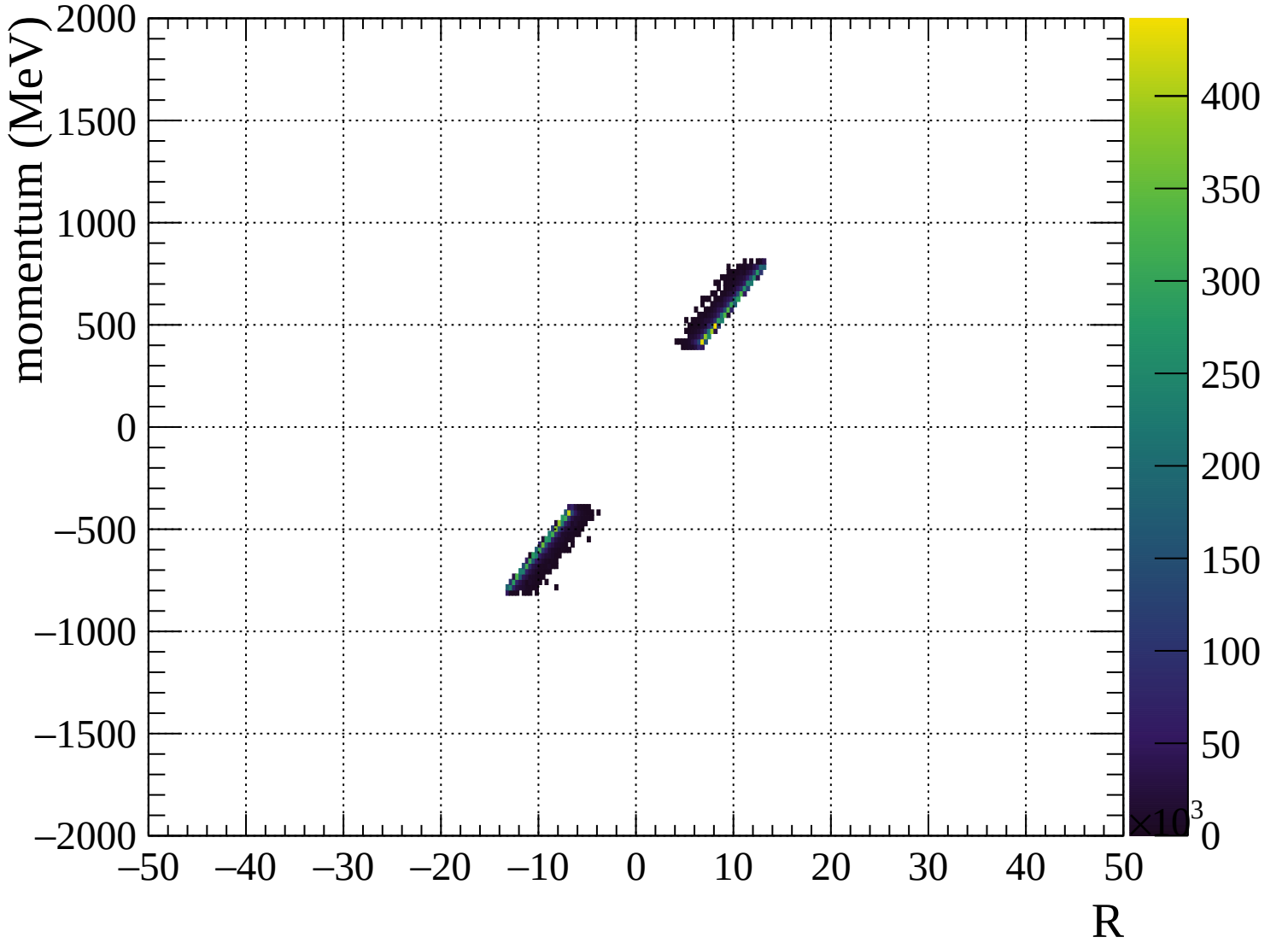


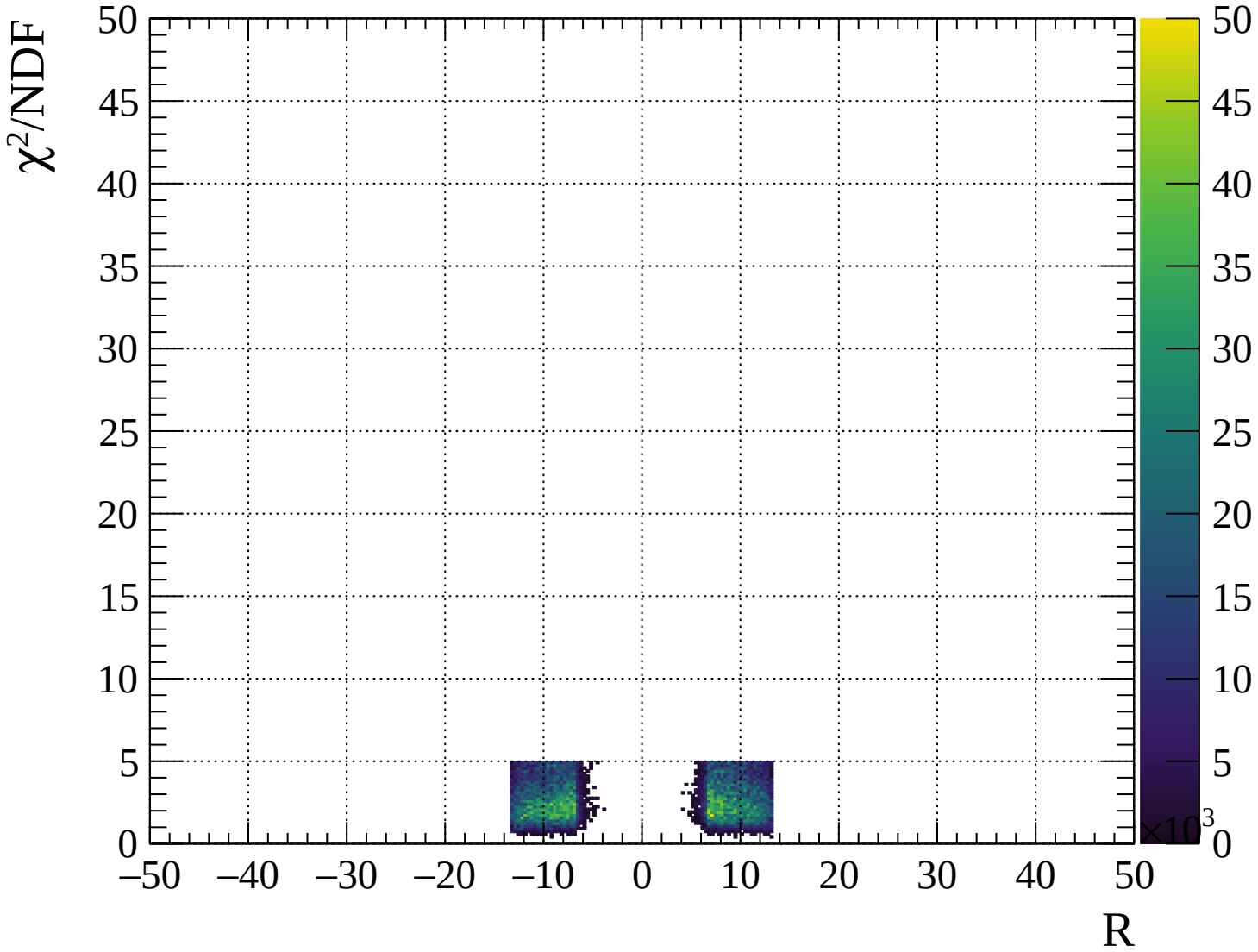






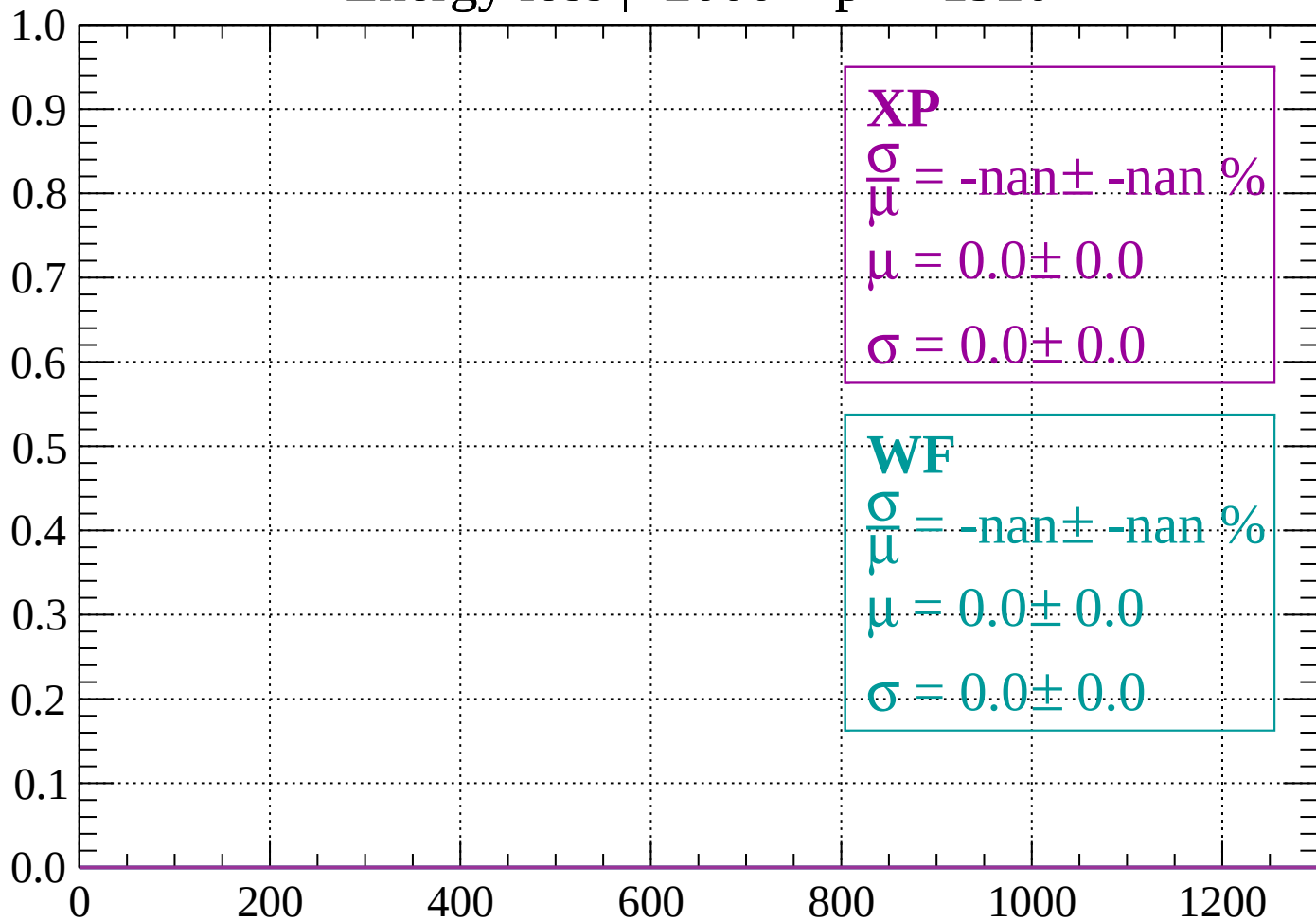






Energy loss | -2000 < p < -1920

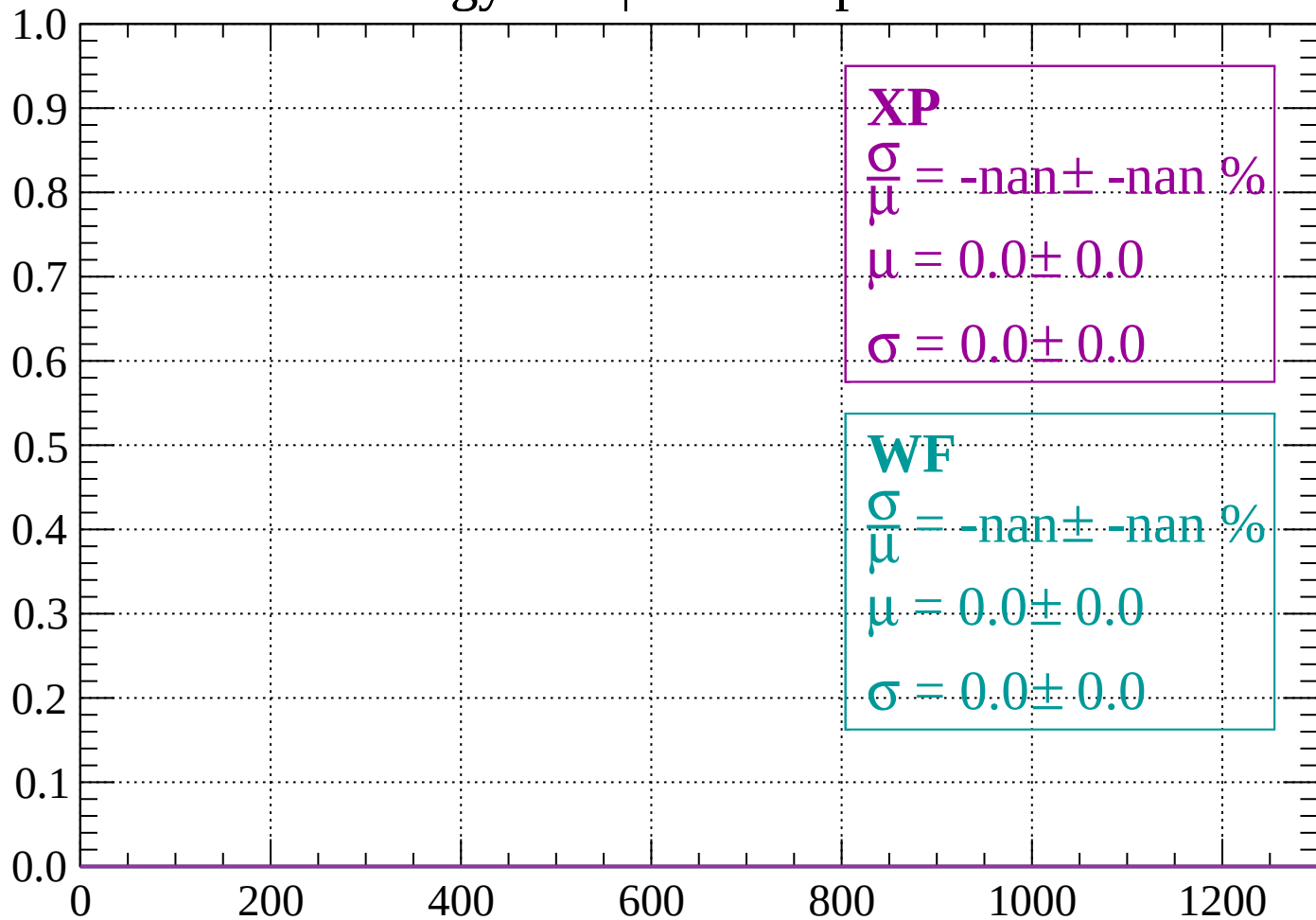
Count



dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1840

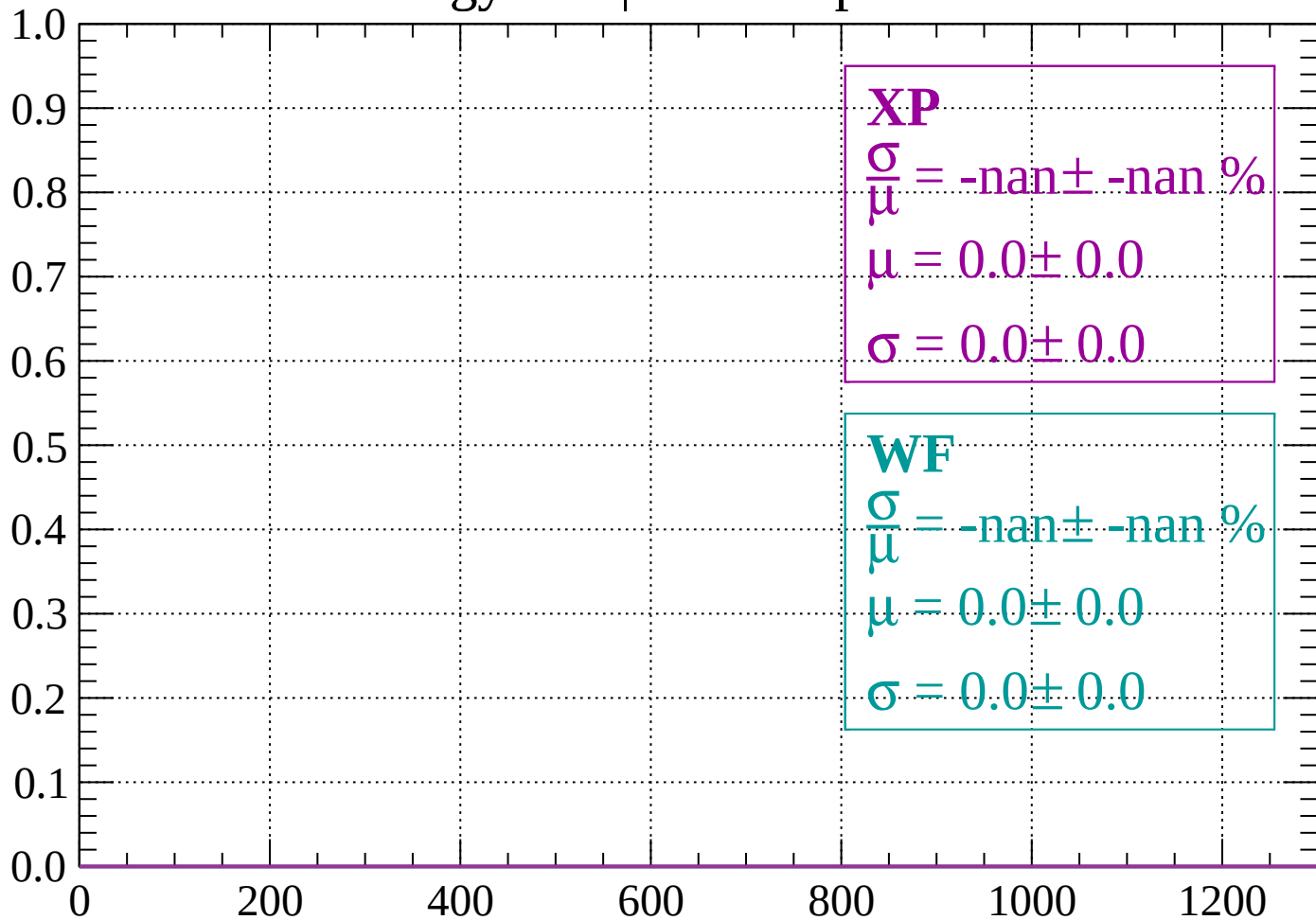
Count



dE/dx (ADC counts/cm)

Energy loss | -1840 < p < -1760

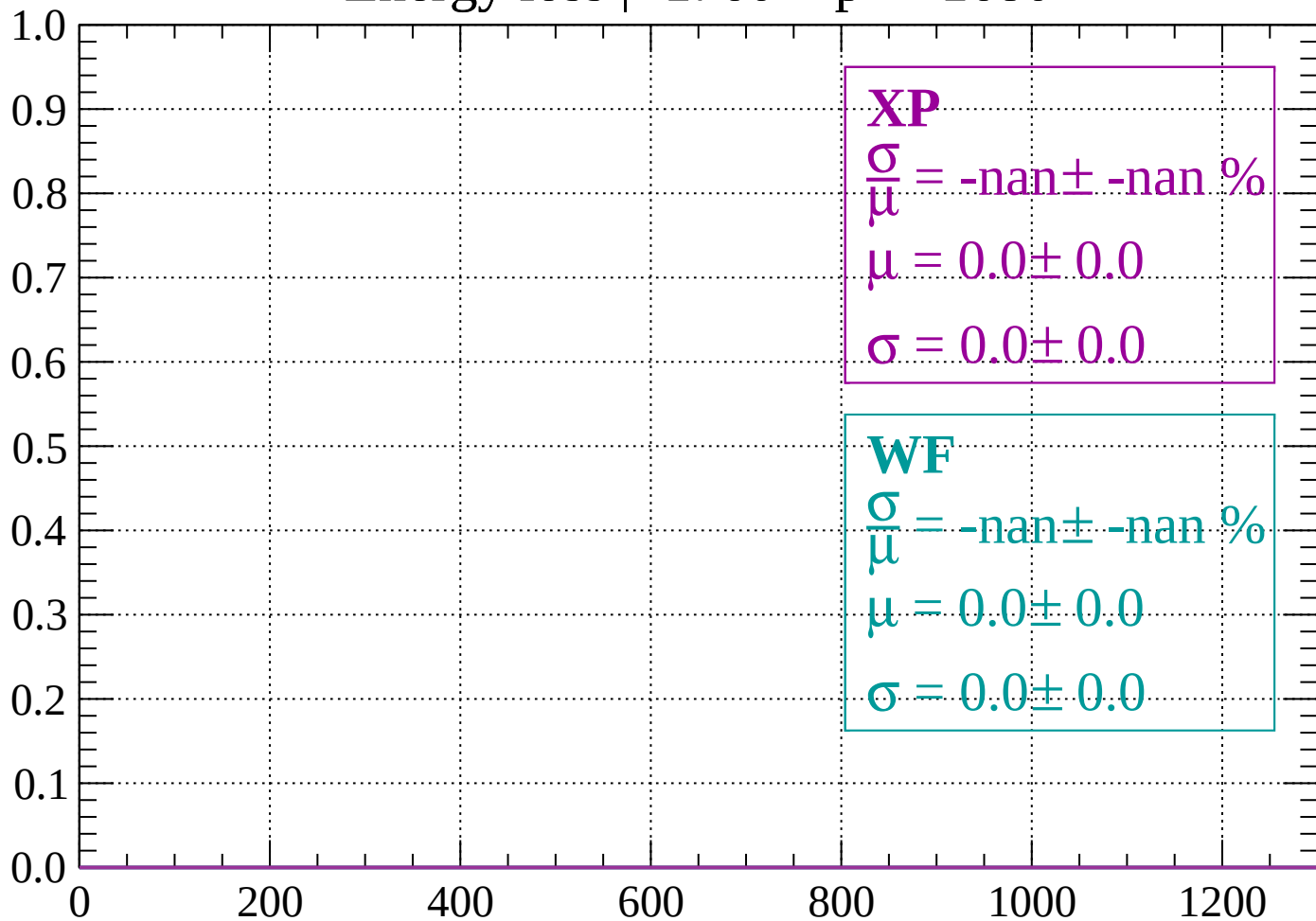
Count



dE/dx (ADC counts/cm)

Energy loss | -1760 < p < -1680

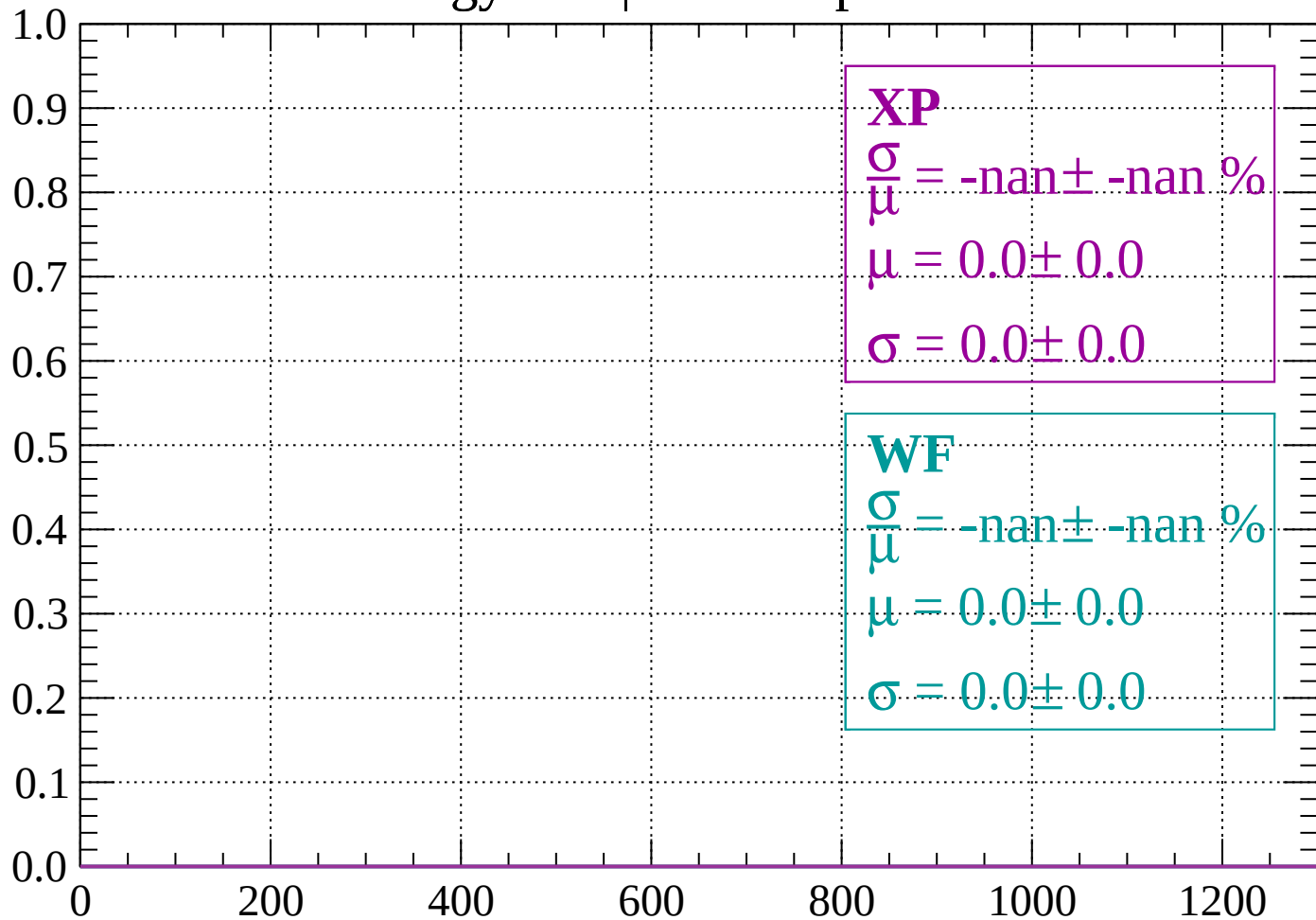
Count



dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1600

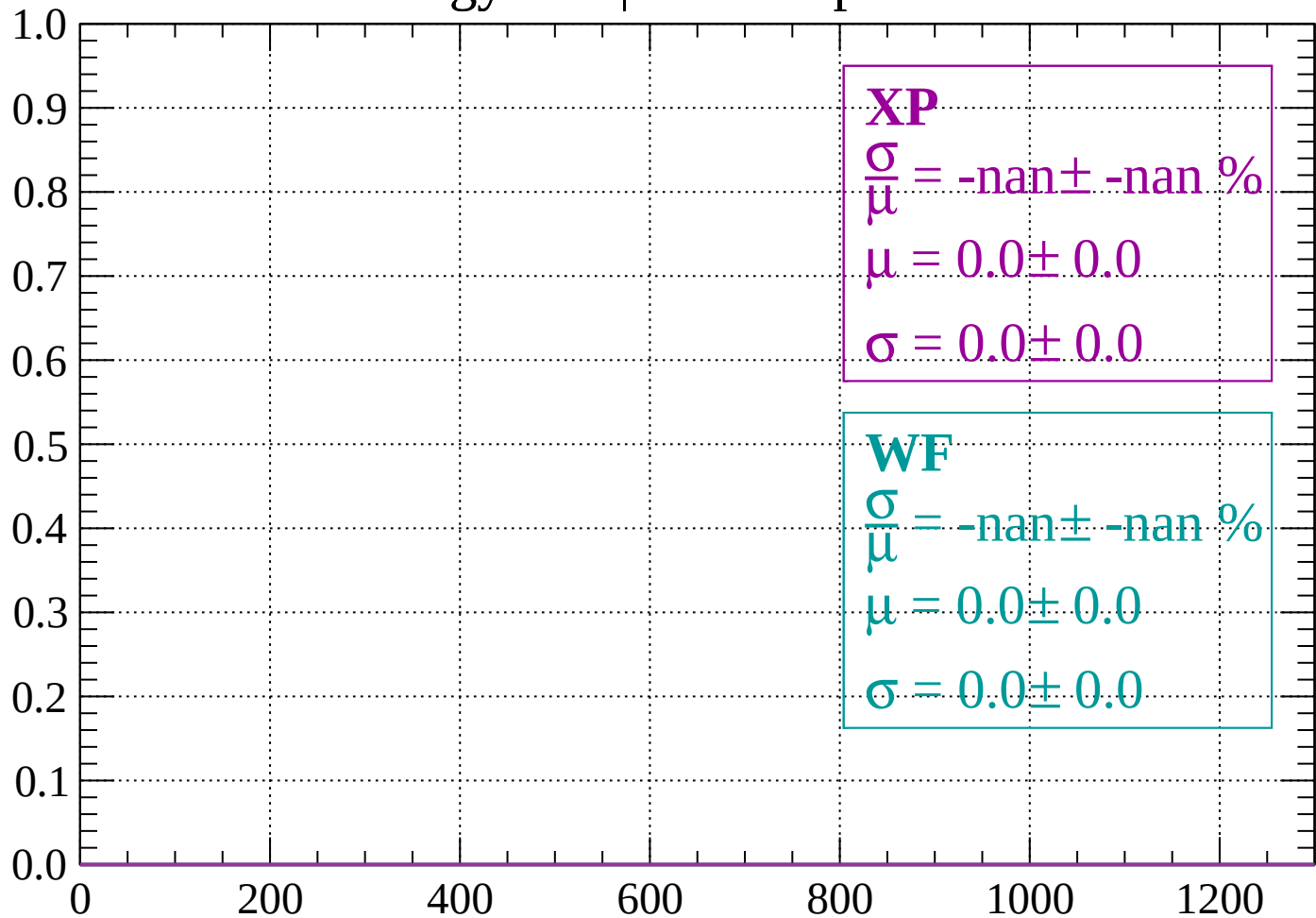
Count



dE/dx (ADC counts/cm)

Energy loss | $-1600 < p < -1520$

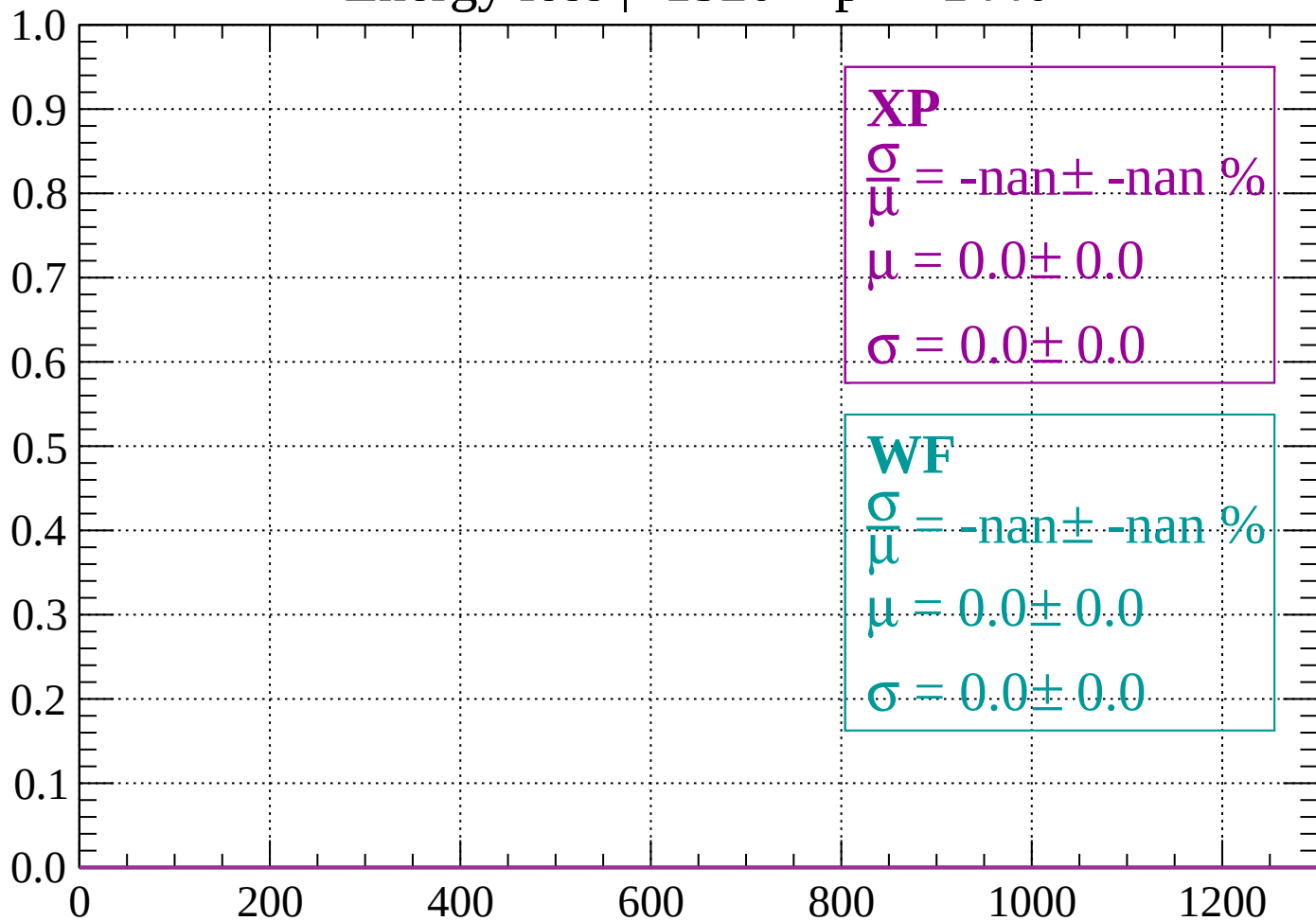
Count



dE/dx (ADC counts/cm)

Energy loss | -1520 < p < -1440

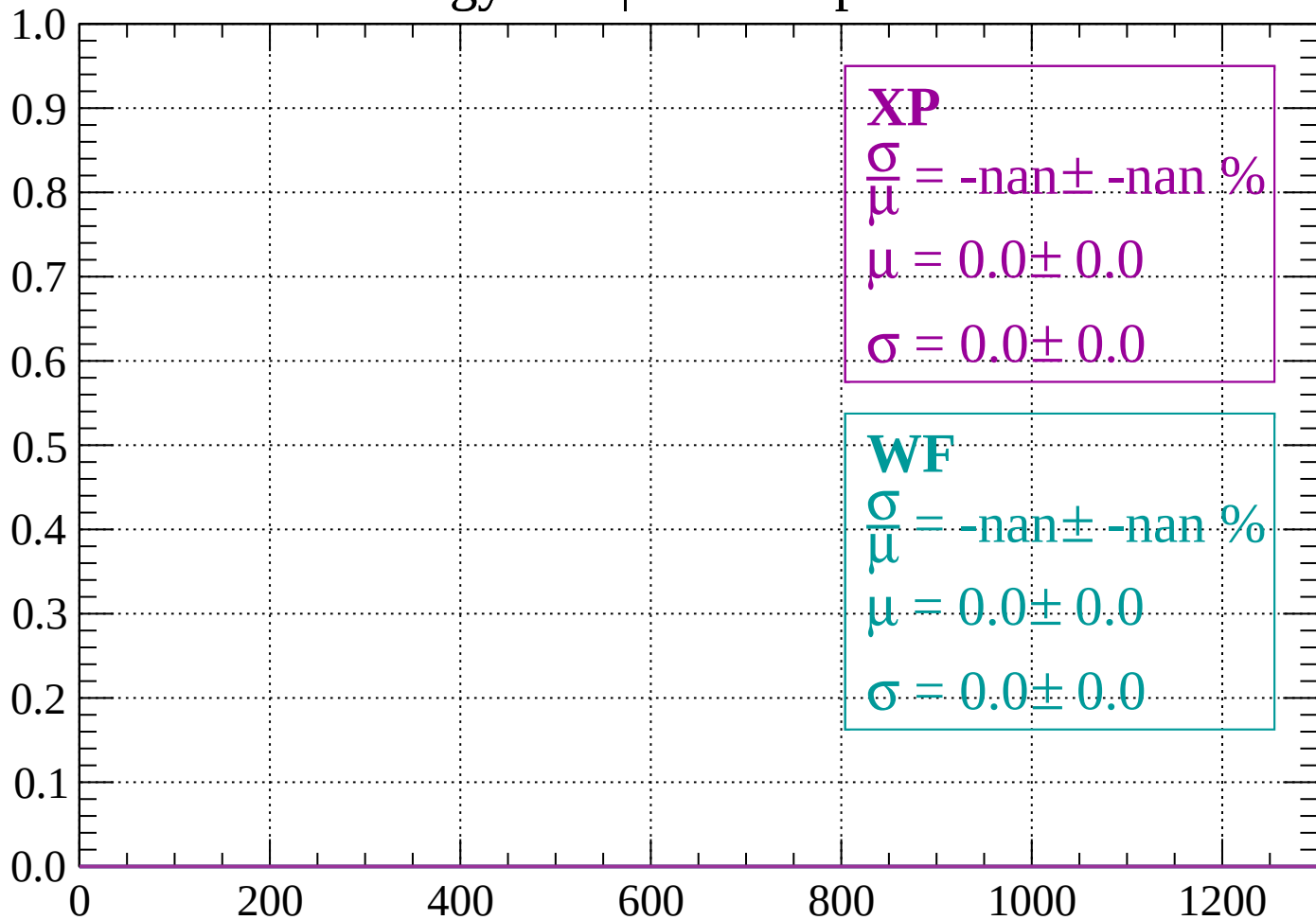
Count



dE/dx (ADC counts/cm)

Energy loss | -1440 < p < -1360

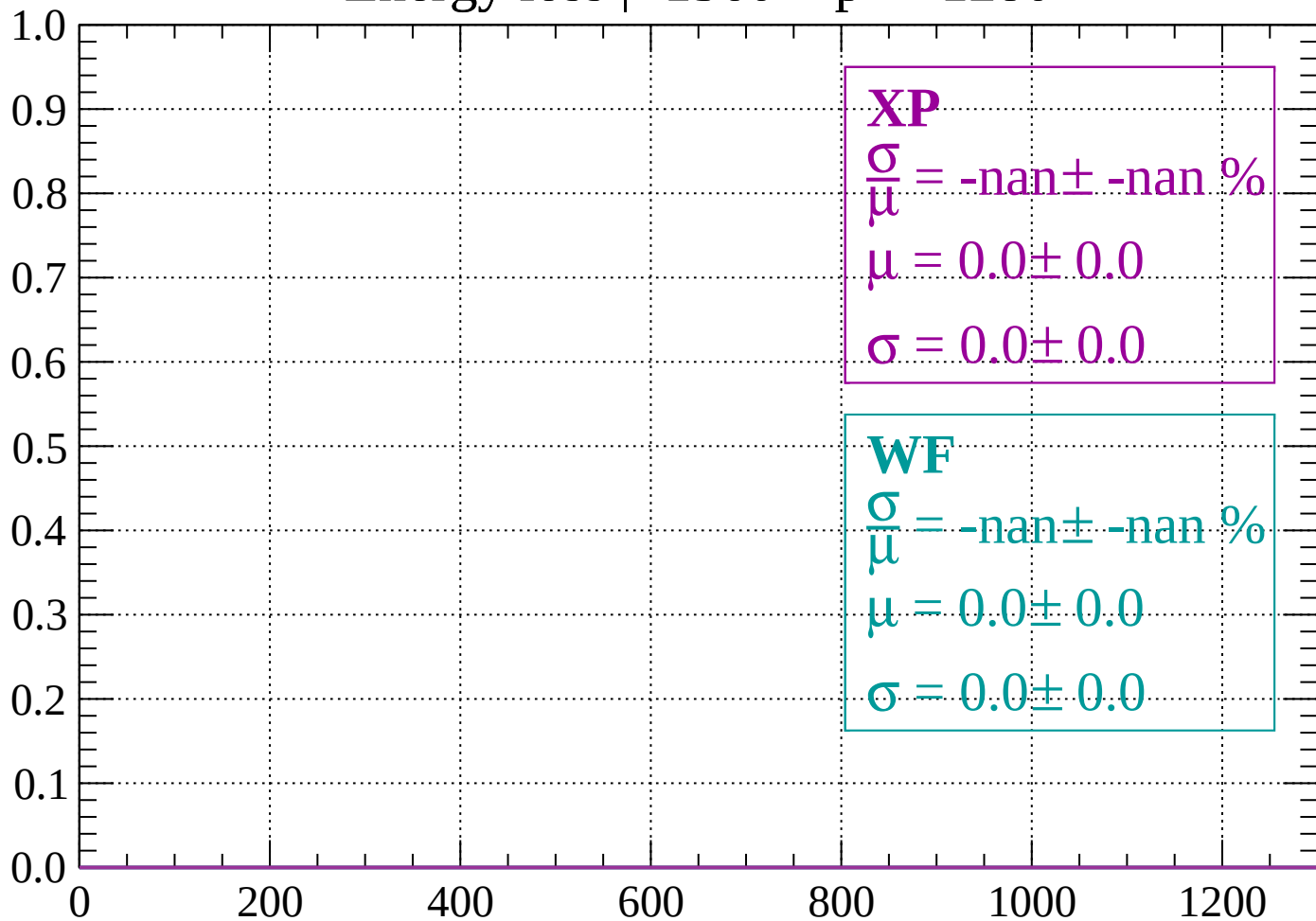
Count



dE/dx (ADC counts/cm)

Energy loss | -1360 < p < -1280

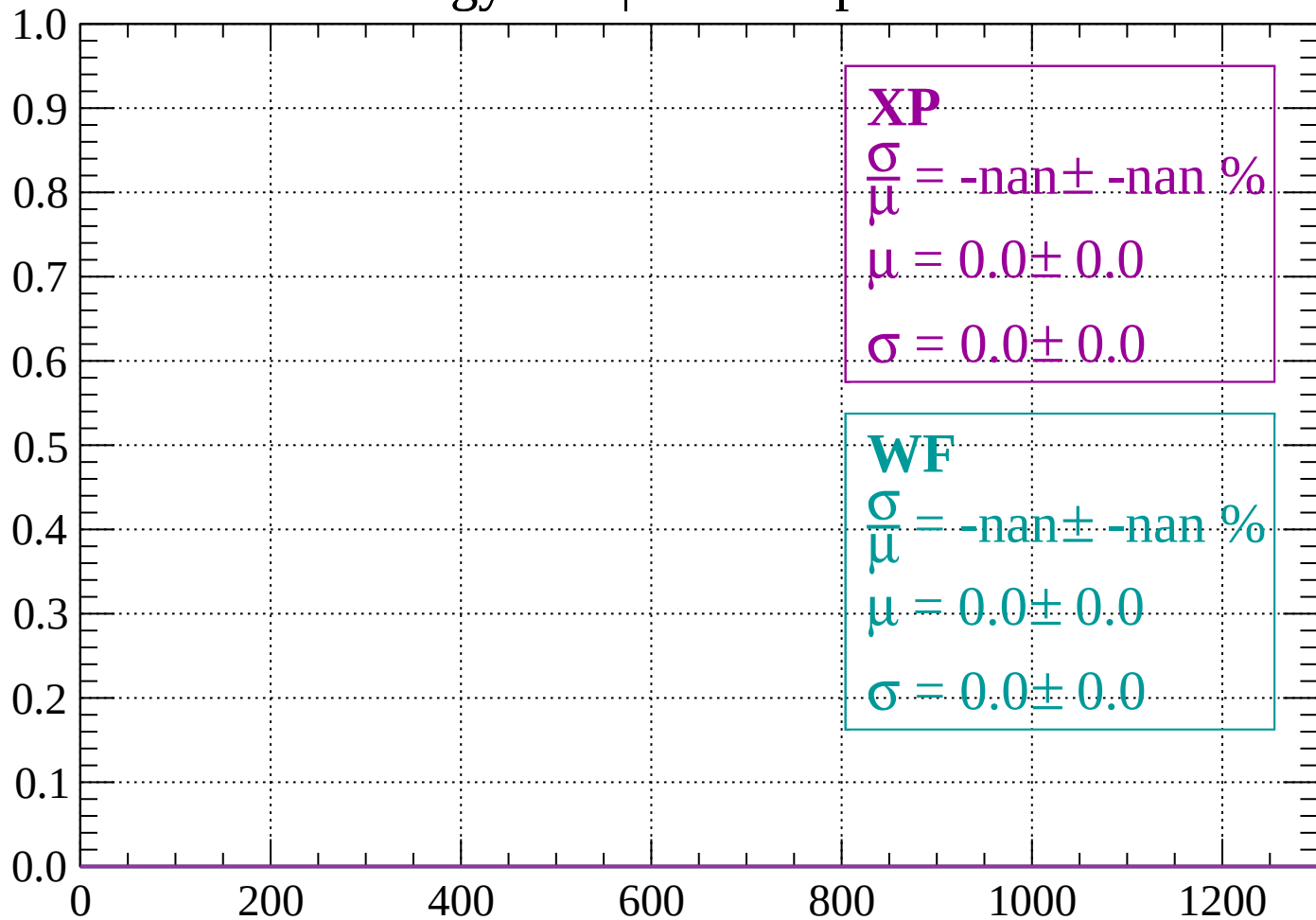
Count



dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1200$

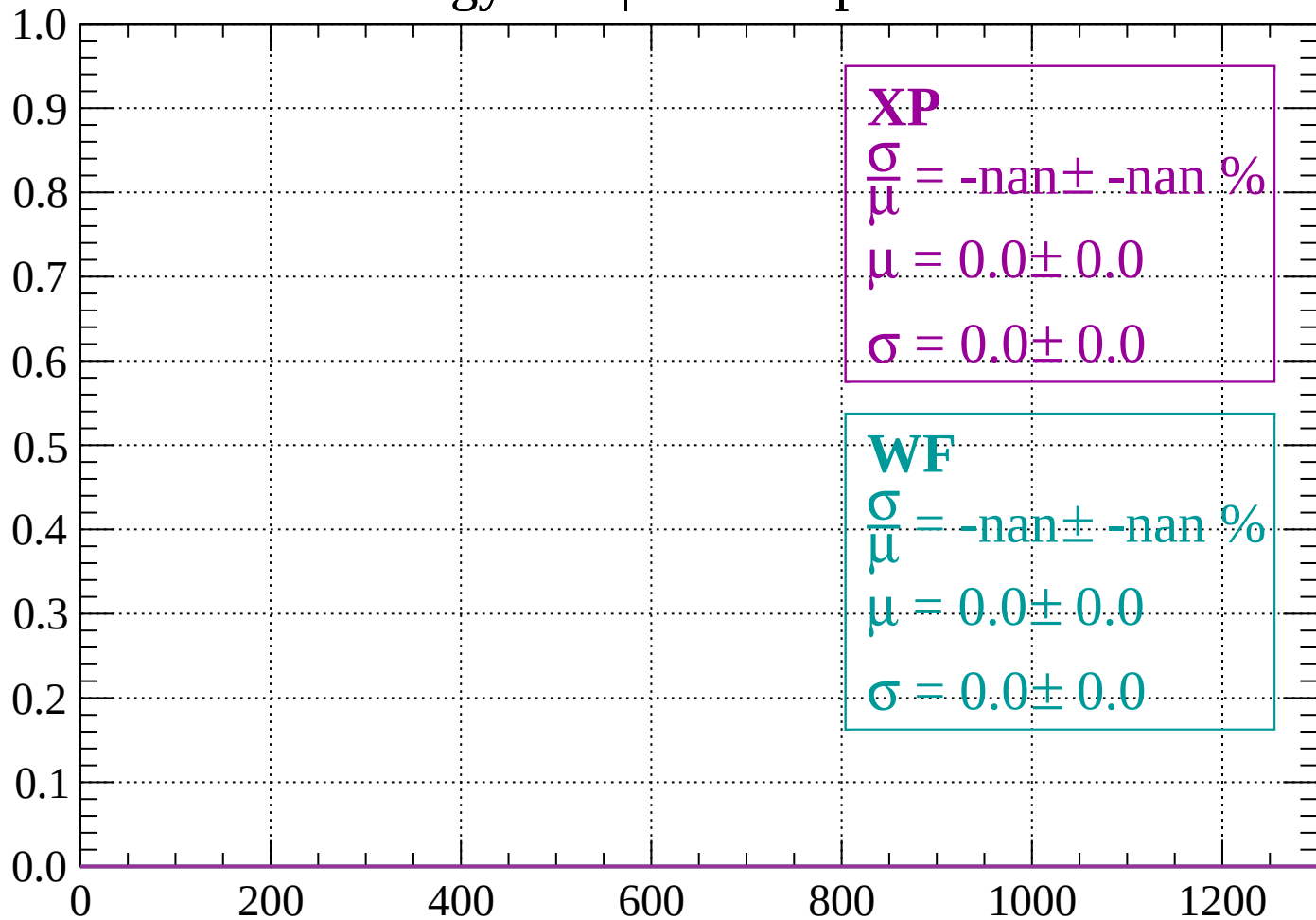
Count



dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1120$

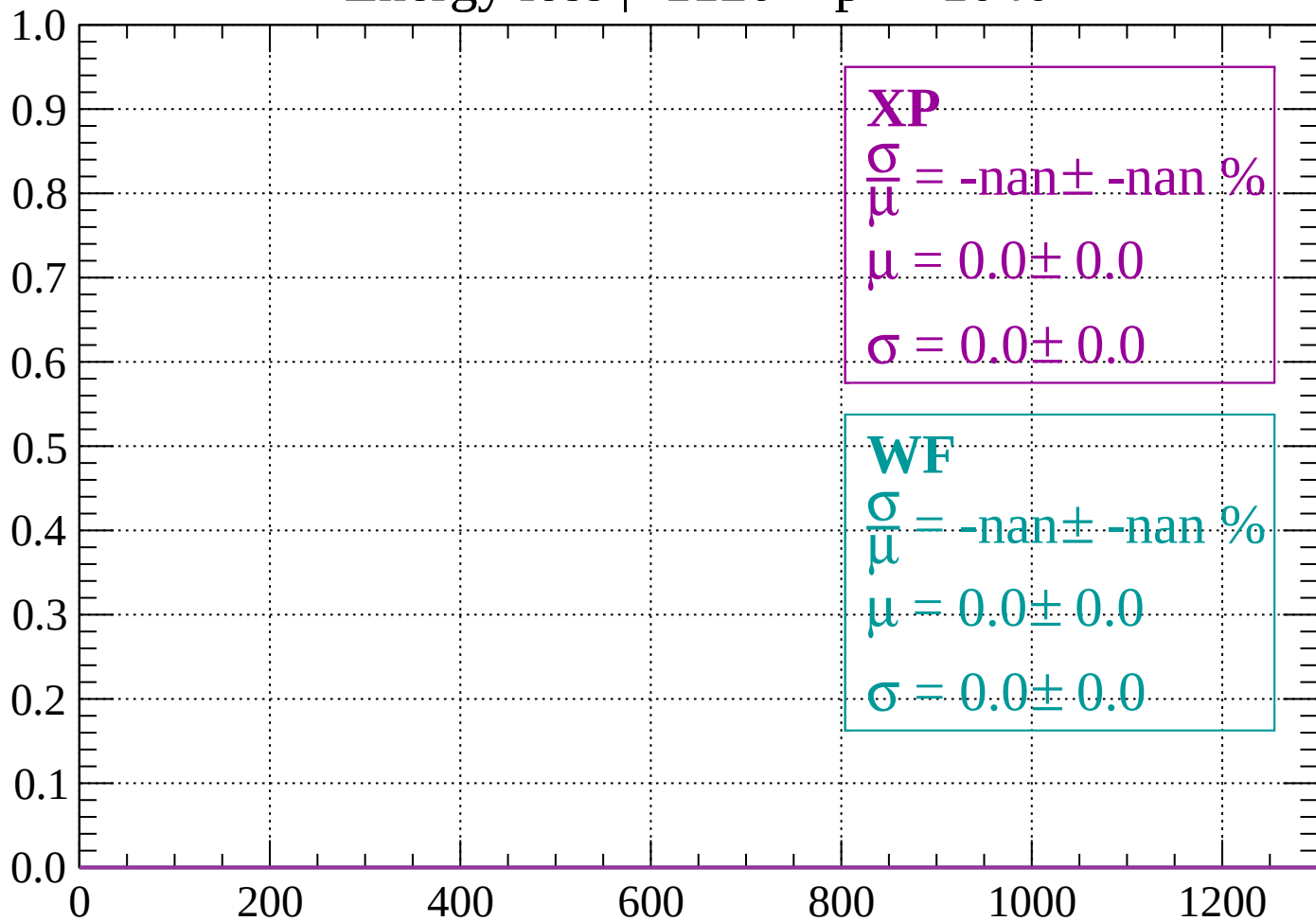
Count



dE/dx (ADC counts/cm)

Energy loss | -1120 < p < -1040

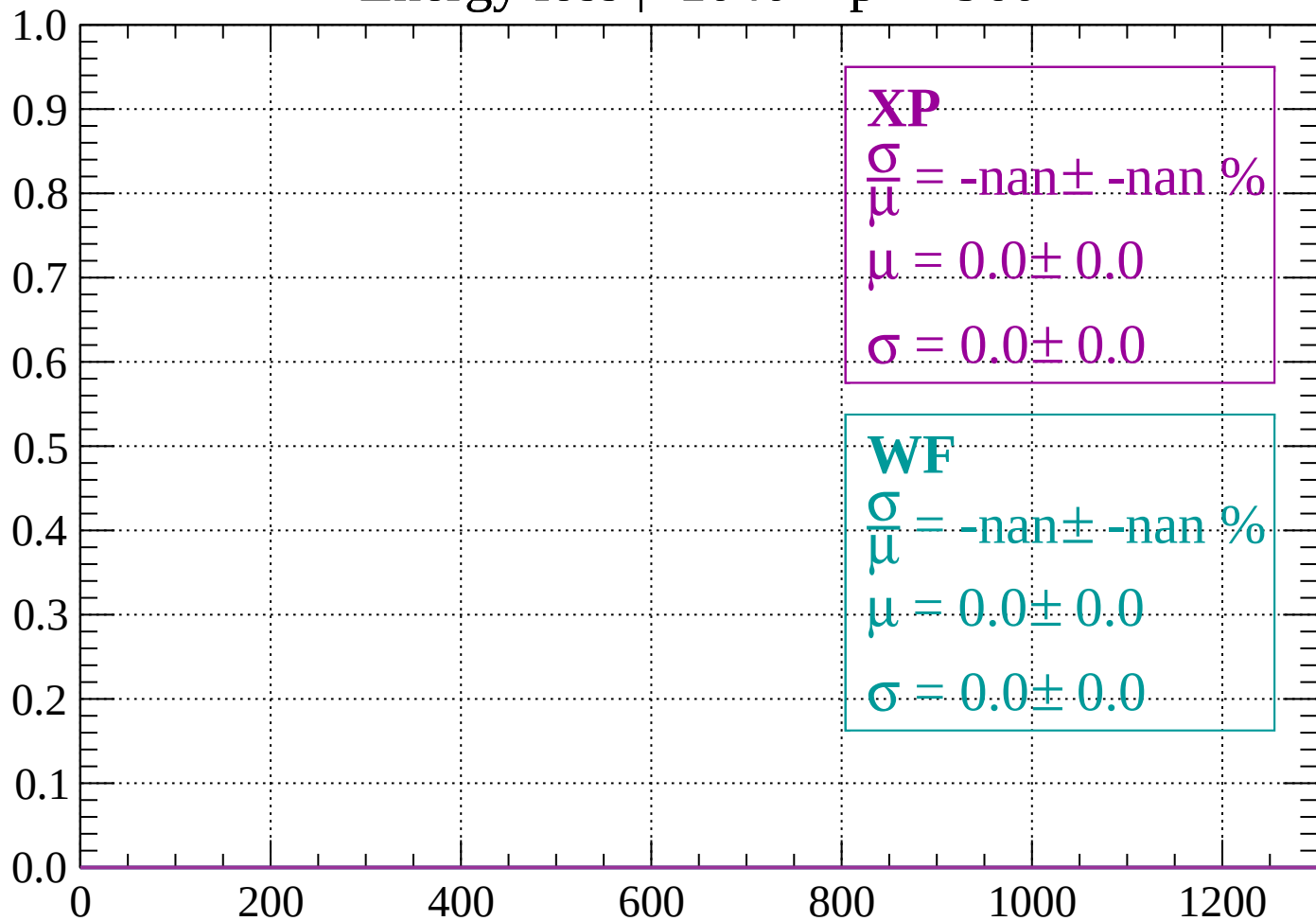
Count



dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -960$

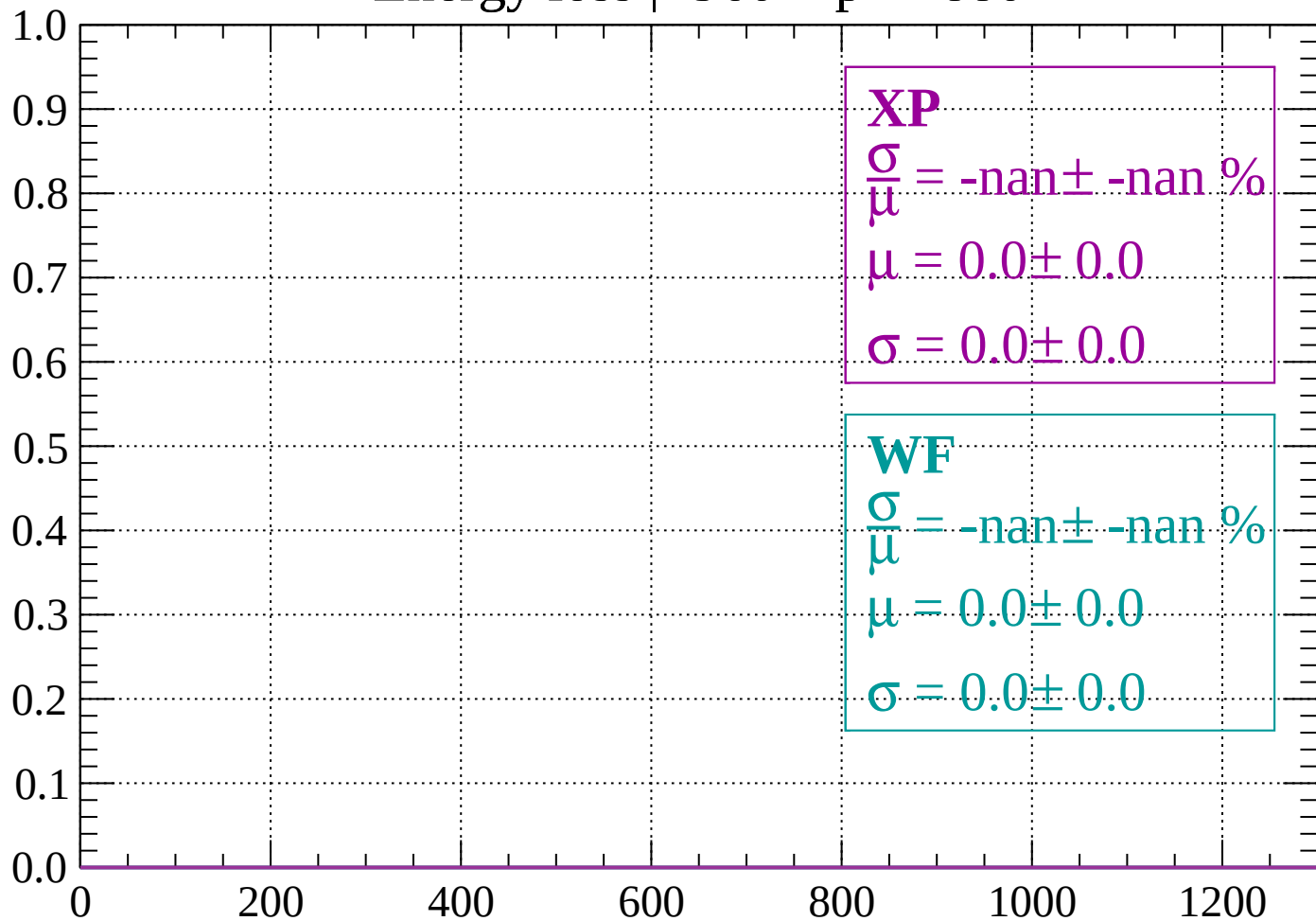
Count



dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -880$

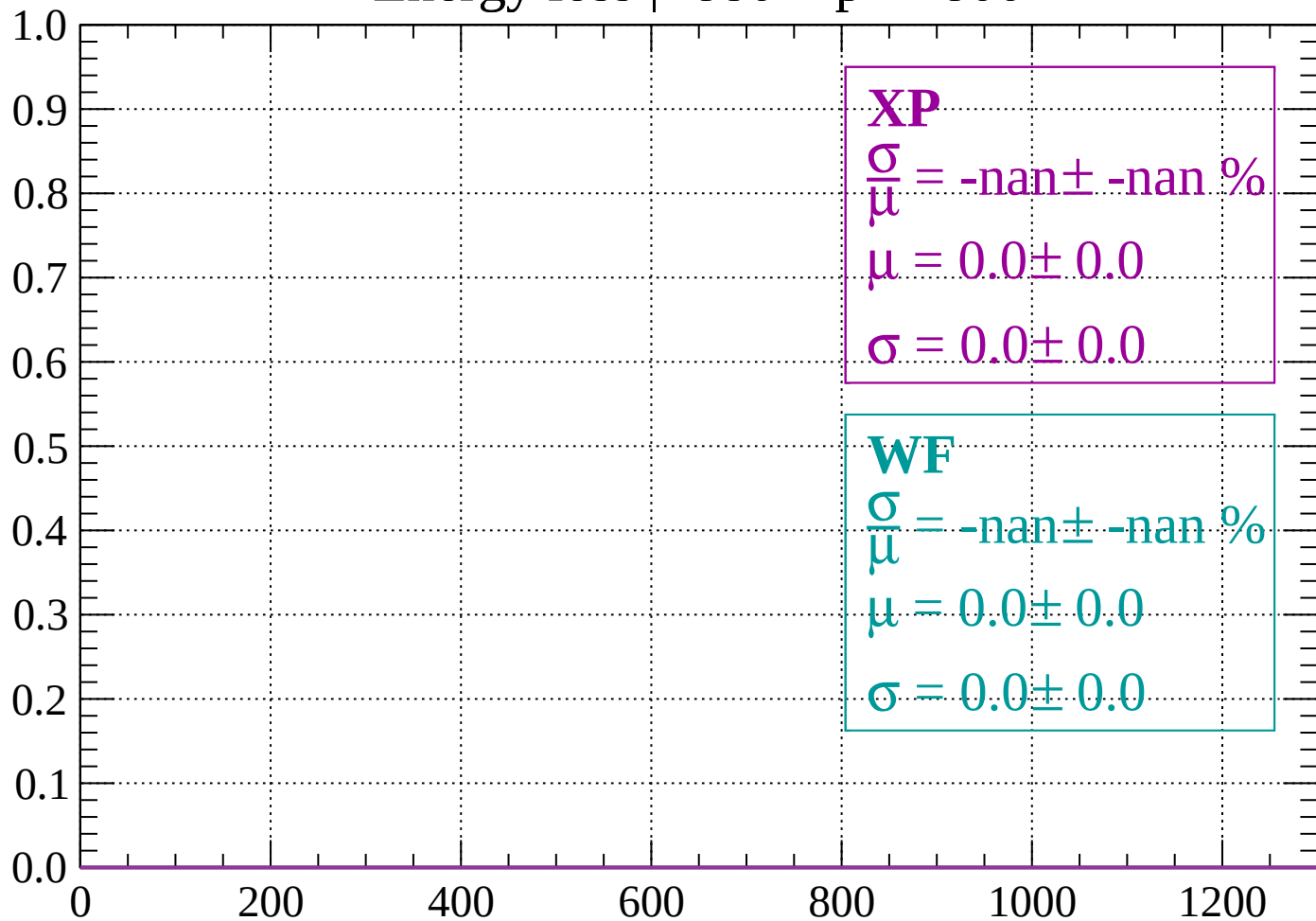
Count



dE/dx (ADC counts/cm)

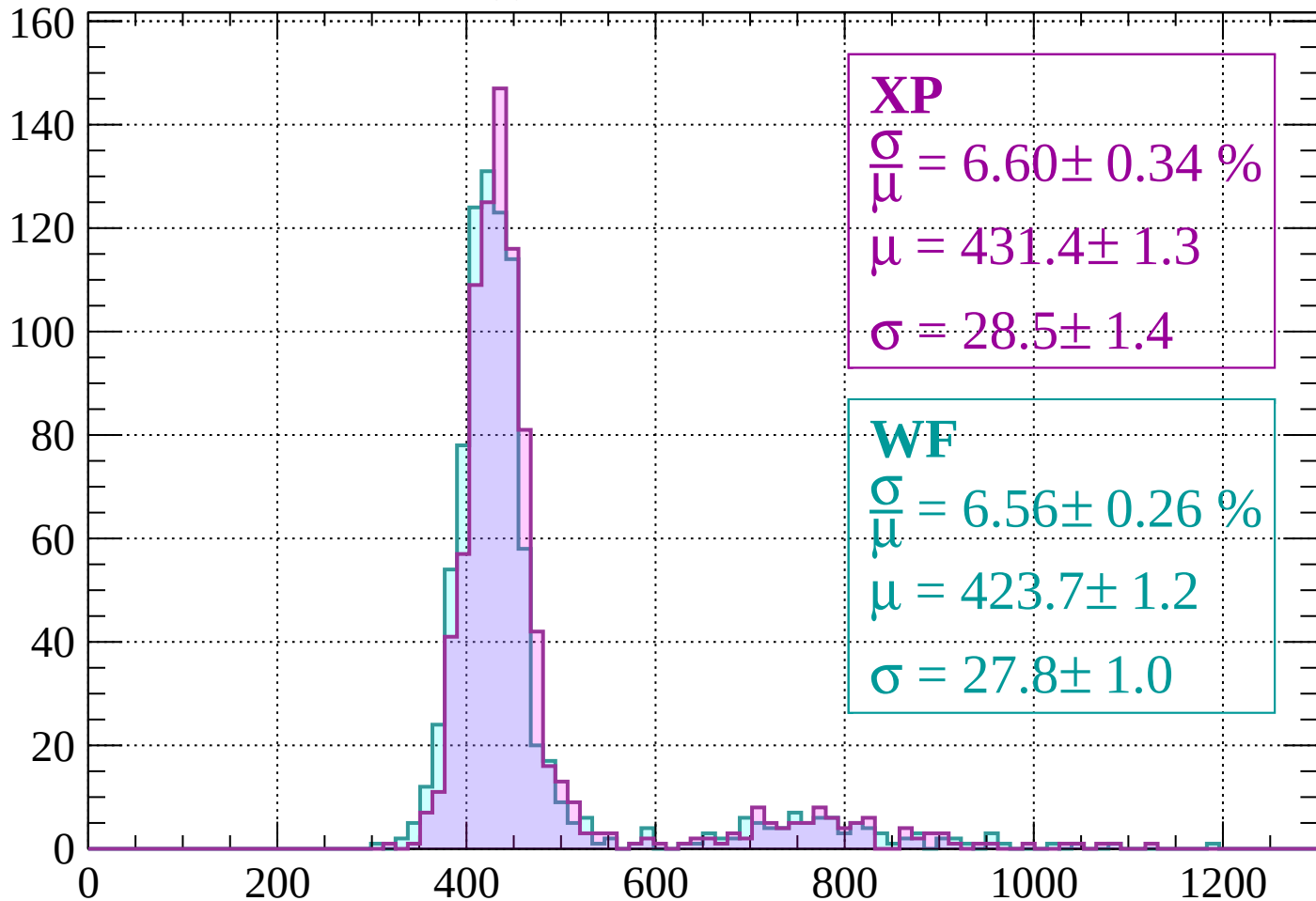
Energy loss | $-880 < p < -800$

Count



Energy loss | $-800 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 6.60 \pm 0.34 \%$$

$$\mu = 431.4 \pm 1.3$$

$$\sigma = 28.5 \pm 1.4$$

WF

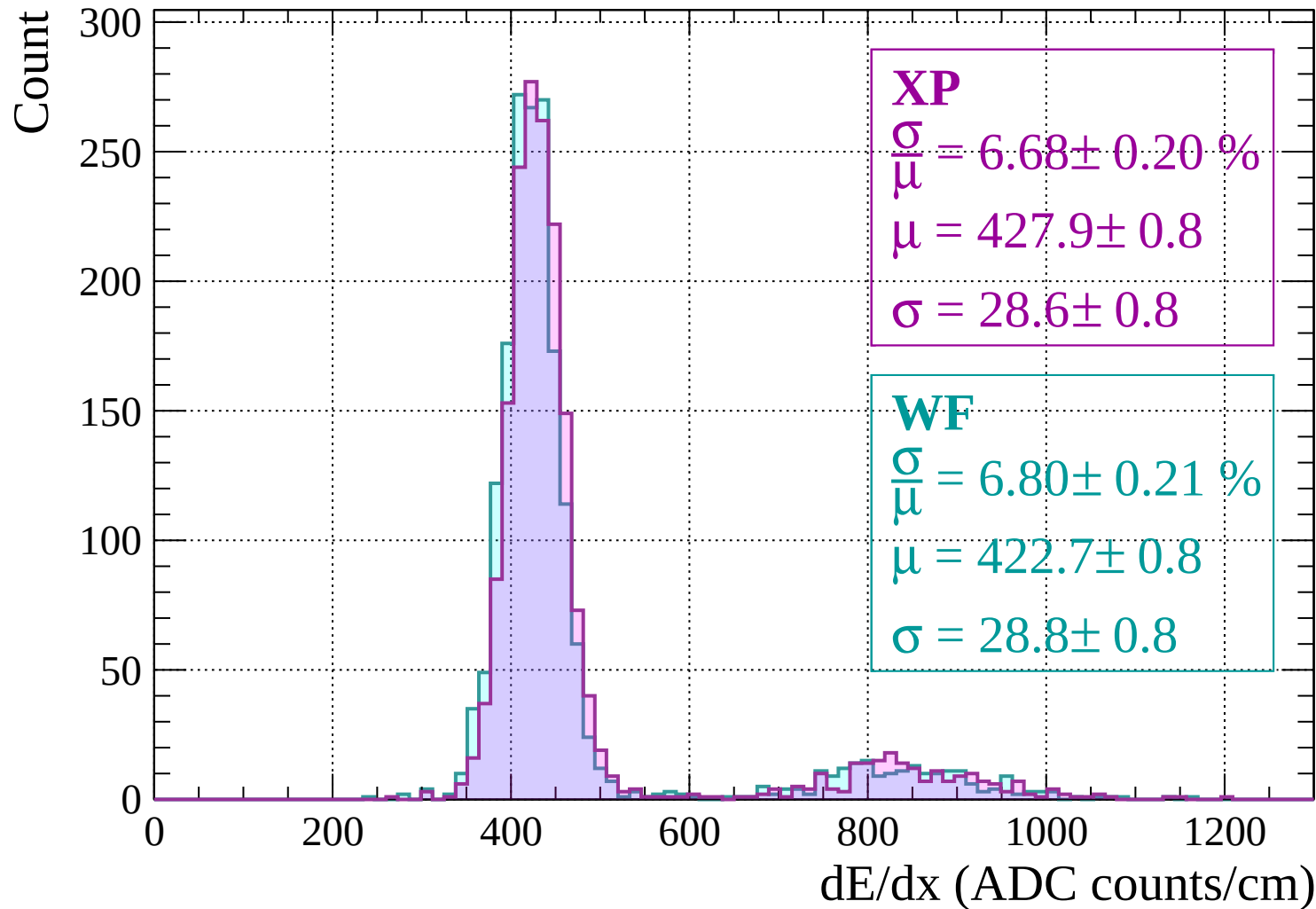
$$\frac{\sigma}{\mu} = 6.56 \pm 0.26 \%$$

$$\mu = 423.7 \pm 1.2$$

$$\sigma = 27.8 \pm 1.0$$

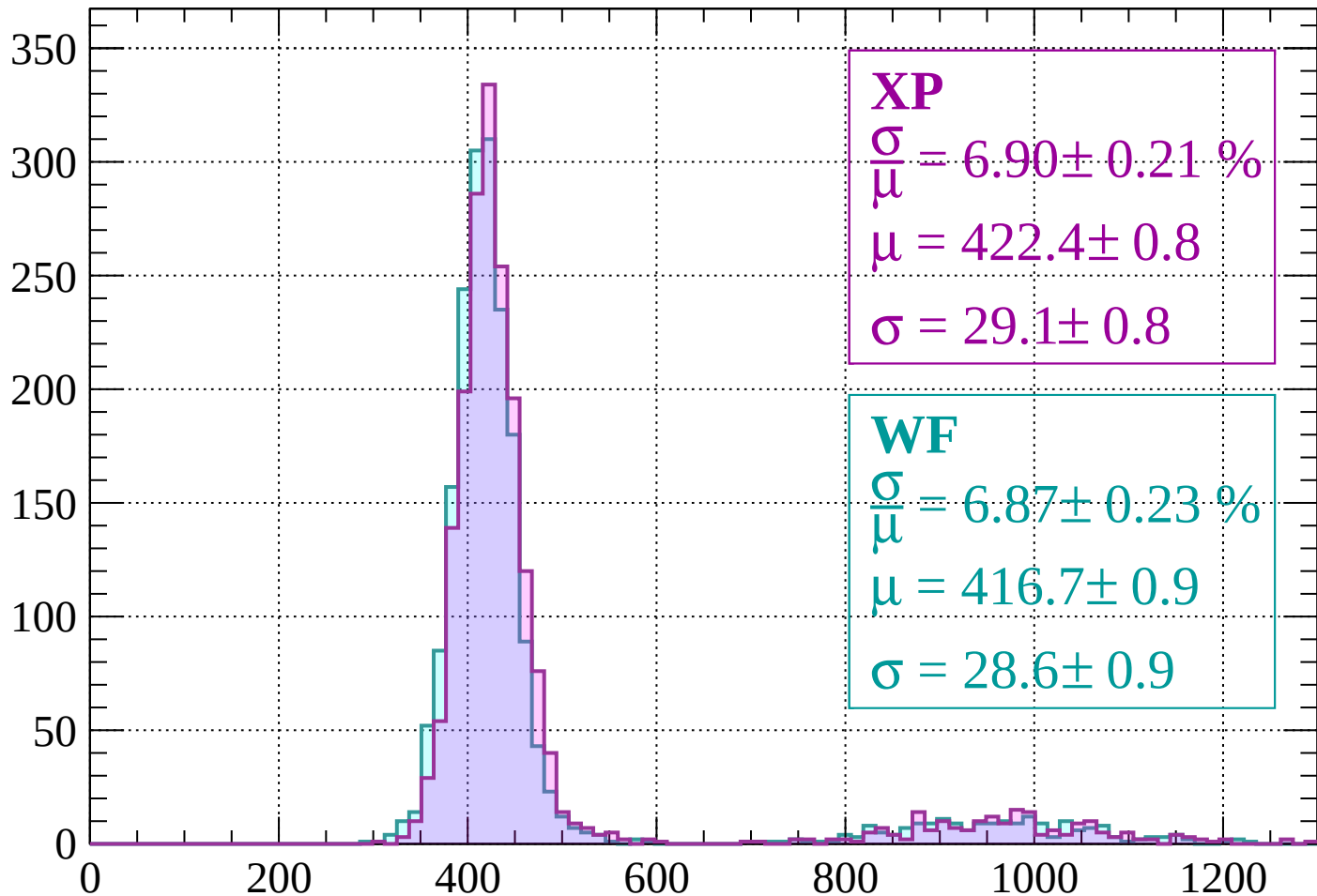
dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -640$



Energy loss | $-640 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 6.90 \pm 0.21 \%$$

$$\mu = 422.4 \pm 0.8$$

$$\sigma = 29.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.87 \pm 0.23 \%$$

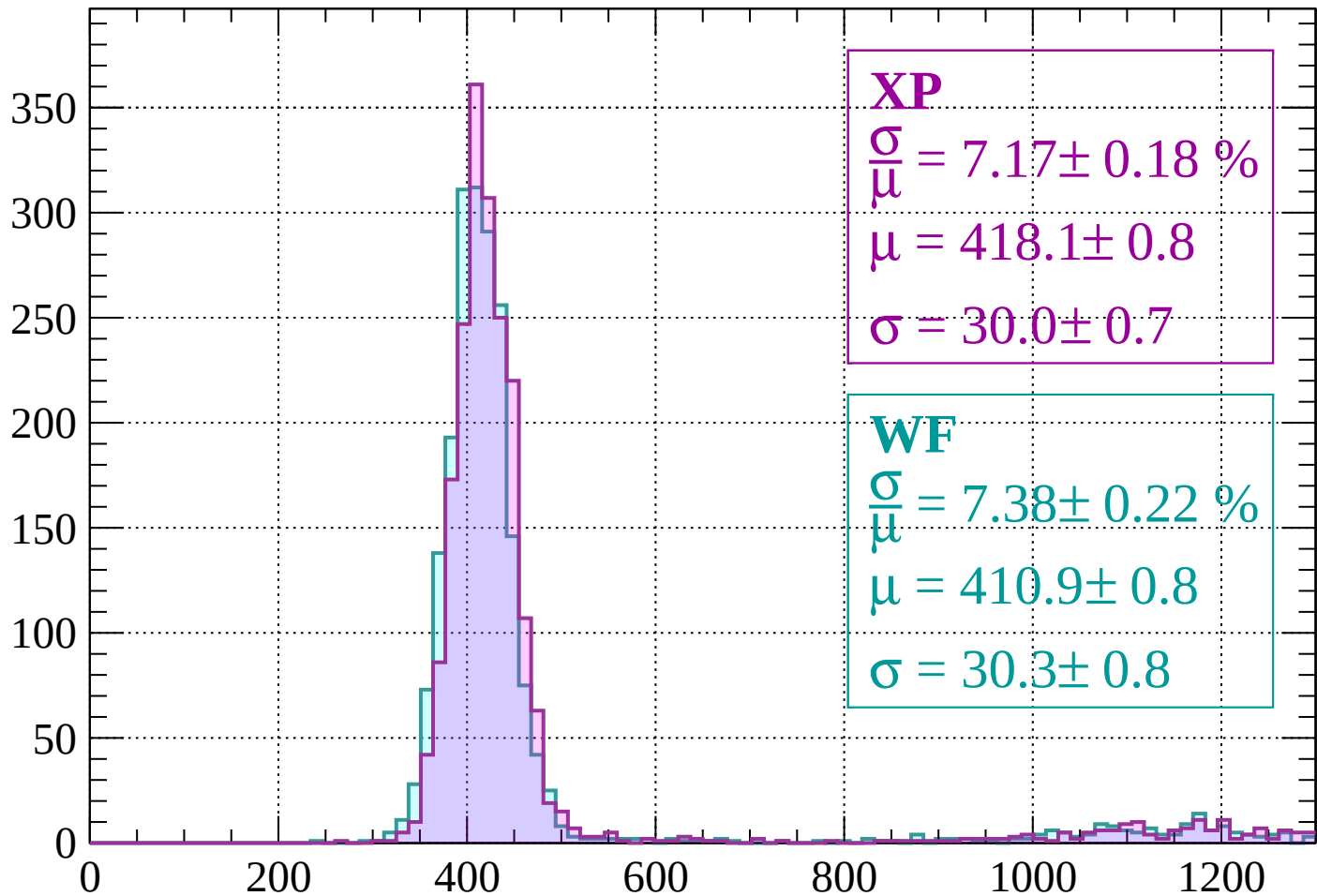
$$\mu = 416.7 \pm 0.9$$

$$\sigma = 28.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 7.17 \pm 0.18 \%$$

$$\mu = 418.1 \pm 0.8$$

$$\sigma = 30.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.38 \pm 0.22 \%$$

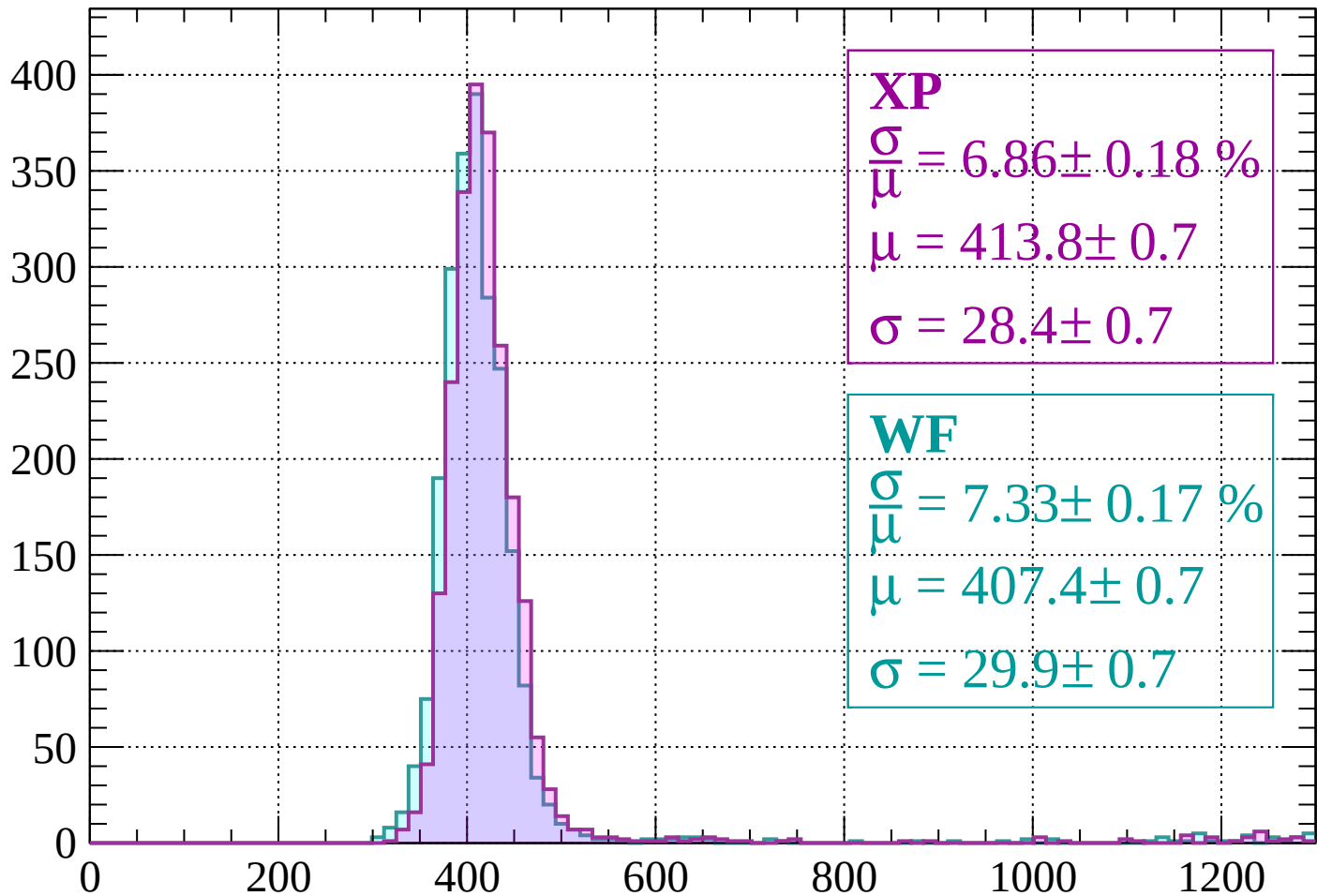
$$\mu = 410.9 \pm 0.8$$

$$\sigma = 30.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 6.86 \pm 0.18 \%$$

$$\mu = 413.8 \pm 0.7$$

$$\sigma = 28.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.33 \pm 0.17 \%$$

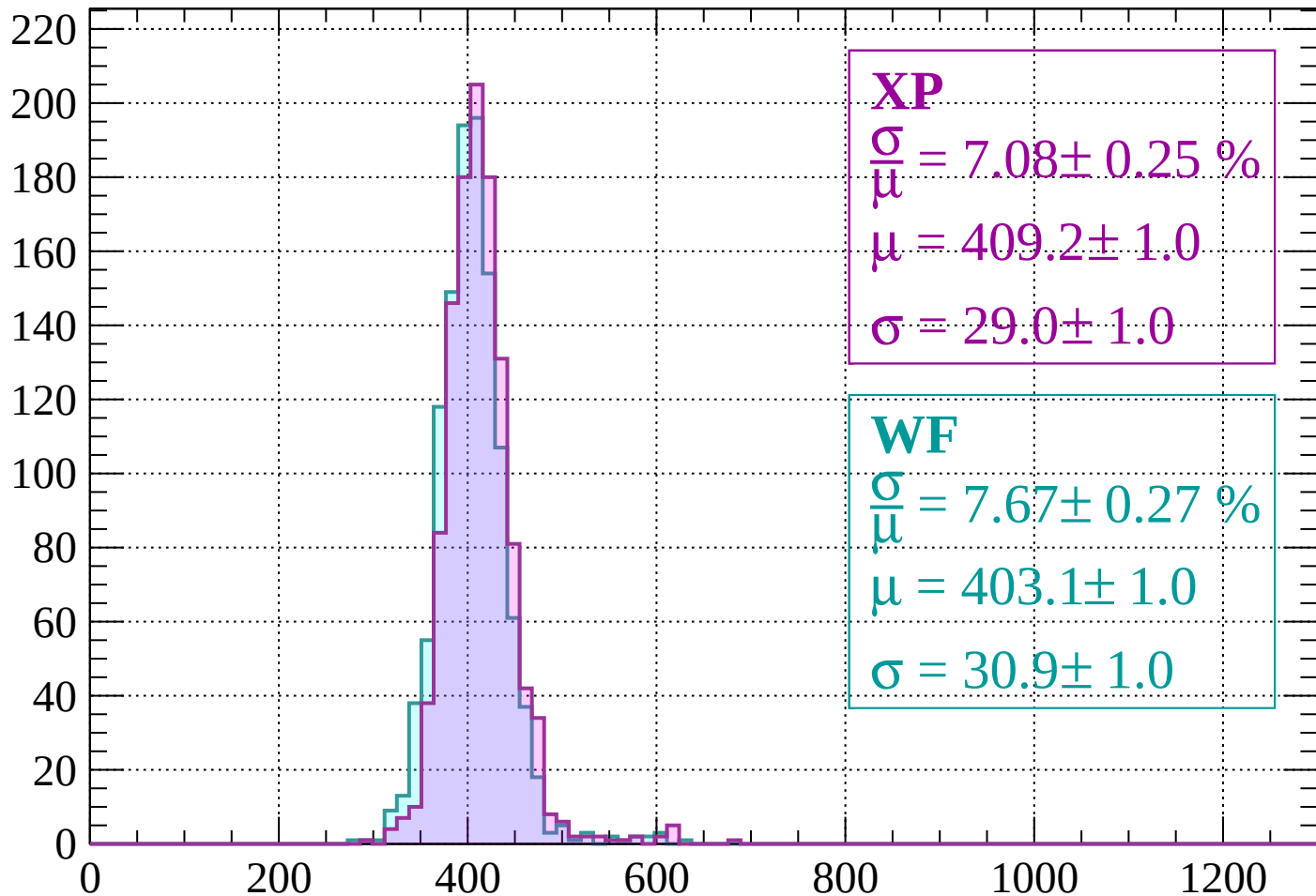
$$\mu = 407.4 \pm 0.7$$

$$\sigma = 29.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 7.08 \pm 0.25 \%$$

$$\mu = 409.2 \pm 1.0$$

$$\sigma = 29.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.67 \pm 0.27 \%$$

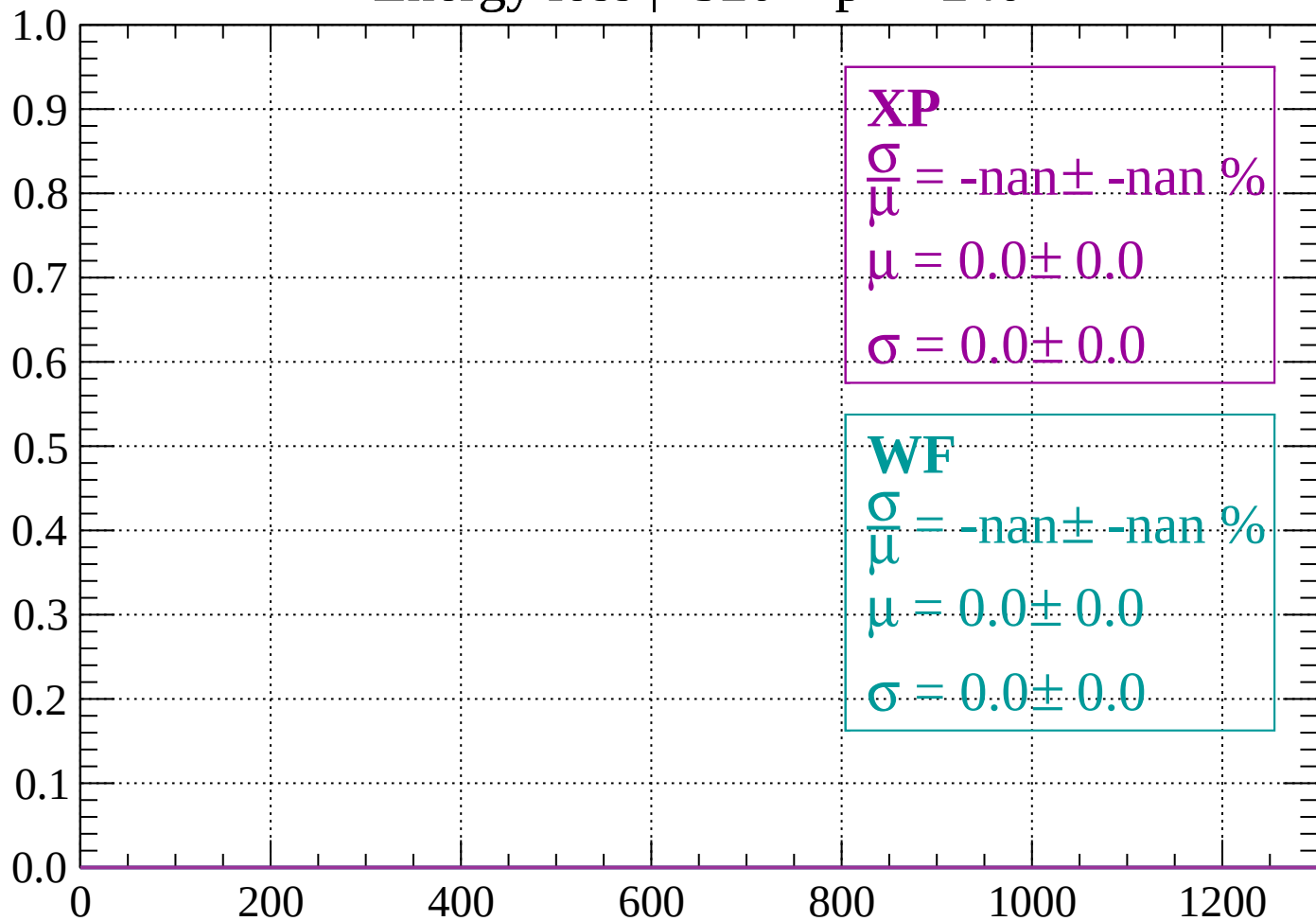
$$\mu = 403.1 \pm 1.0$$

$$\sigma = 30.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -240$

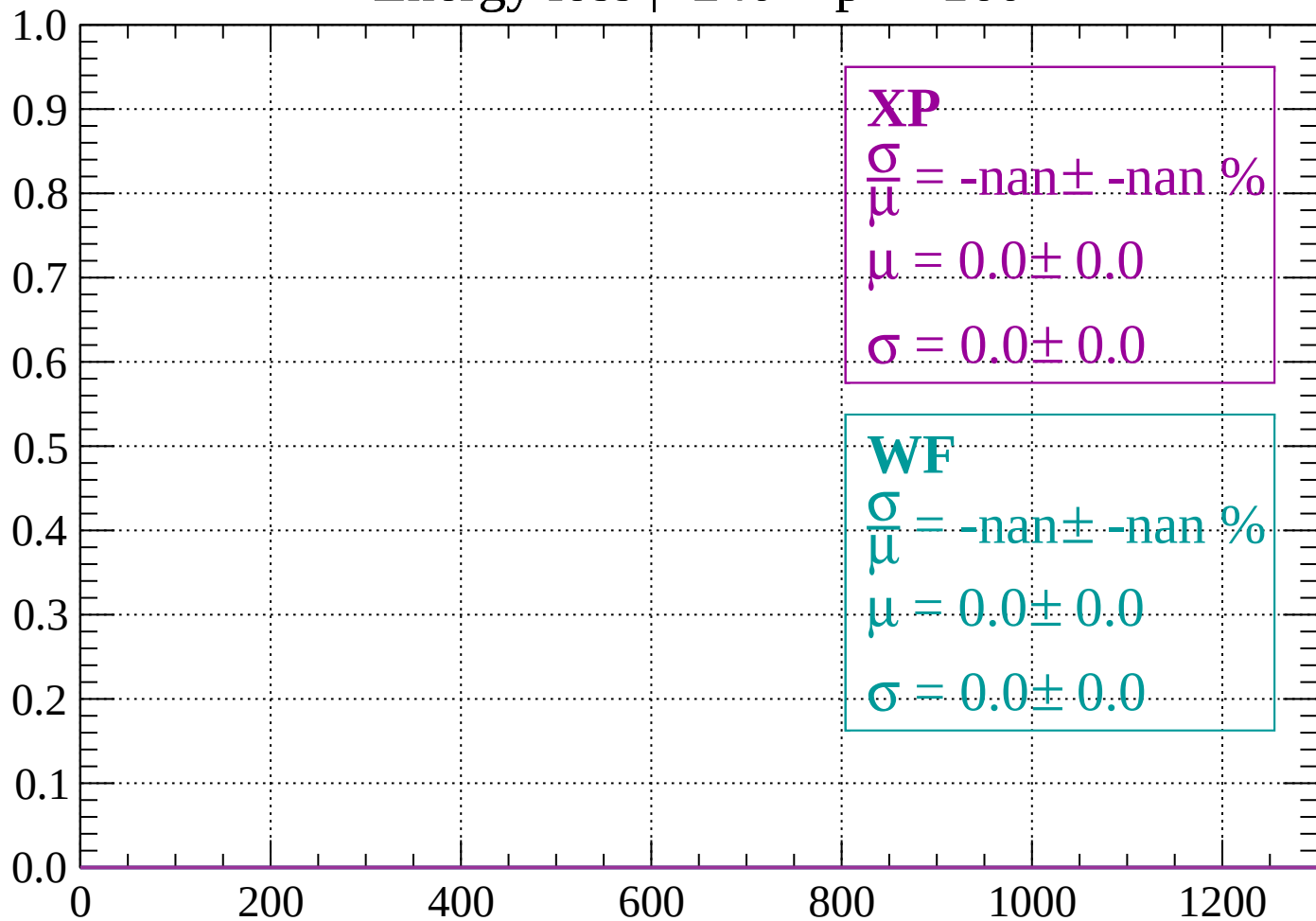
Count



dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -160$

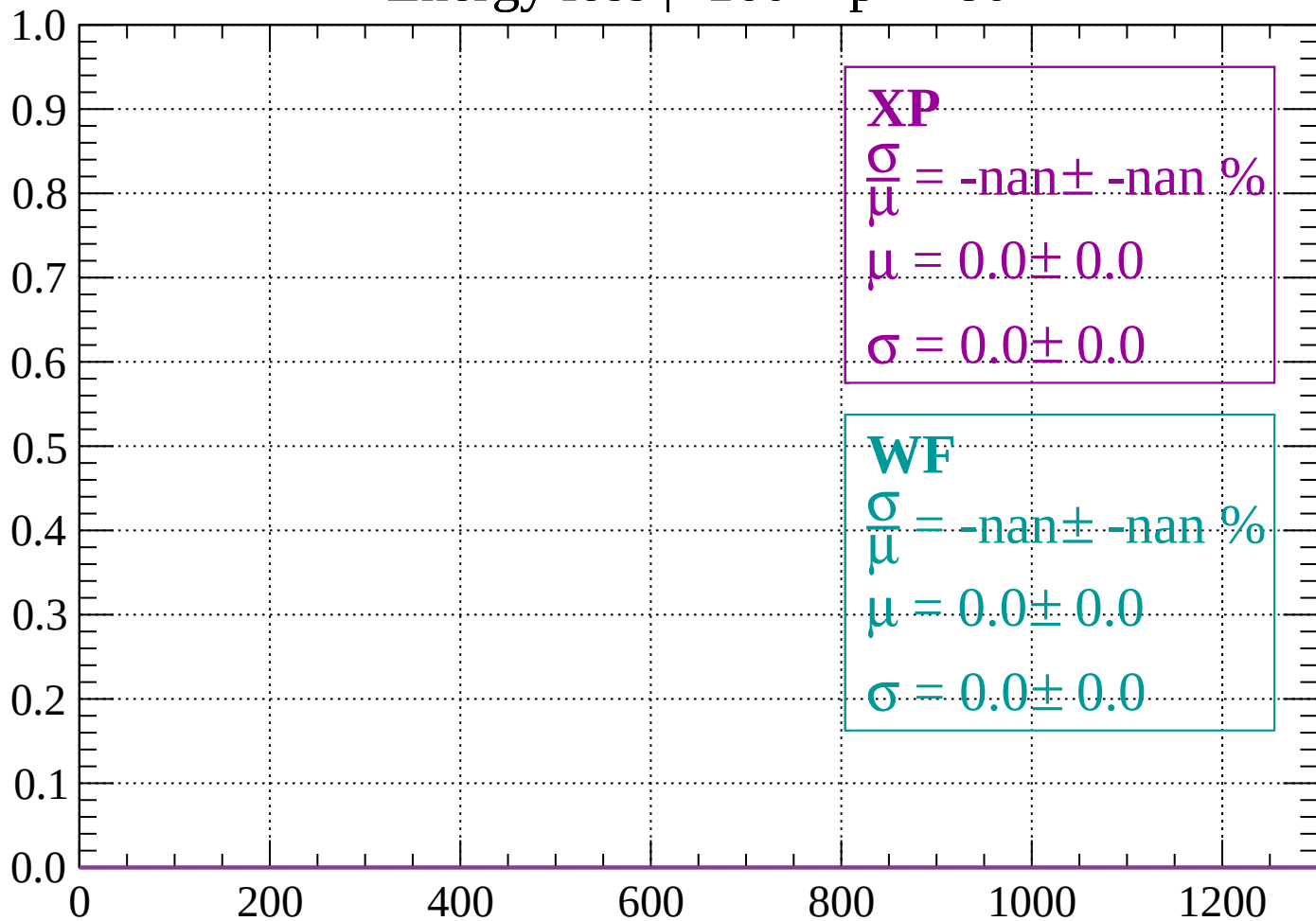
Count



dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -80$

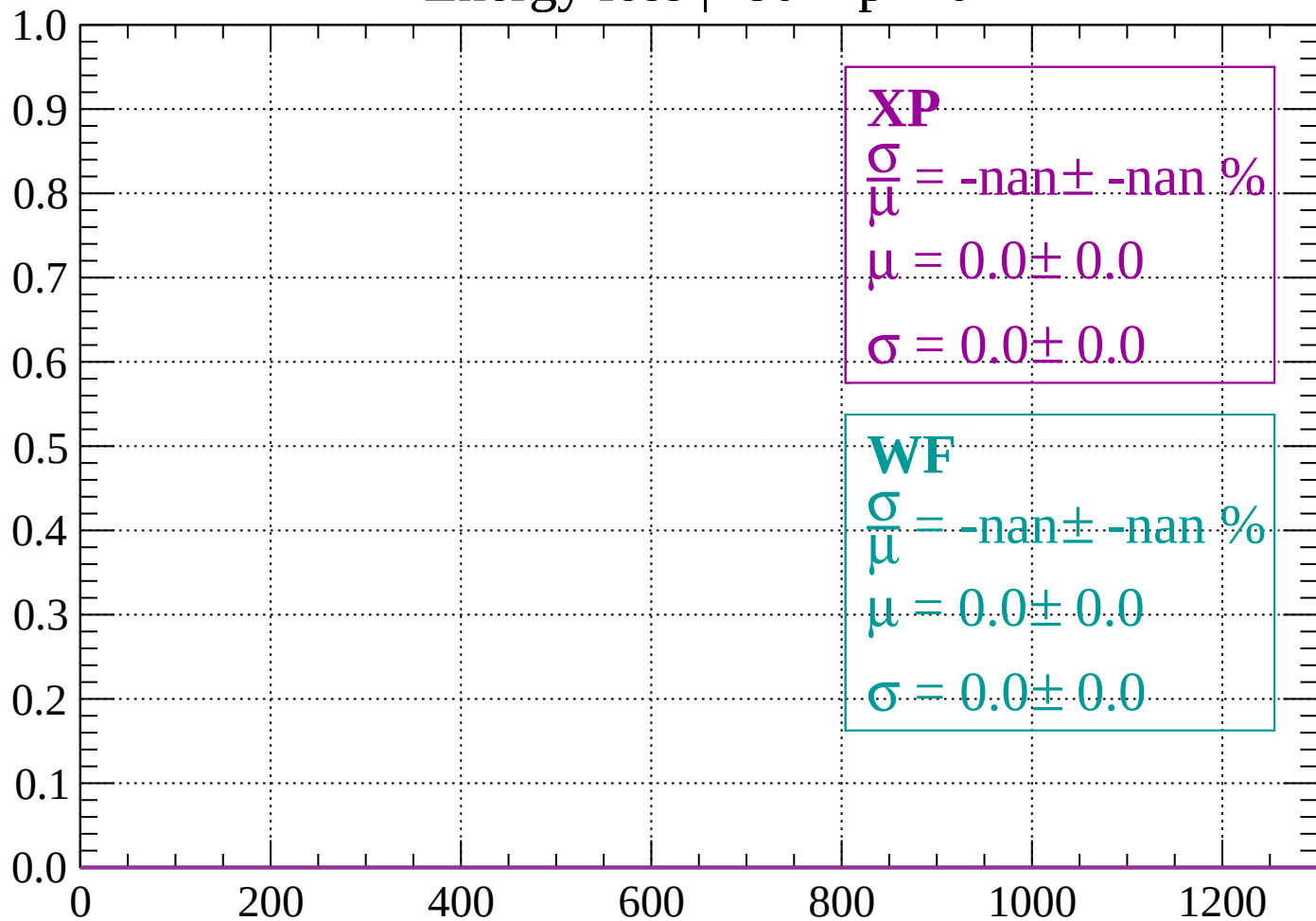
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < p < 0$

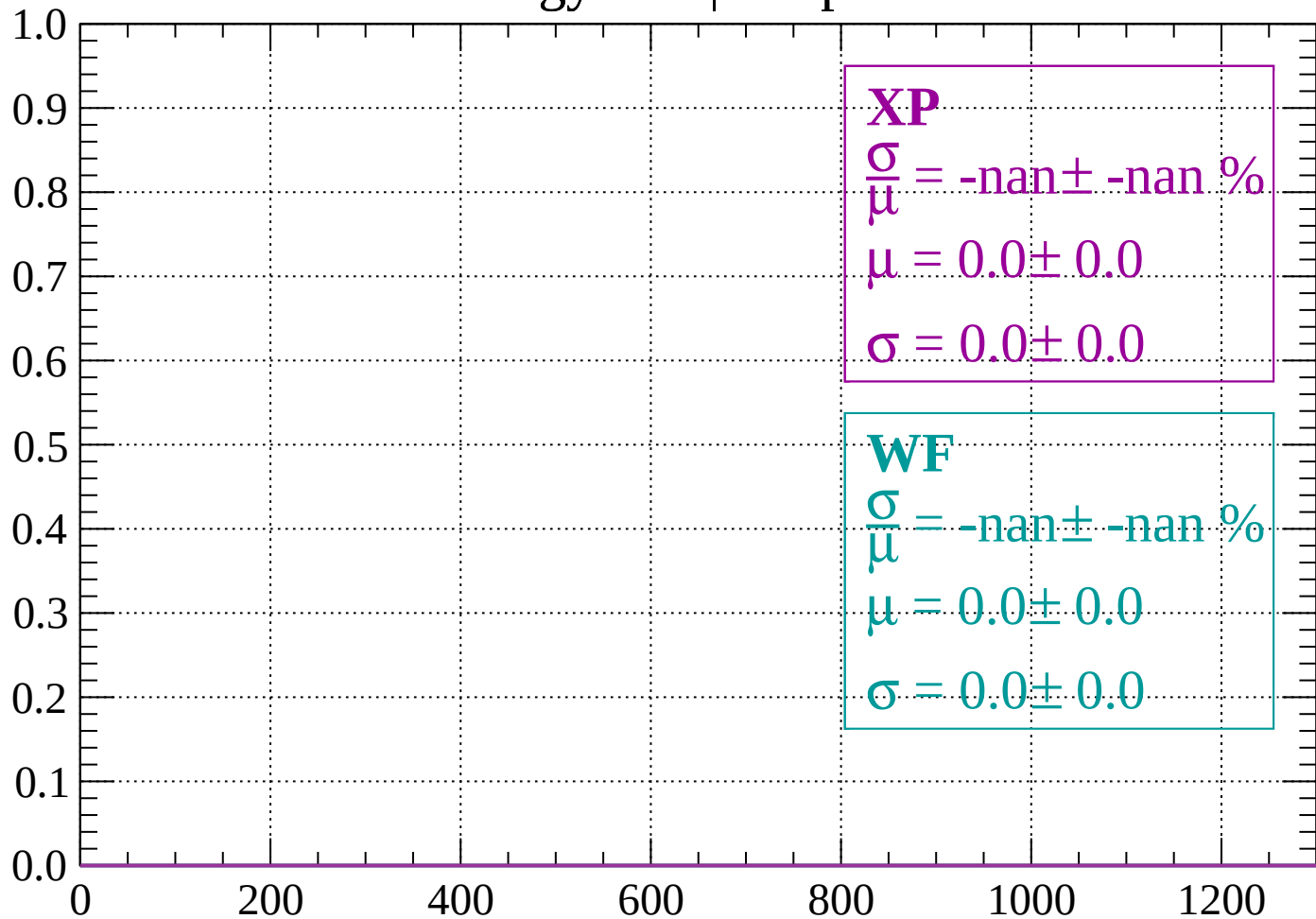
Count



dE/dx (ADC counts/cm)

Energy loss | $0 < p < 80$

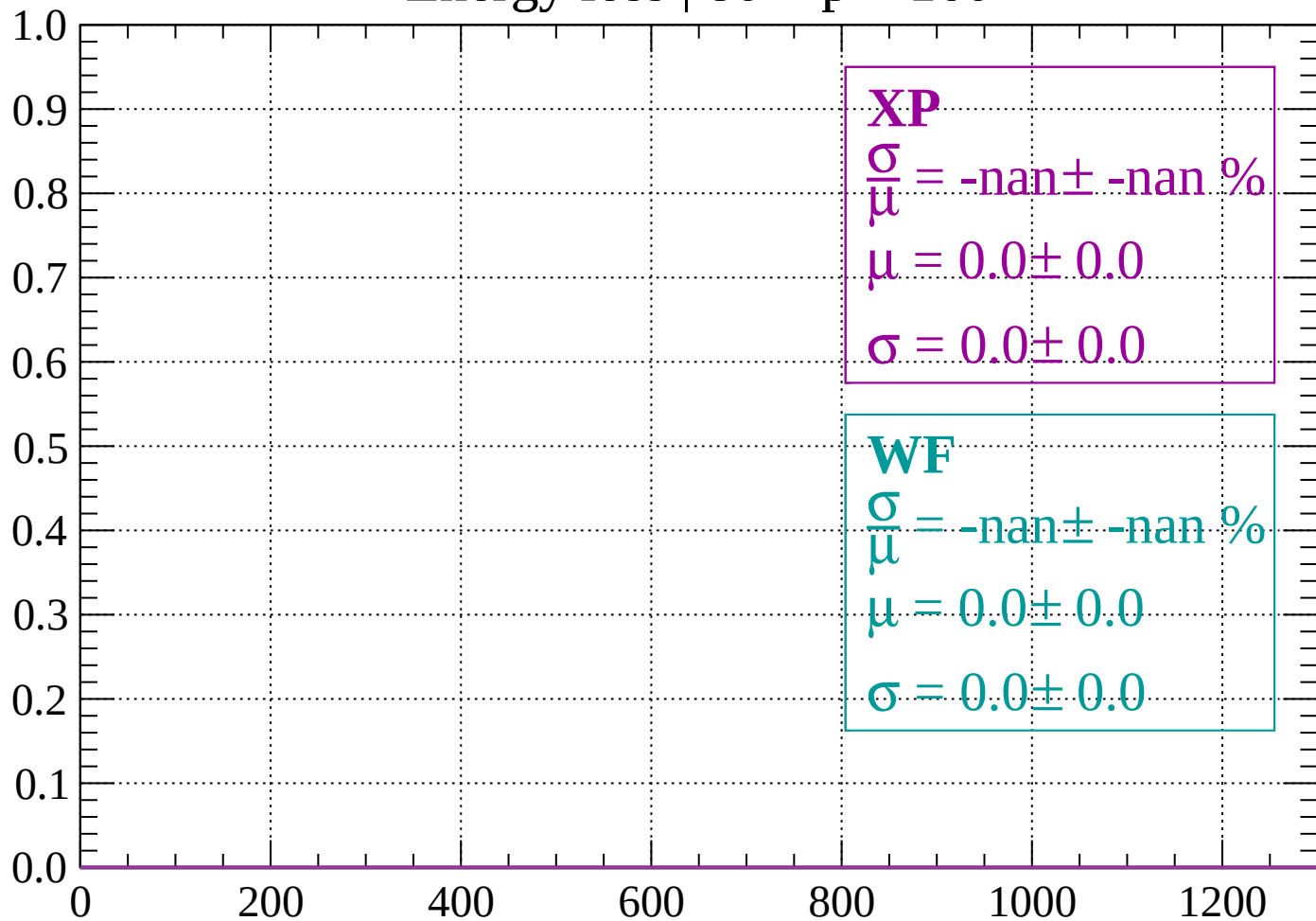
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < p < 160$

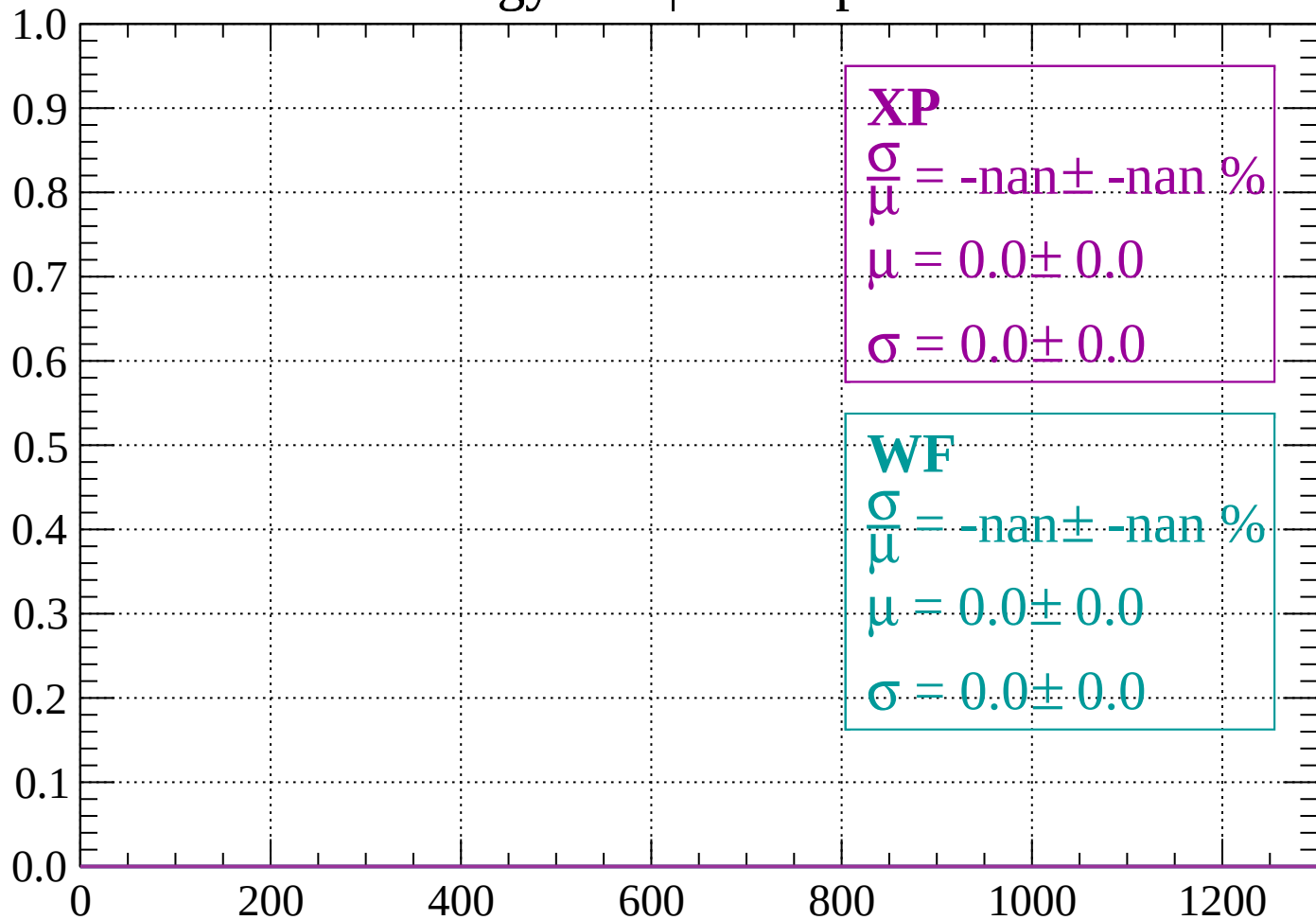
Count



dE/dx (ADC counts/cm)

Energy loss | $160 < p < 240$

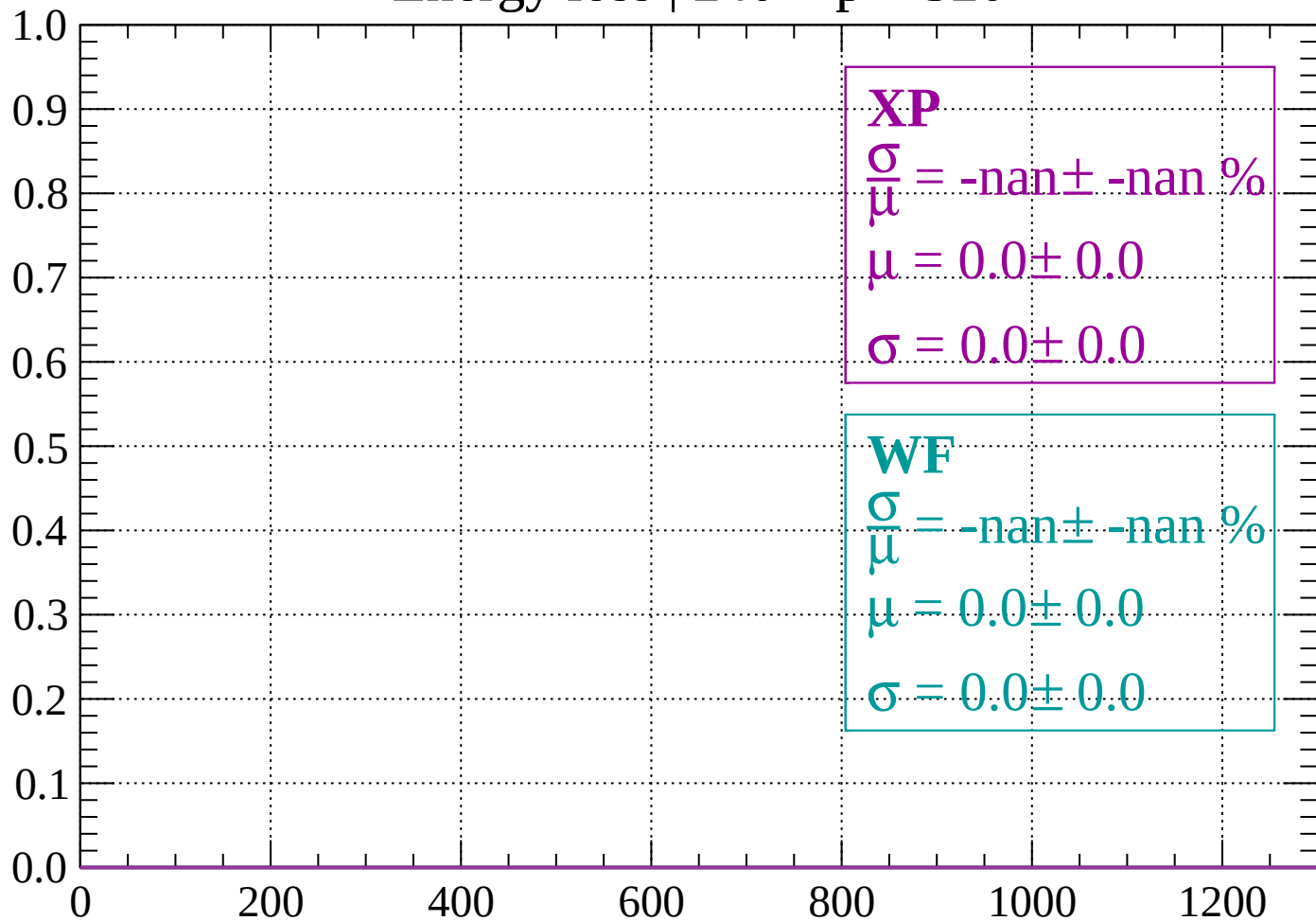
Count



dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

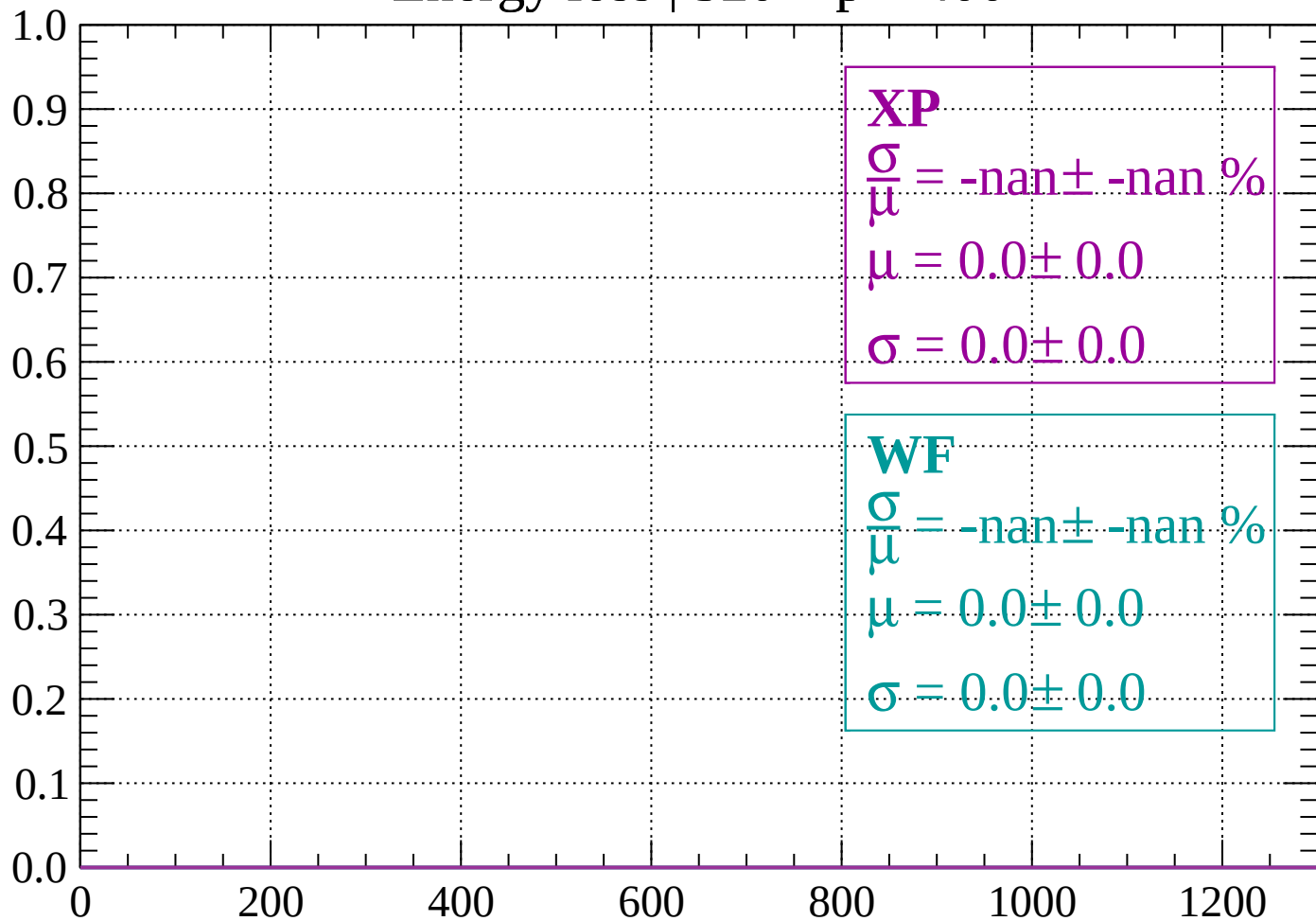
Count



dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

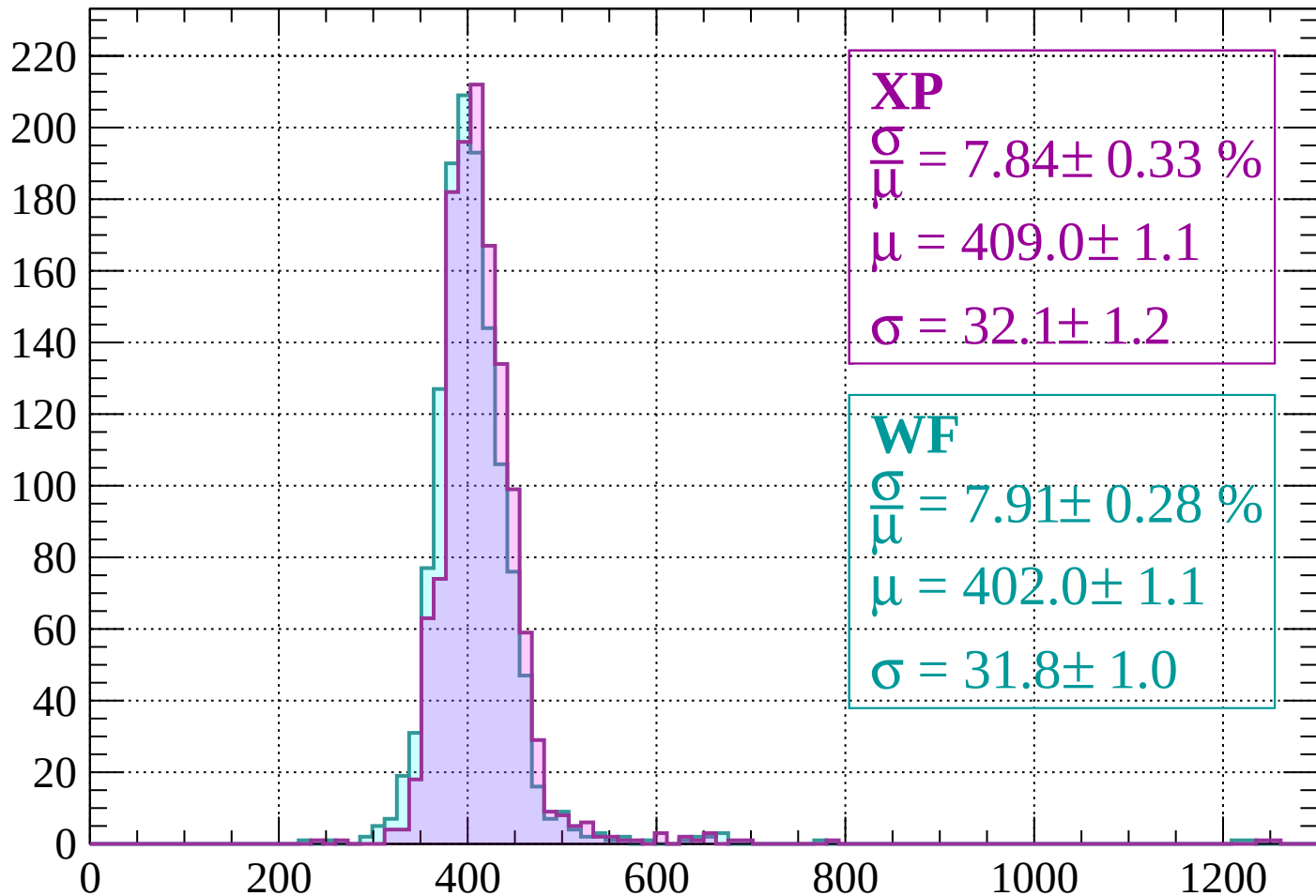
Count



dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.33 \%$$

$$\mu = 409.0 \pm 1.1$$

$$\sigma = 32.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.91 \pm 0.28 \%$$

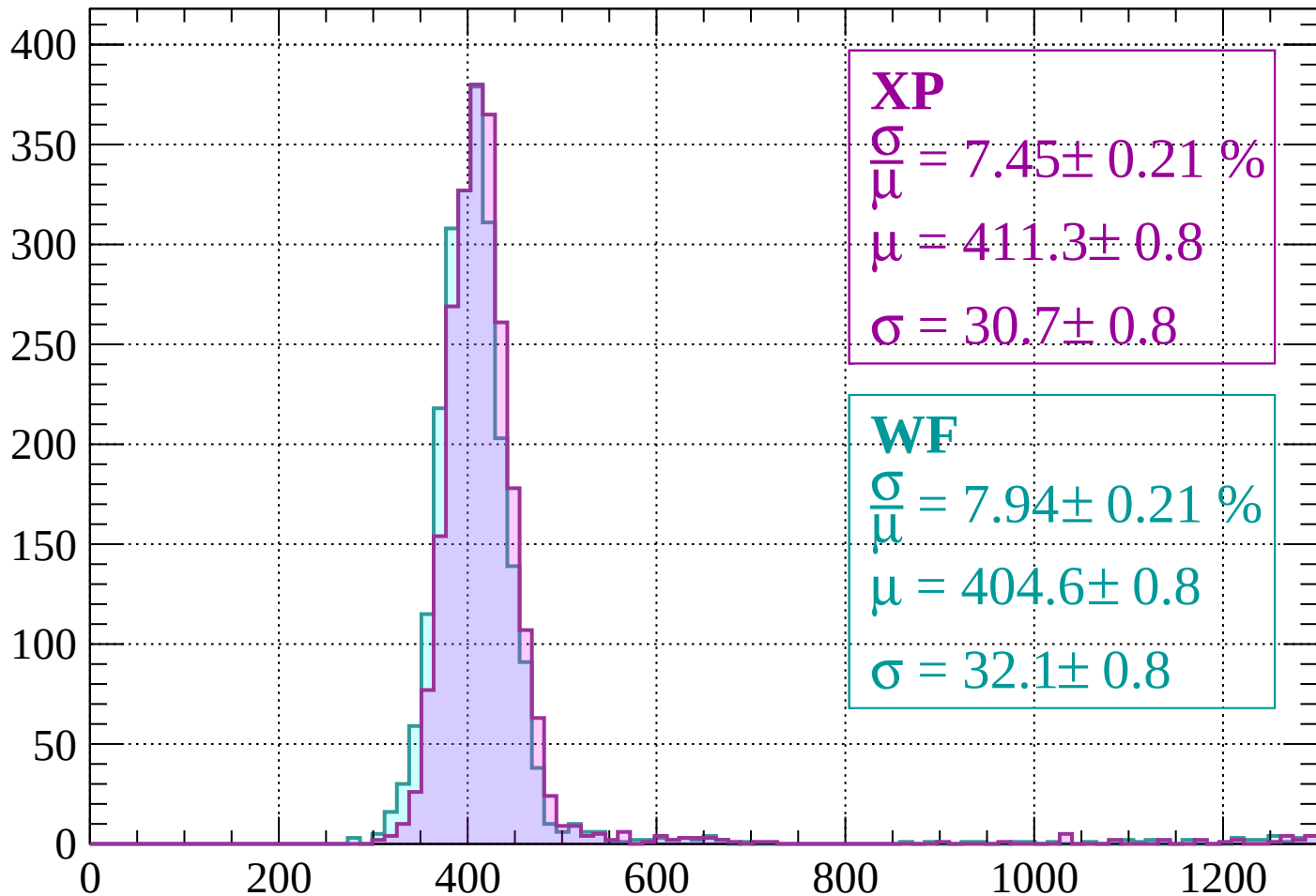
$$\mu = 402.0 \pm 1.1$$

$$\sigma = 31.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.45 \pm 0.21 \%$$

$$\mu = 411.3 \pm 0.8$$

$$\sigma = 30.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.94 \pm 0.21 \%$$

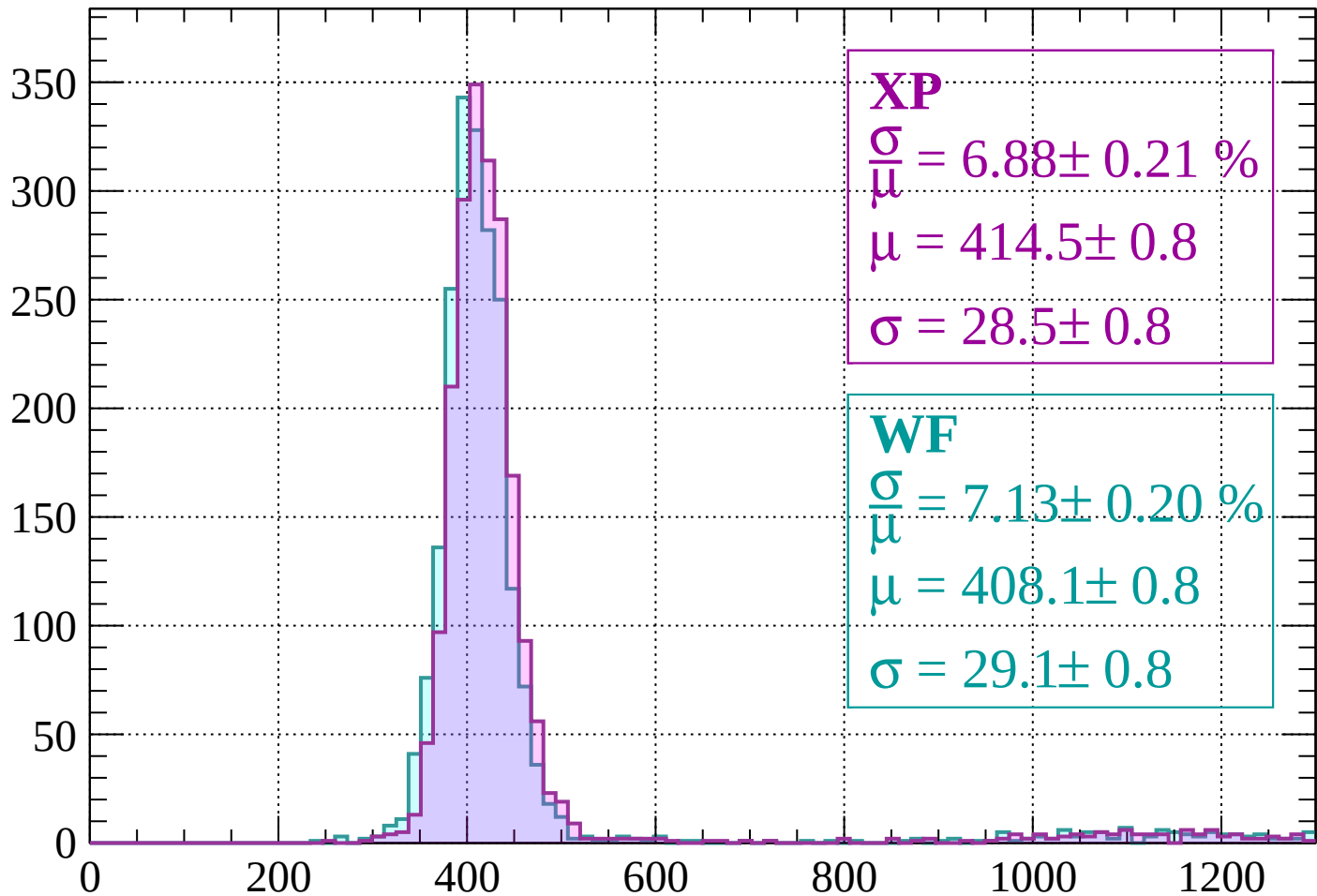
$$\mu = 404.6 \pm 0.8$$

$$\sigma = 32.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 6.88 \pm 0.21 \%$$

$$\mu = 414.5 \pm 0.8$$

$$\sigma = 28.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.13 \pm 0.20 \%$$

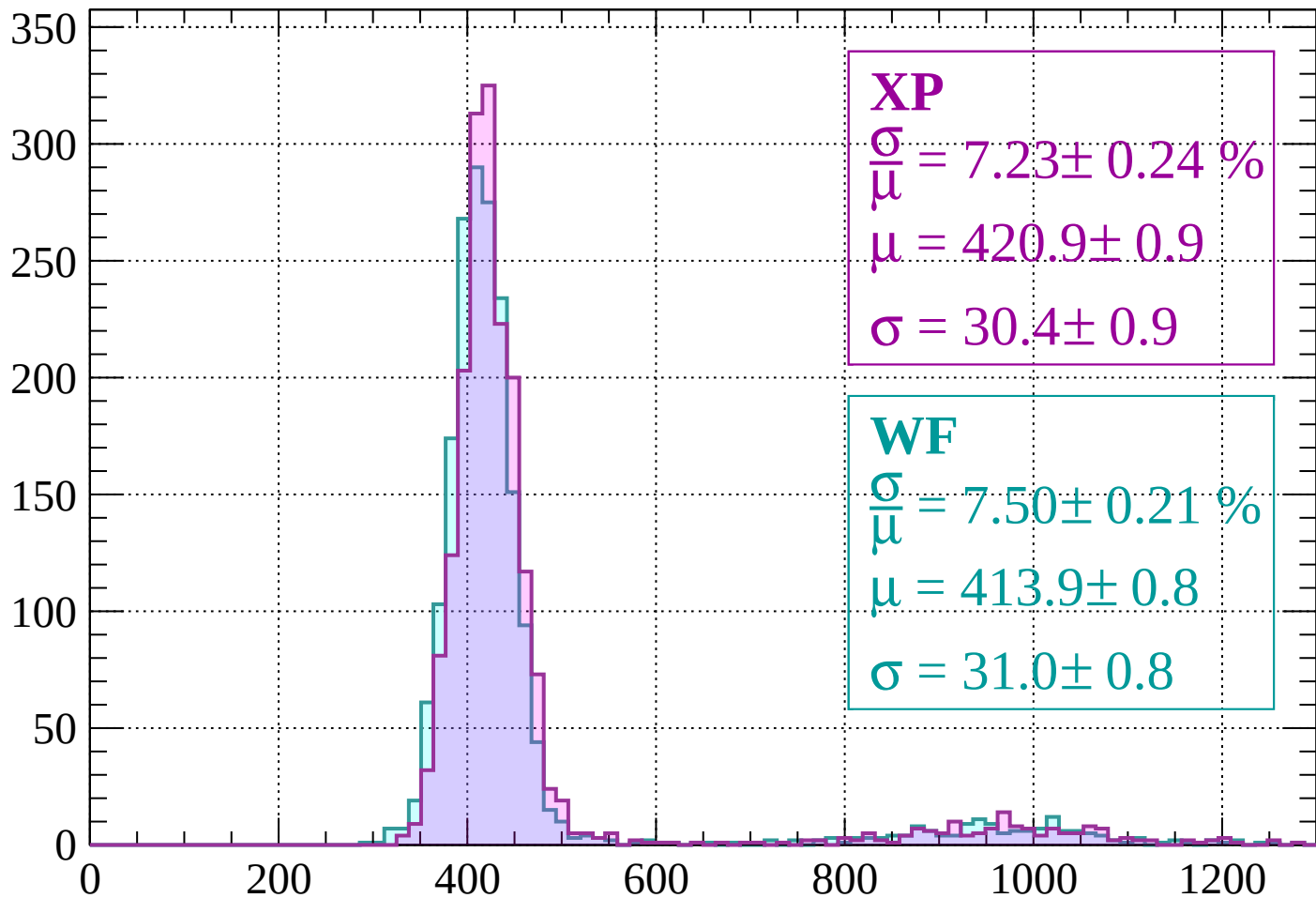
$$\mu = 408.1 \pm 0.8$$

$$\sigma = 29.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$

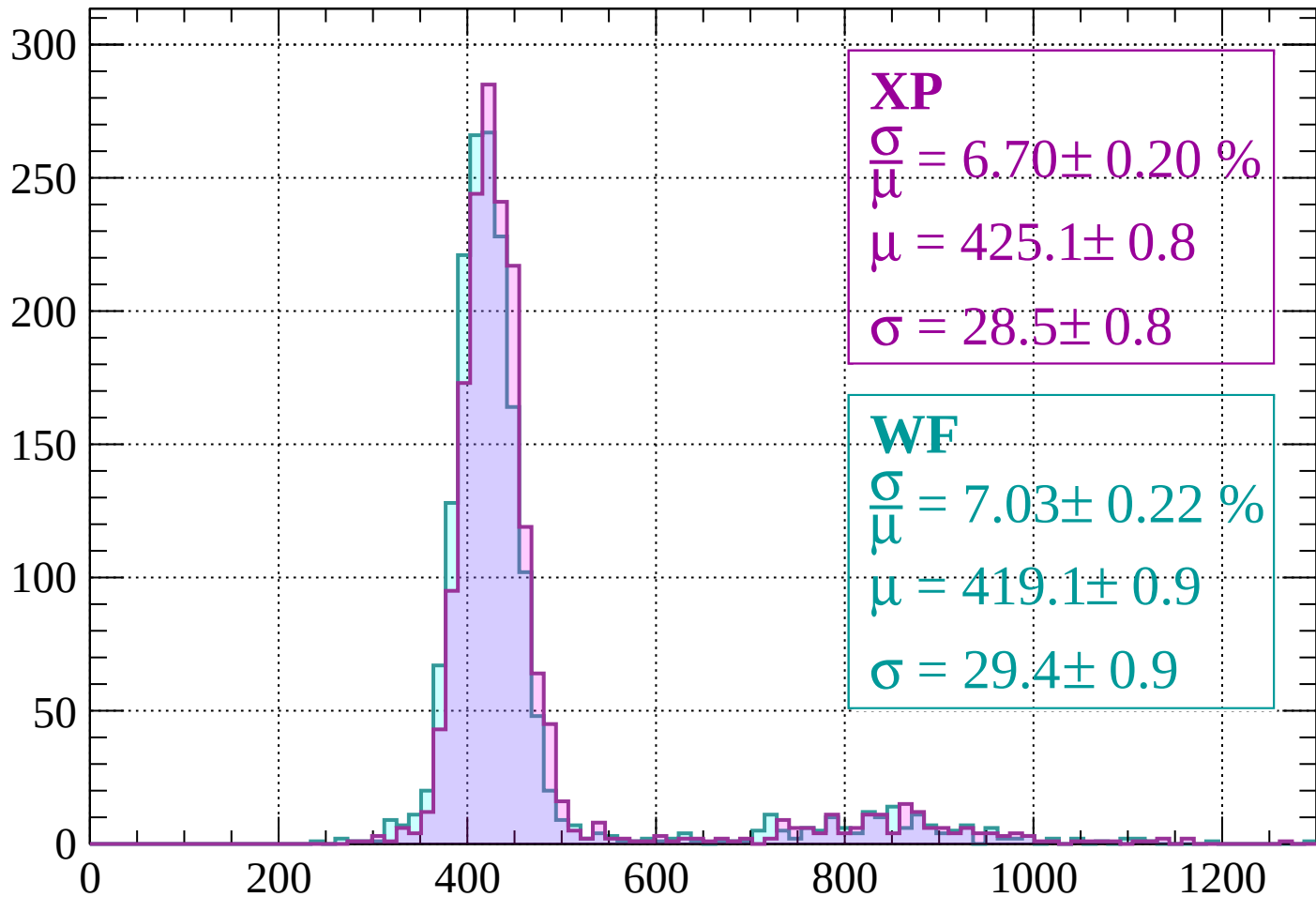
Count



dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 6.70 \pm 0.20 \%$$

$$\mu = 425.1 \pm 0.8$$

$$\sigma = 28.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.03 \pm 0.22 \%$$

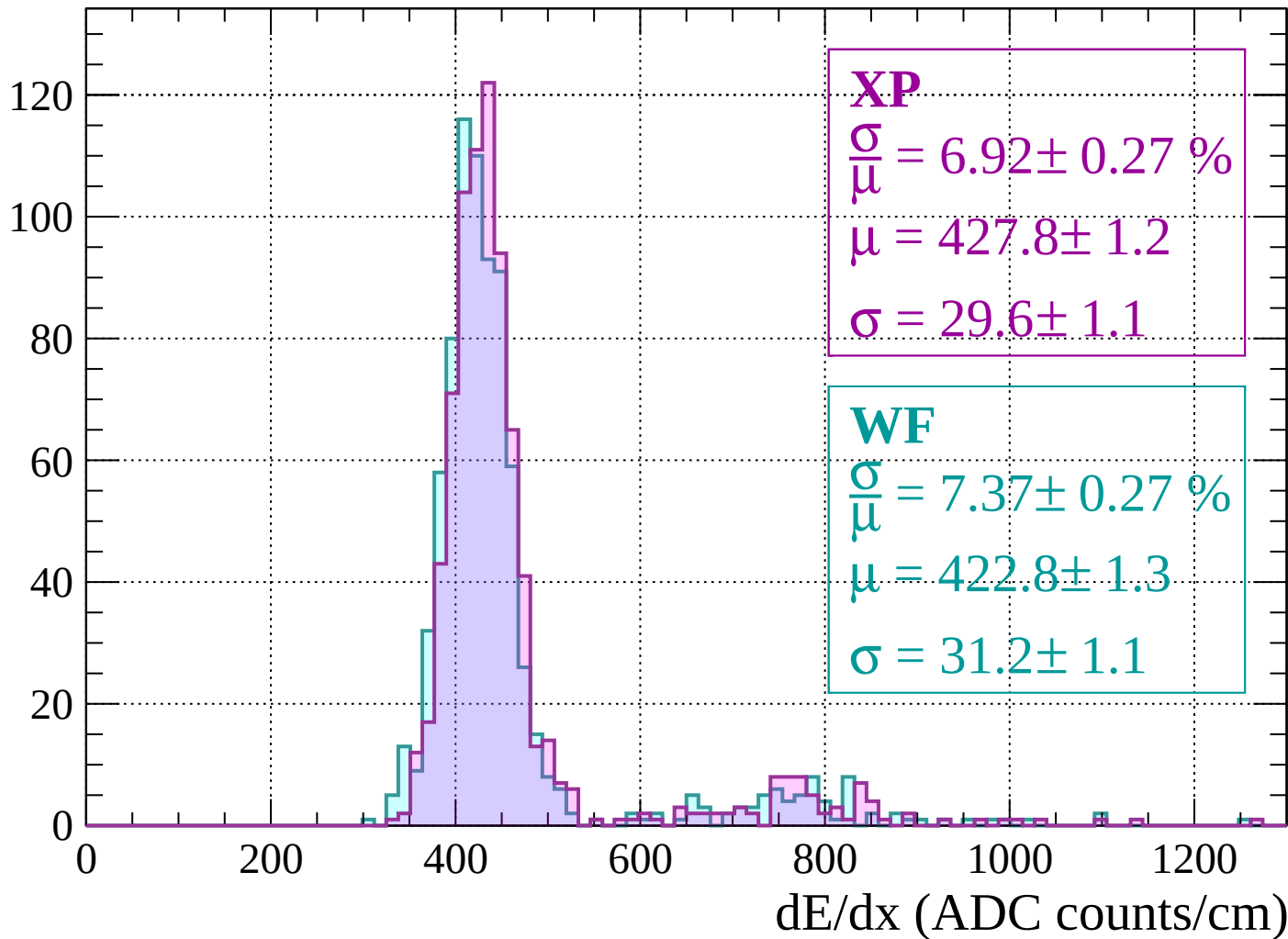
$$\mu = 419.1 \pm 0.9$$

$$\sigma = 29.4 \pm 0.9$$

dE/dx (ADC counts/cm)

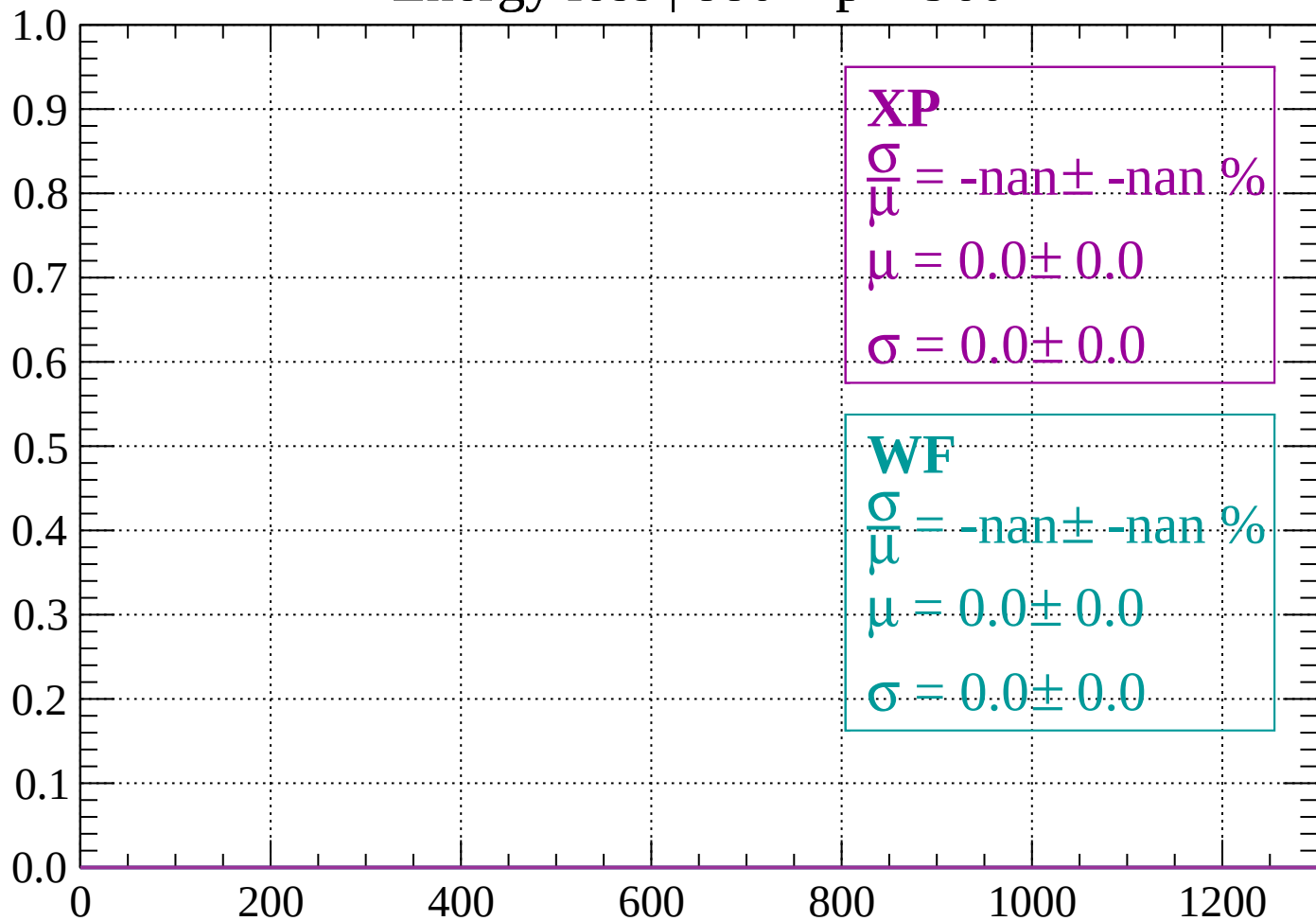
Energy loss | $800 < p < 880$

Count



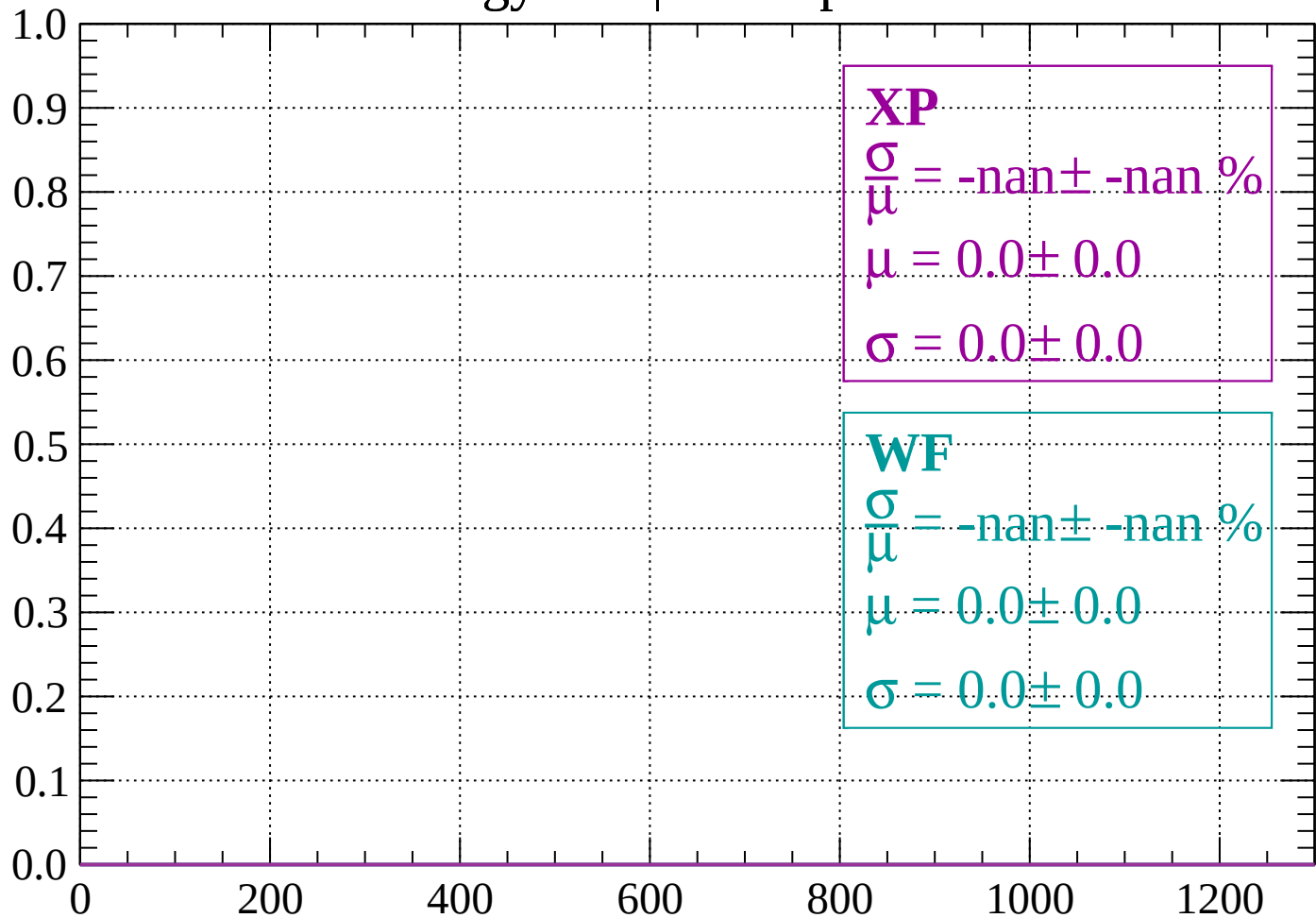
Energy loss | $880 < p < 960$

Count



Energy loss | $960 < p < 1040$

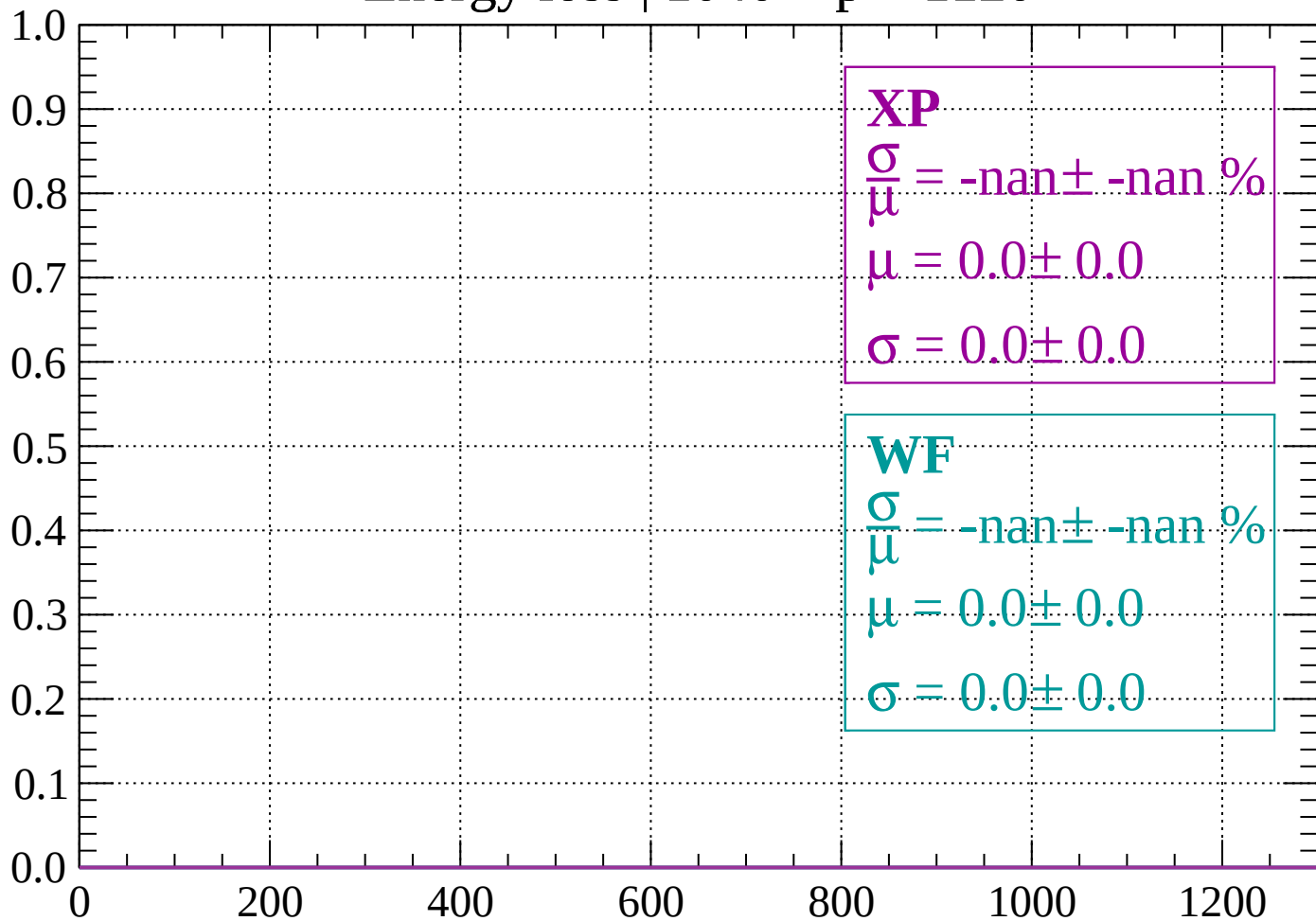
Count



dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

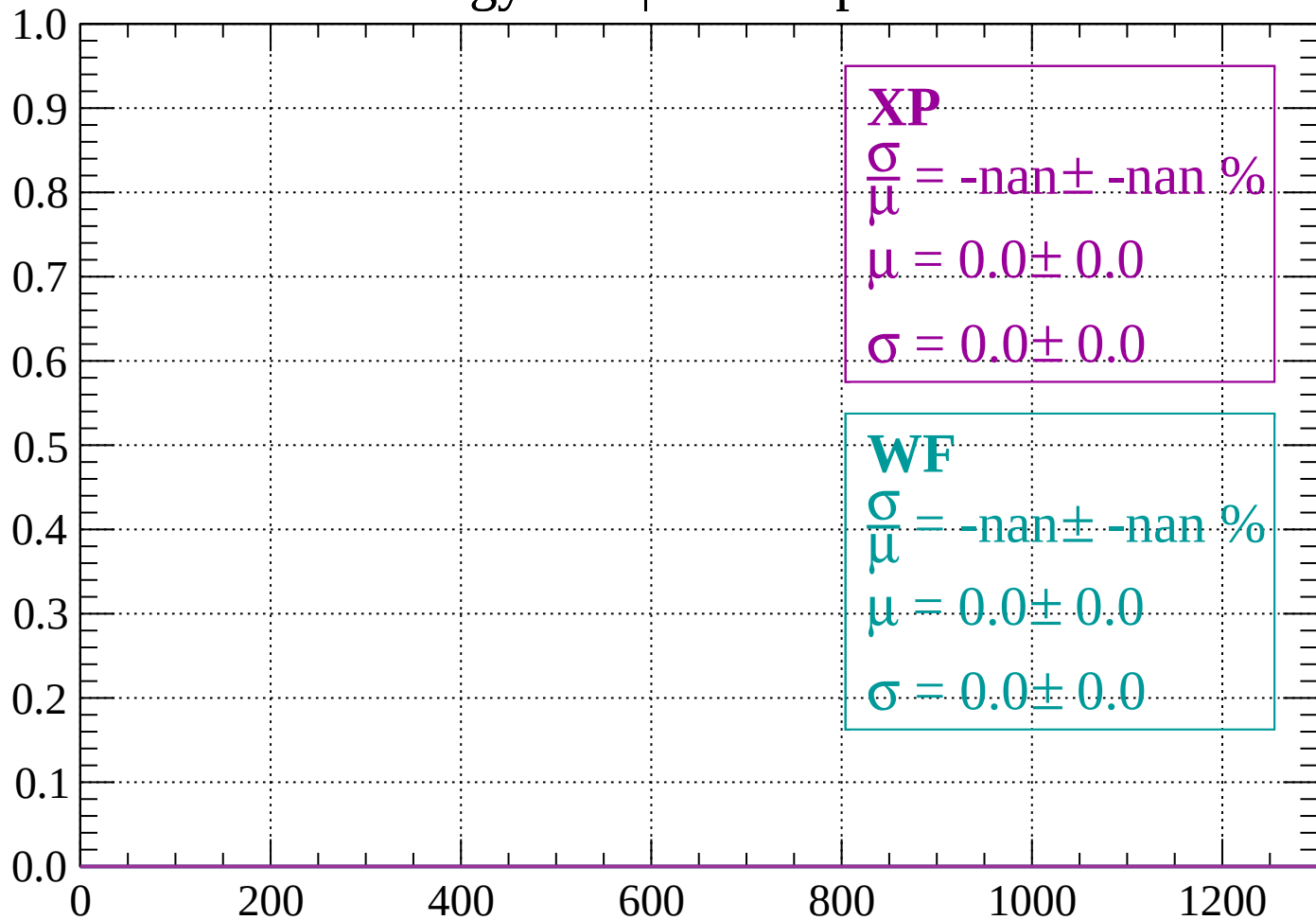
Count



dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

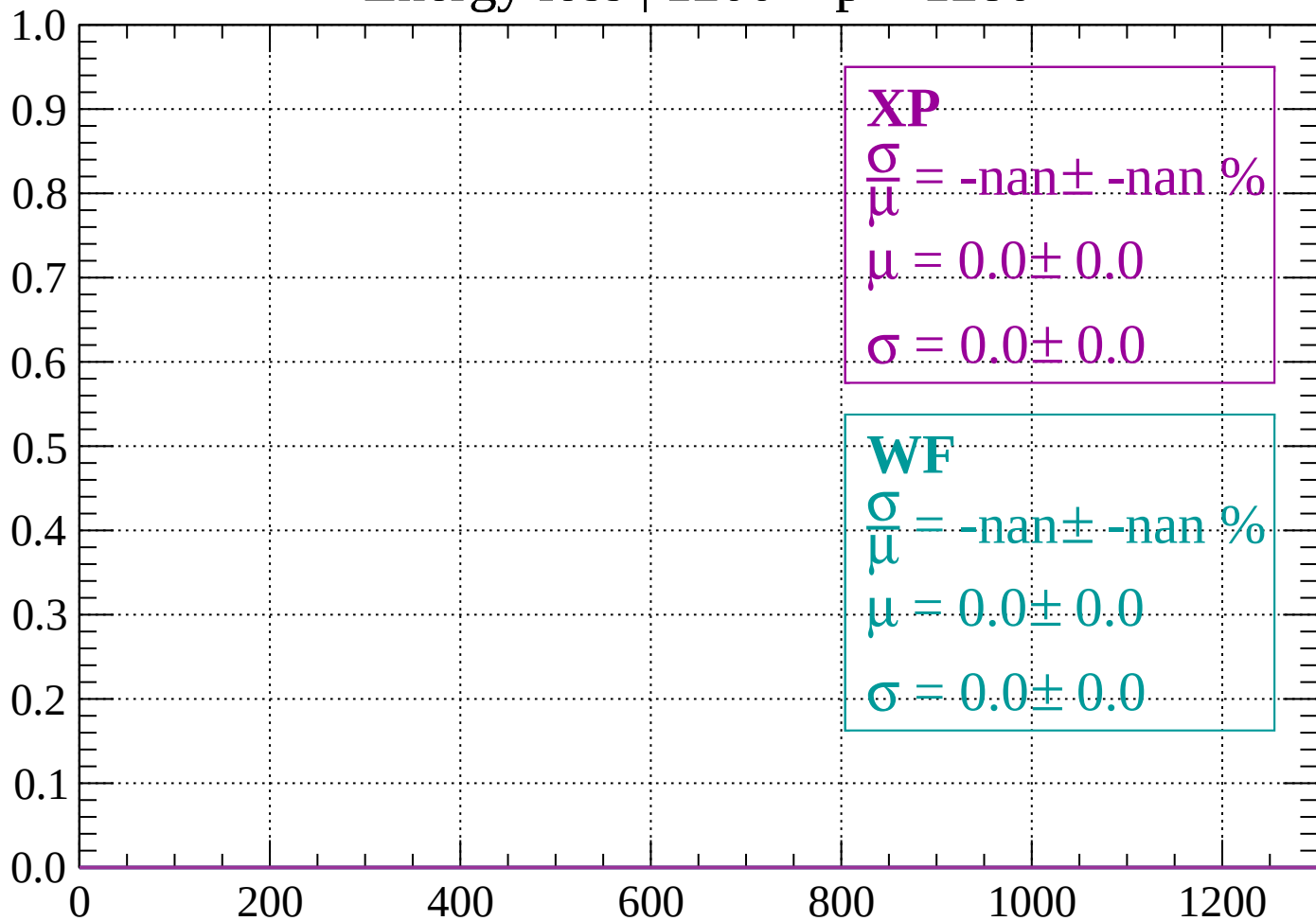
Count



dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

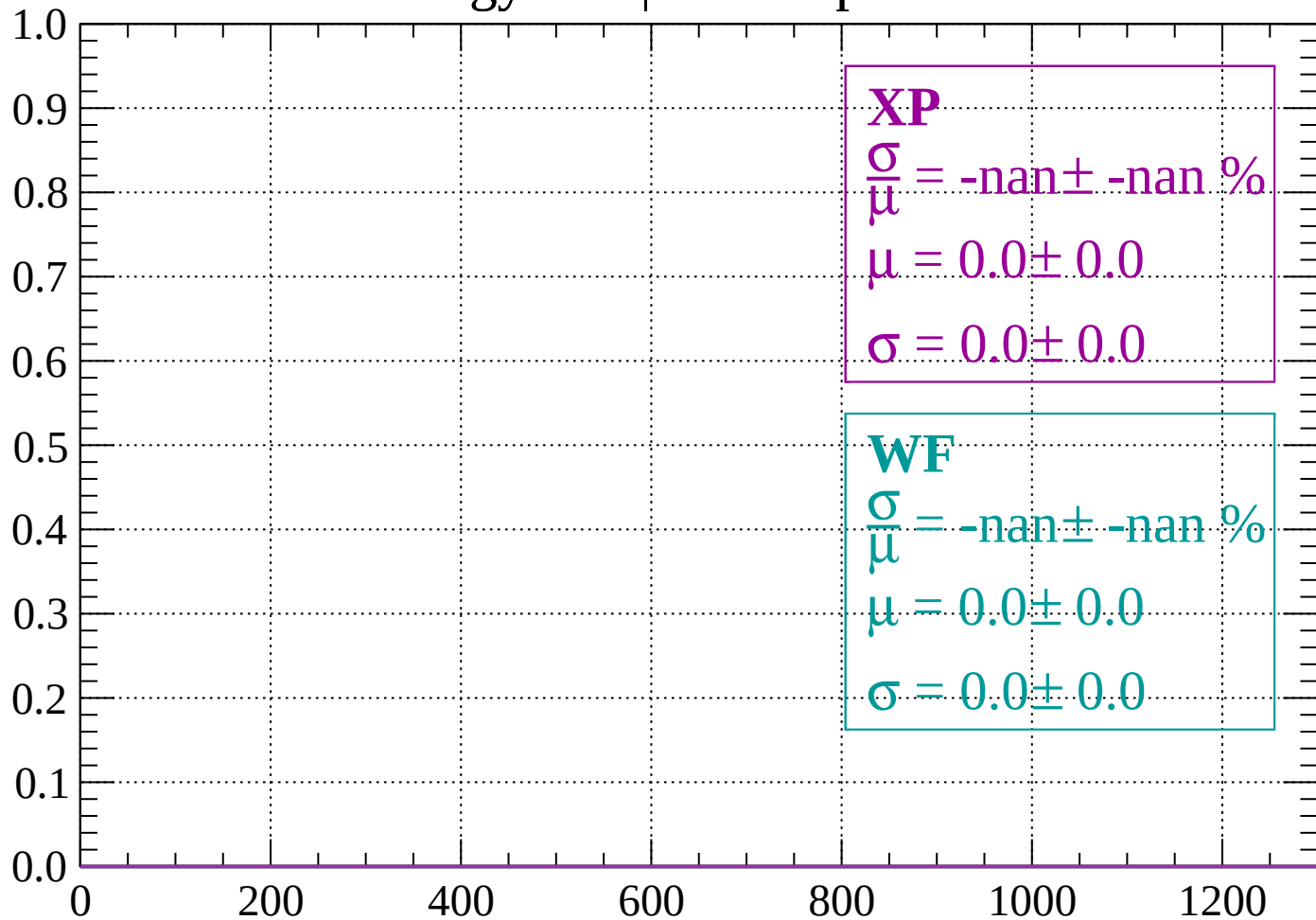
Count



dE/dx (ADC counts/cm)

Energy loss | 1280 < p < 1360

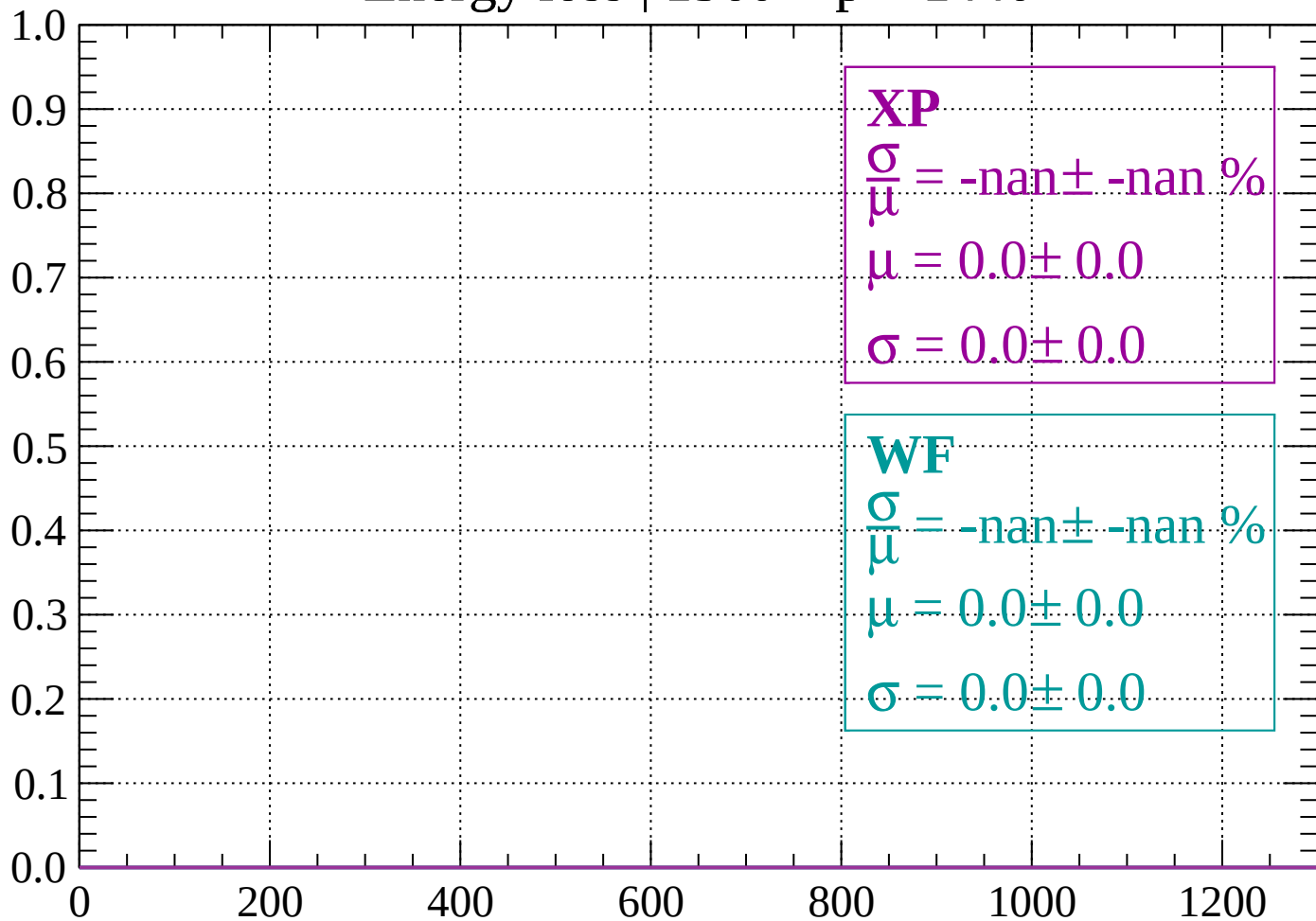
Count



dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

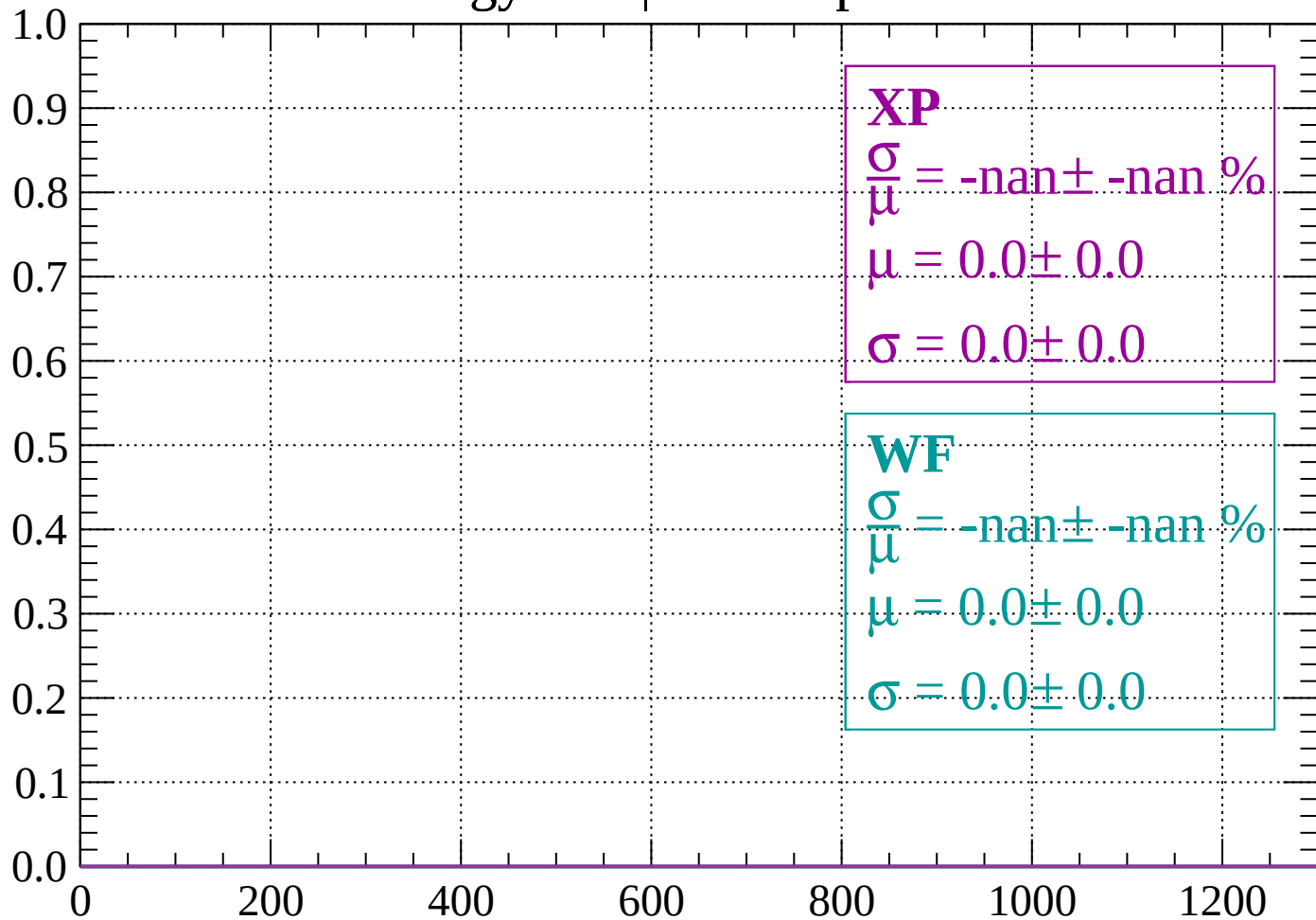
Count



dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

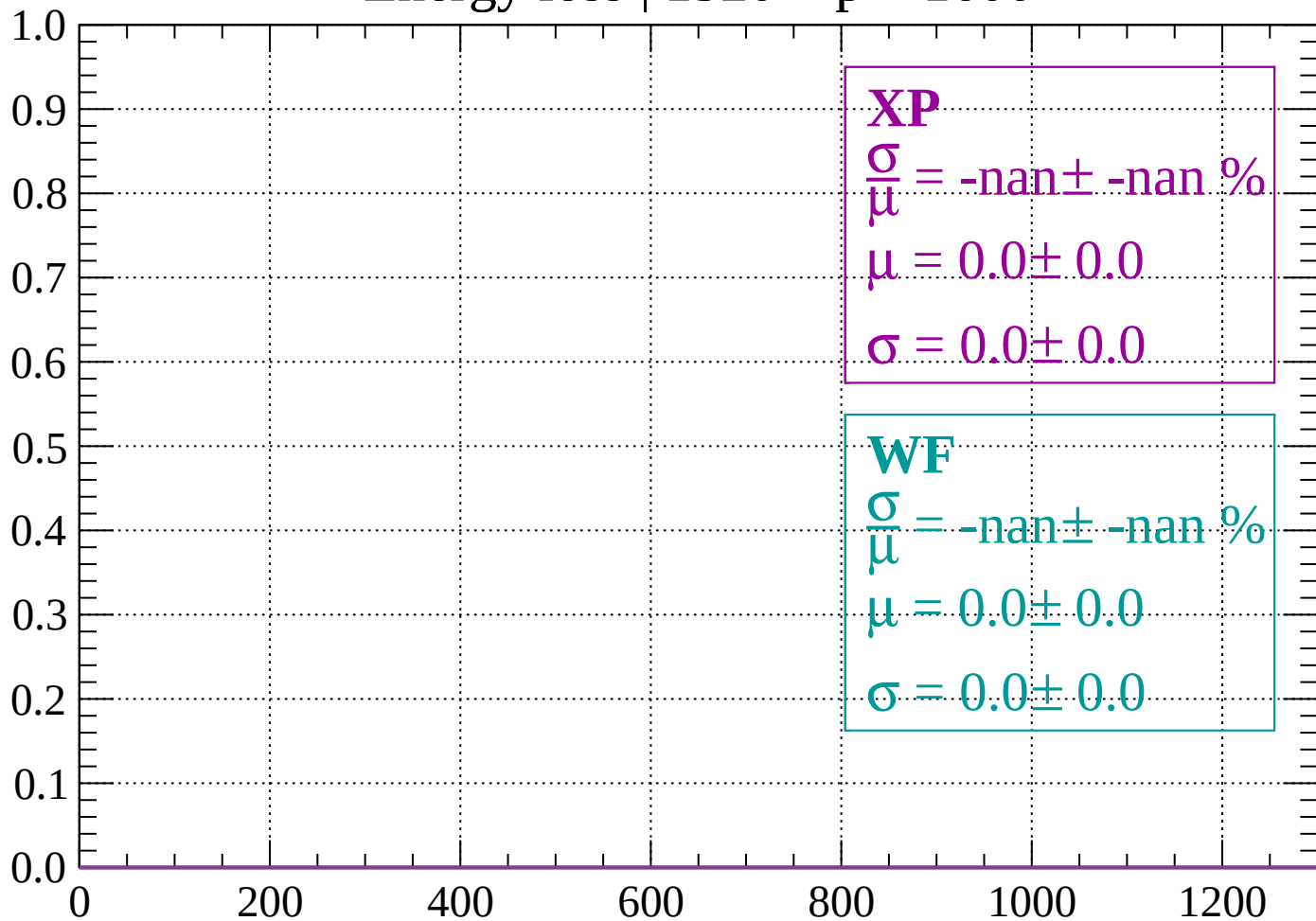
Count



dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

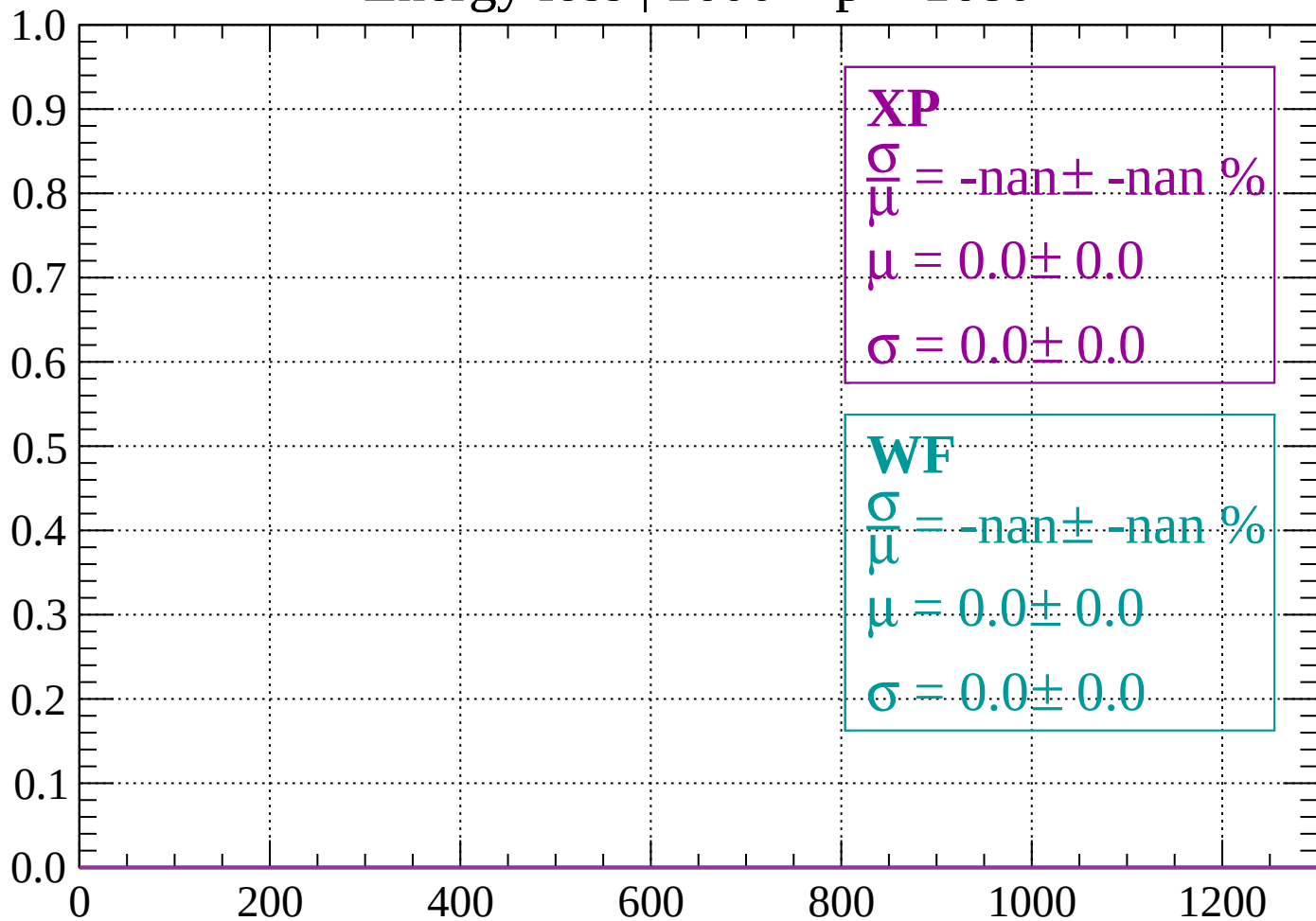
Count



dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

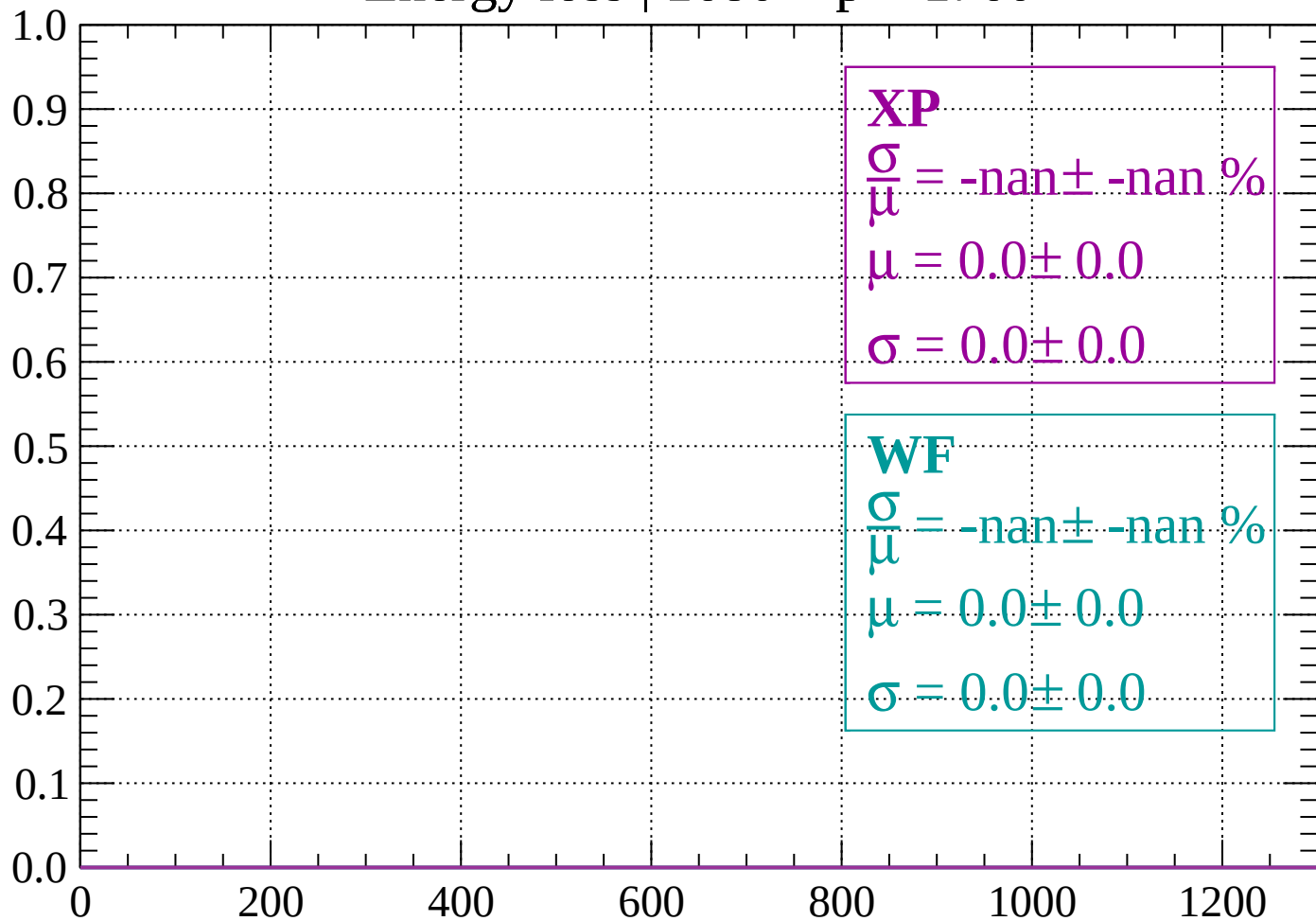
Count



dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

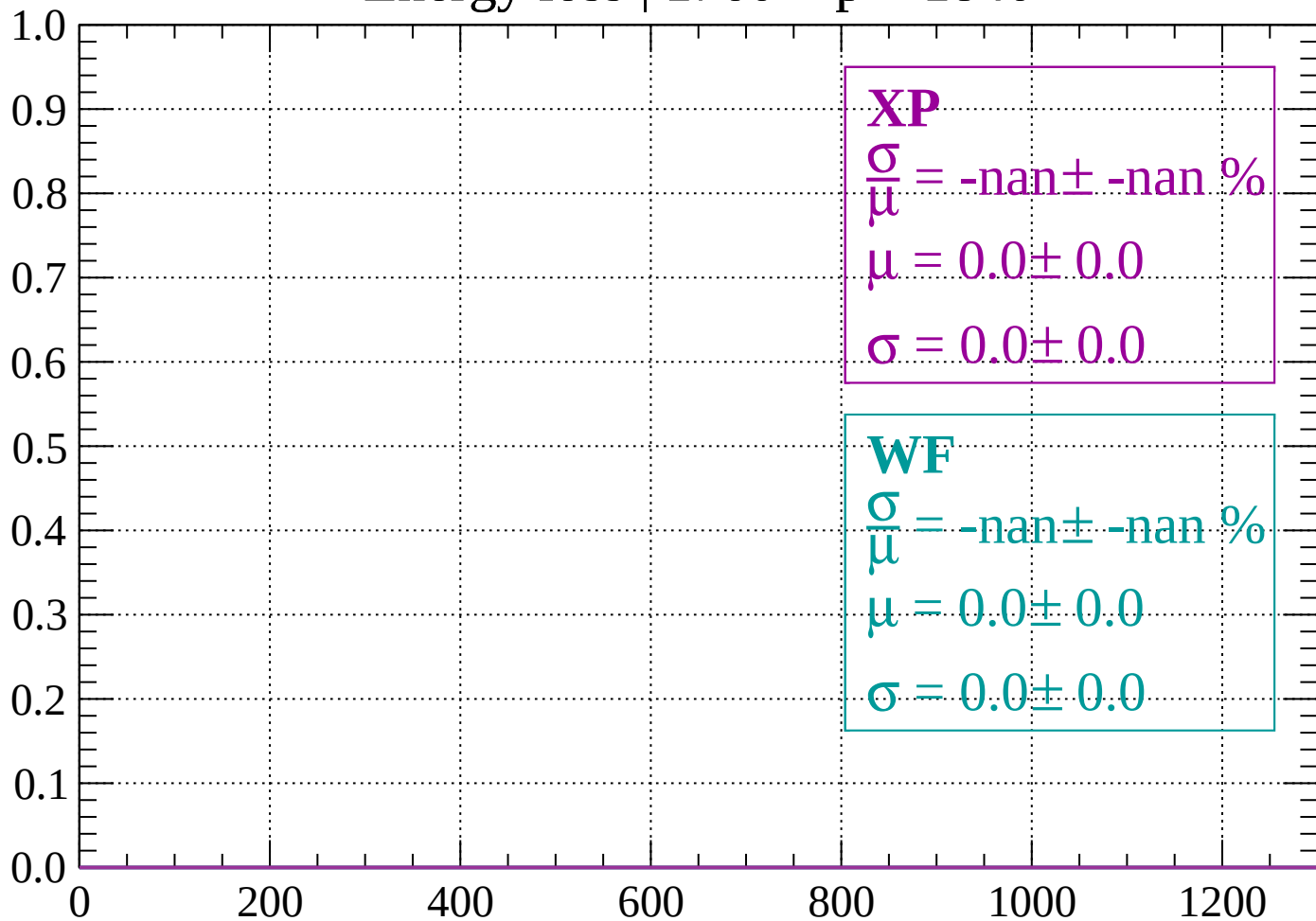
Count



dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1840$

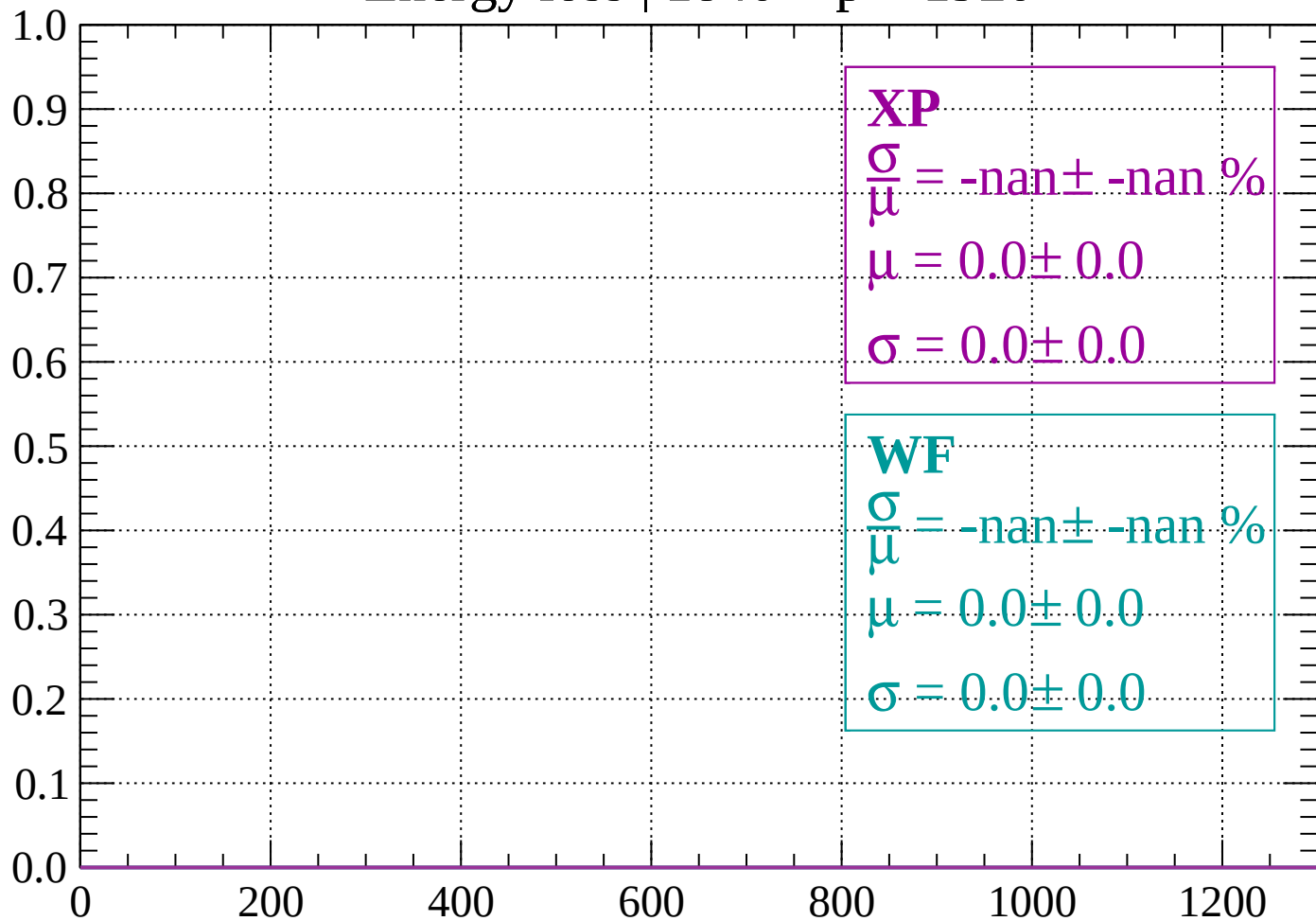
Count



dE/dx (ADC counts/cm)

Energy loss | 1840 < p < 1920

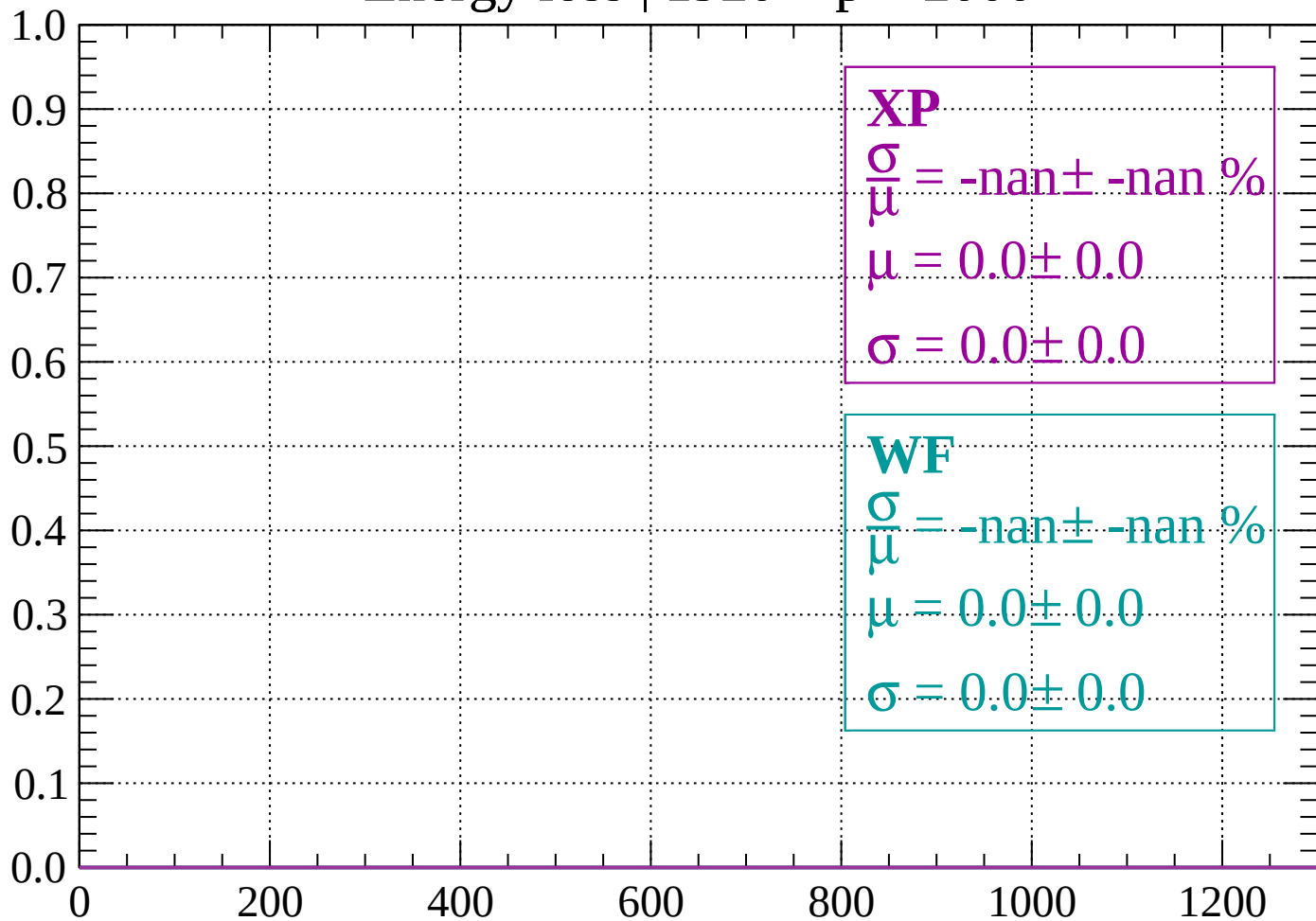
Count



dE/dx (ADC counts/cm)

Energy loss | 1920 < p < 2000

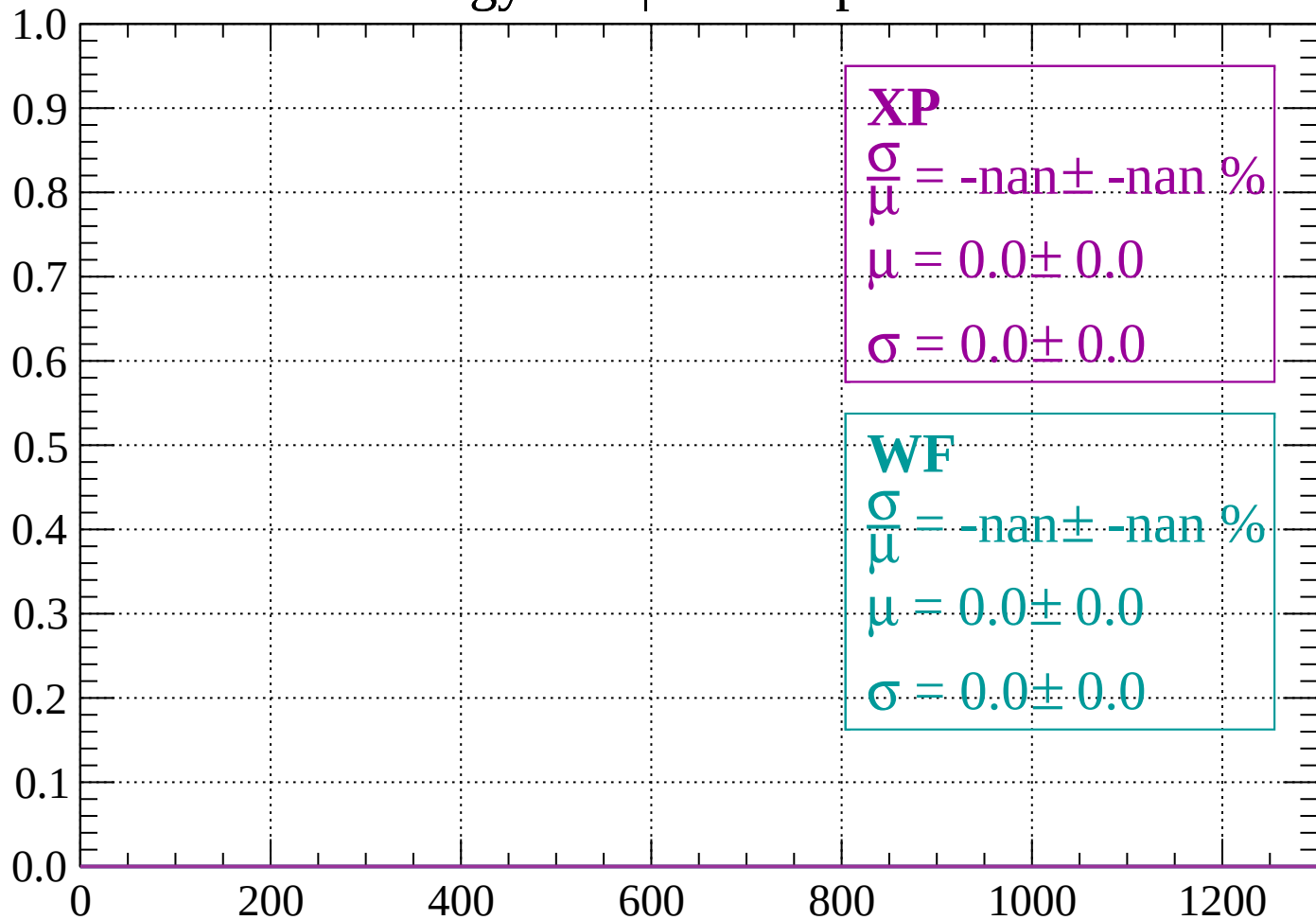
Count



dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2080$

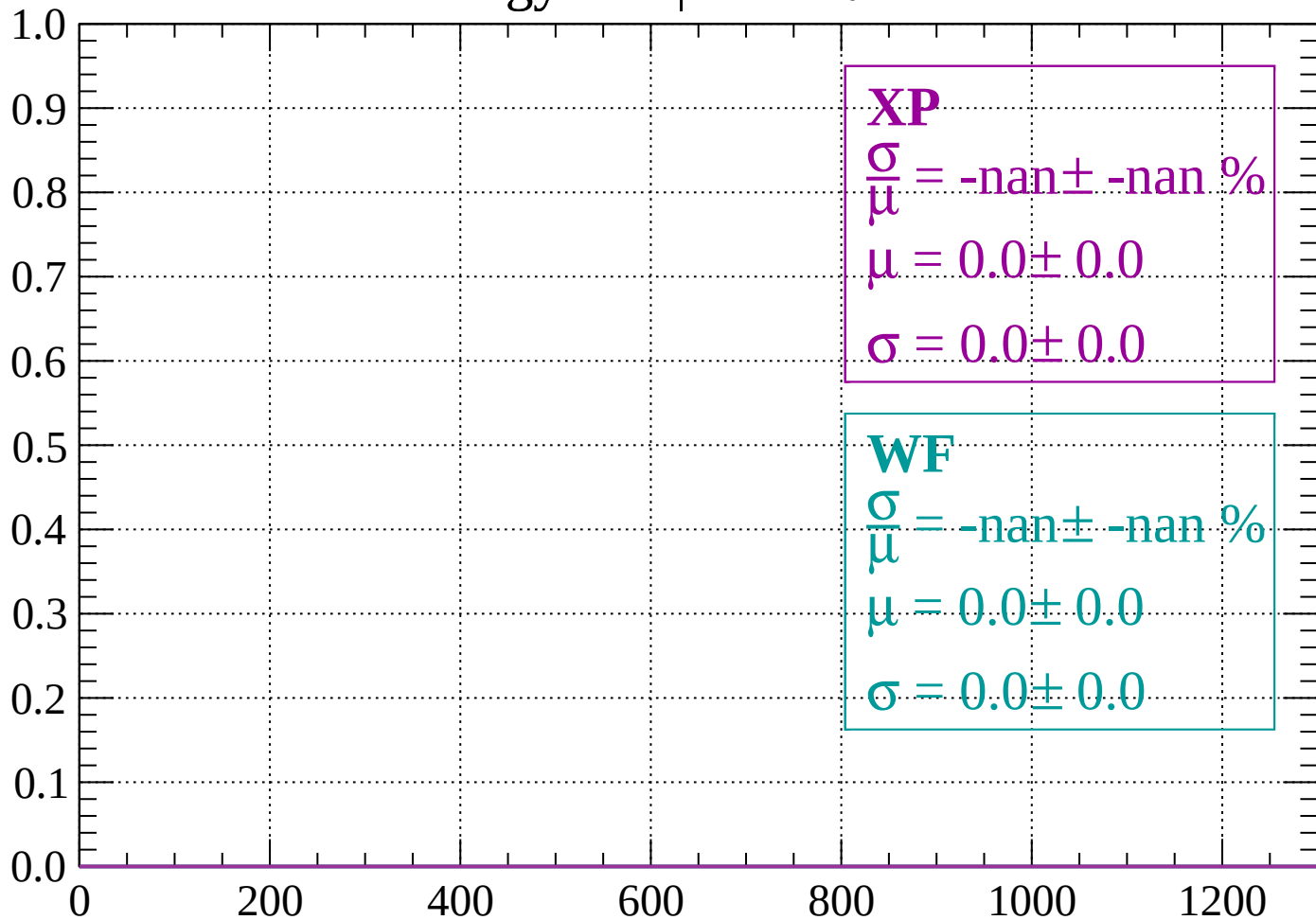
Count



dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

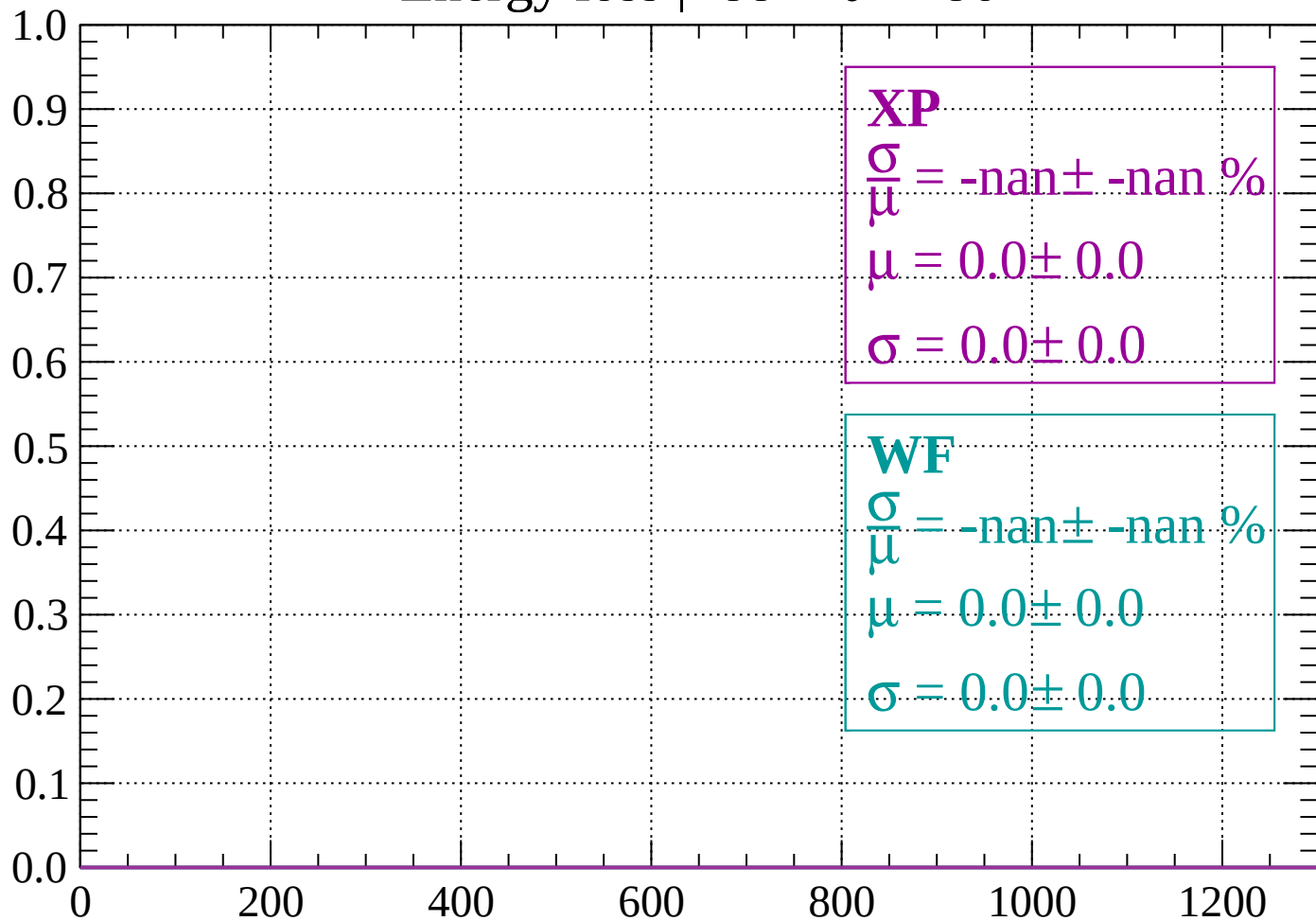
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

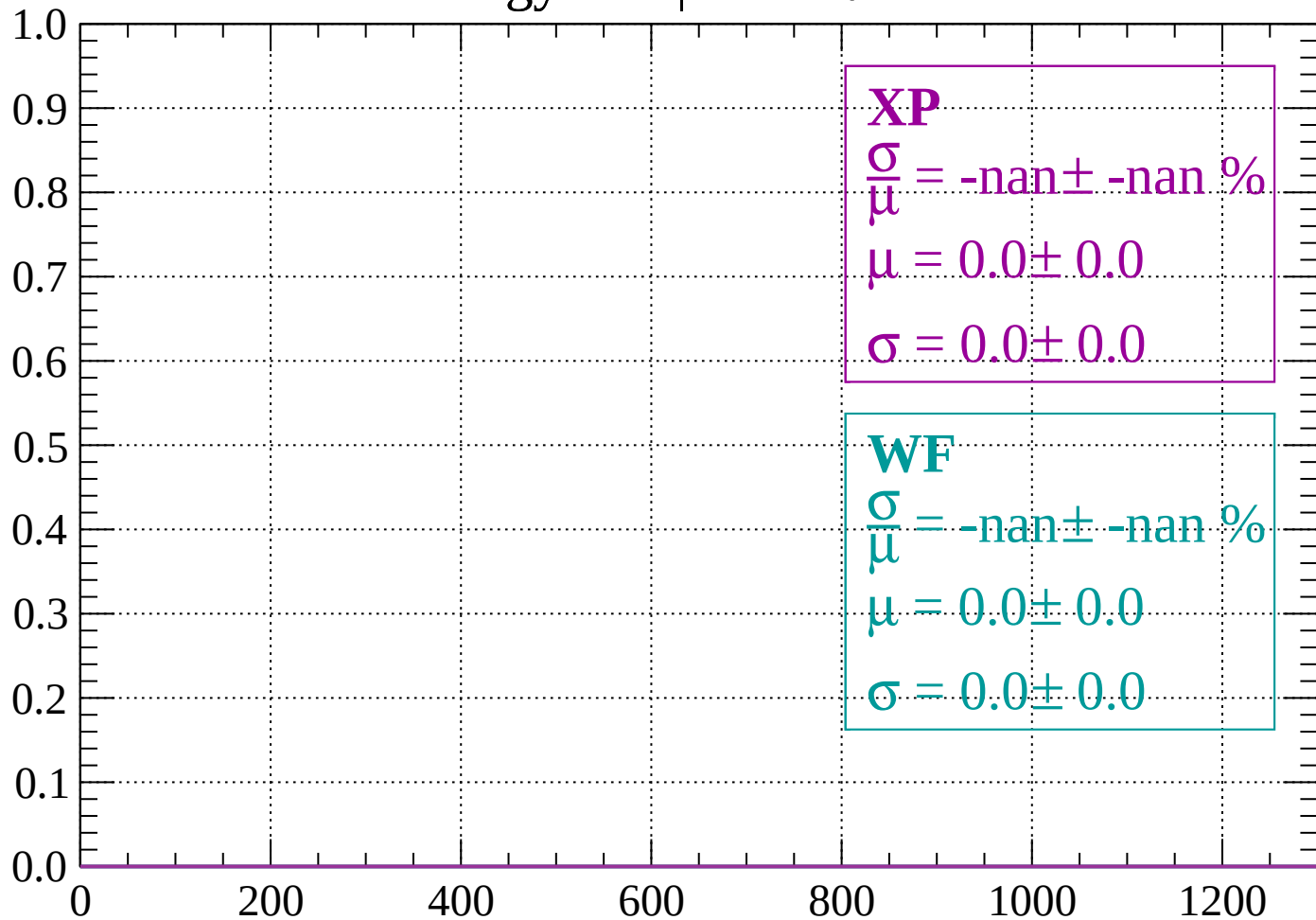
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

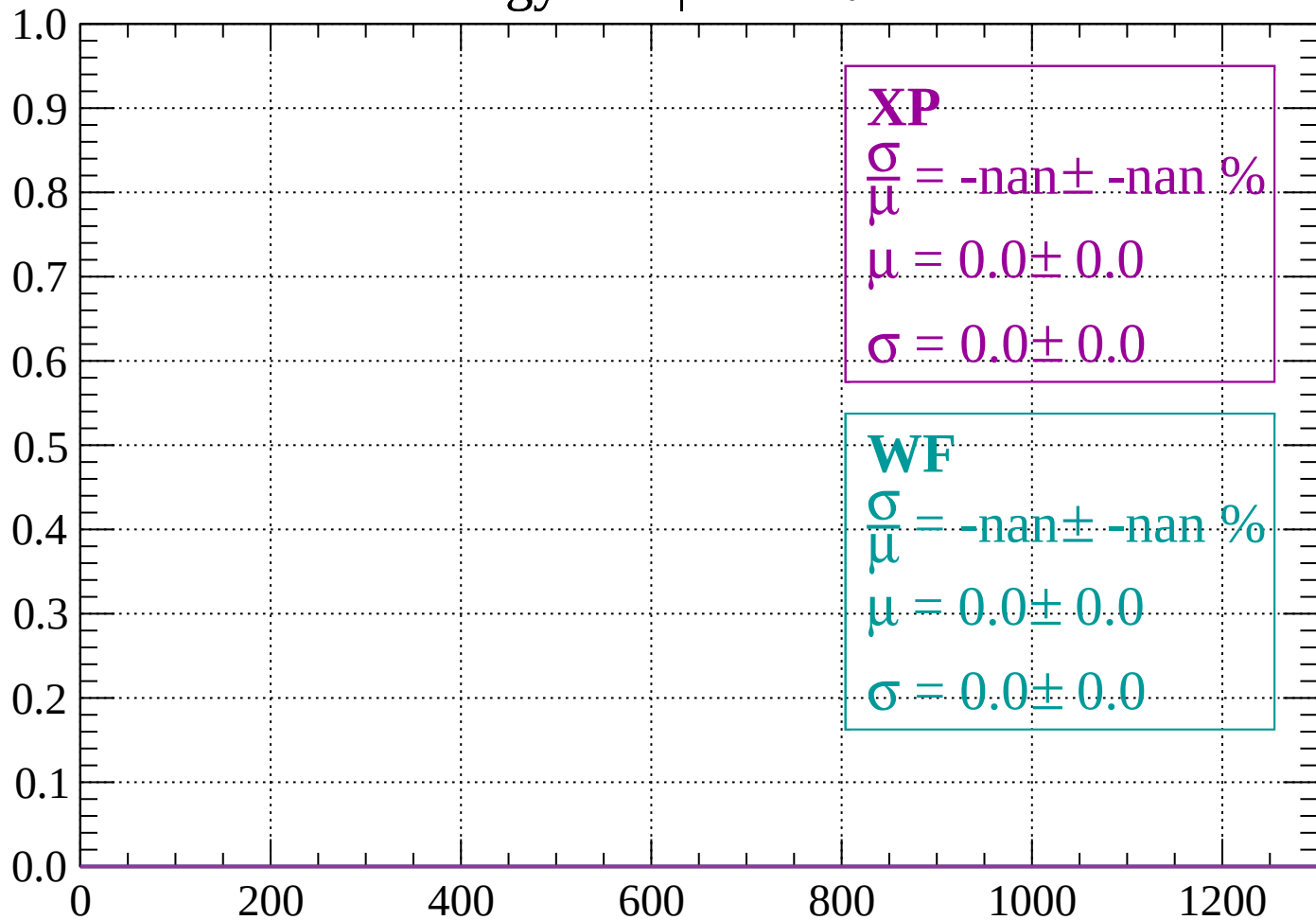
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

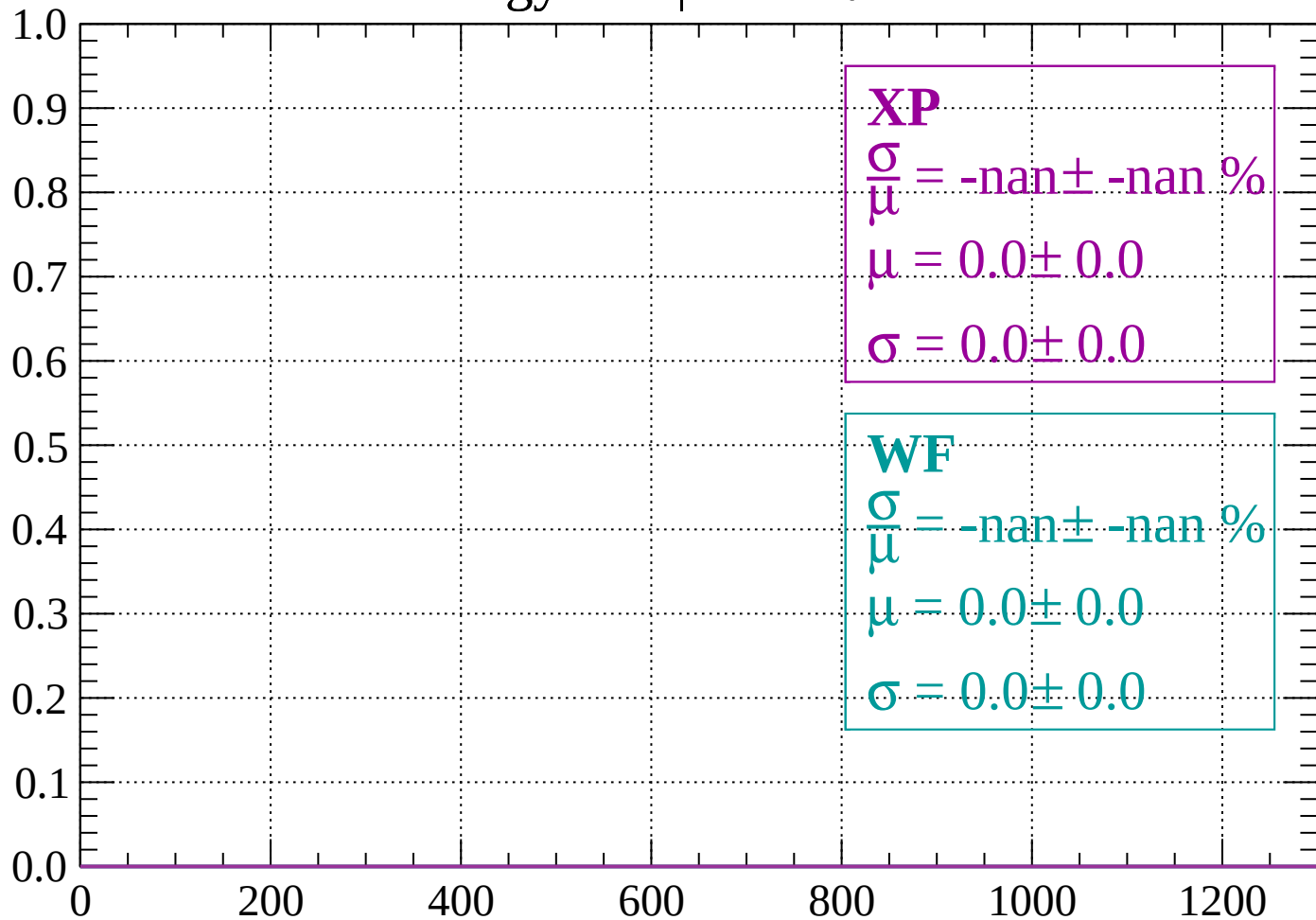
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

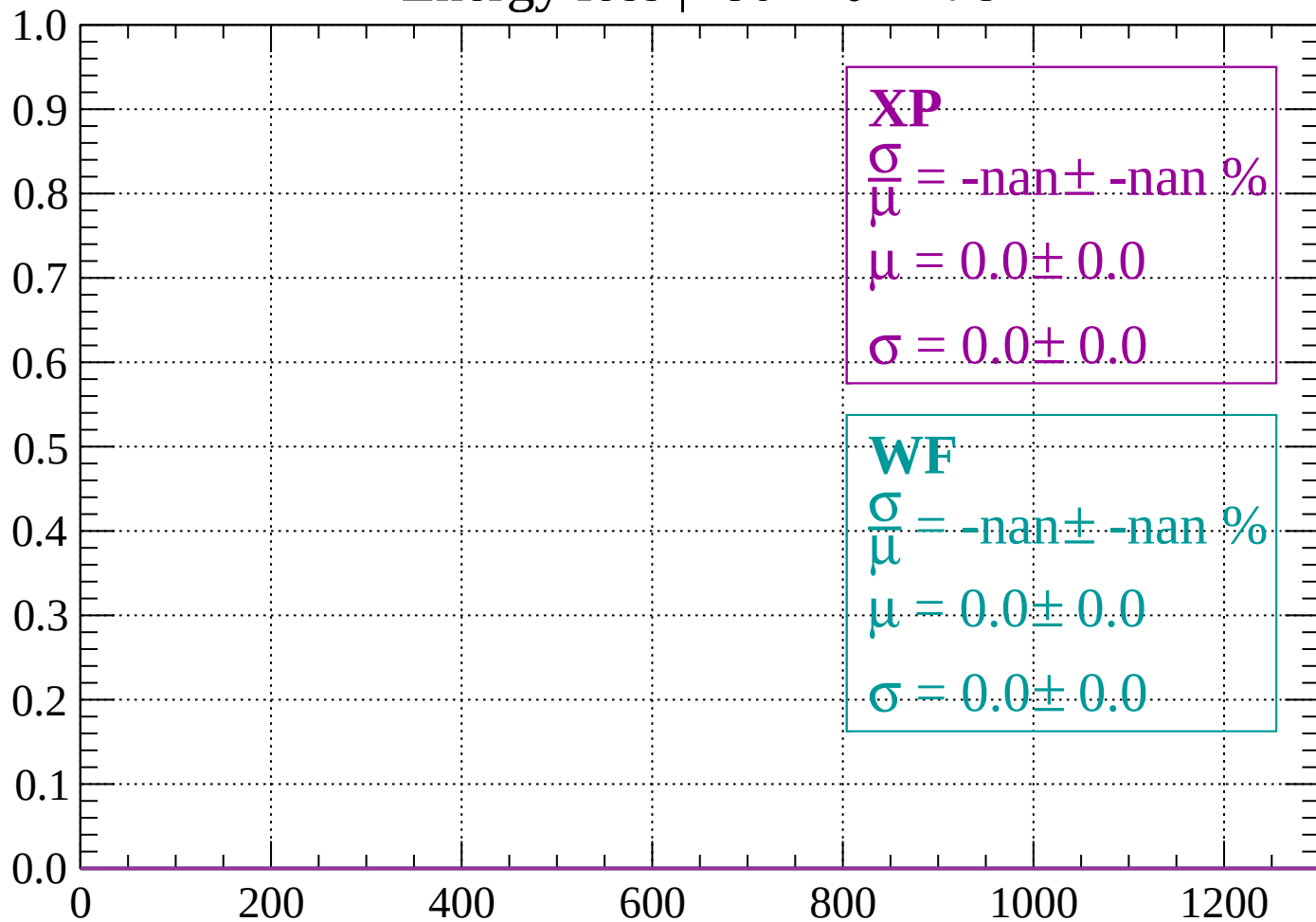
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

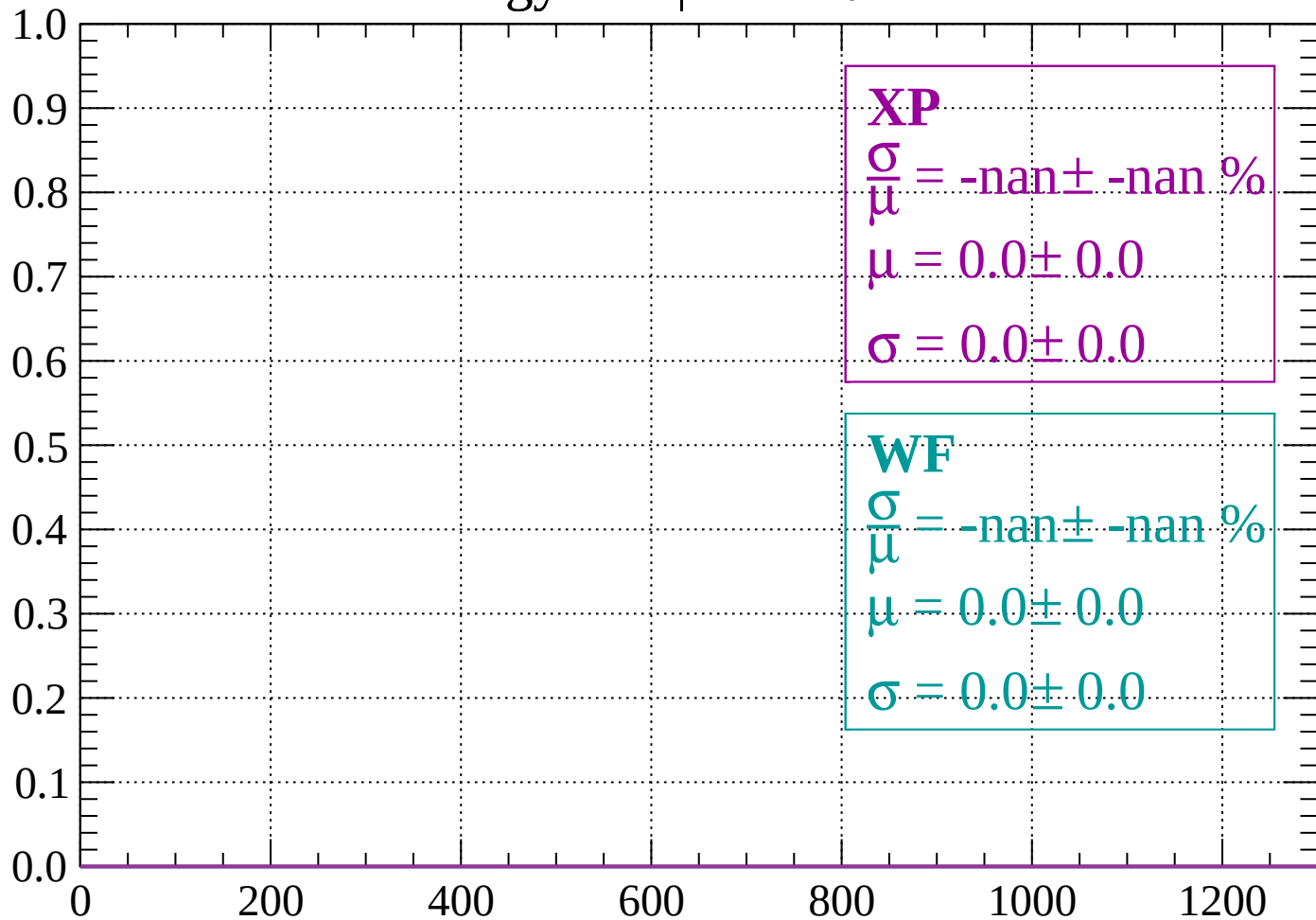
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

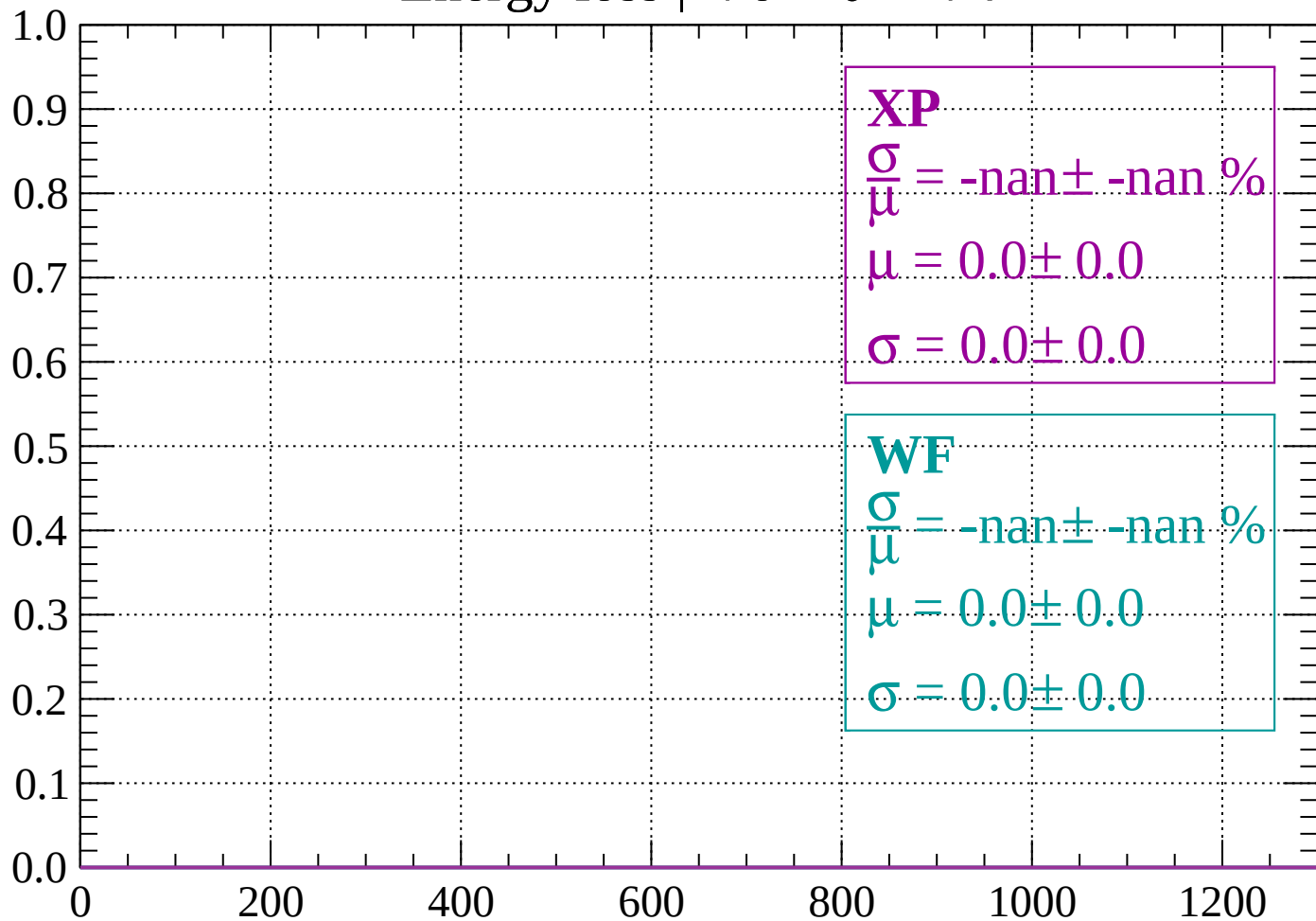
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

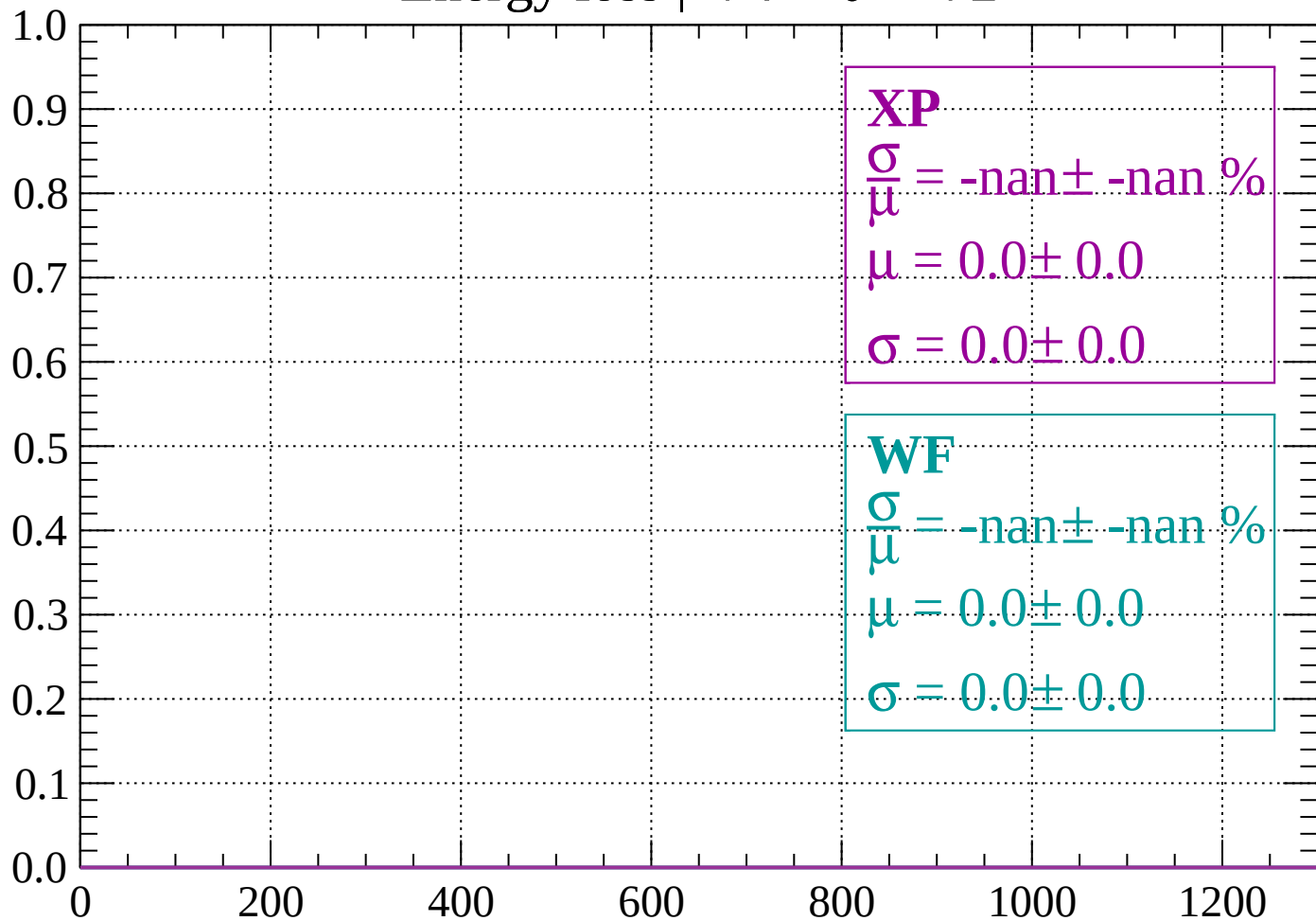
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

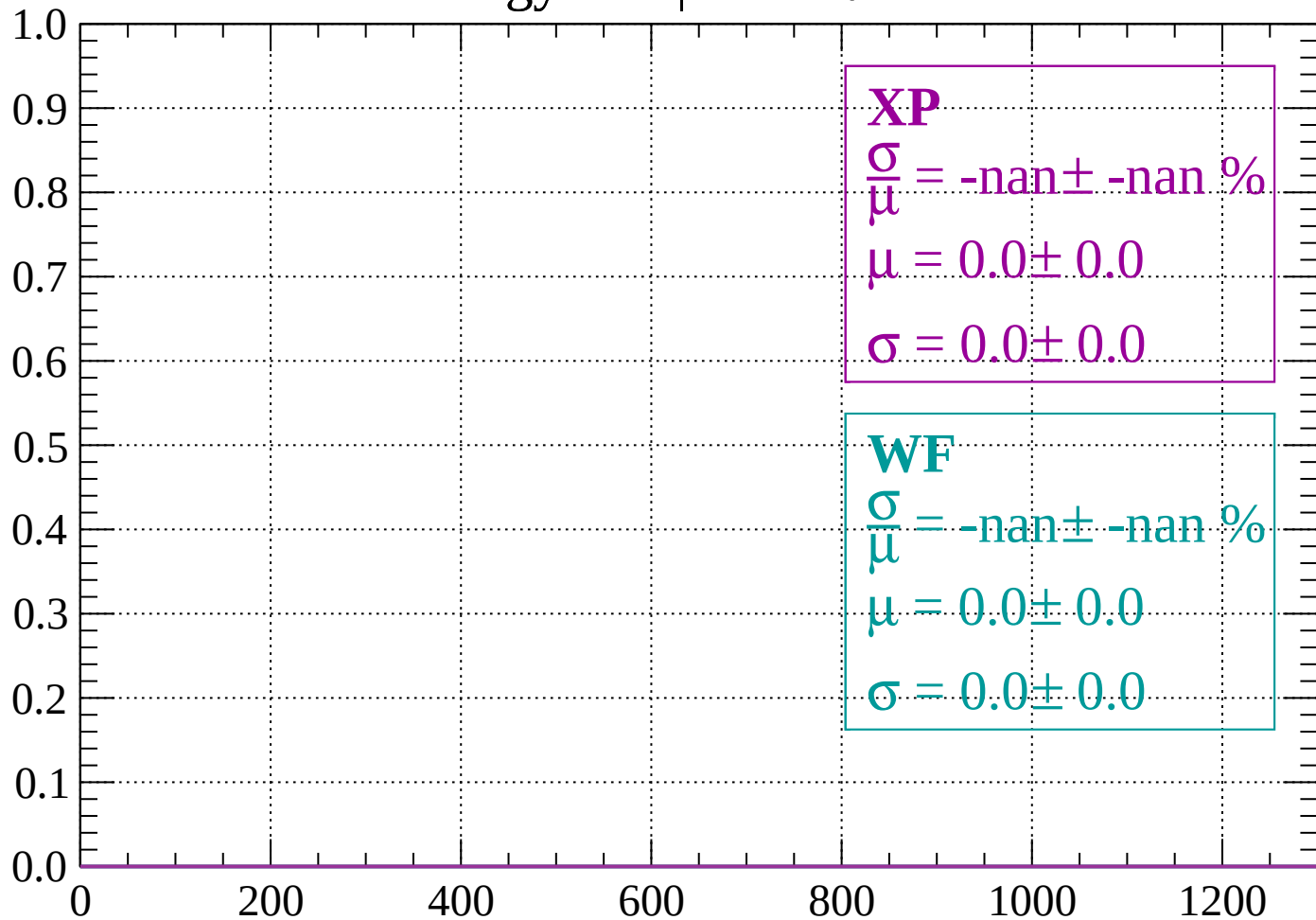
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

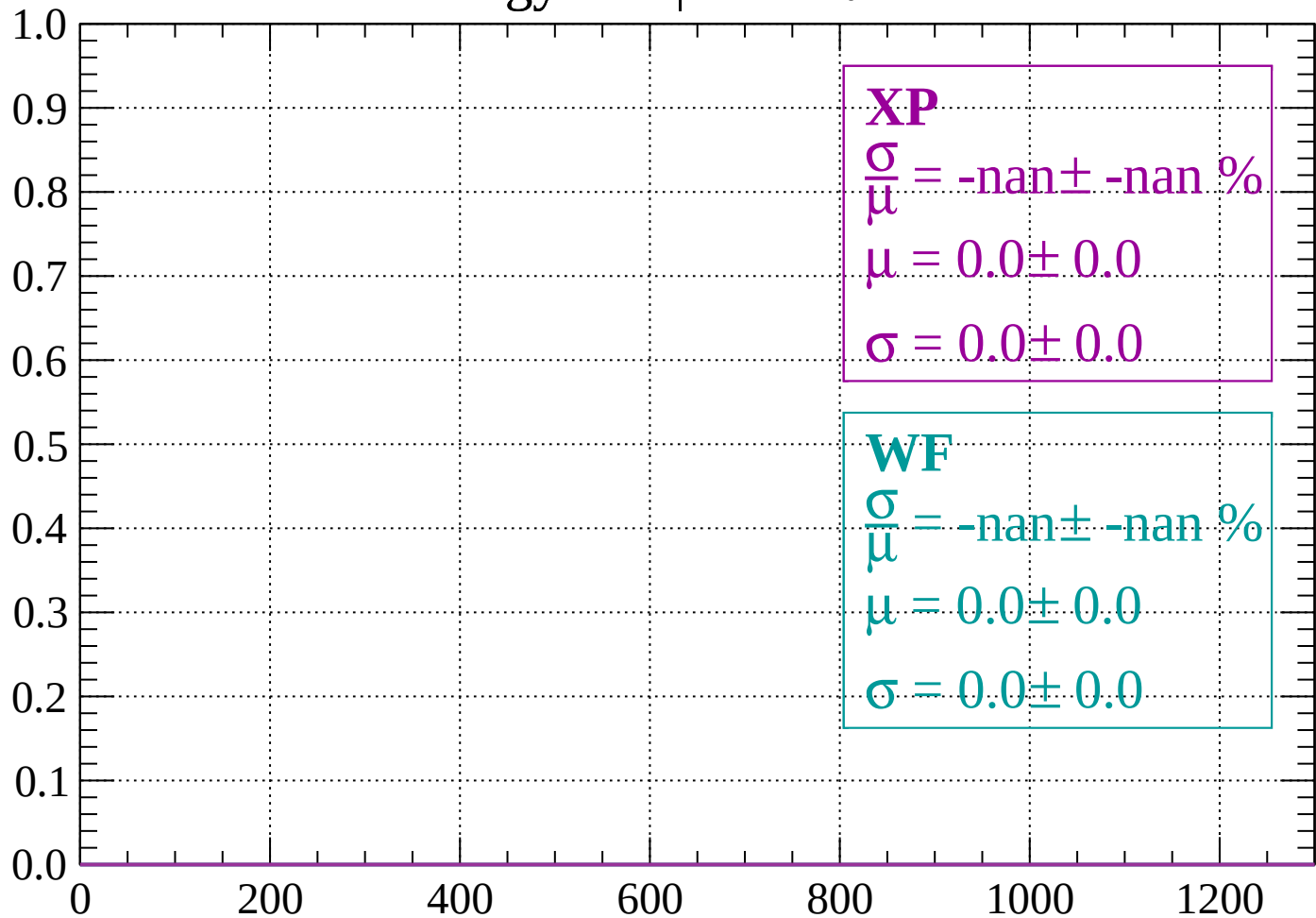
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

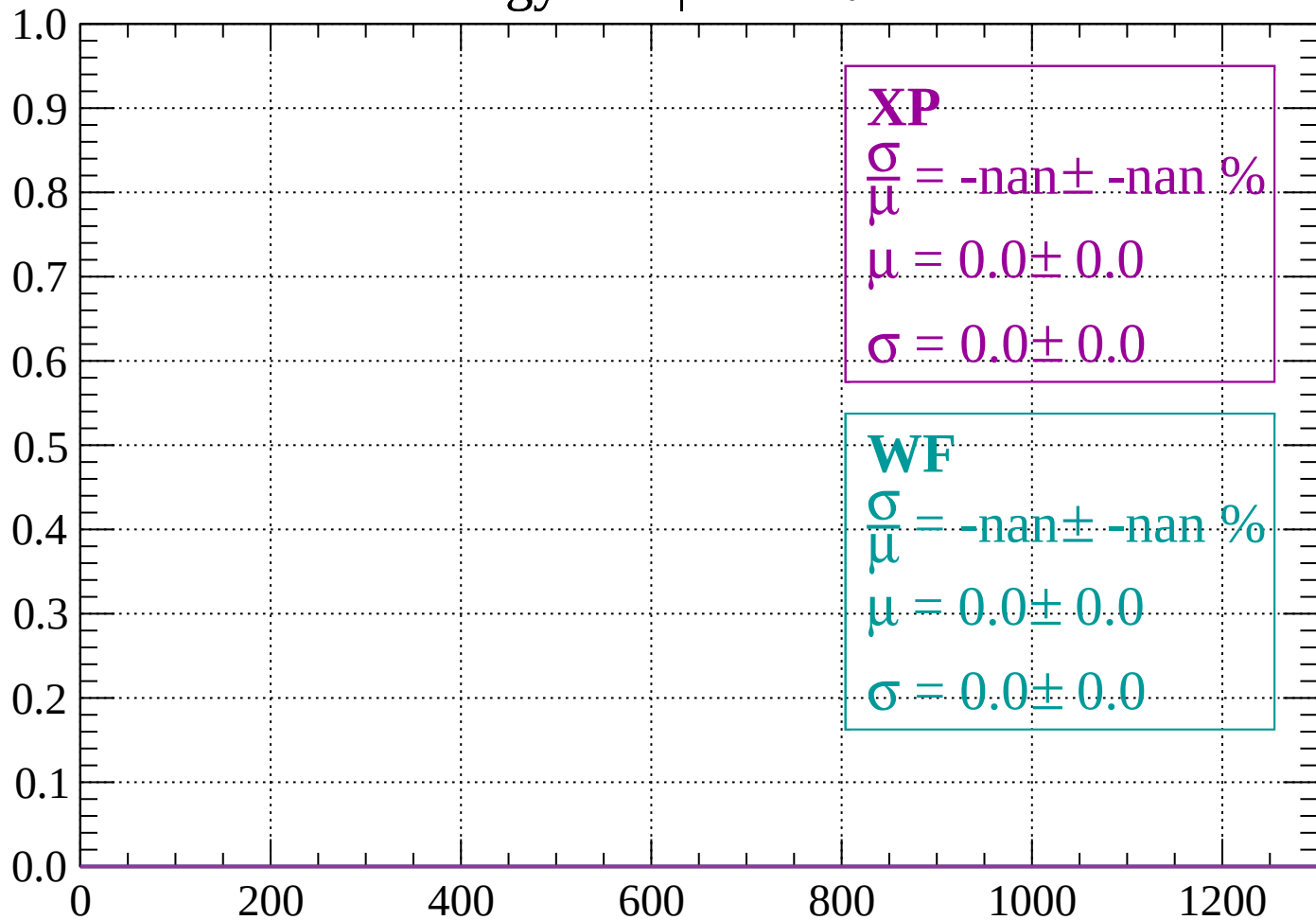
Count



dE/dx (ADC counts/cm)

Energy loss | $-68 < \theta < -66$

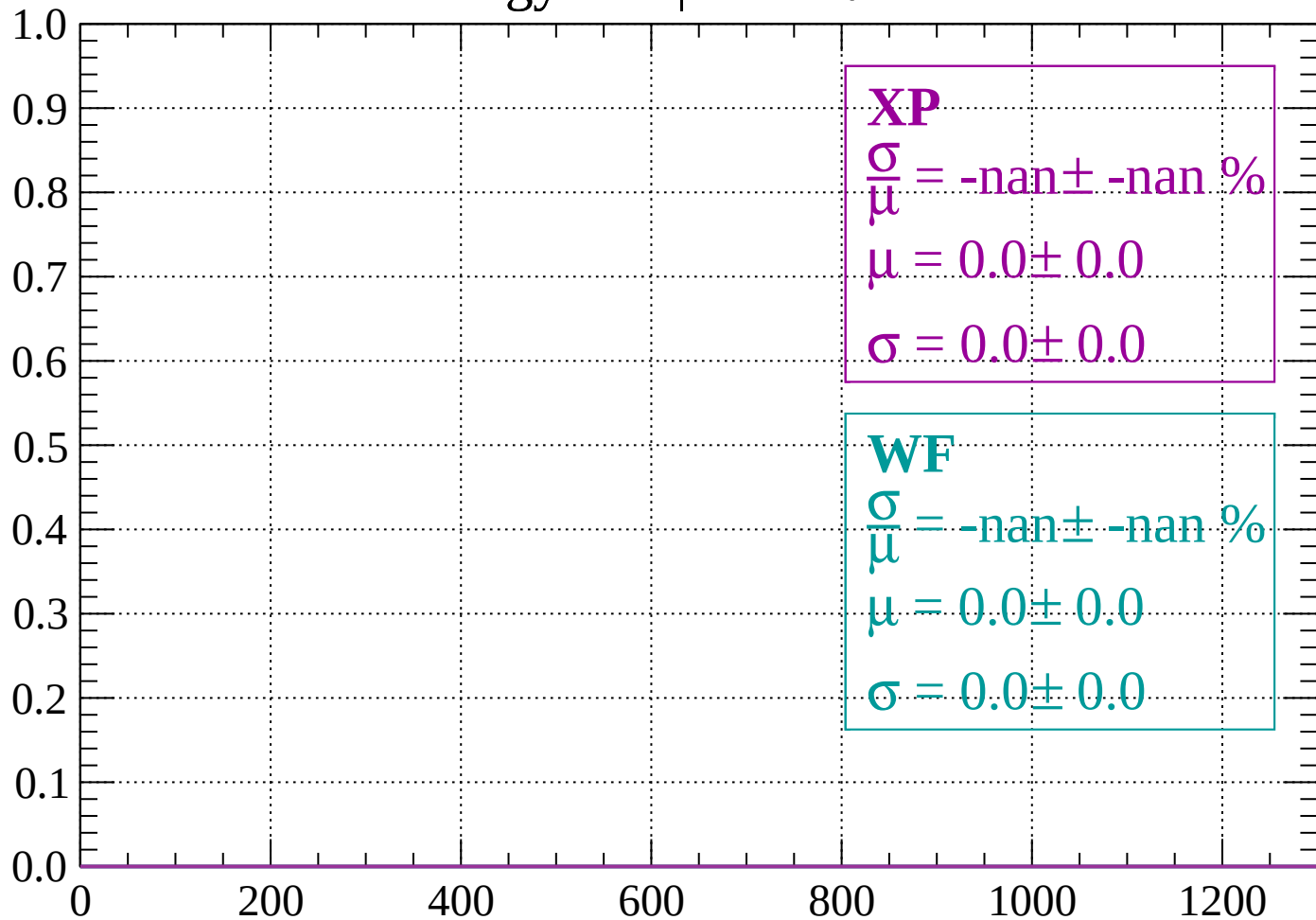
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -64$

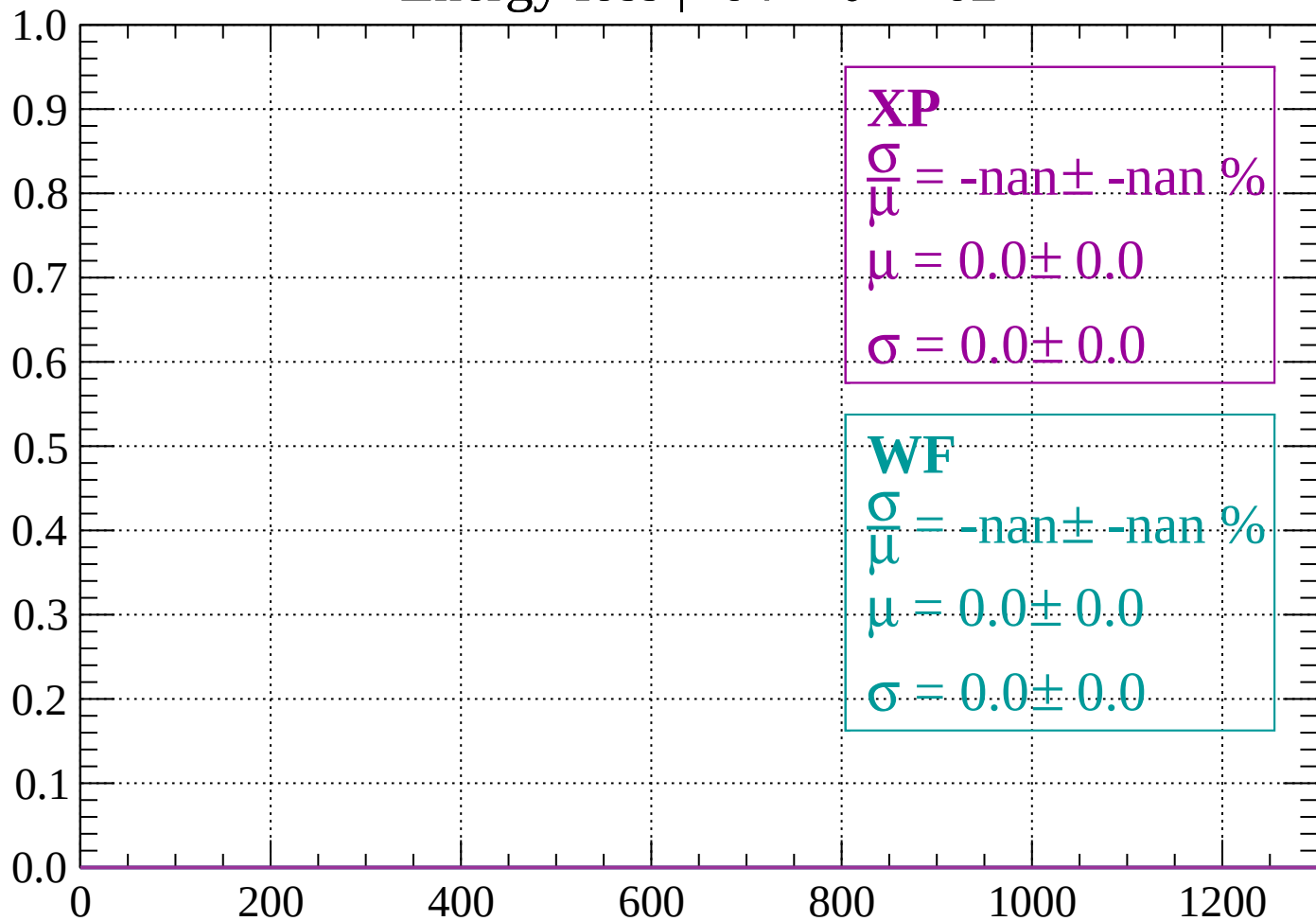
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

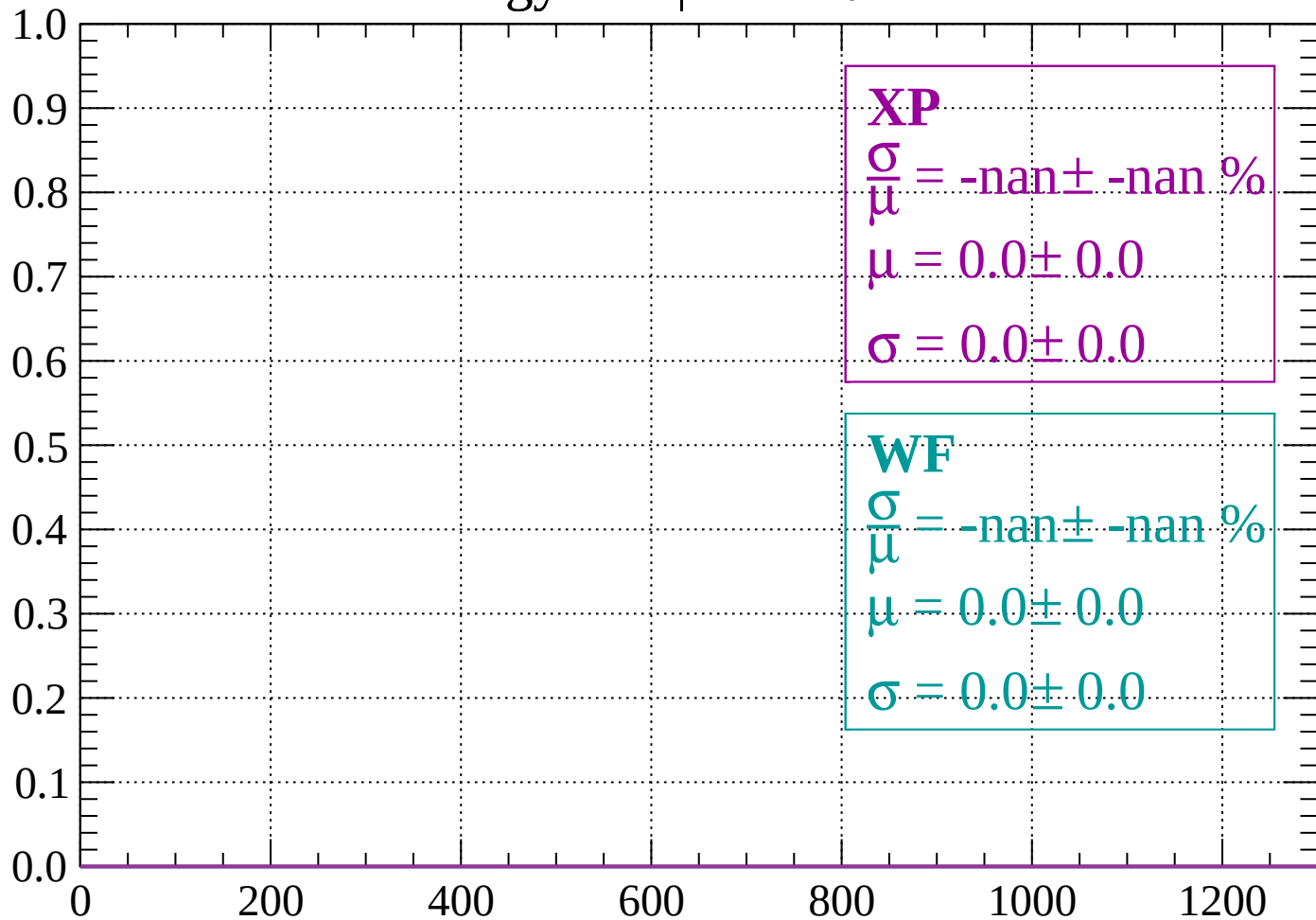
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

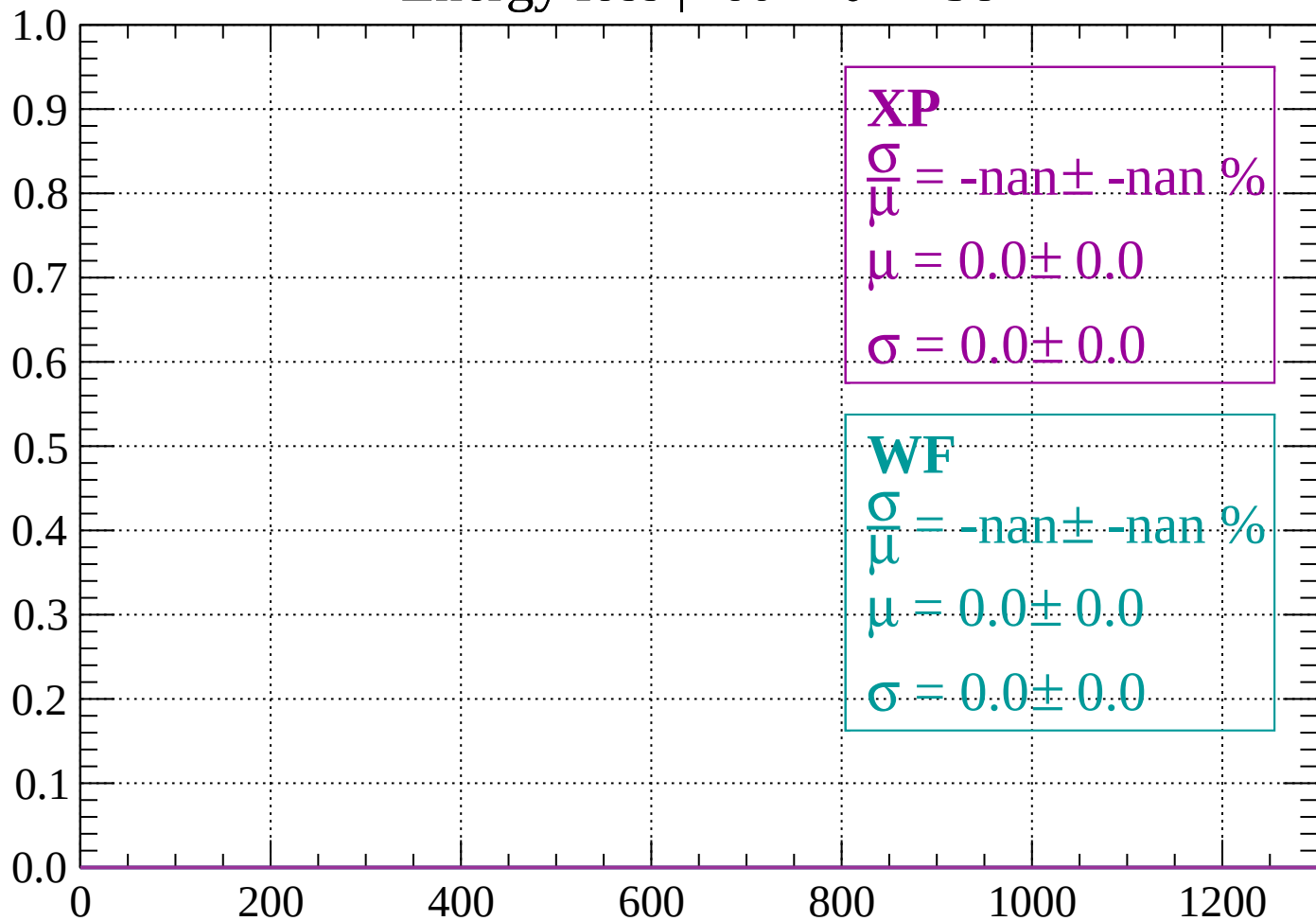
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

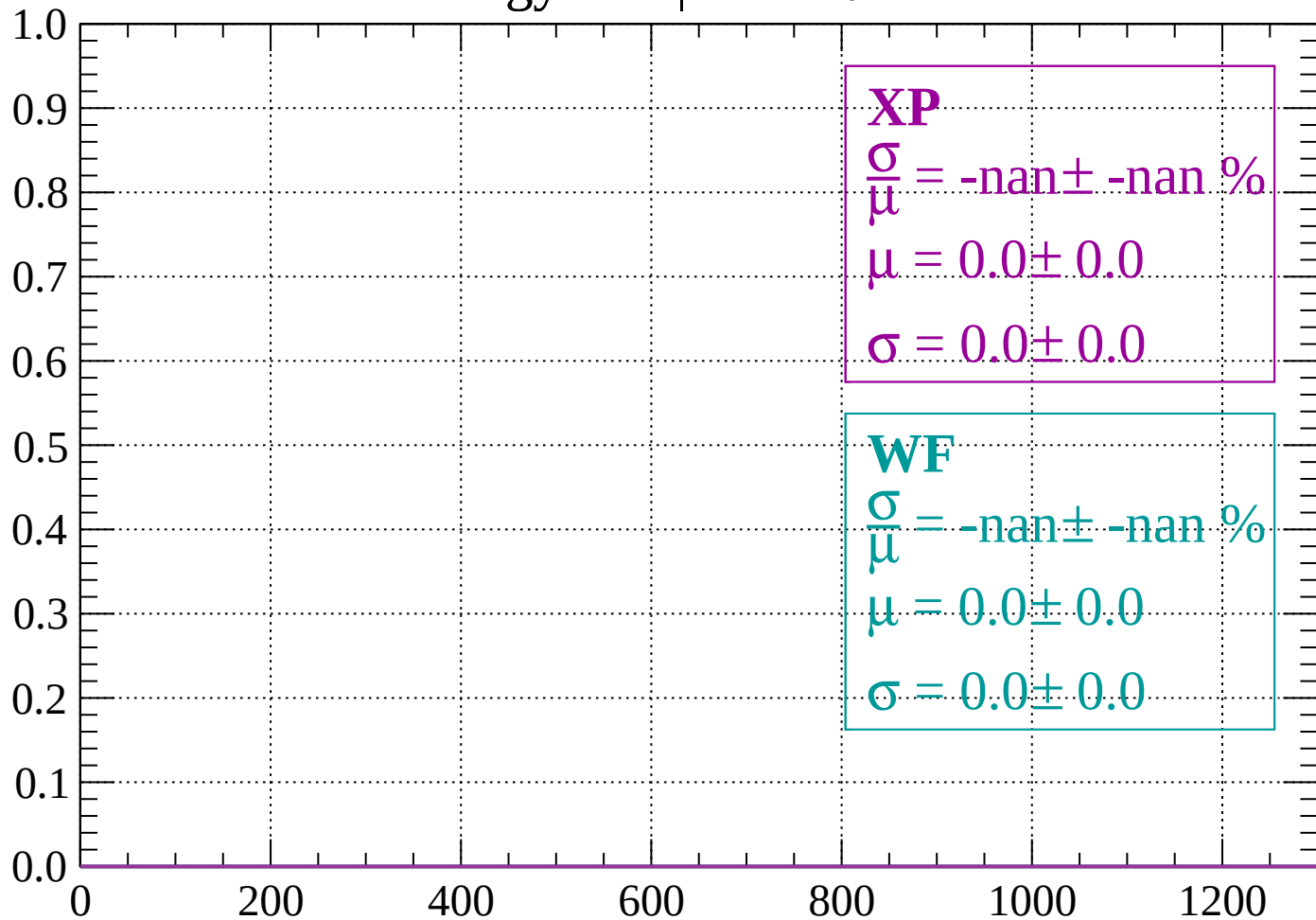
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

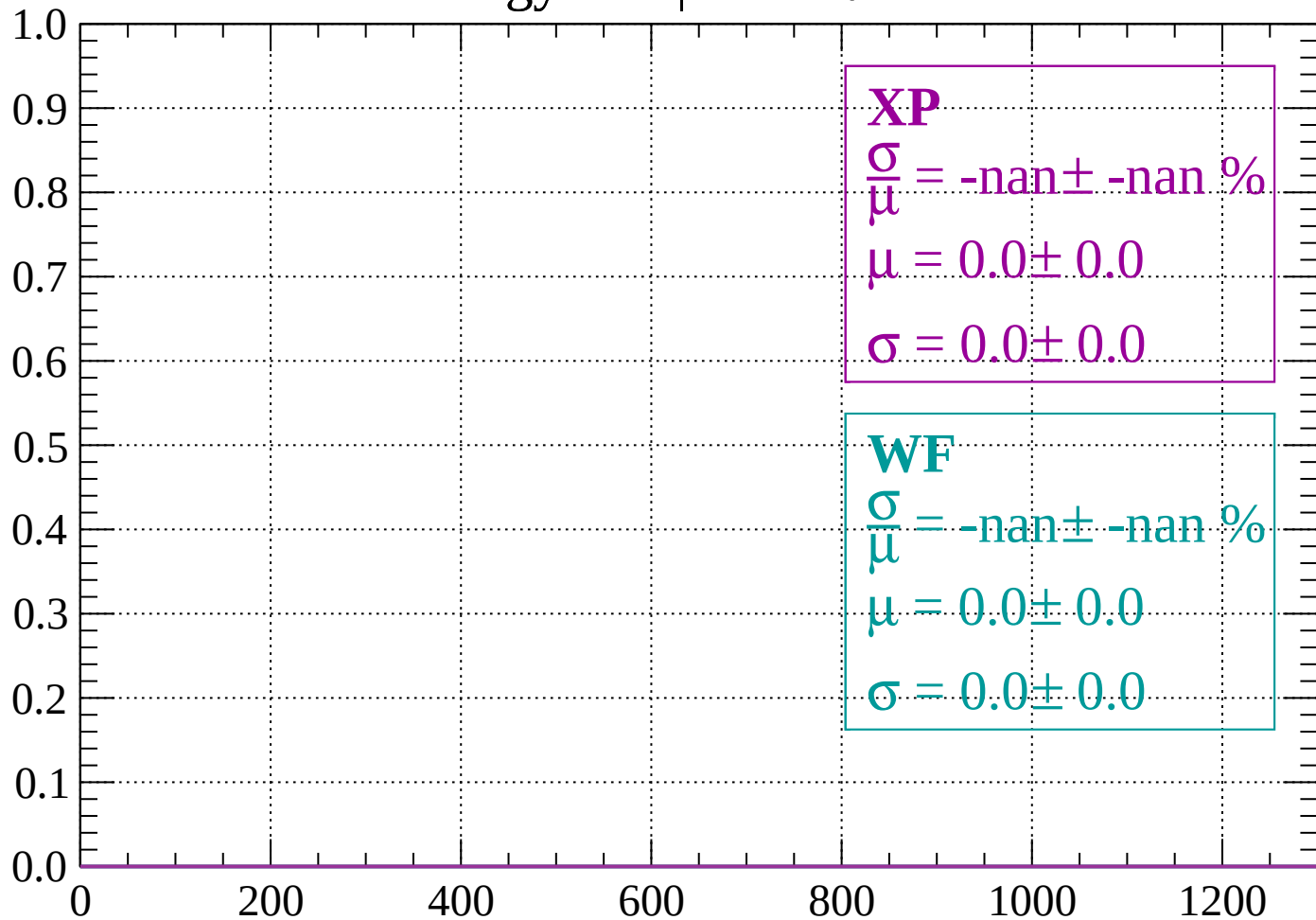
Count



dE/dx (ADC counts/cm)

Energy loss | $-56 < \theta < -54$

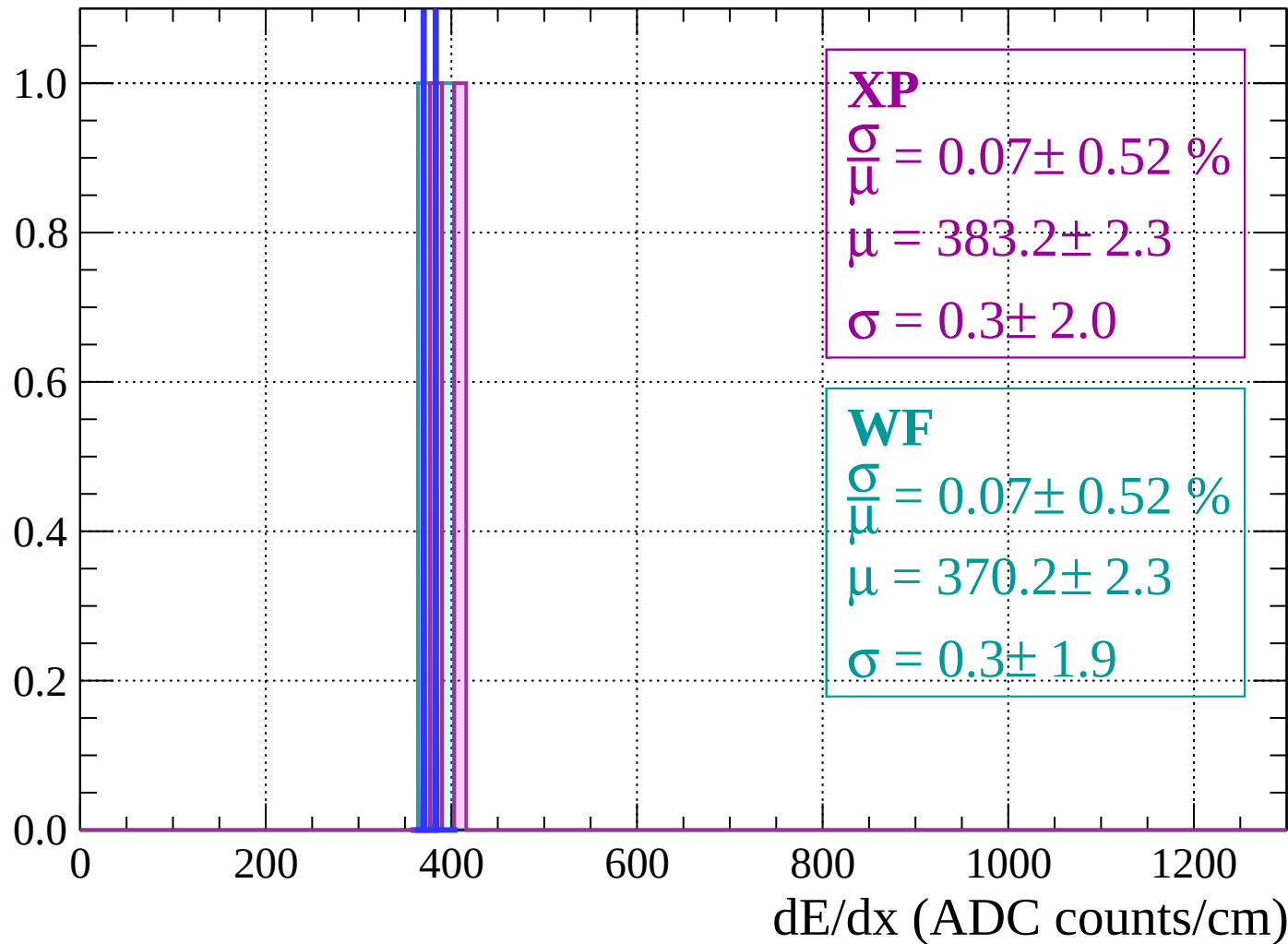
Count



dE/dx (ADC counts/cm)

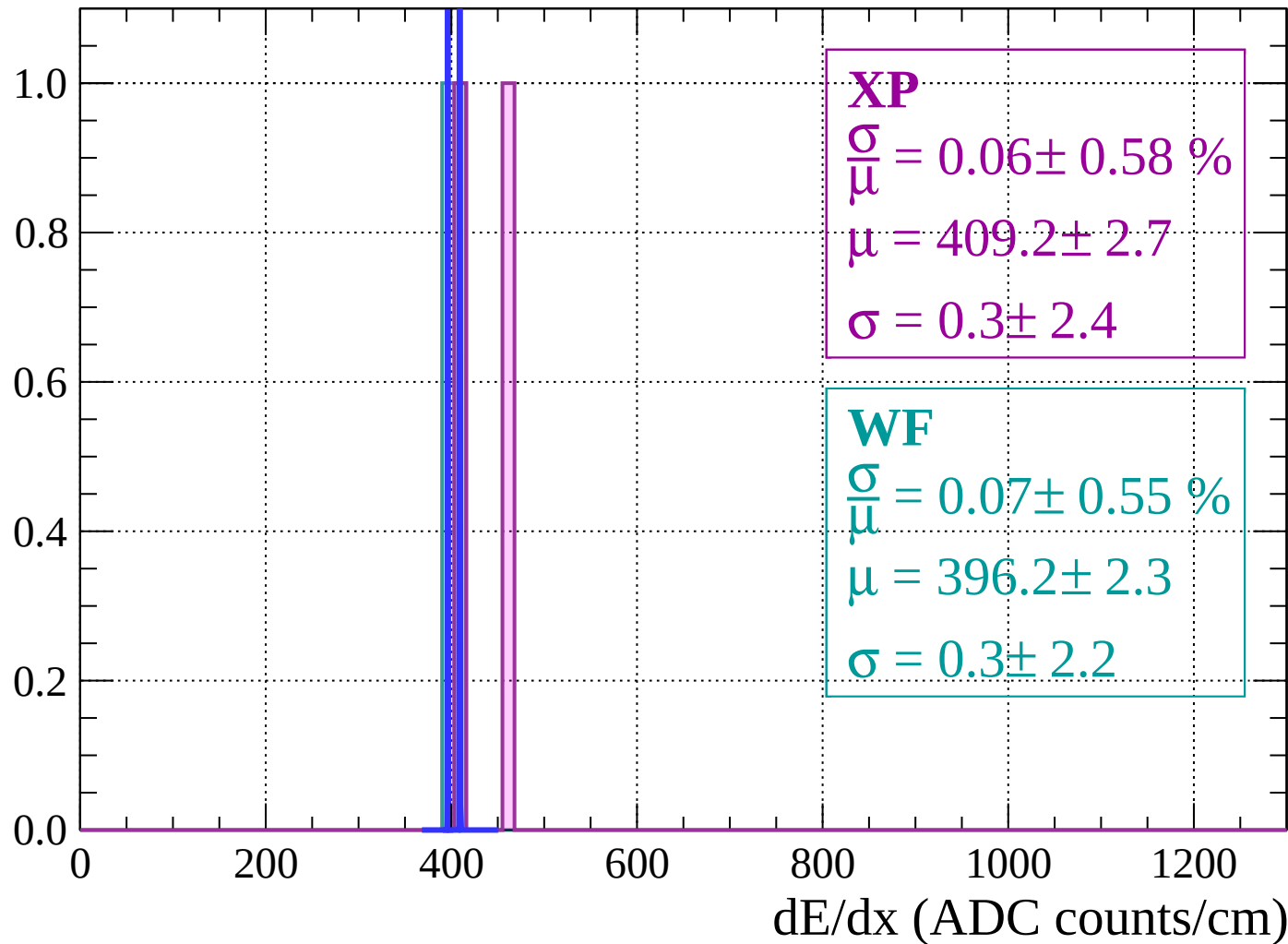
Energy loss | $-54 < \theta < -52$

Count



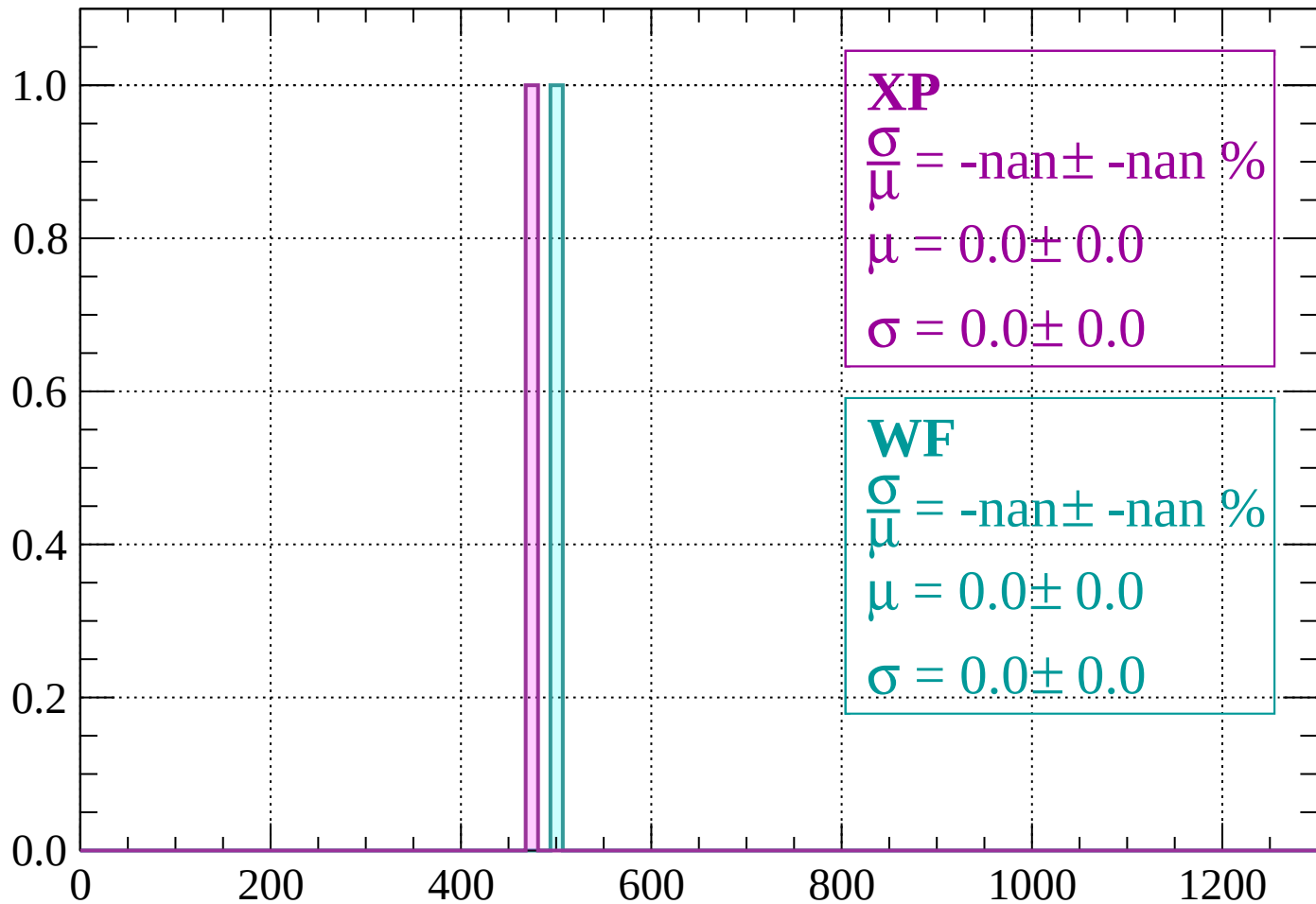
Energy loss | $-52 < \theta < -50$

Count



Energy loss | $-50 < \theta < -48$

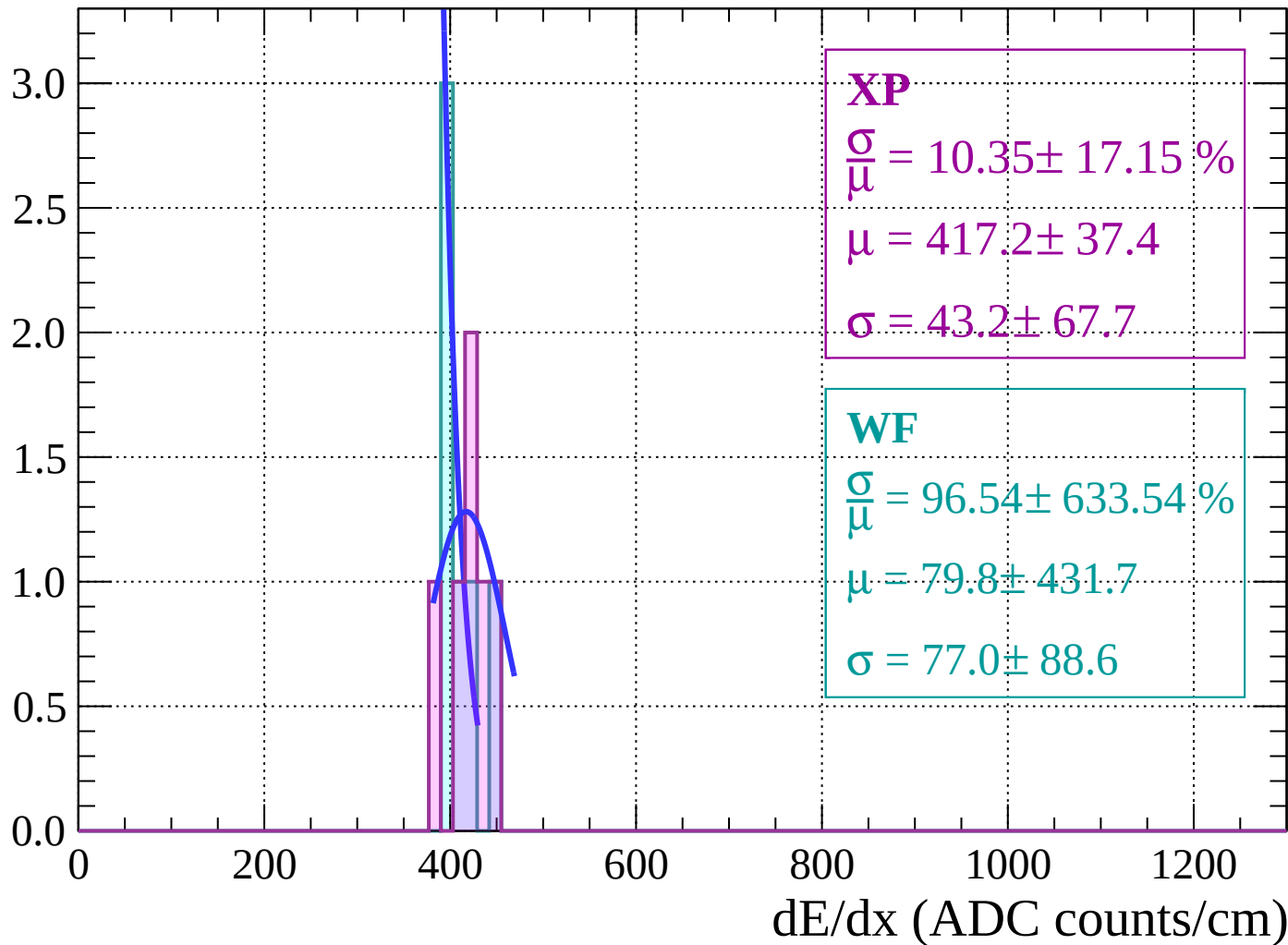
Count



dE/dx (ADC counts/cm)

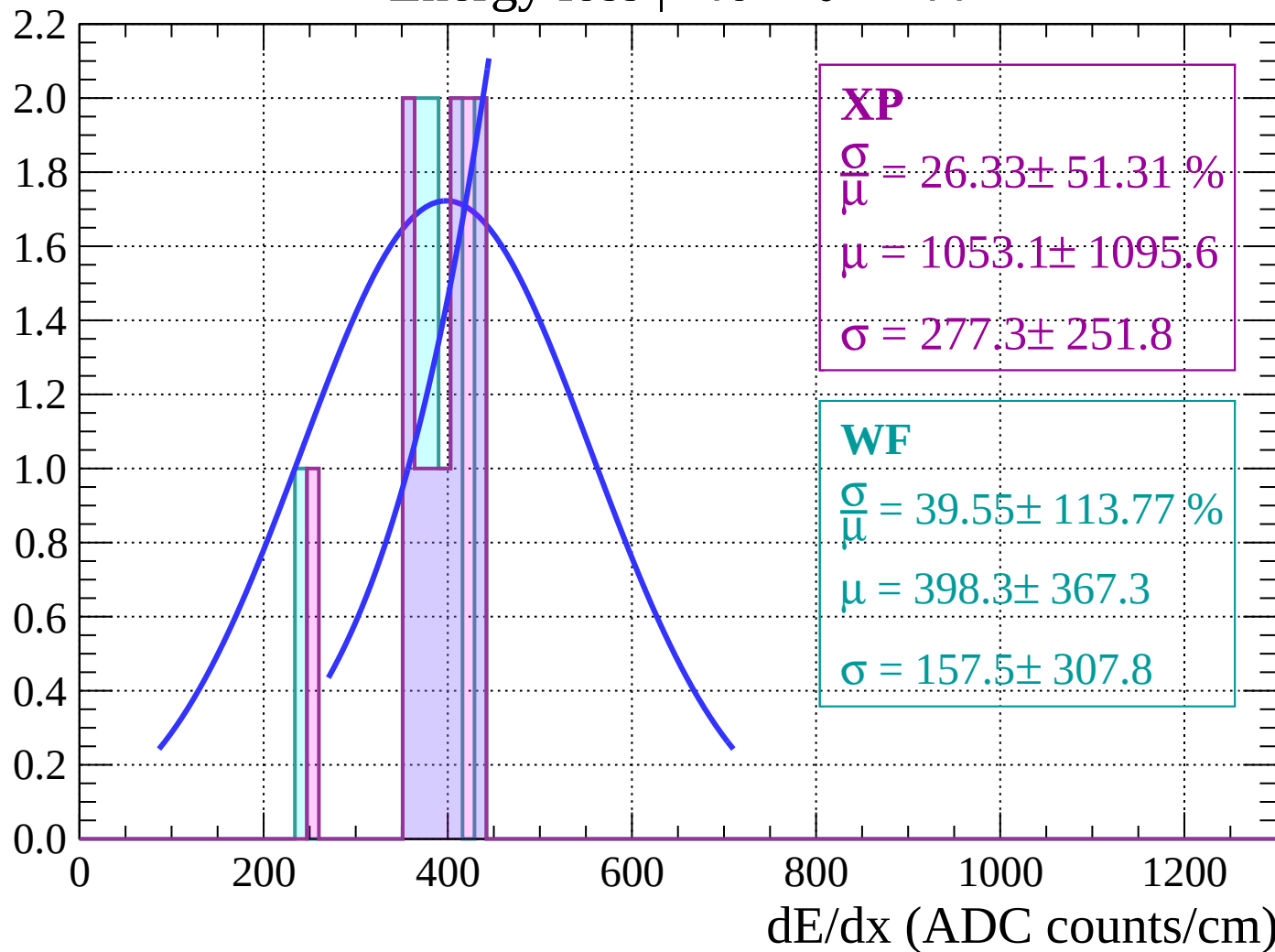
Energy loss | $-48 < \theta < -46$

Count

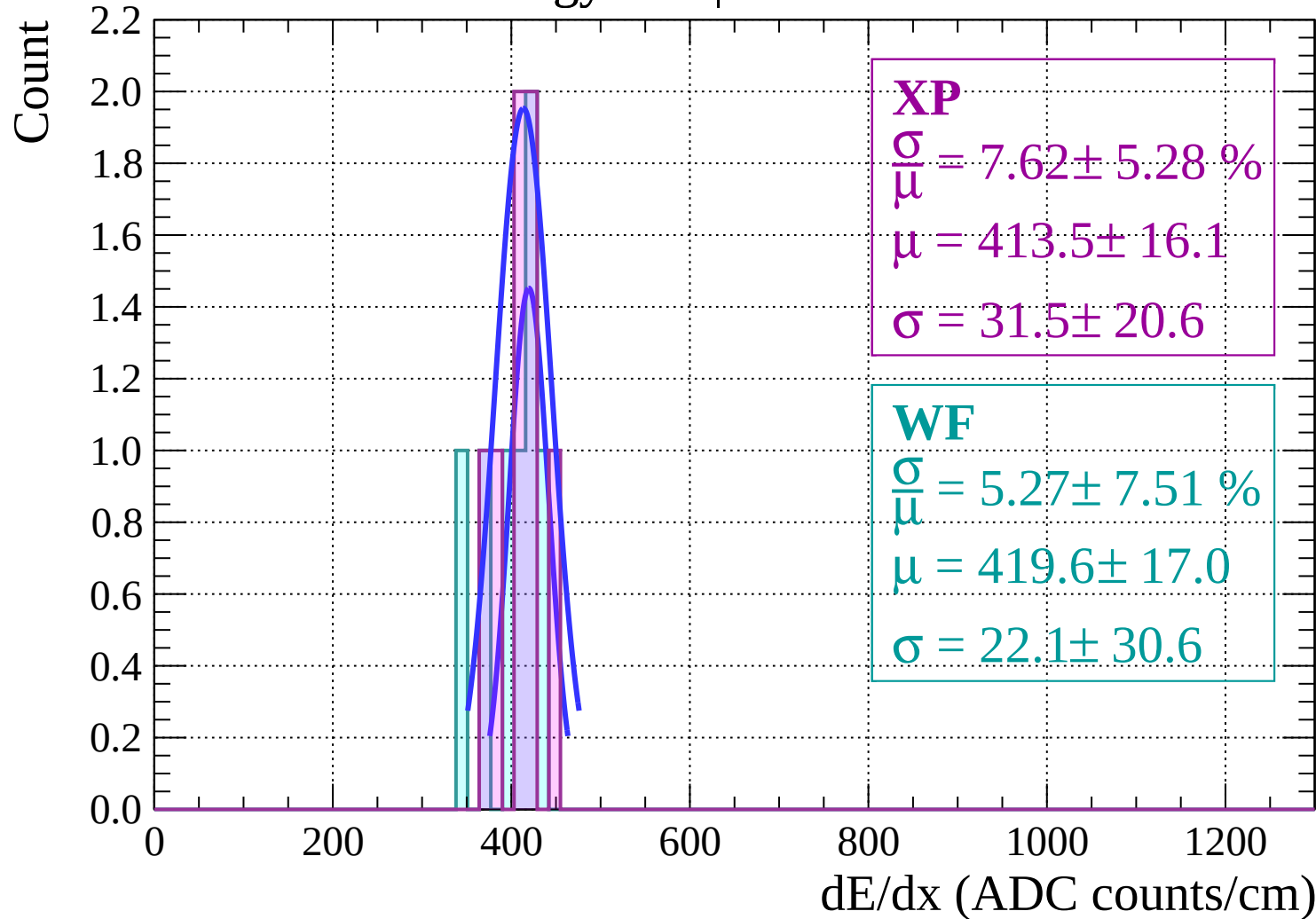


Energy loss | $-46 < \theta < -44$

Count

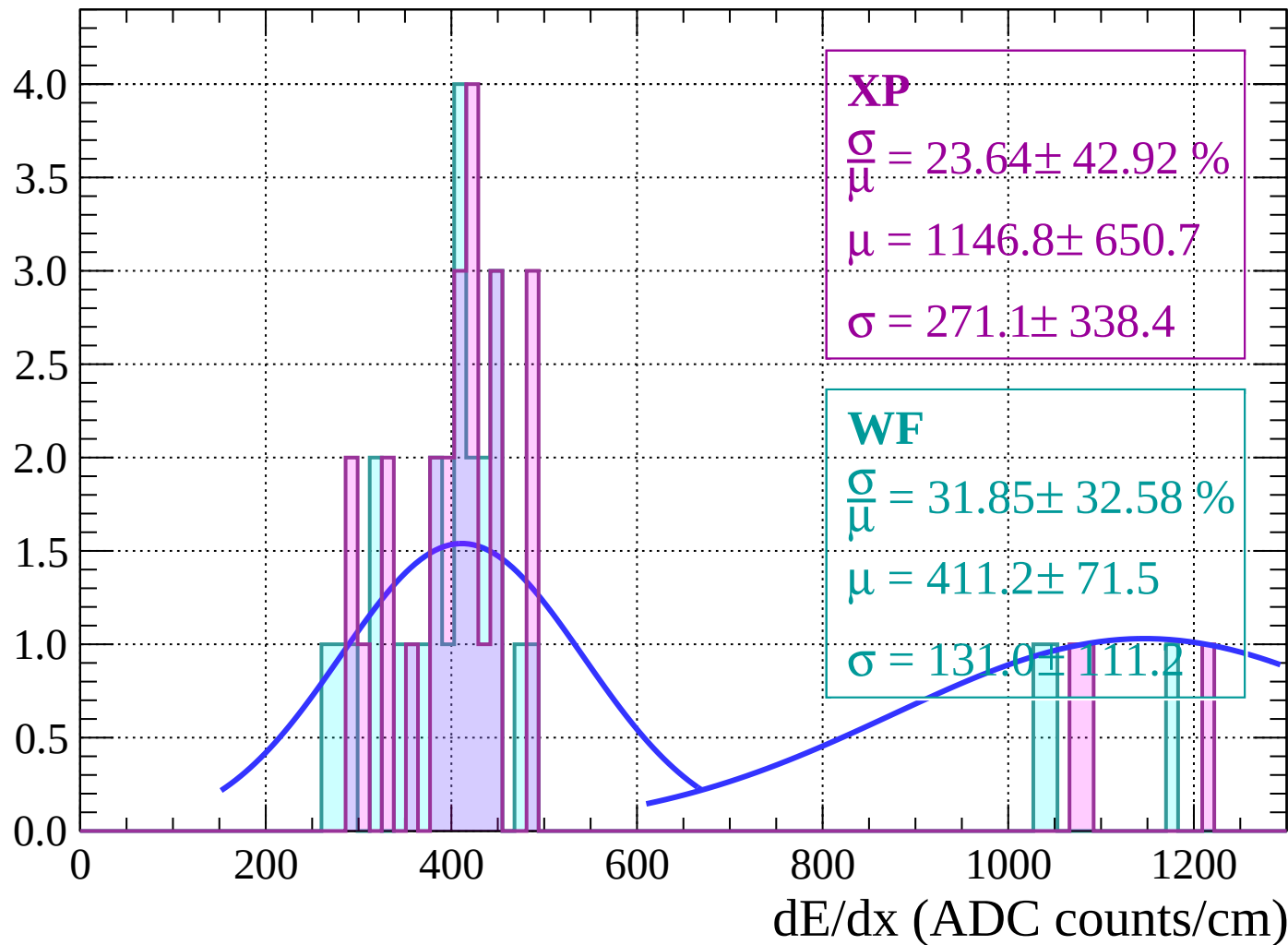


Energy loss | $-44 < \theta < -42$



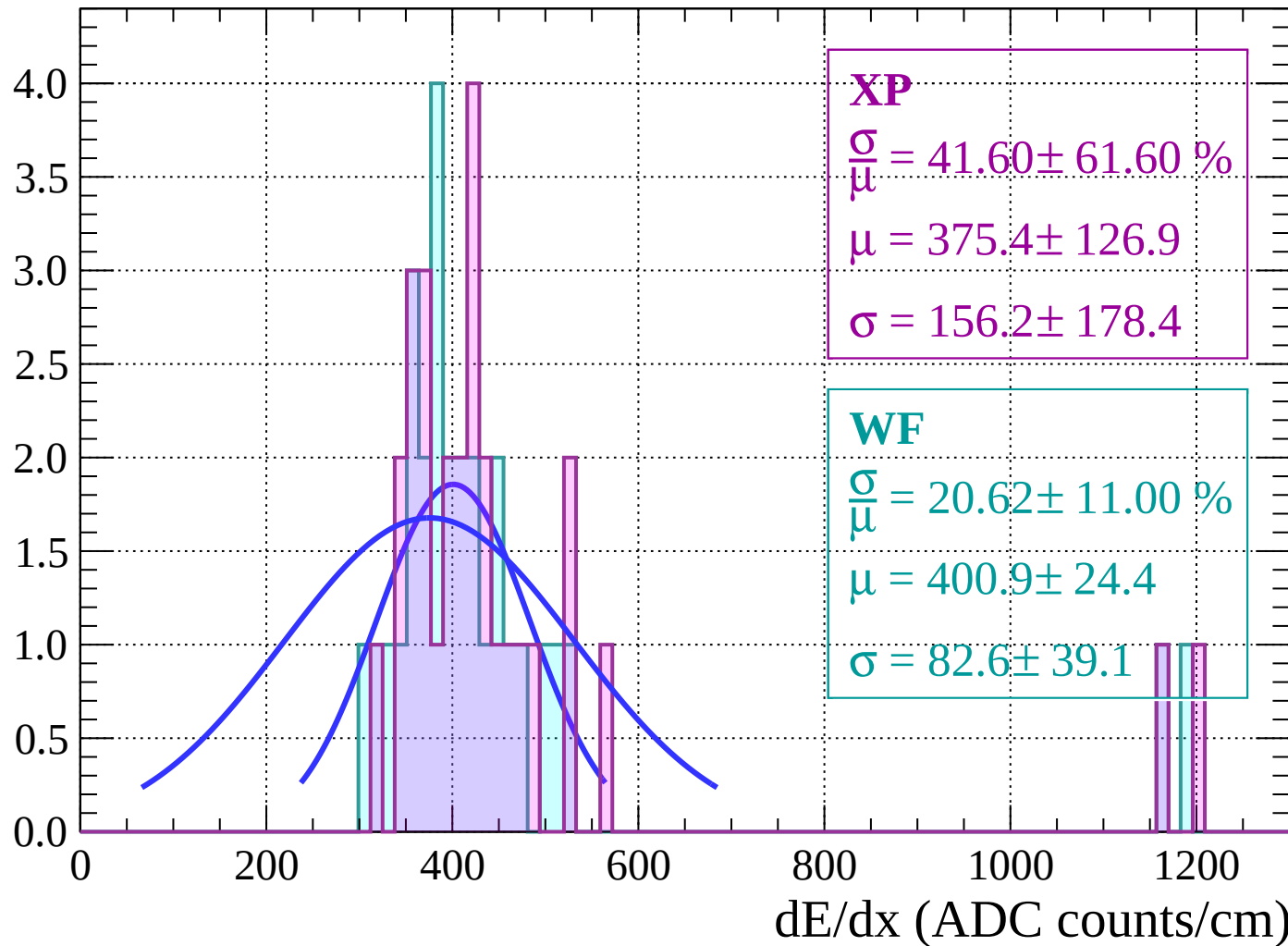
Energy loss | $-42 < \theta < -40$

Count



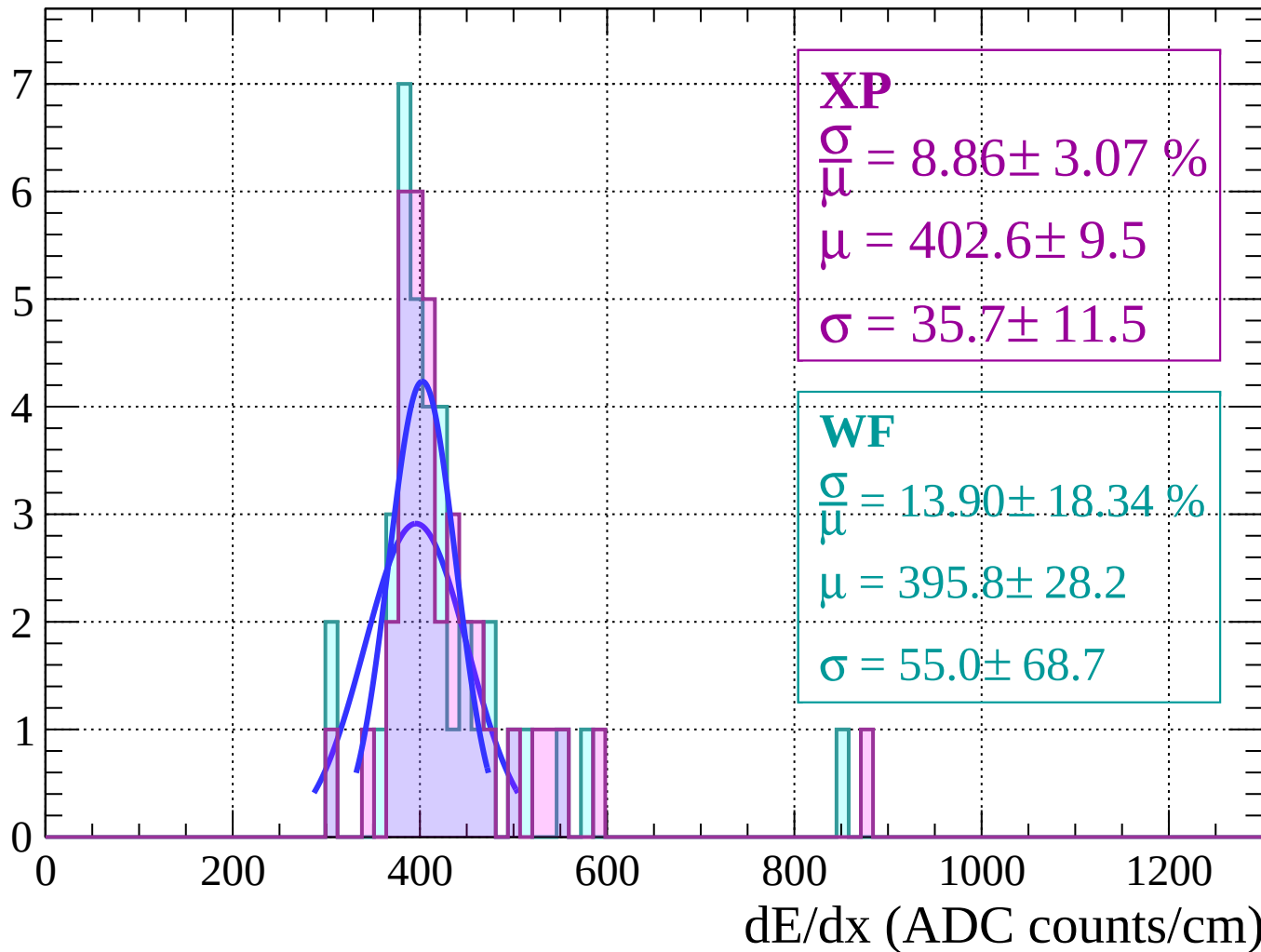
Energy loss | $-40 < \theta < -38$

Count



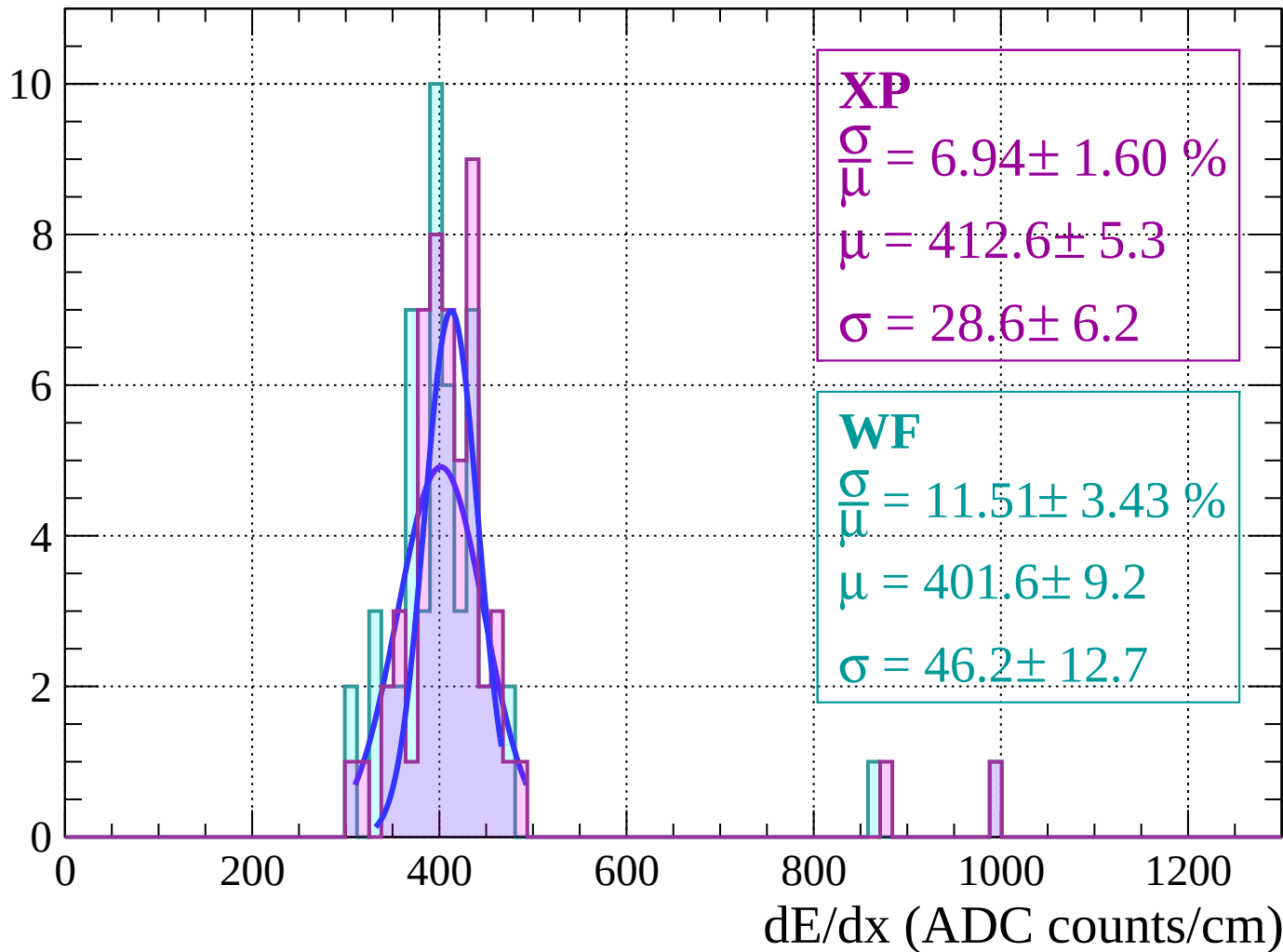
Energy loss $|-38 < \theta < -36$

Count



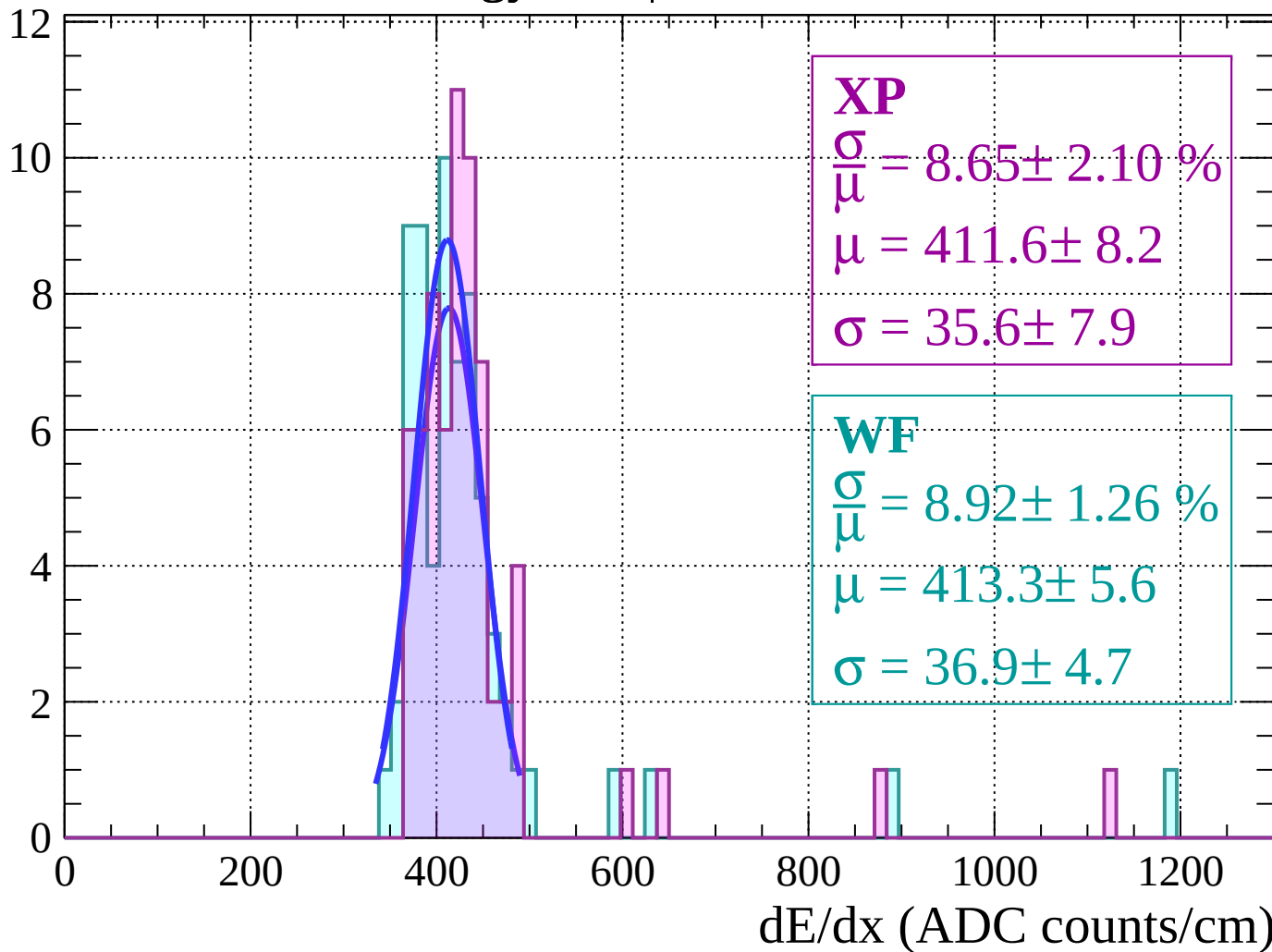
Energy loss $|-36 < \theta < -34$

Count



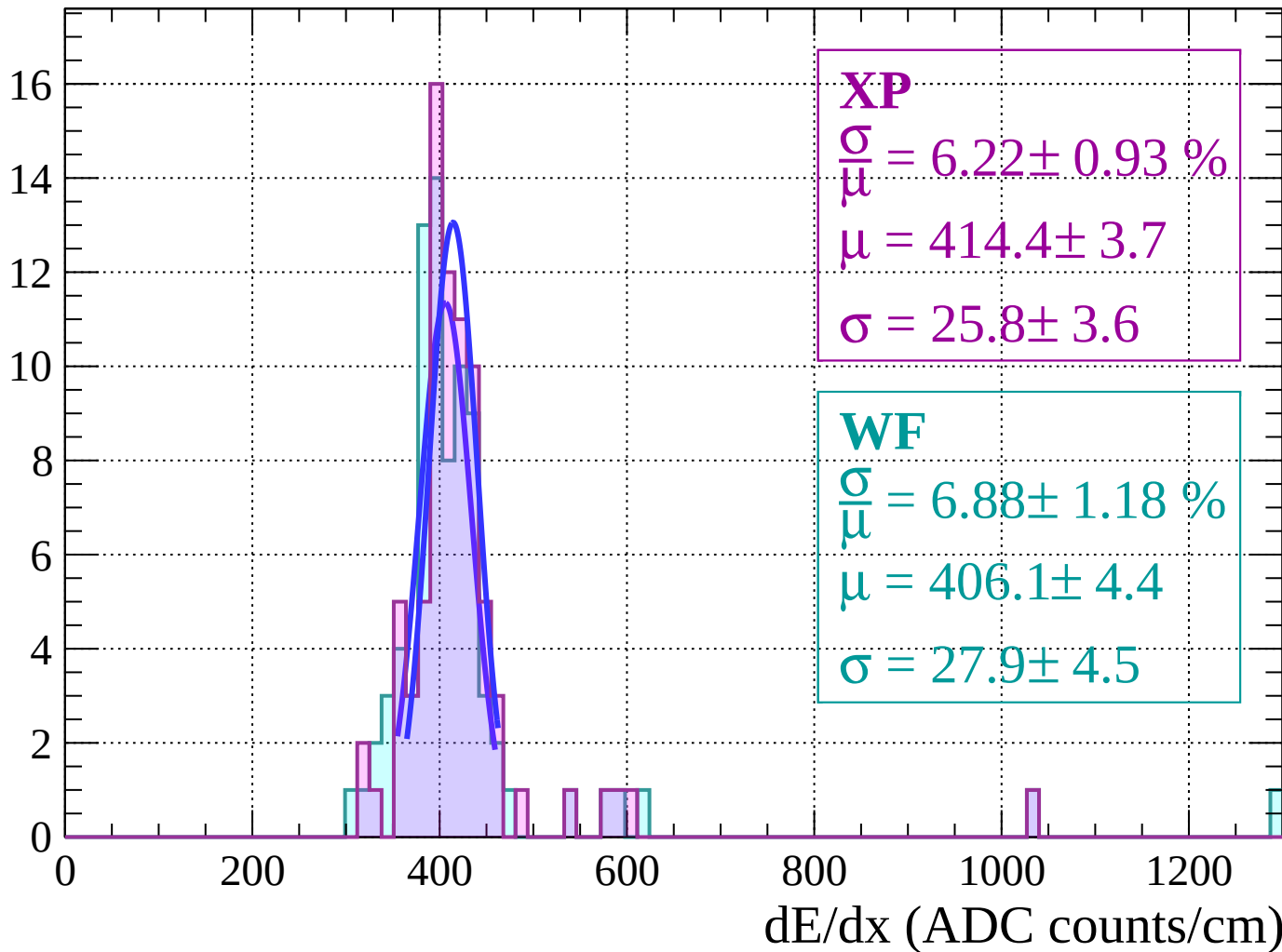
Energy loss $|-34 < \theta < -32$

Count



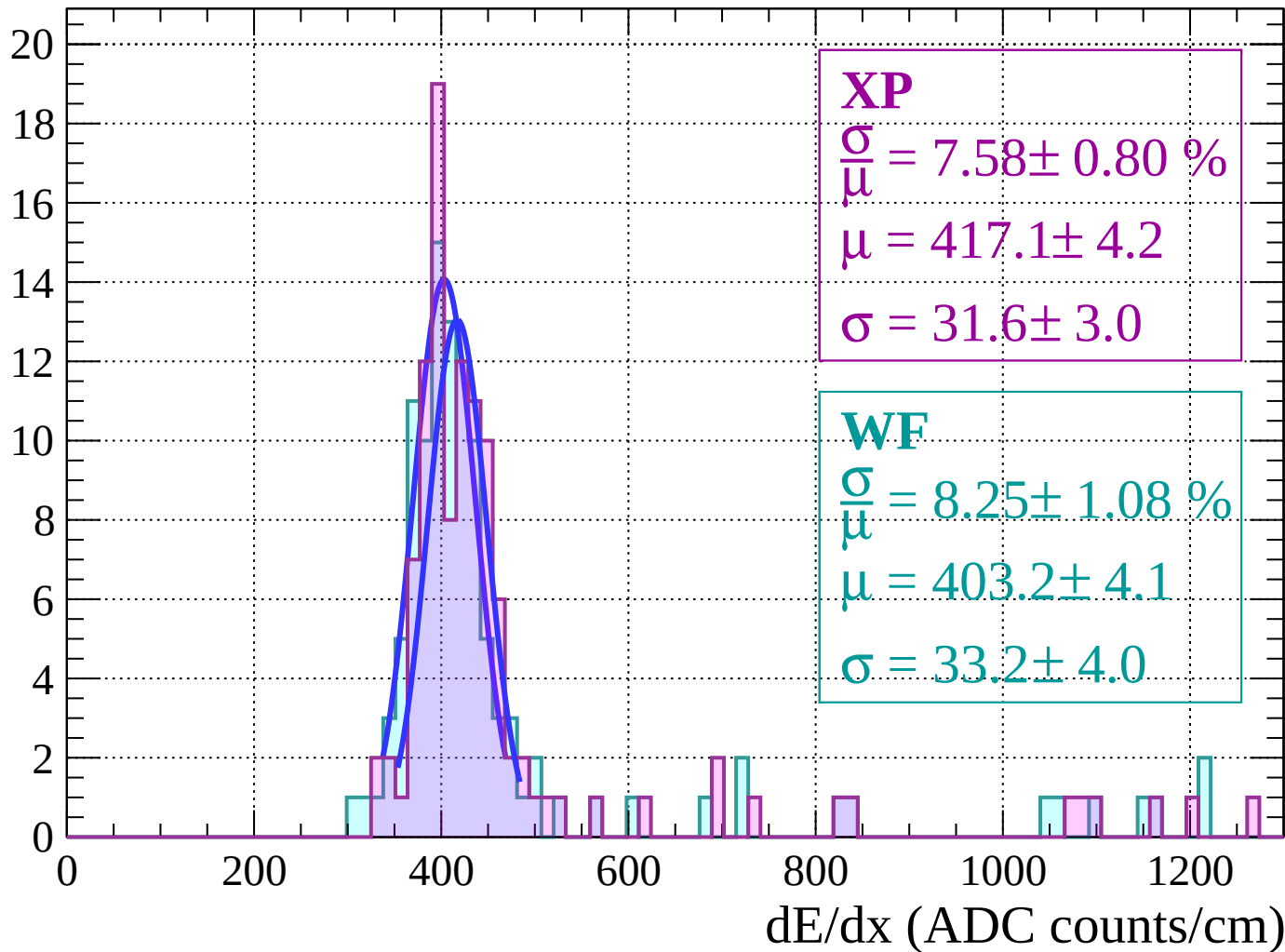
Energy loss | $-32 < \theta < -30$

Count



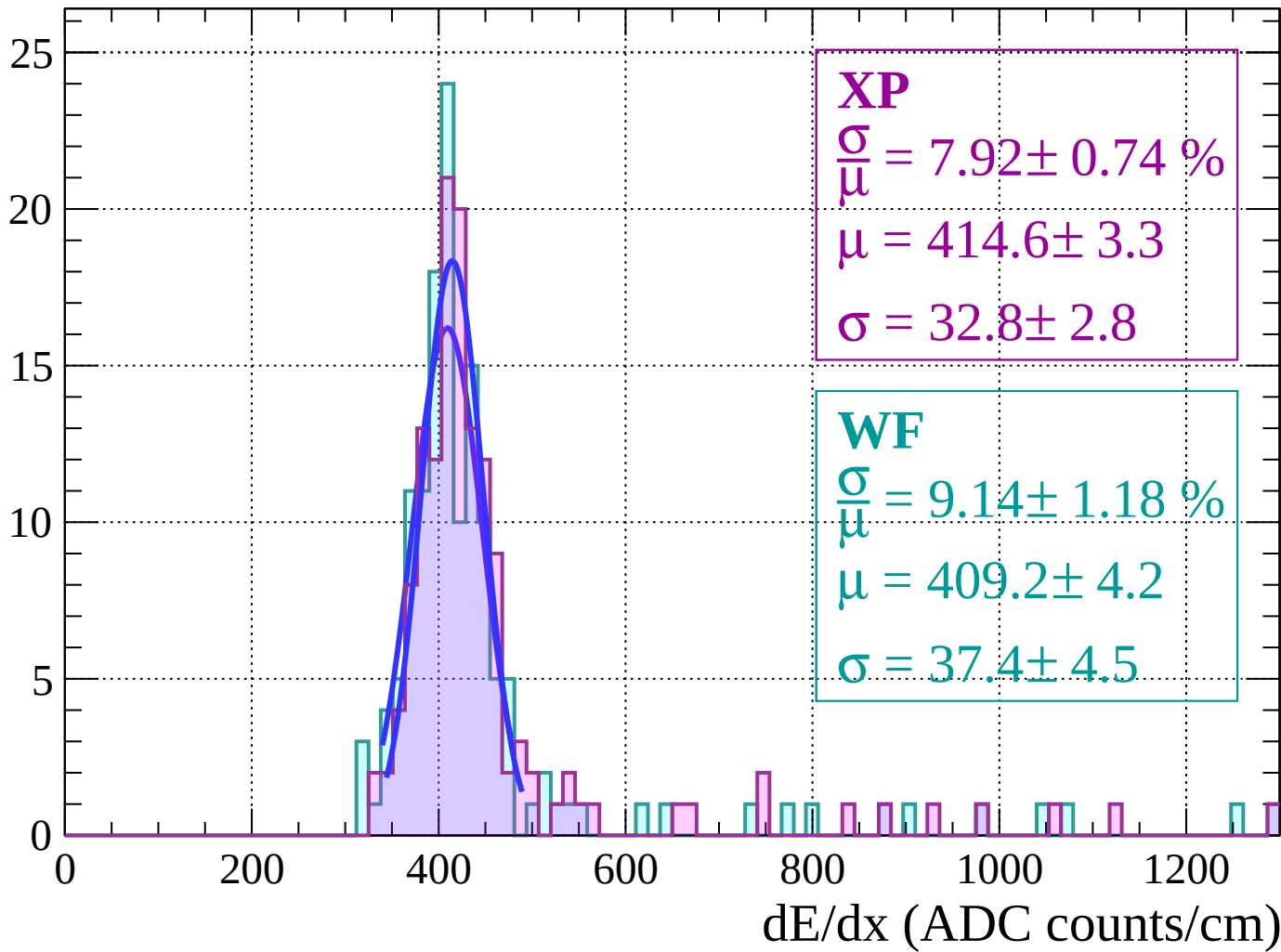
Energy loss | $-30 < \theta < -28$

Count



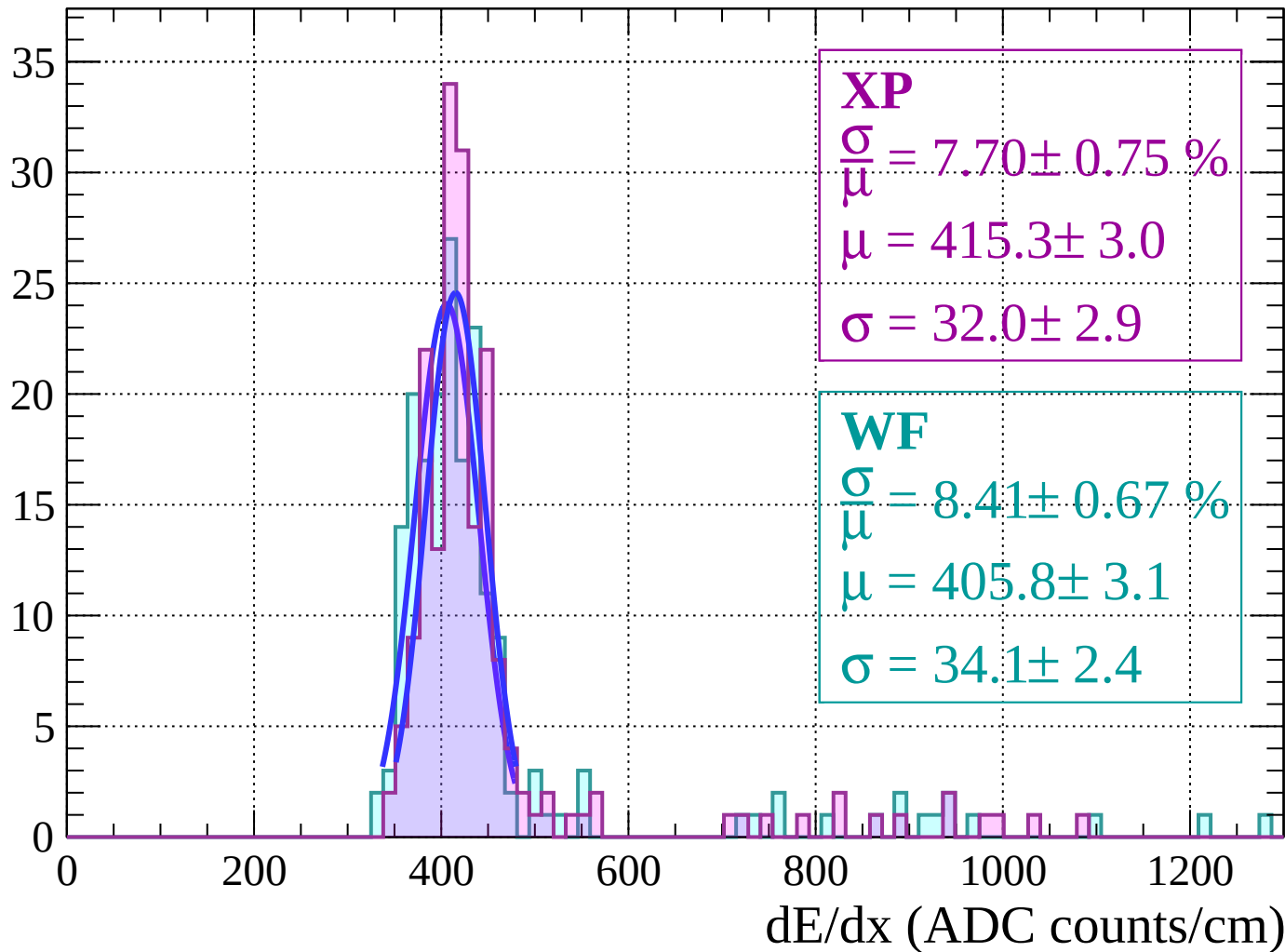
Energy loss $|-28 < \theta < -26$

Count



Energy loss $|-26 < \theta < -24$

Count



Energy loss $|-24 < \theta < -22$

Count

XP

$$\frac{\sigma}{\mu} = 8.85 \pm 0.67 \%$$

$$\mu = 413.6 \pm 2.8$$

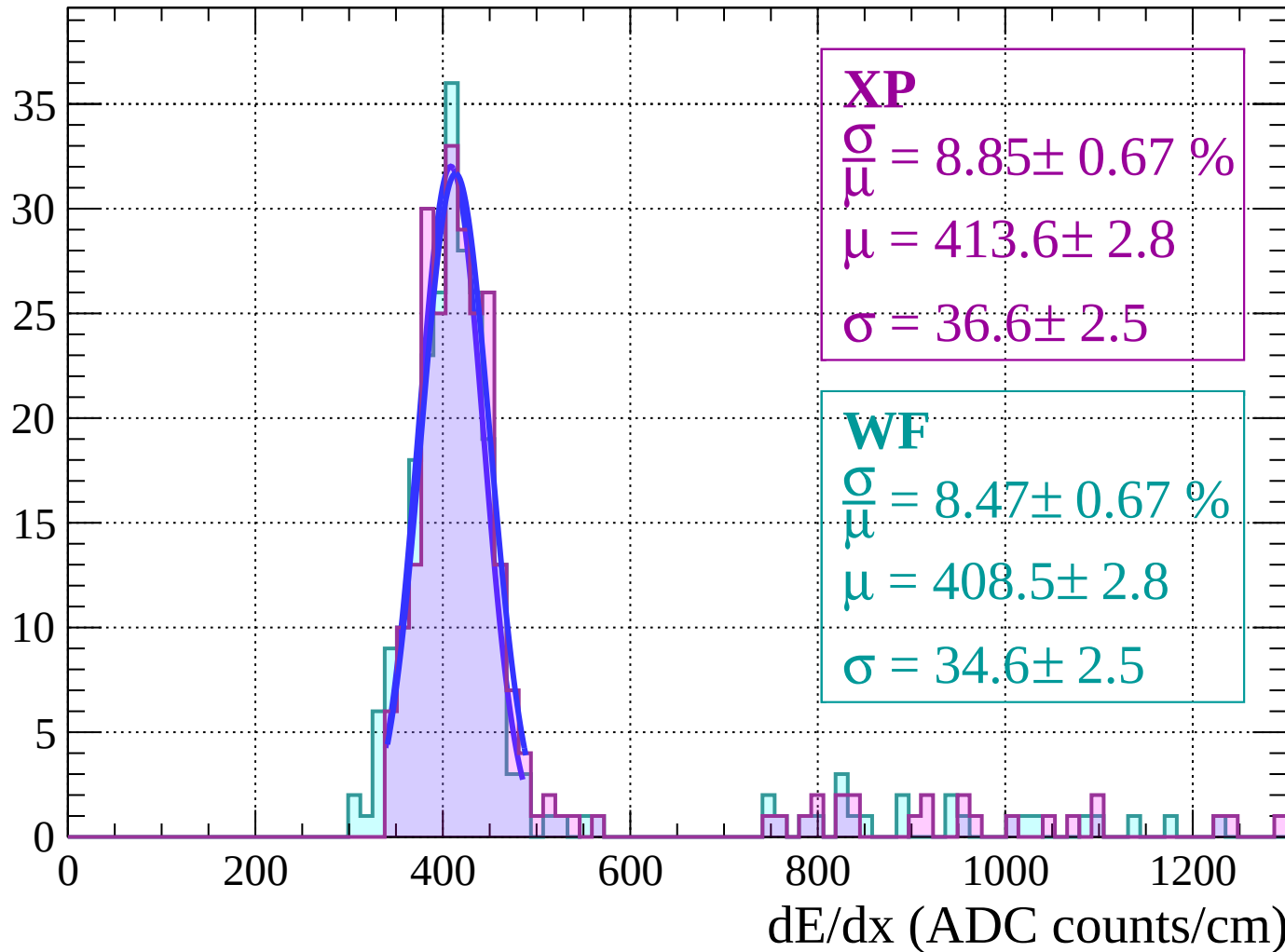
$$\sigma = 36.6 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.67 \%$$

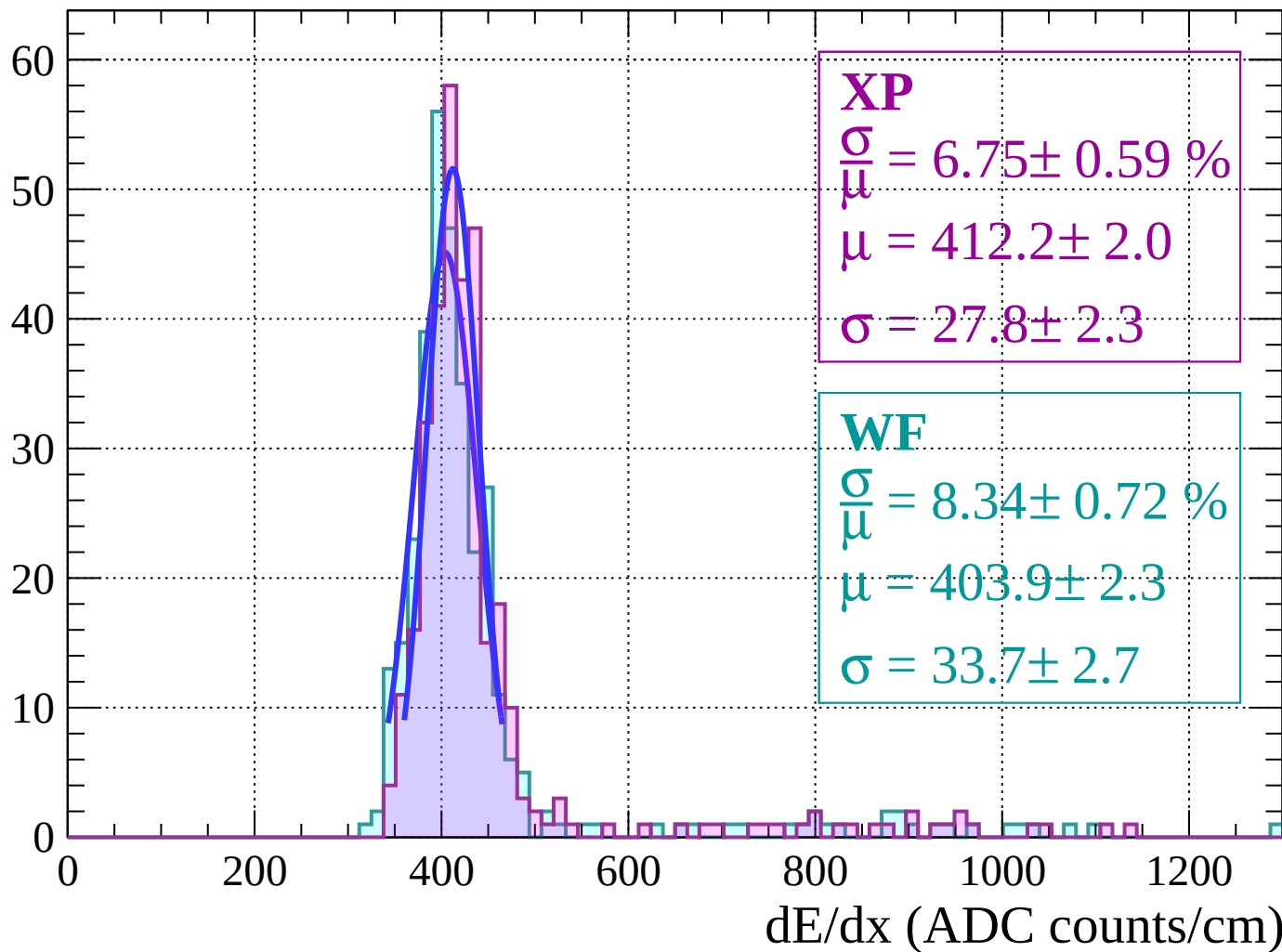
$$\mu = 408.5 \pm 2.8$$

$$\sigma = 34.6 \pm 2.5$$



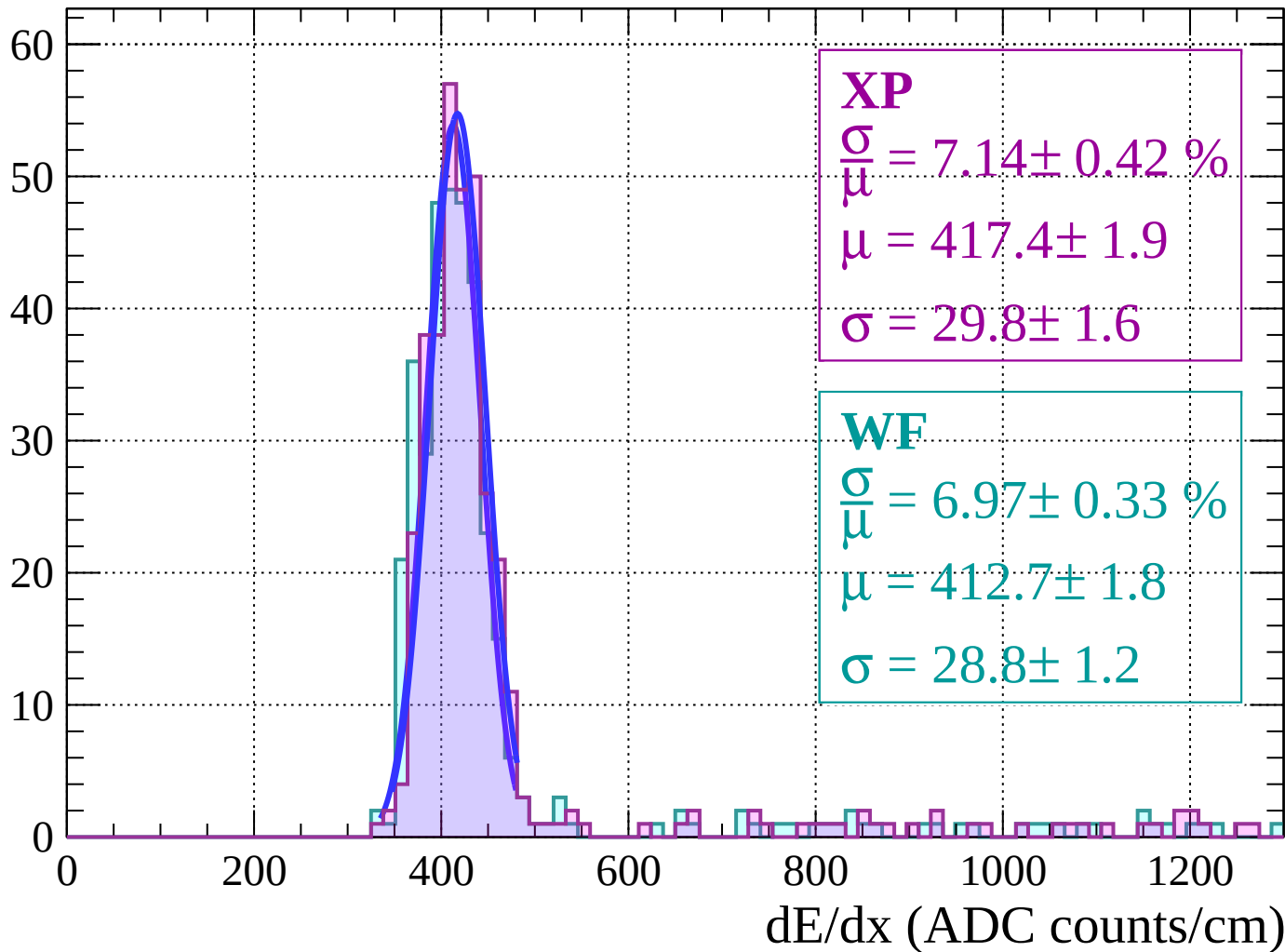
Energy loss | $-22 < \theta < -20$

Count



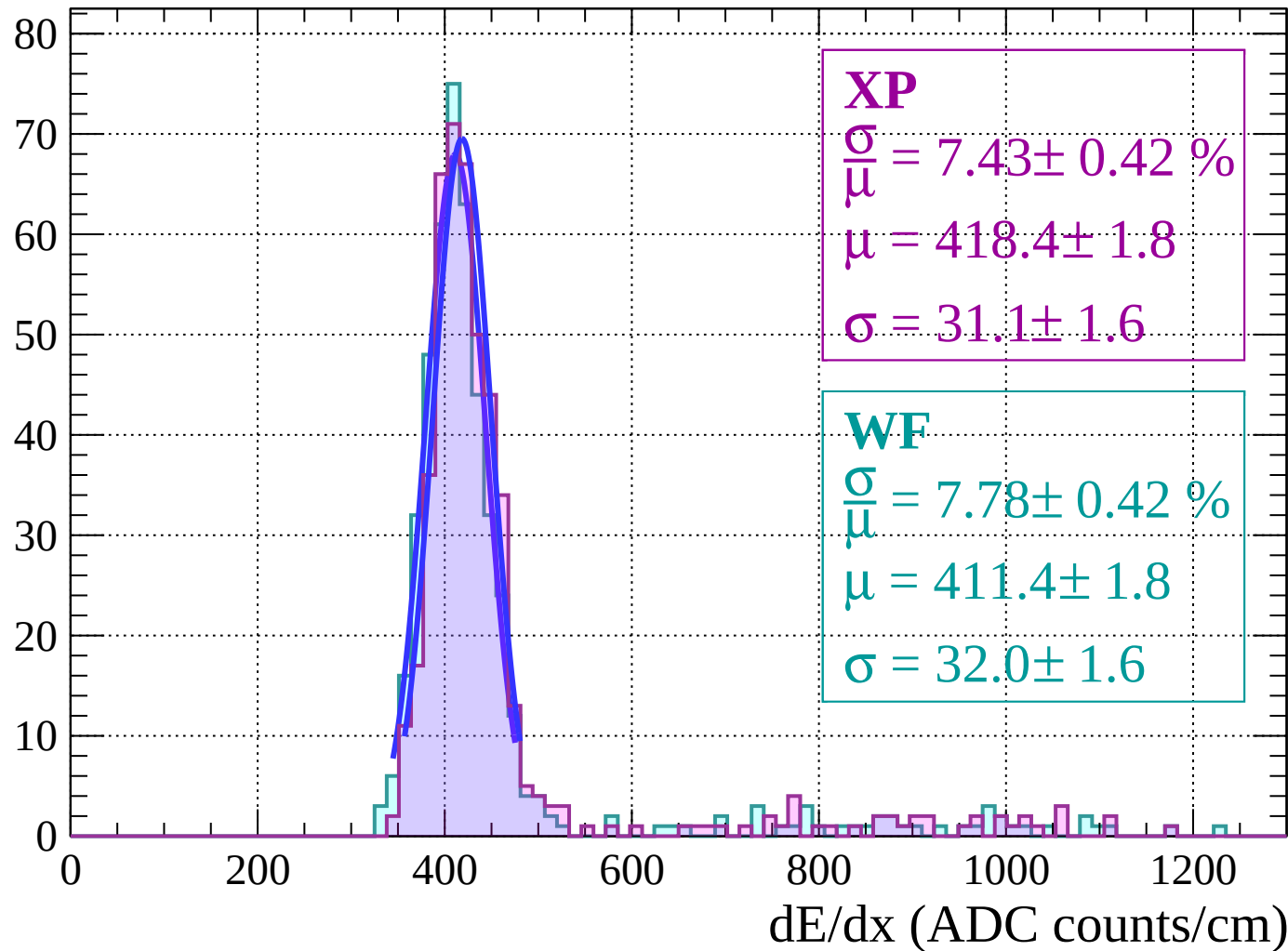
Energy loss | $-20 < \theta < -18$

Count



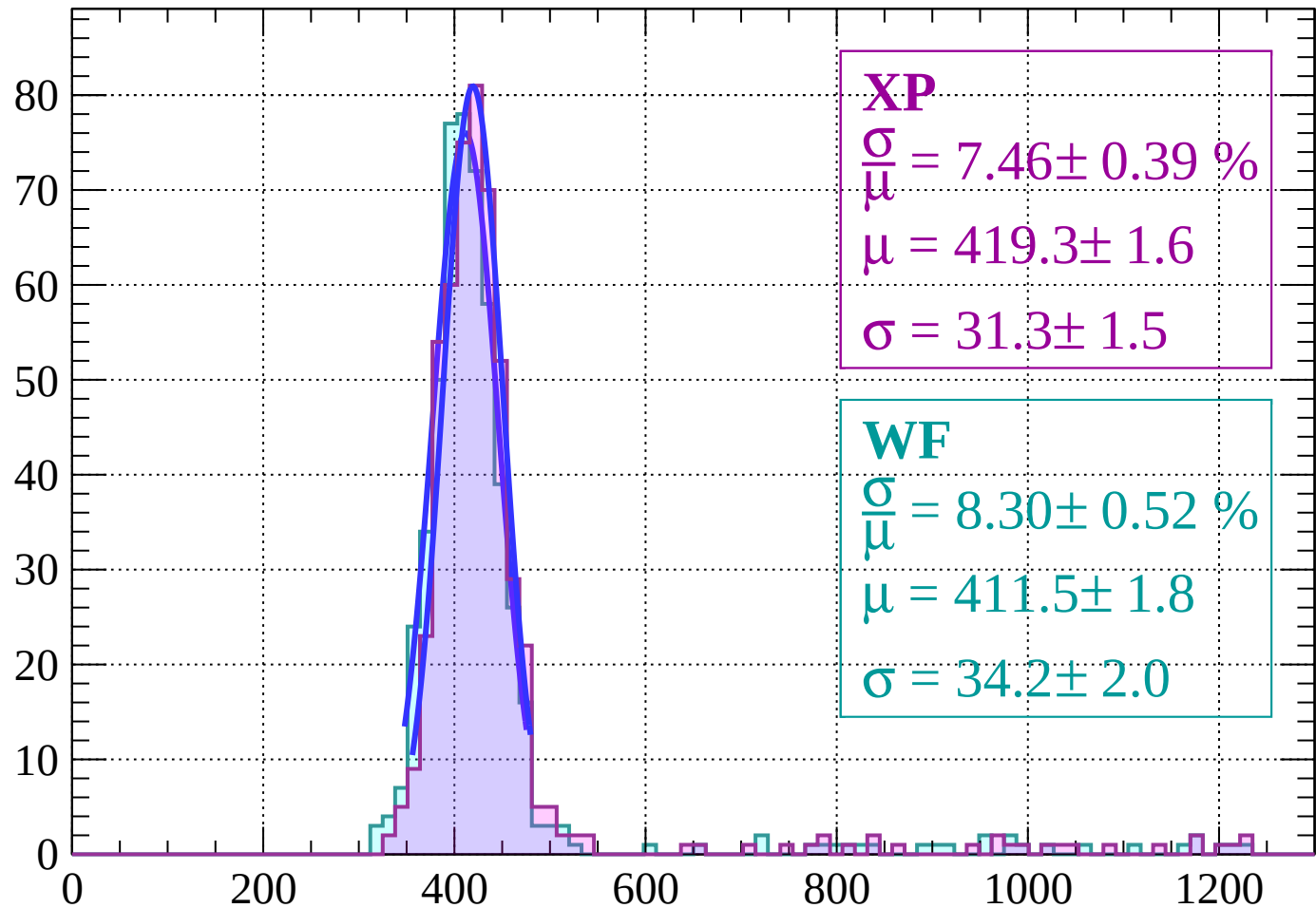
Energy loss | $-18 < \theta < -16$

Count



Energy loss $|-16 < \theta < -14$

Count



XP

$$\frac{\sigma}{\mu} = 7.46 \pm 0.39 \%$$

$$\mu = 419.3 \pm 1.6$$

$$\sigma = 31.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.52 \%$$

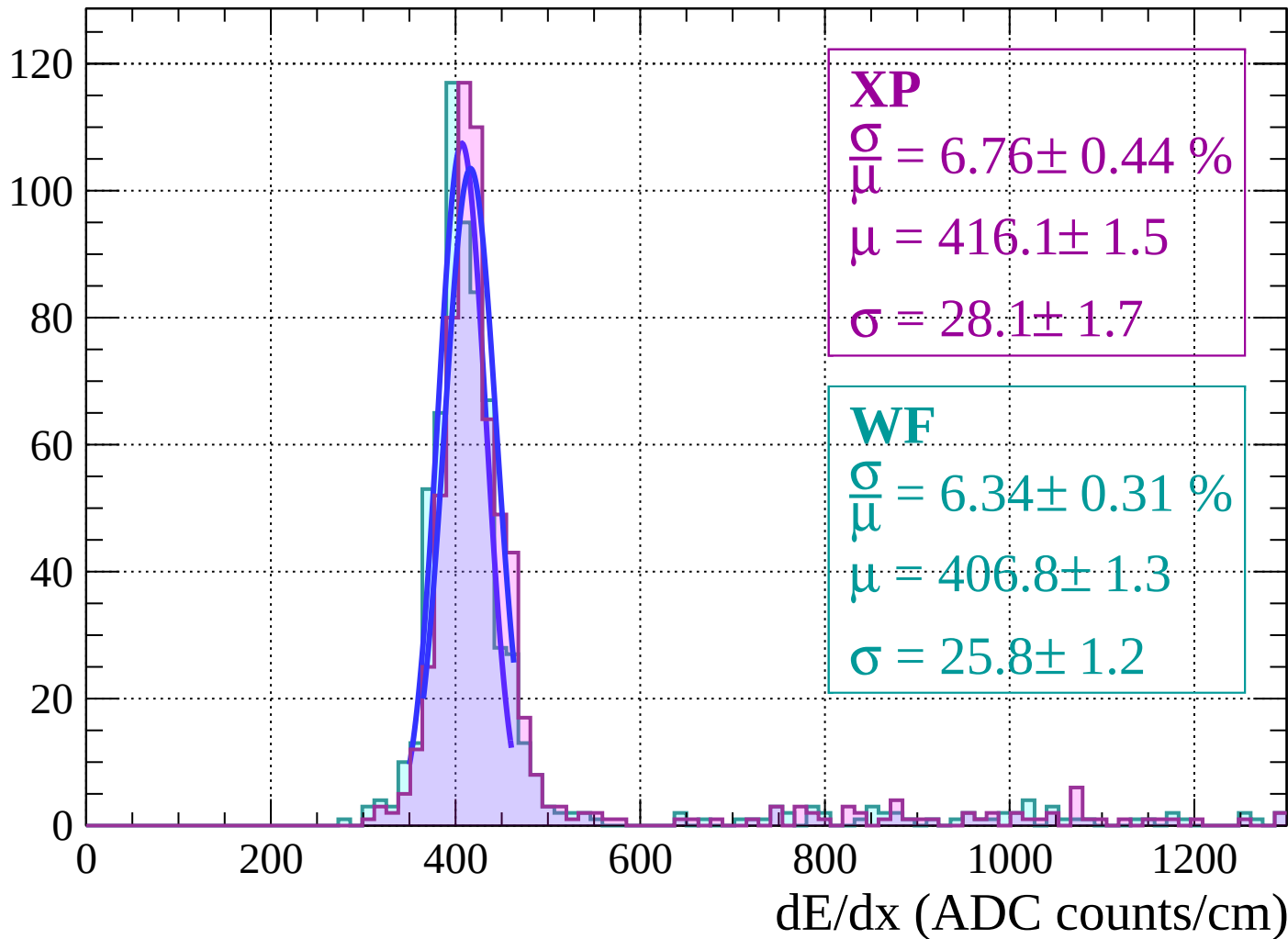
$$\mu = 411.5 \pm 1.8$$

$$\sigma = 34.2 \pm 2.0$$

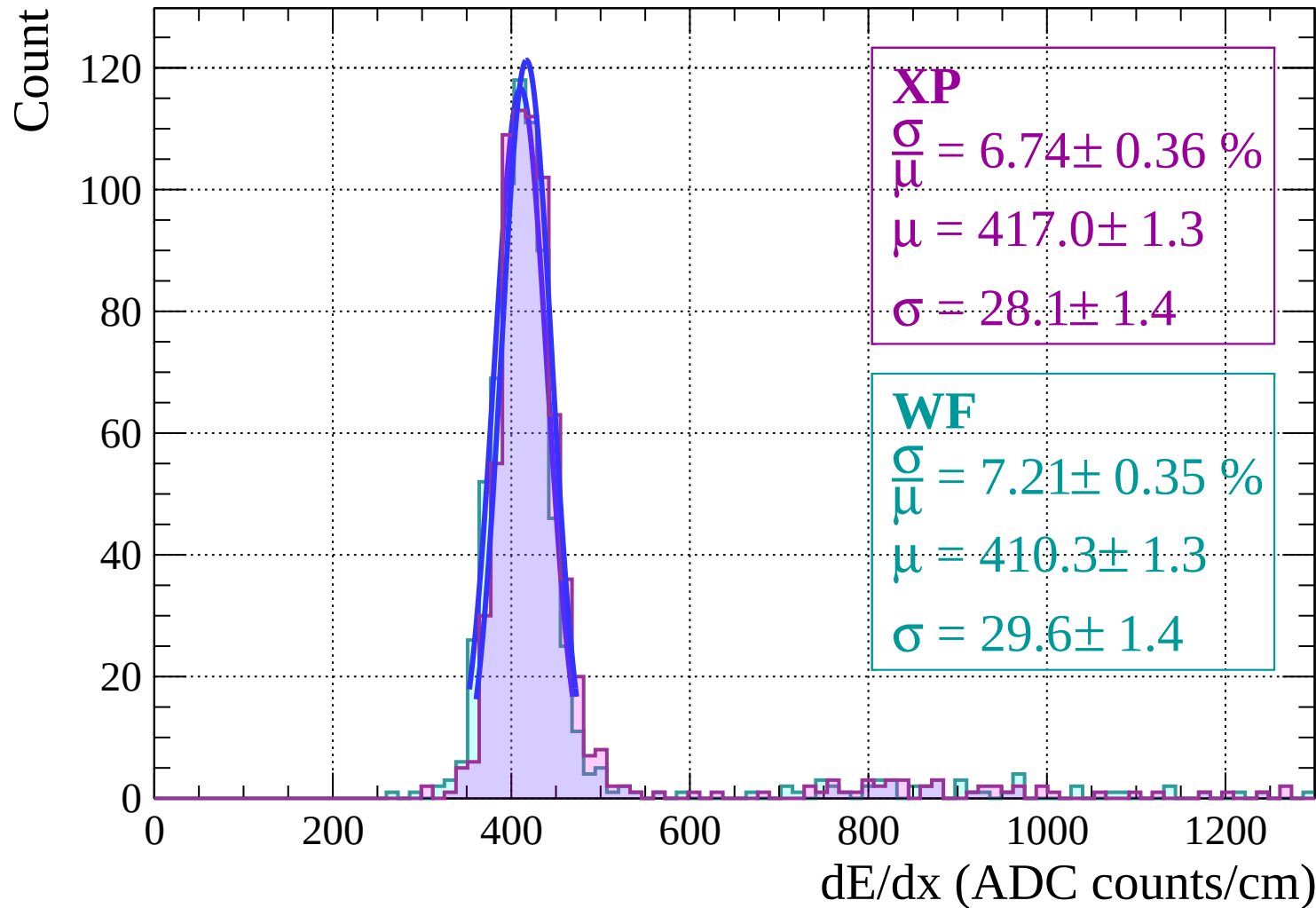
dE/dx (ADC counts/cm)

Energy loss $|-14 < \theta < -12$

Count

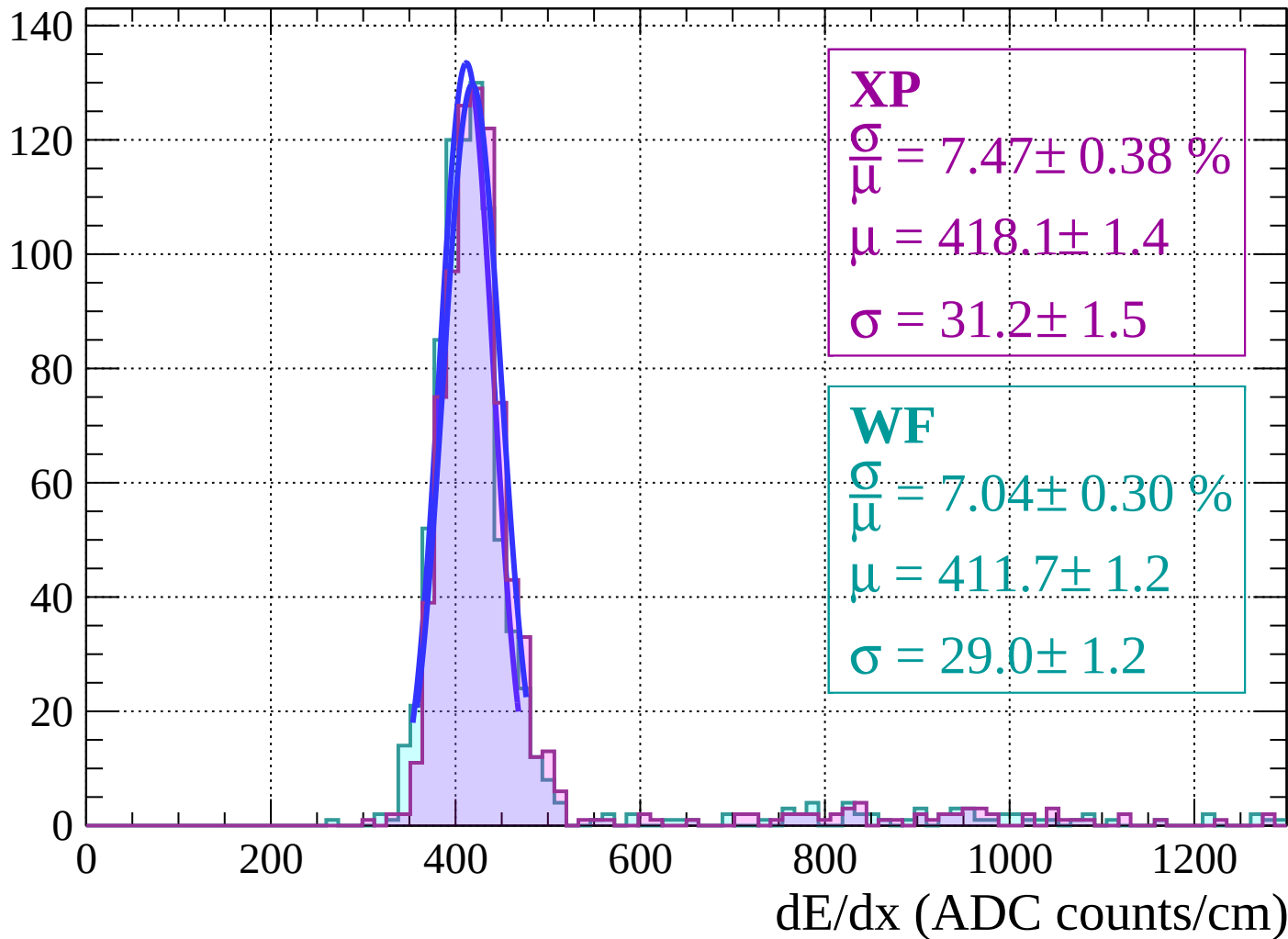


Energy loss | $-12 < \theta < -10$

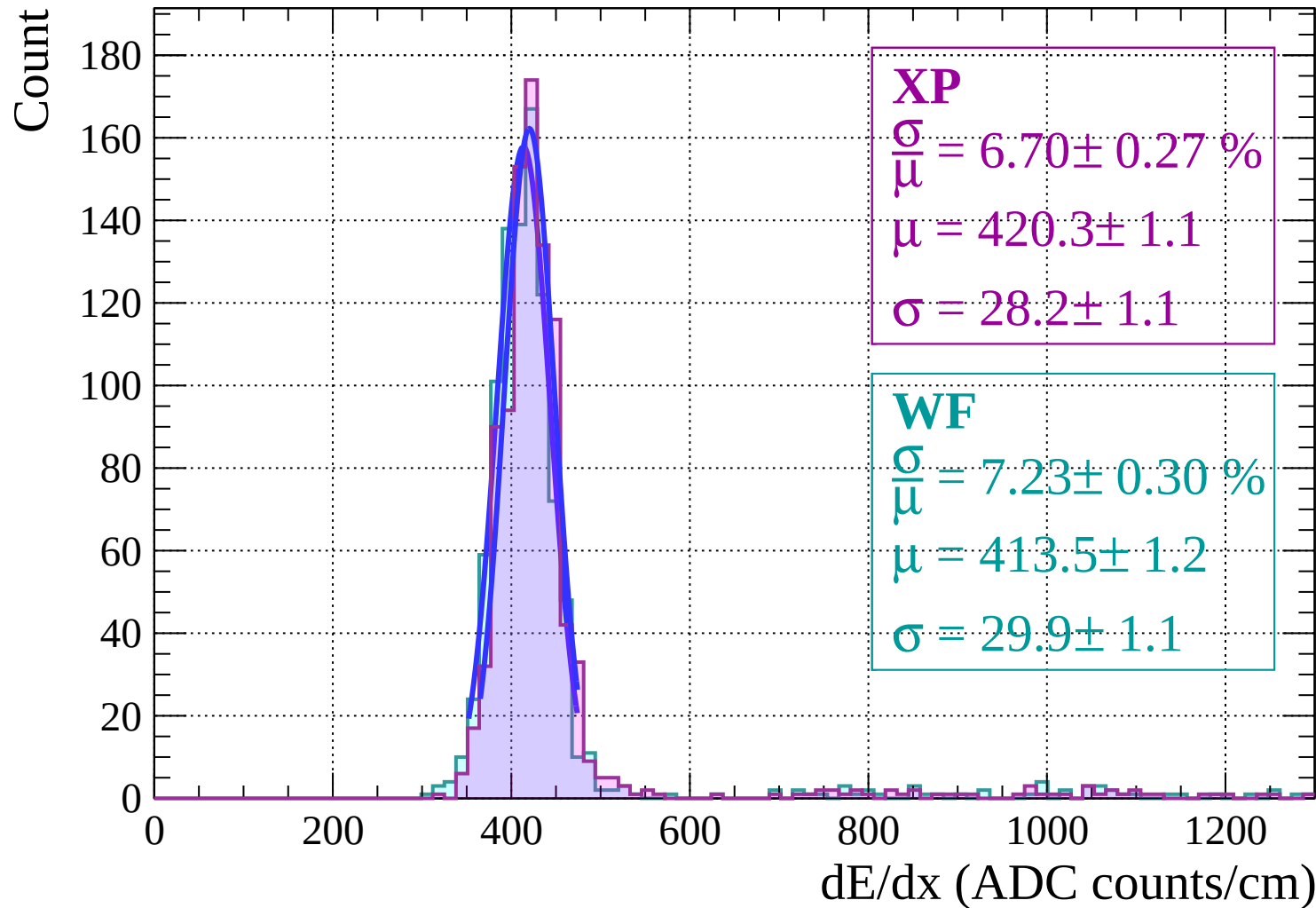


Energy loss | $-10 < \theta < -8$

Count

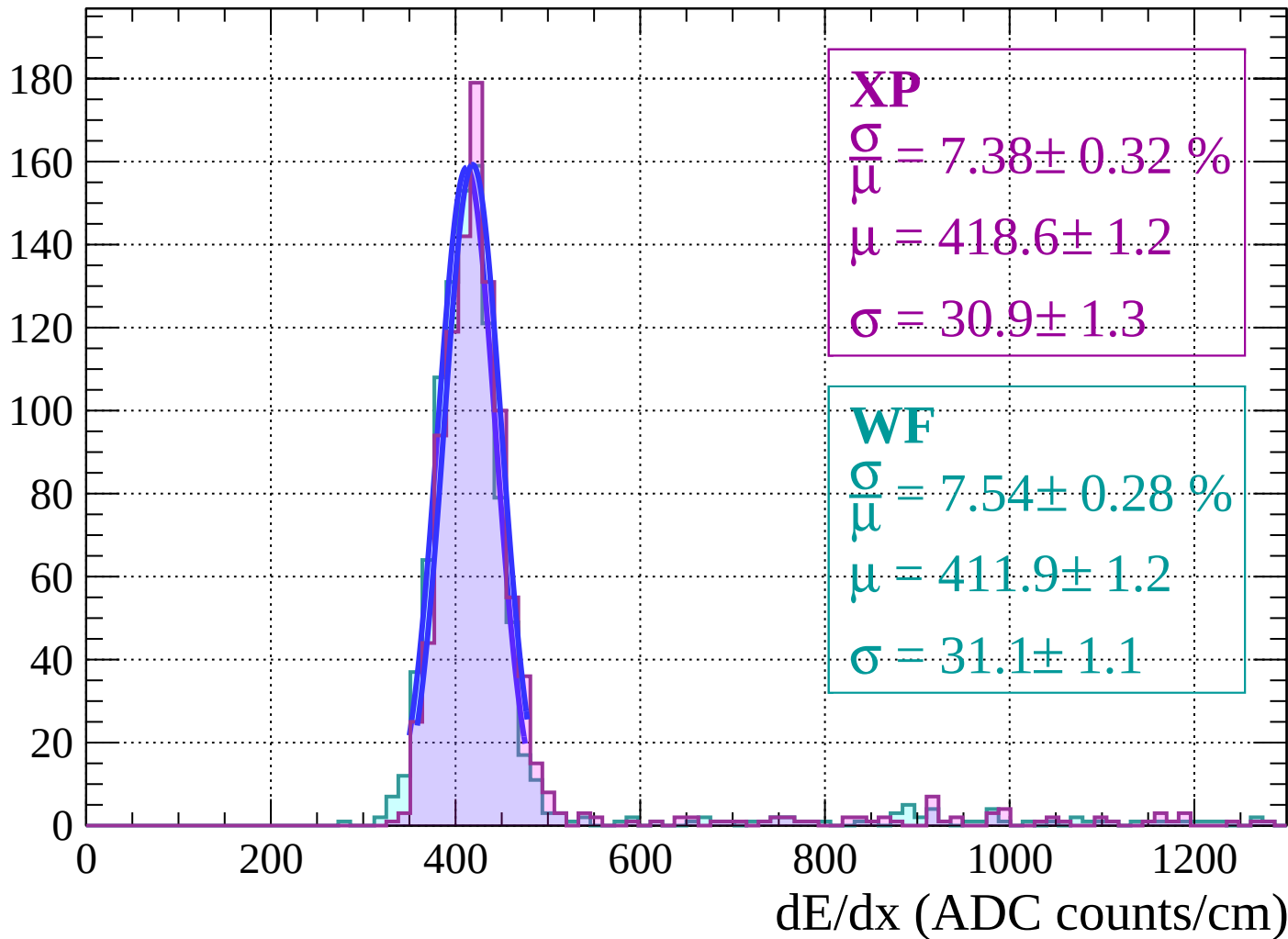


Energy loss | $-8 < \theta < -6$



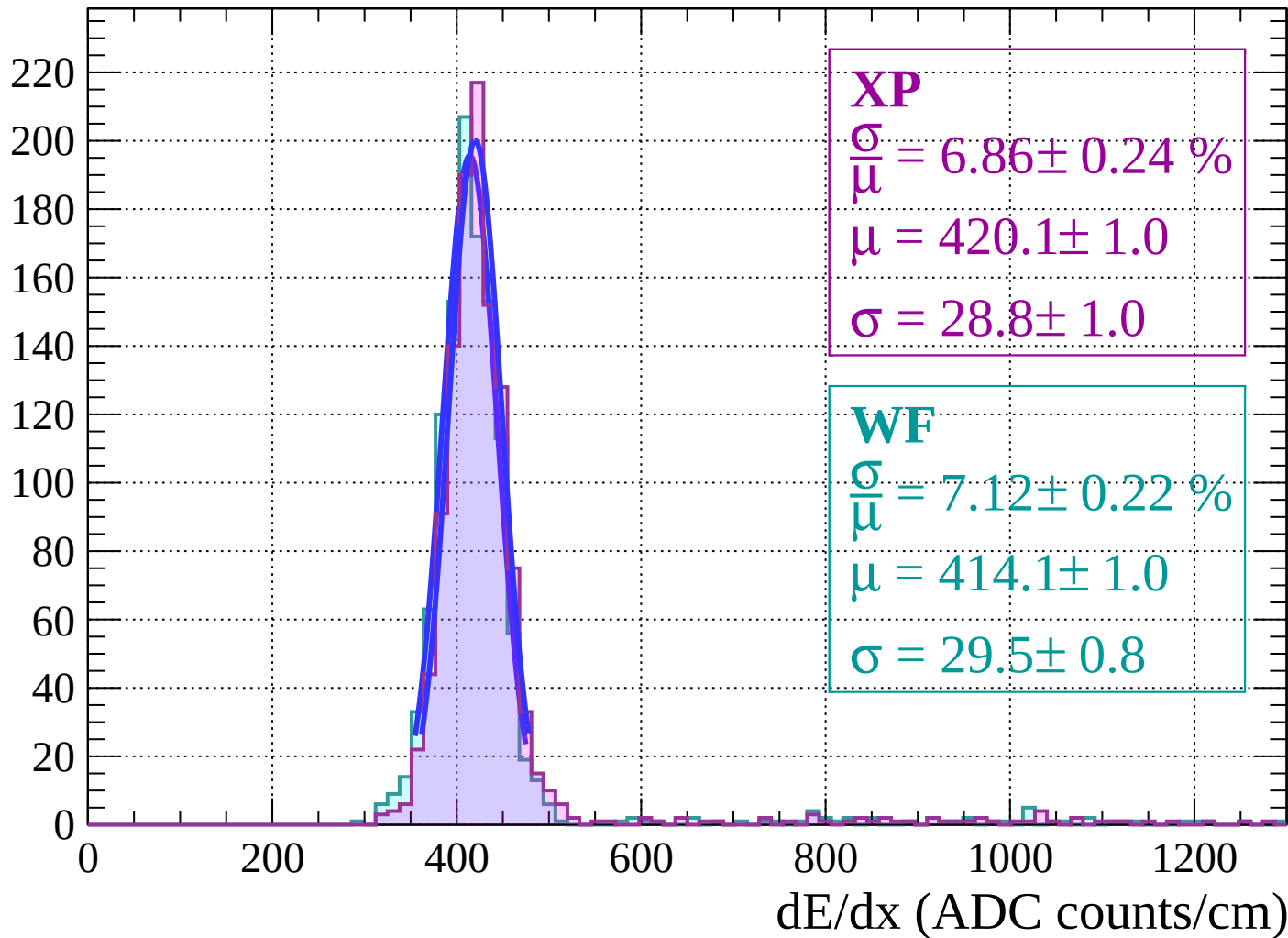
Energy loss | $-6 < \theta < -4$

Count

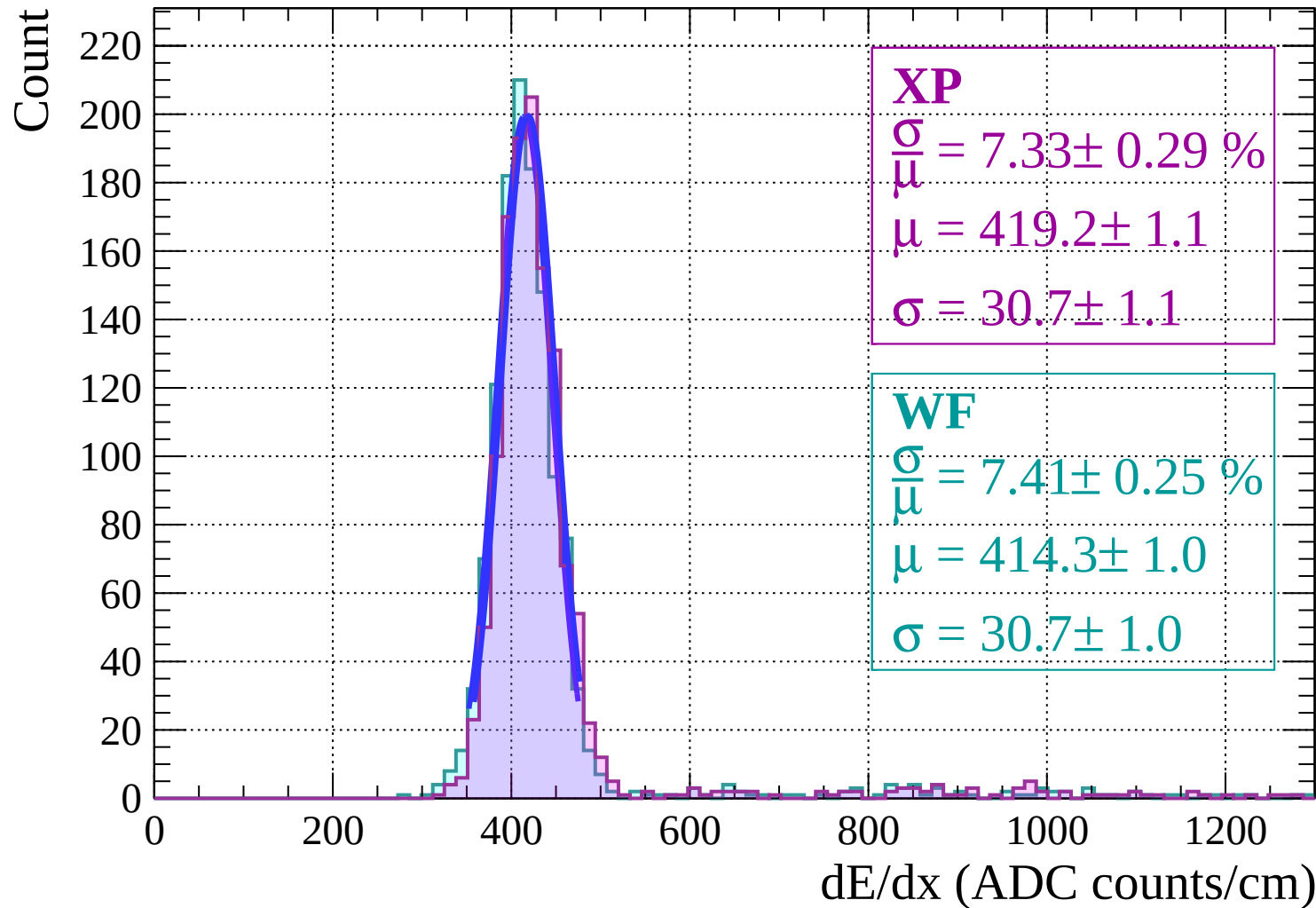


Energy loss | $-4 < \theta < -2$

Count

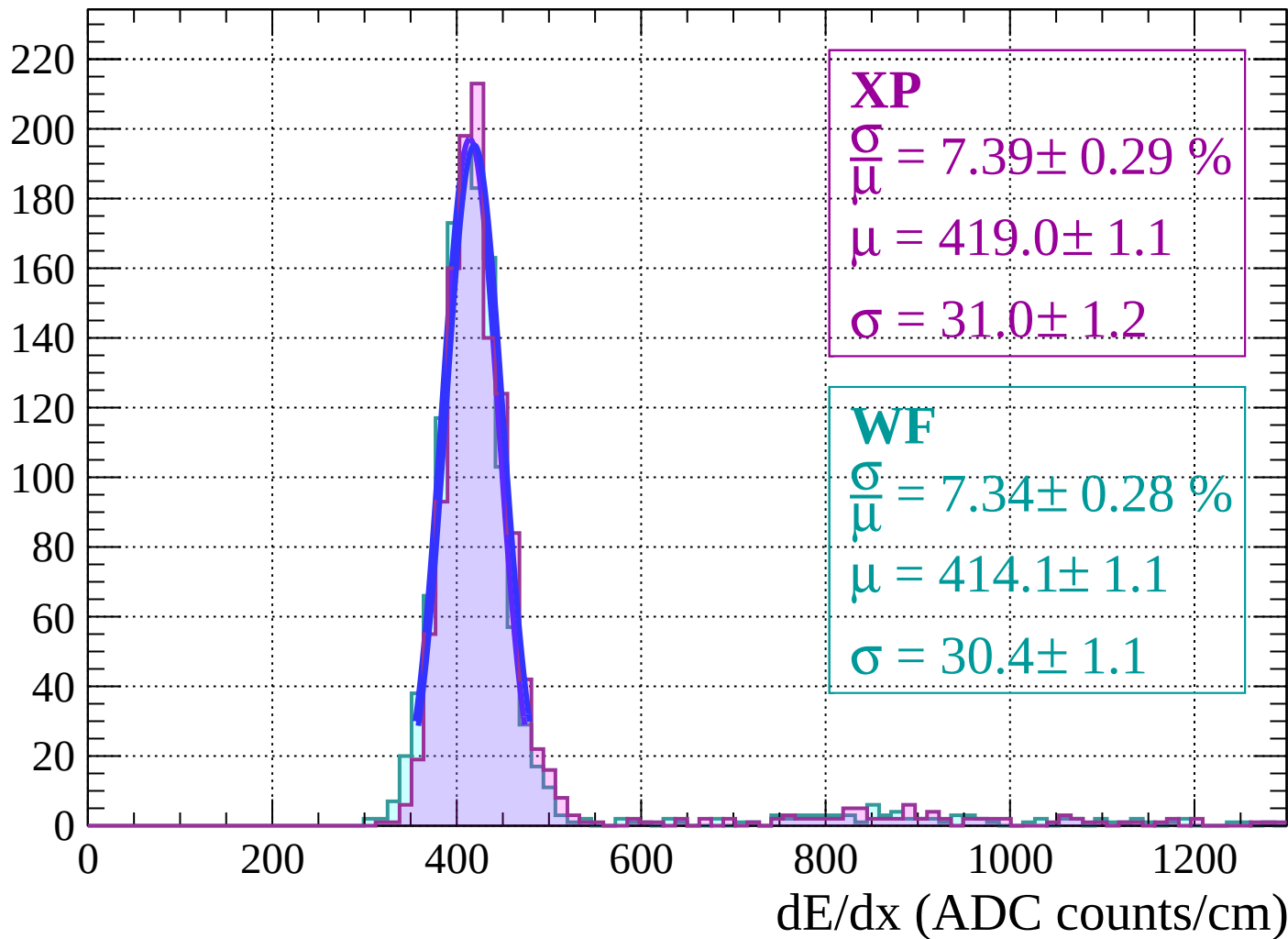


Energy loss | $-2 < \theta < 0$



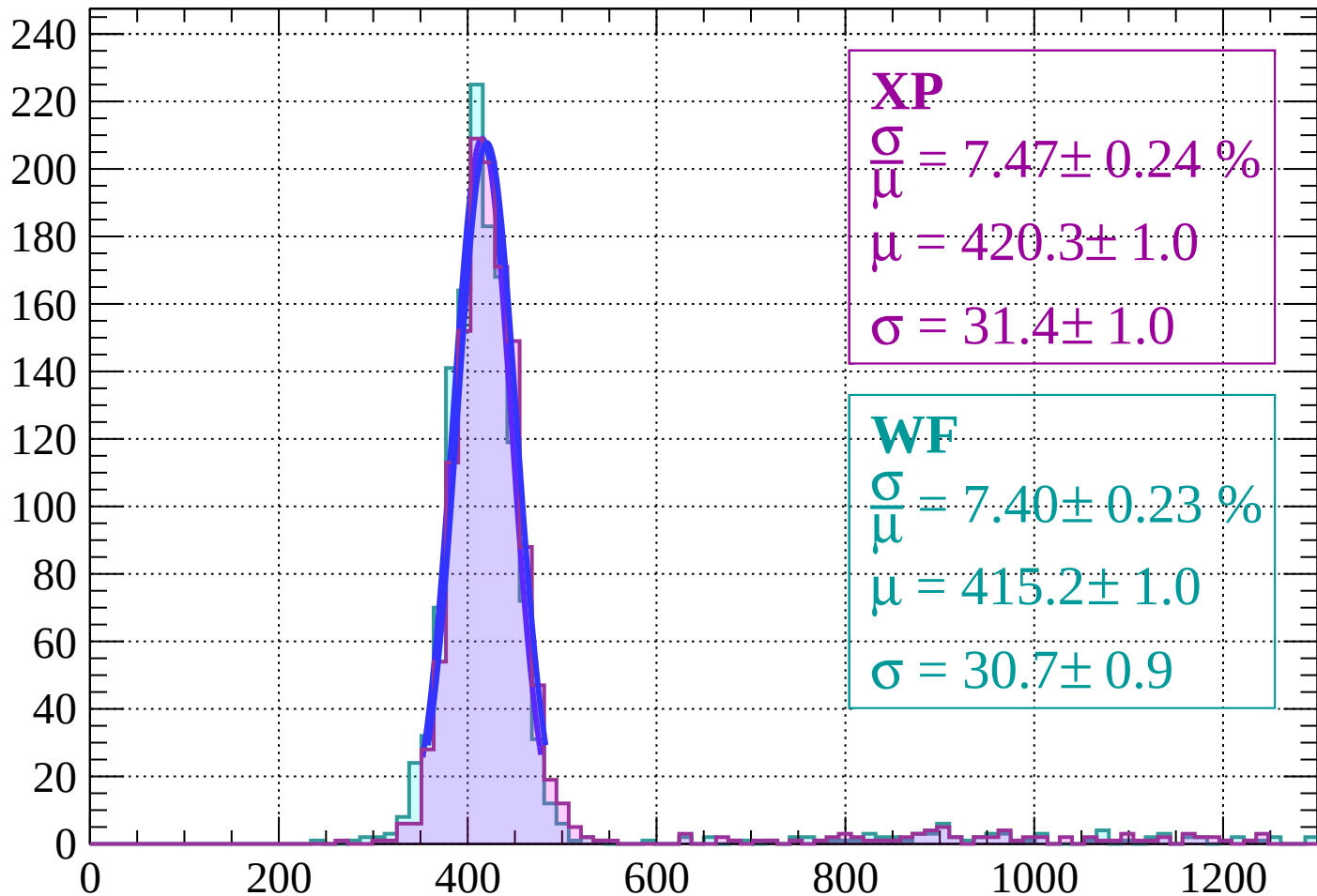
Energy loss | $0 < \theta < 2$

Count



Energy loss | $2 < \theta < 4$

Count



XP

$$\frac{\sigma}{\mu} = 7.47 \pm 0.24 \%$$

$$\mu = 420.3 \pm 1.0$$

$$\sigma = 31.4 \pm 1.0$$

WF

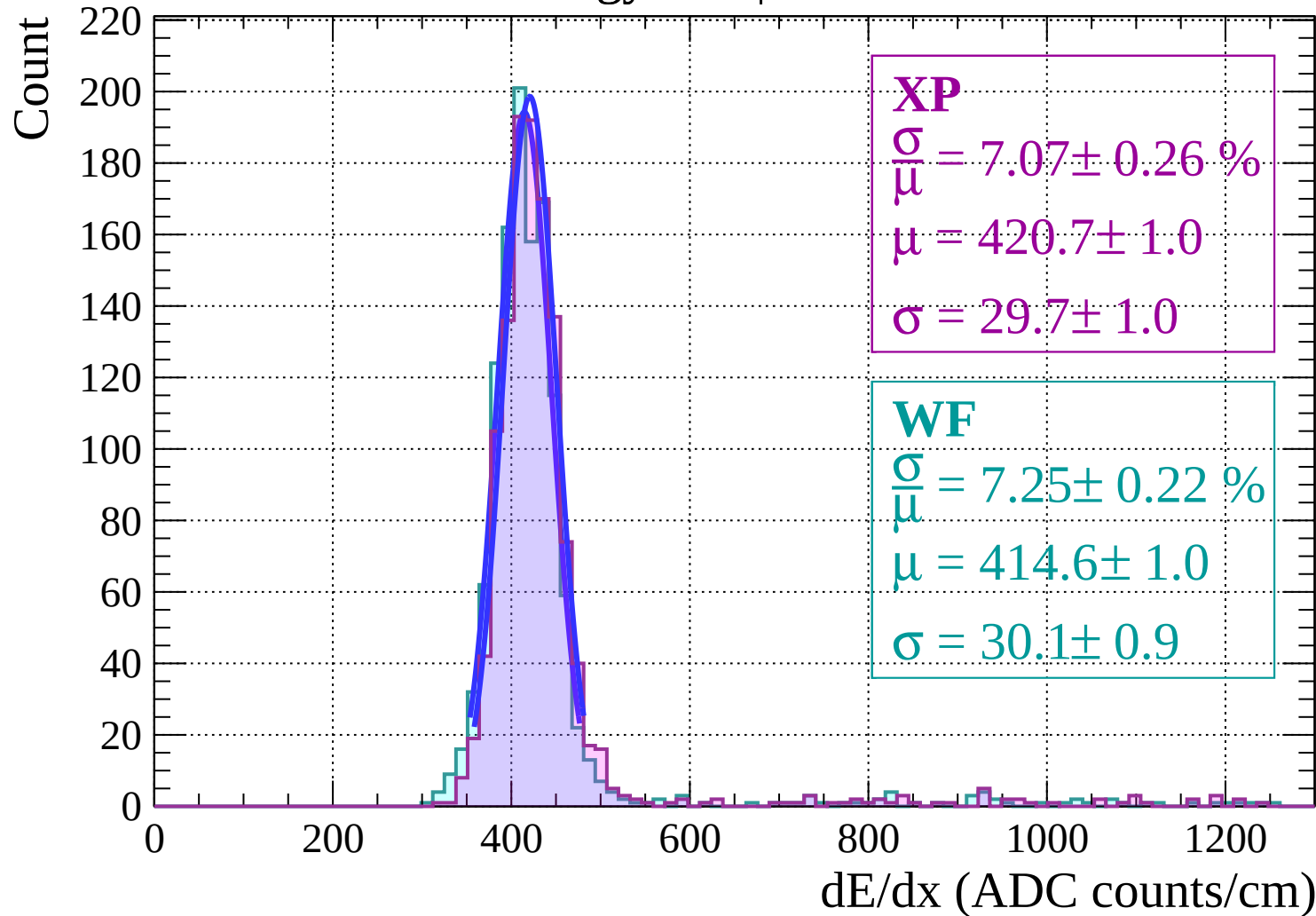
$$\frac{\sigma}{\mu} = 7.40 \pm 0.23 \%$$

$$\mu = 415.2 \pm 1.0$$

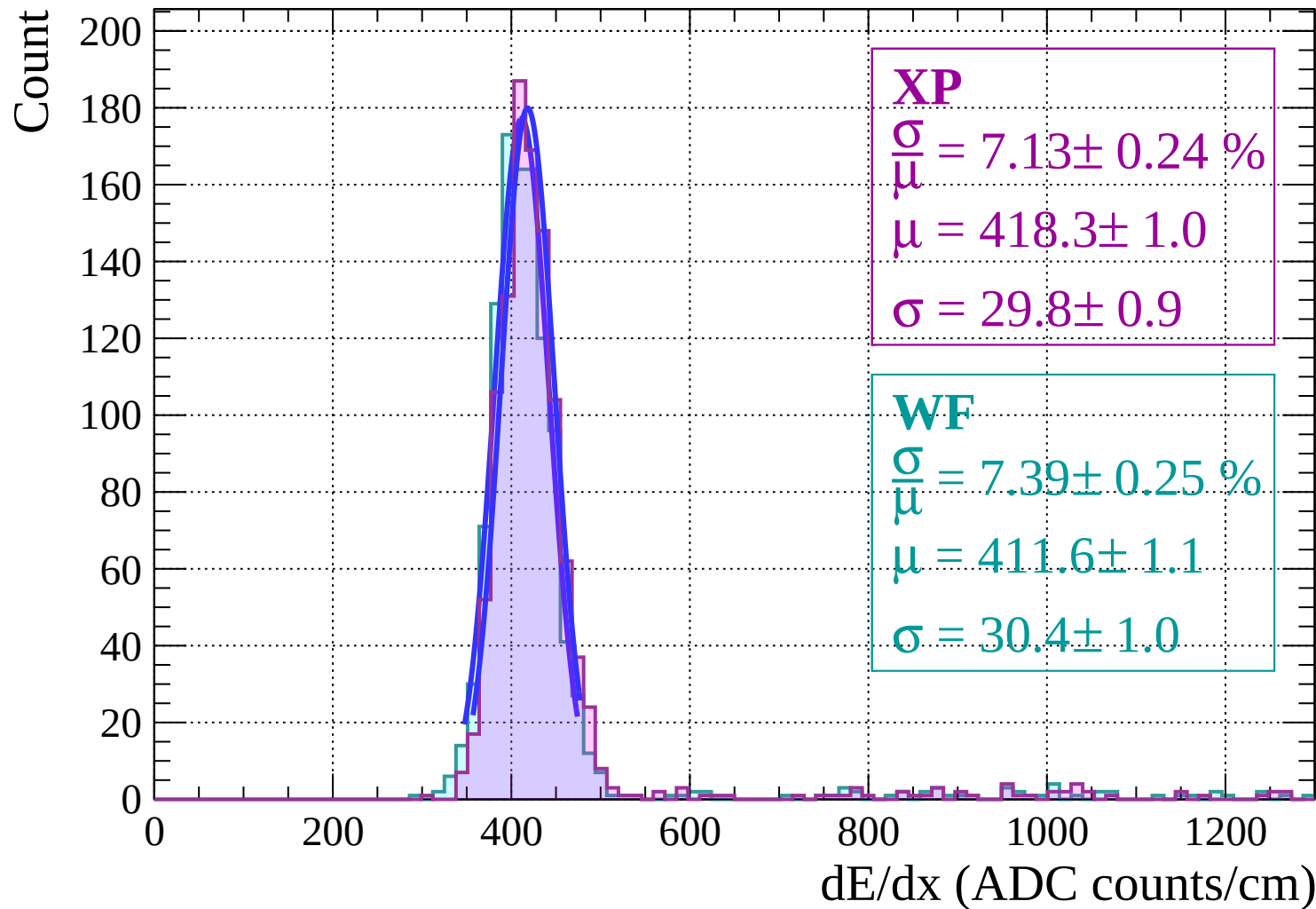
$$\sigma = 30.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $4 < \theta < 6$

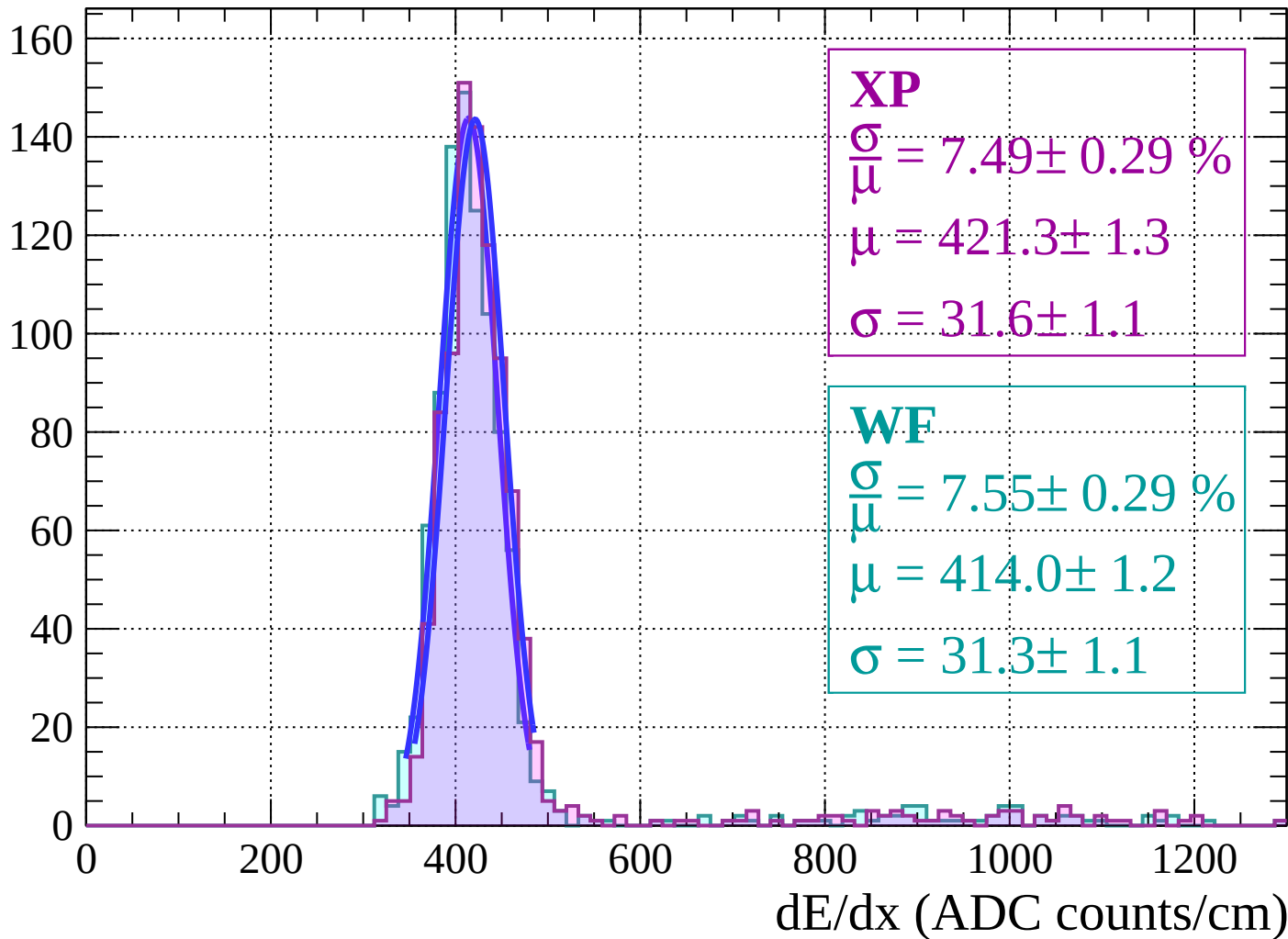


Energy loss | $6 < \theta < 8$



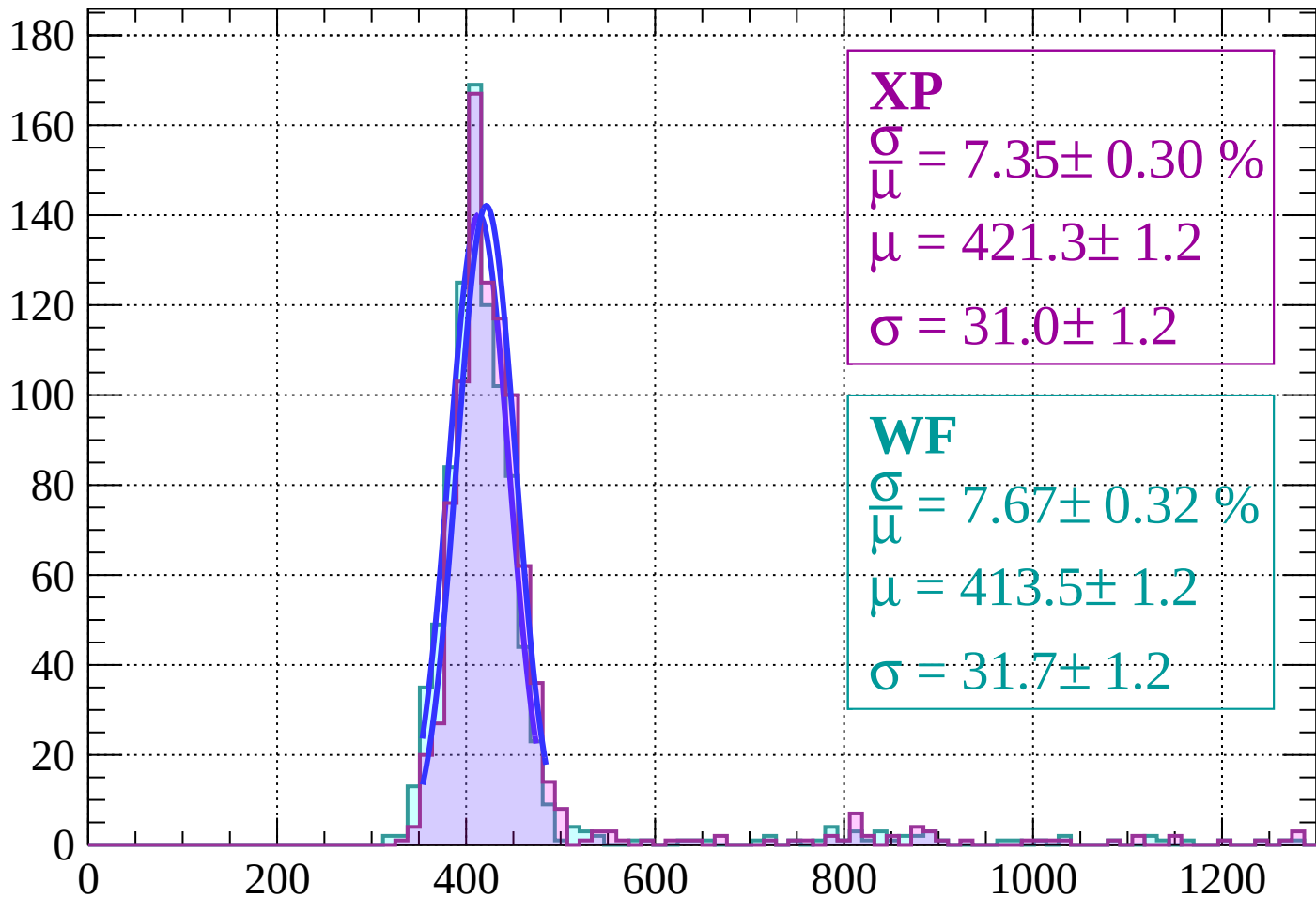
Energy loss | $8 < \theta < 10$

Count



Energy loss | $10 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.30 \%$$

$$\mu = 421.3 \pm 1.2$$

$$\sigma = 31.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.67 \pm 0.32 \%$$

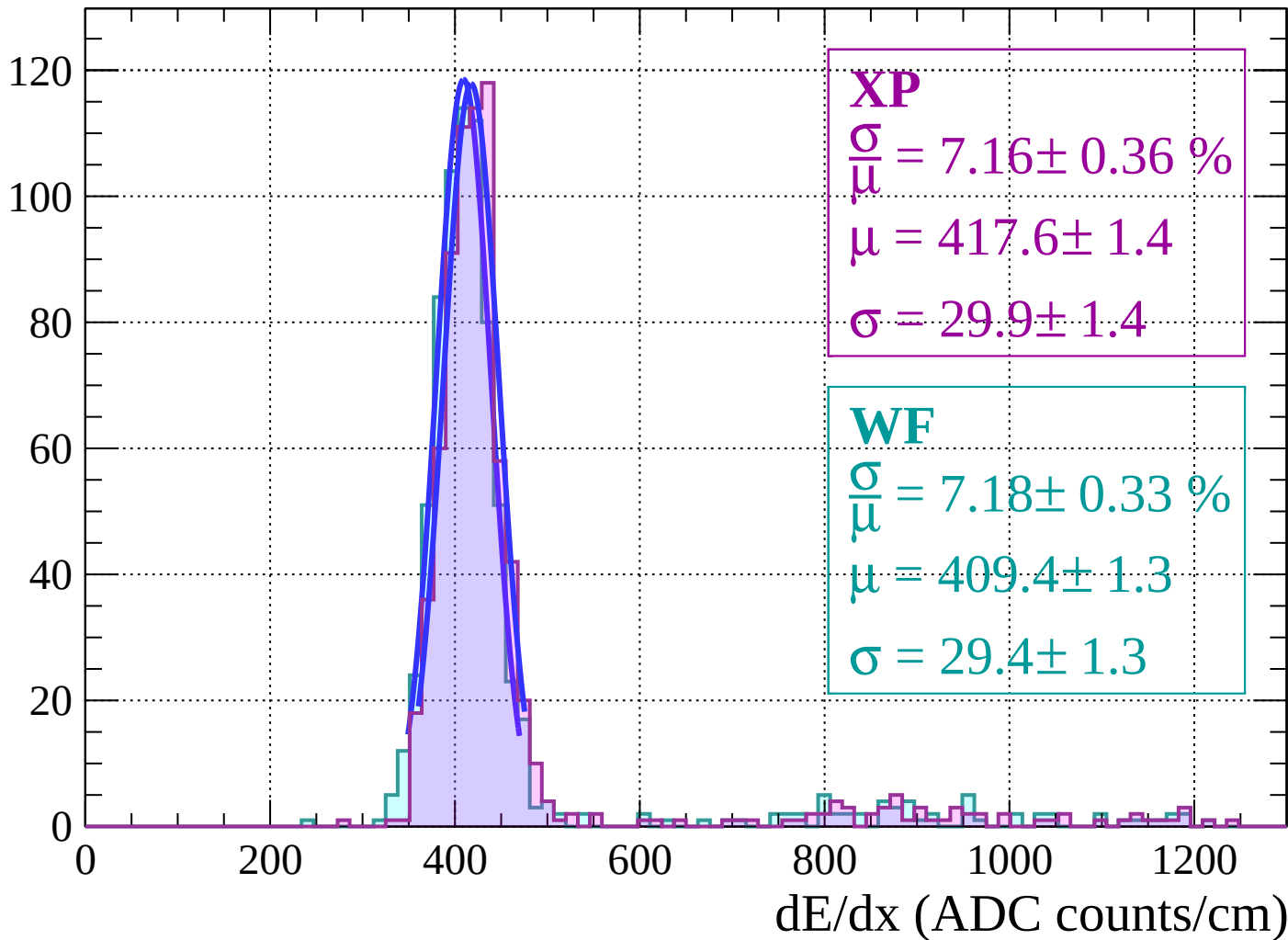
$$\mu = 413.5 \pm 1.2$$

$$\sigma = 31.7 \pm 1.2$$

dE/dx (ADC counts/cm)

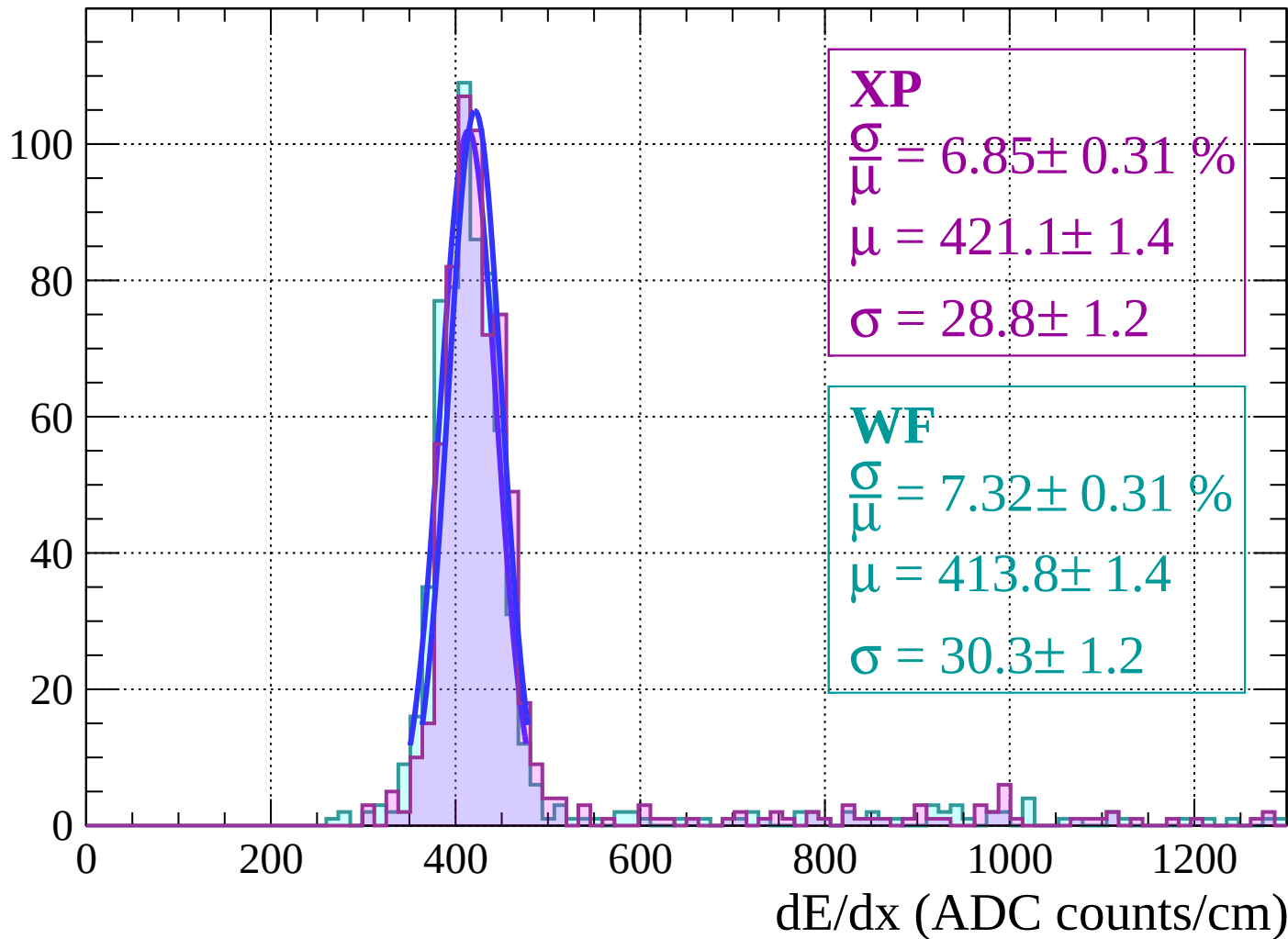
Energy loss | $12 < \theta < 14$

Count



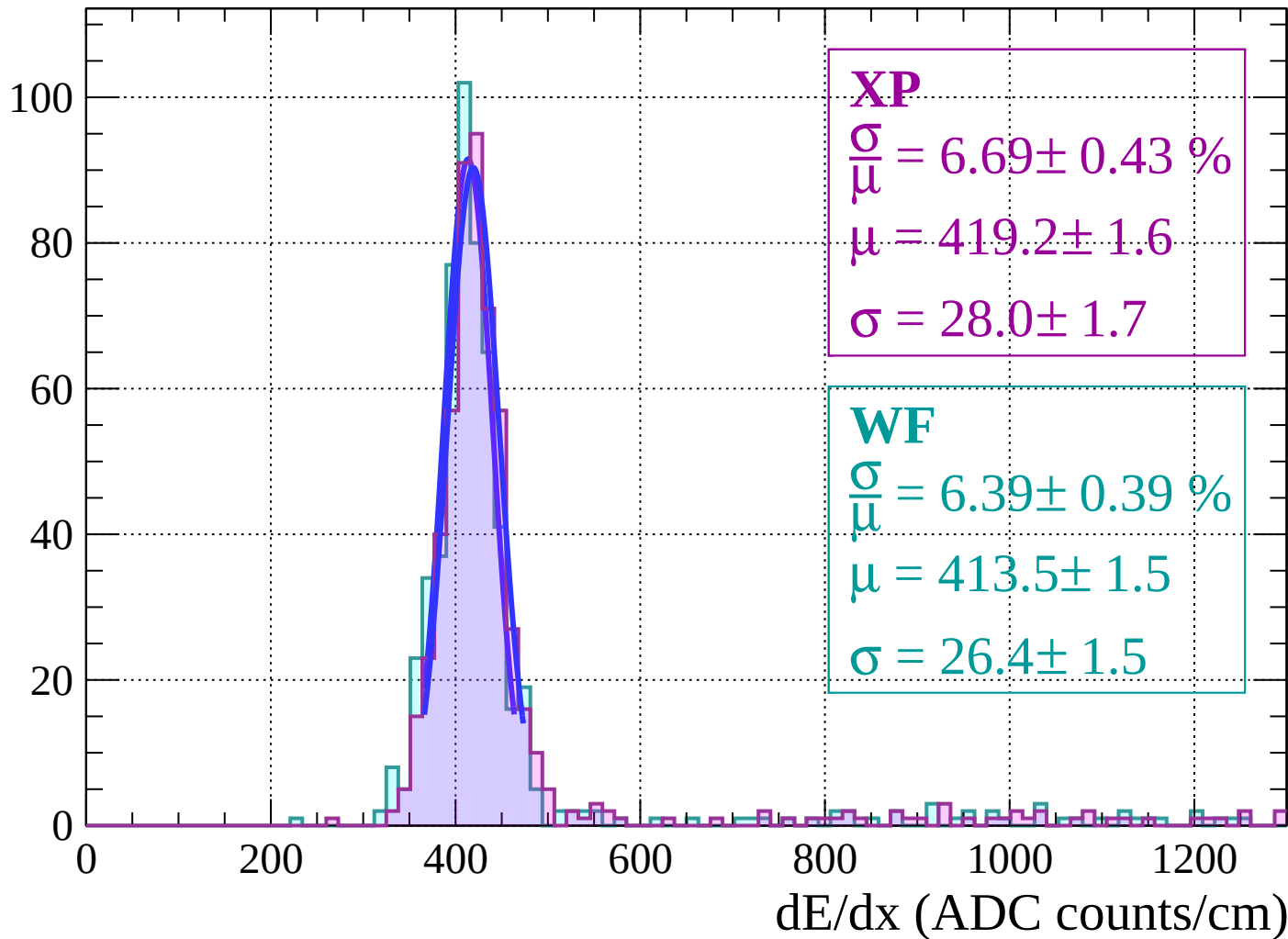
Energy loss | $14 < \theta < 16$

Count



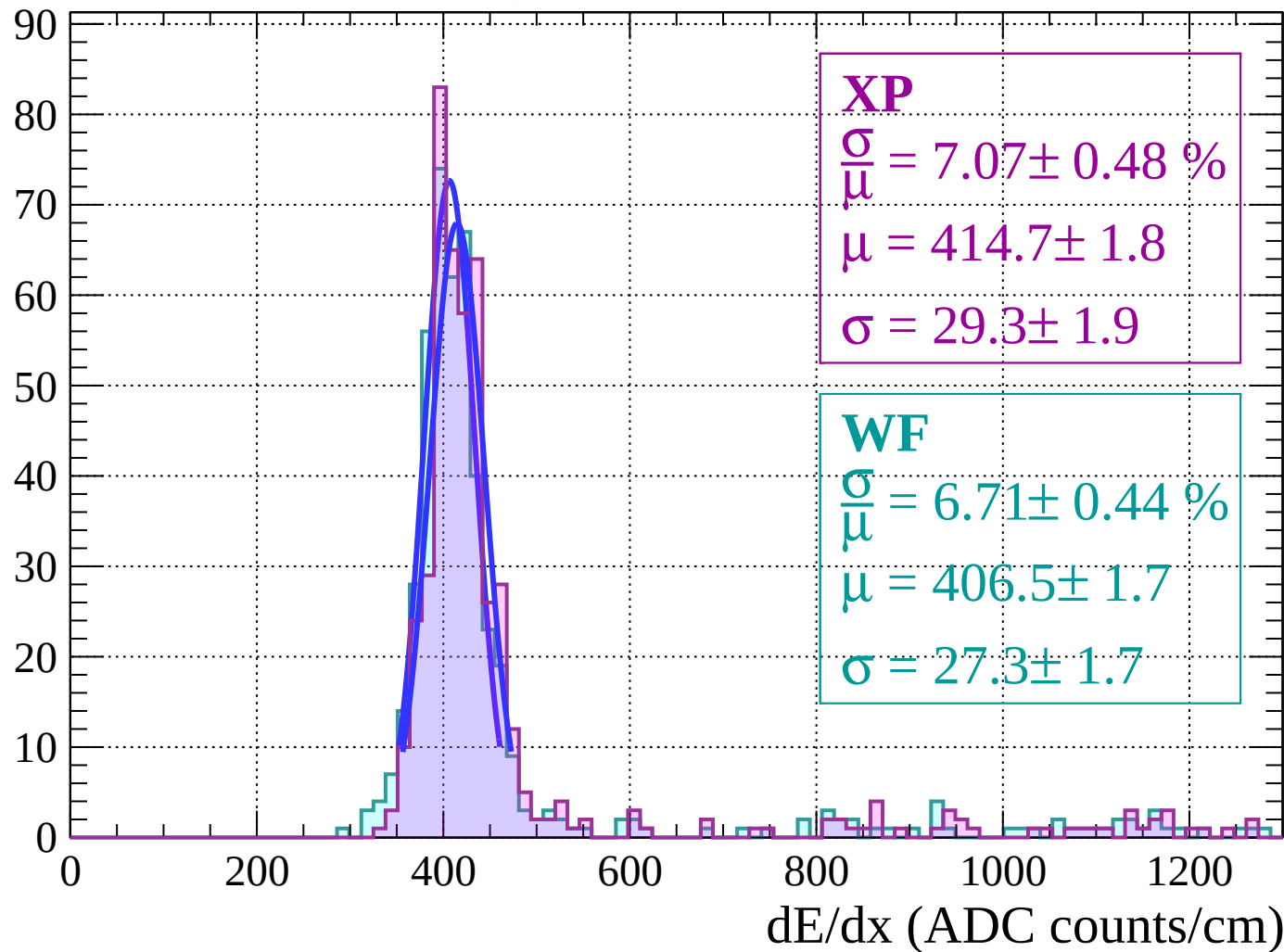
Energy loss | $16 < \theta < 18$

Count



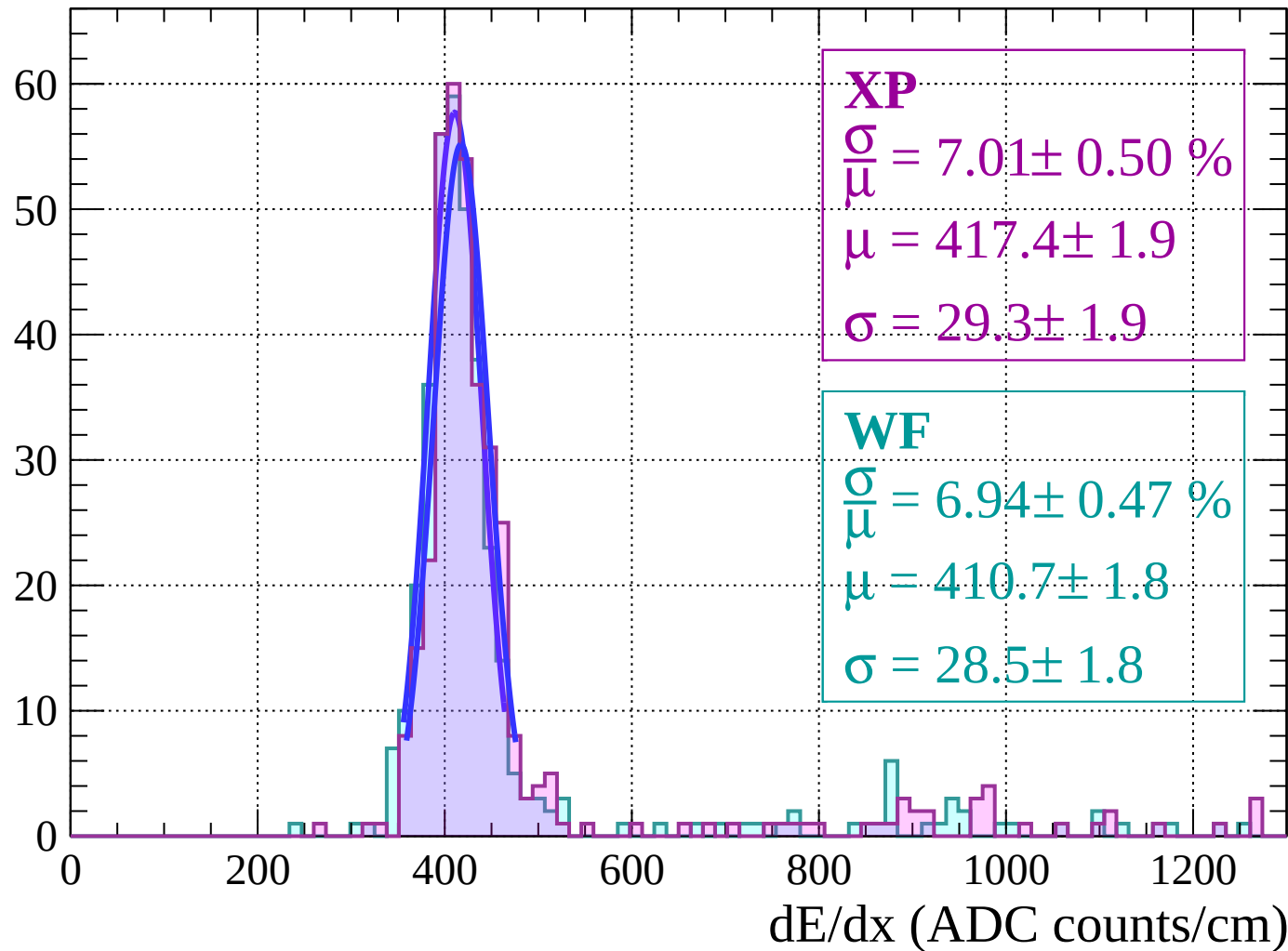
Energy loss | $18 < \theta < 20$

Count



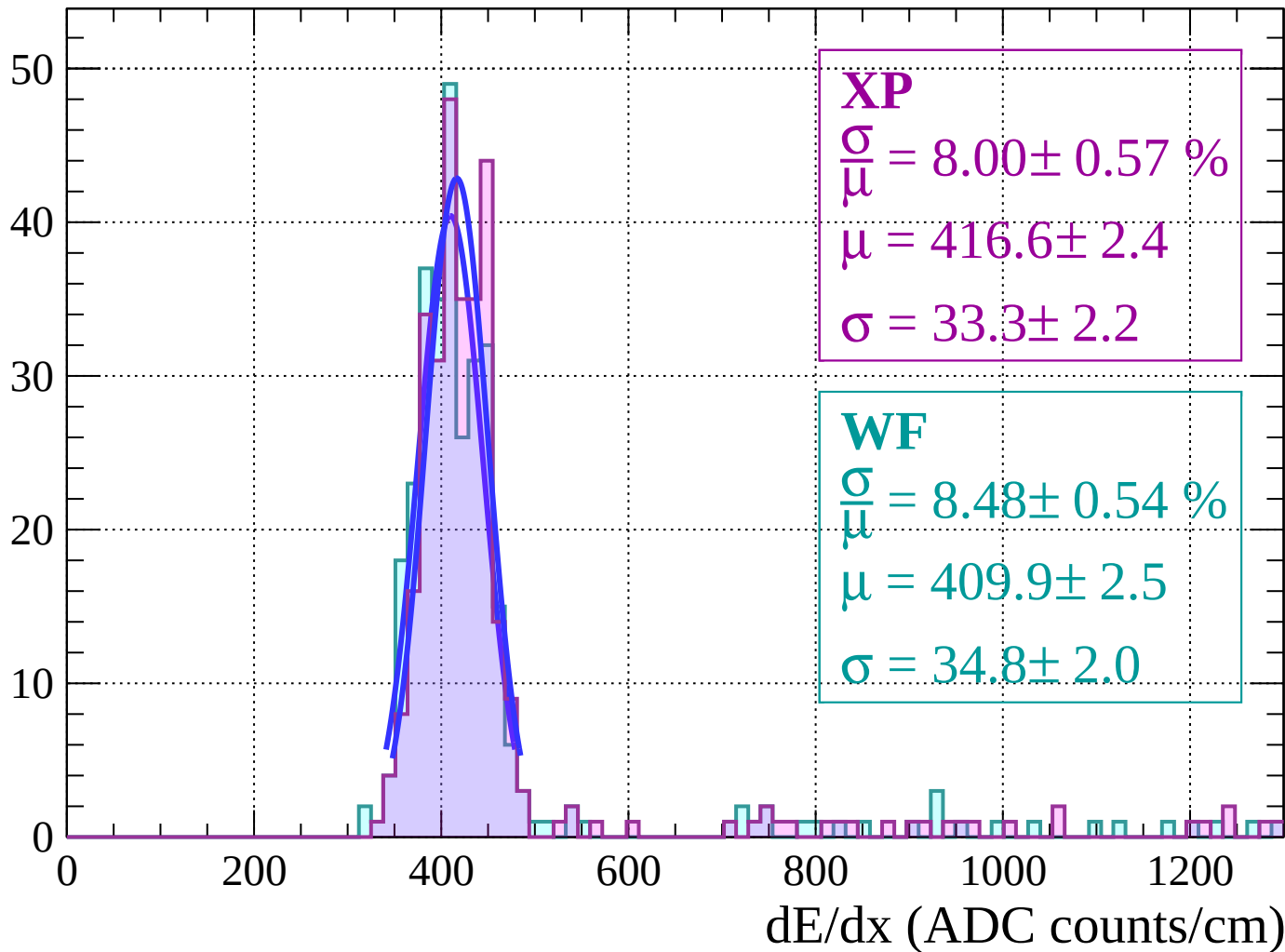
Energy loss | $20 < \theta < 22$

Count



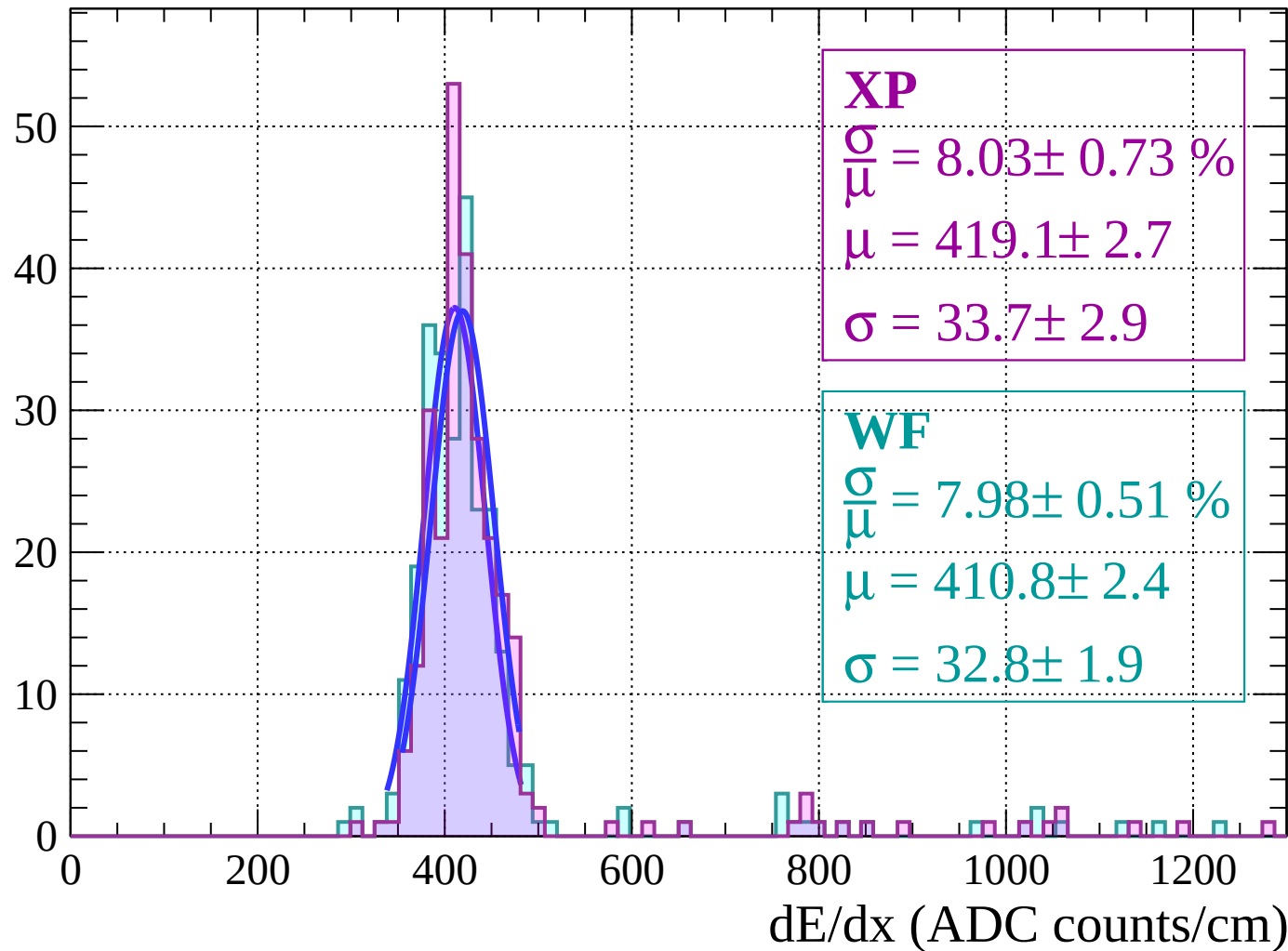
Energy loss | $22 < \theta < 24$

Count



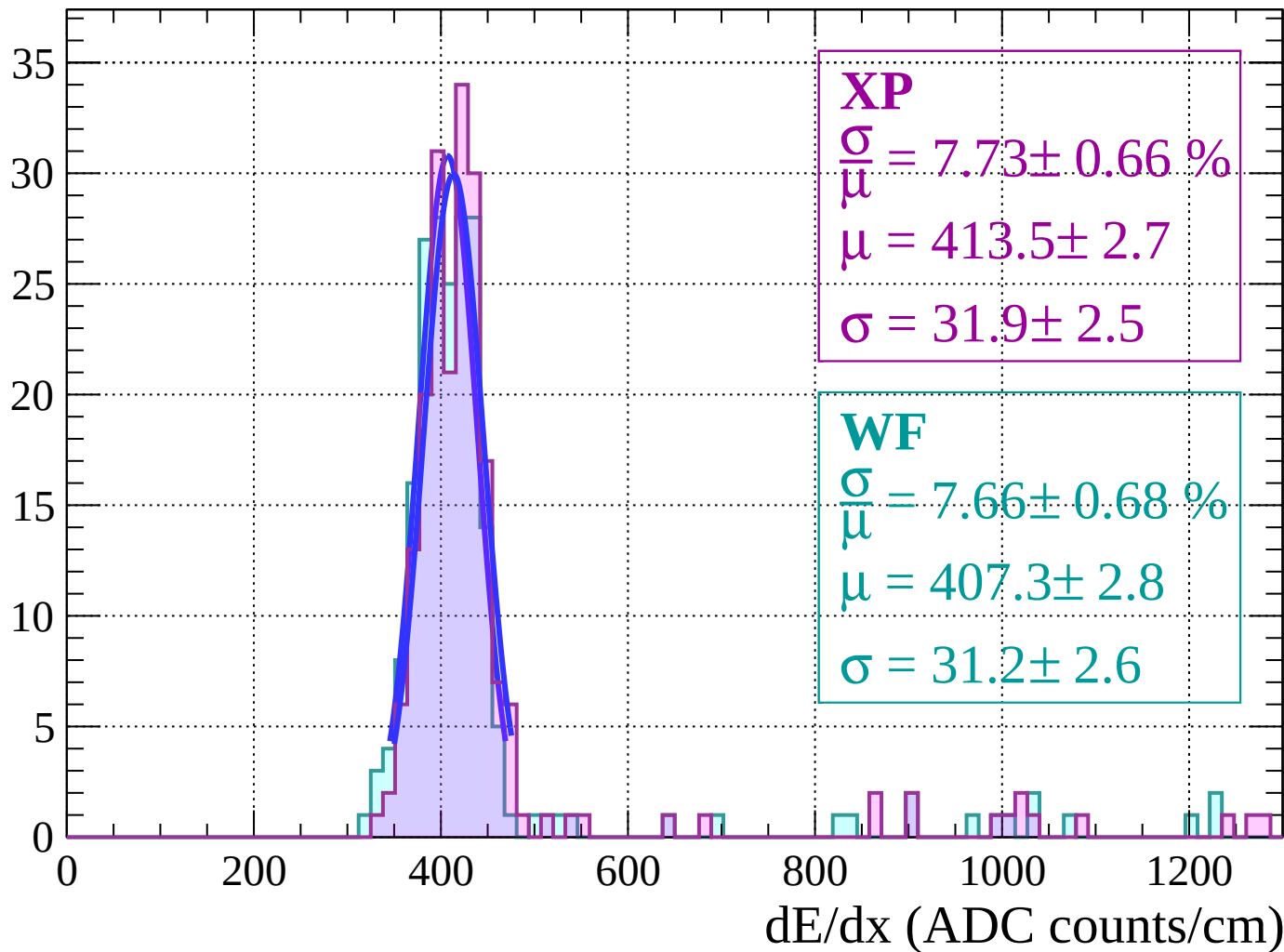
Energy loss | $24 < \theta < 26$

Count



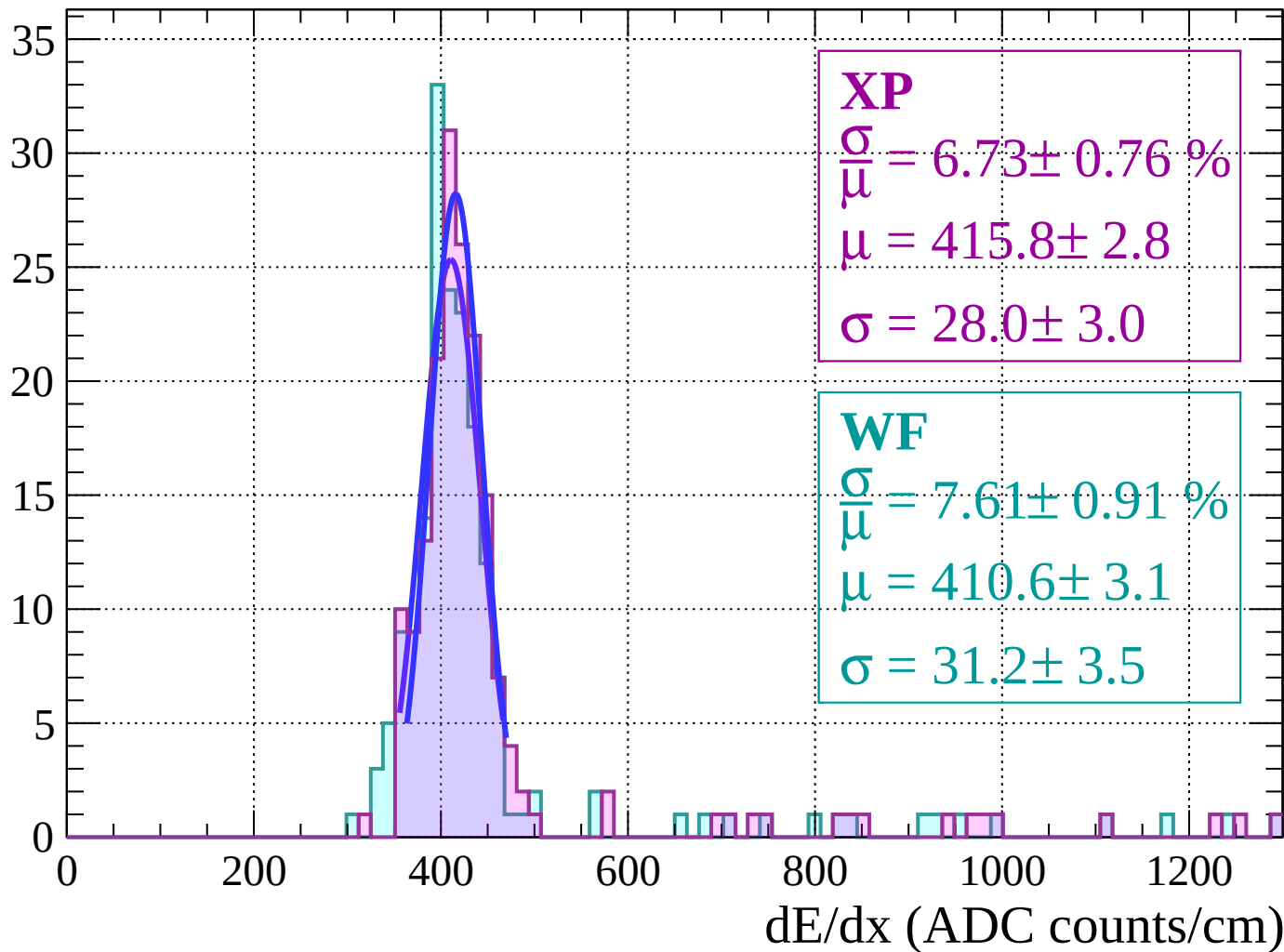
Energy loss | $26 < \theta < 28$

Count



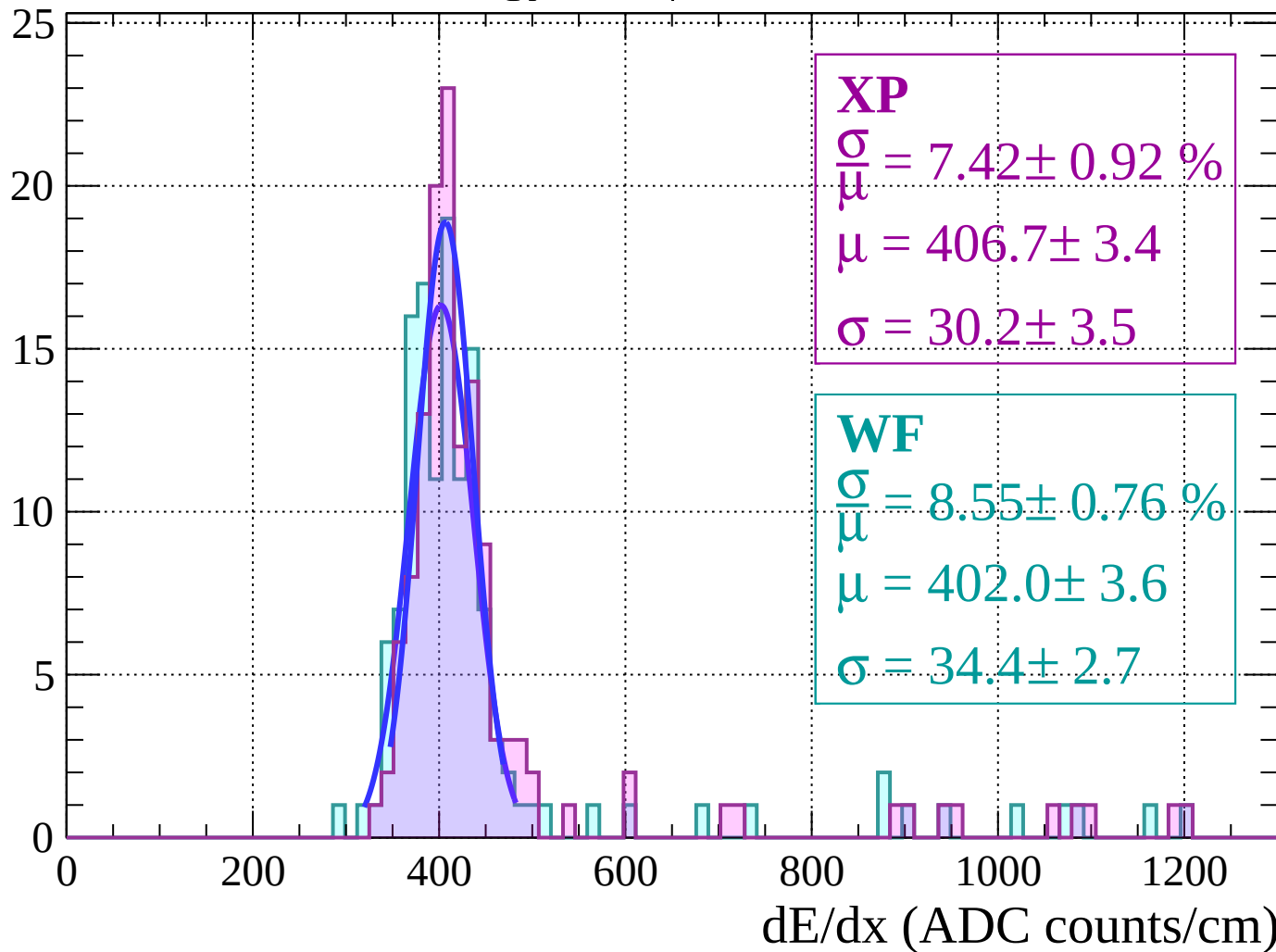
Energy loss | $28 < \theta < 30$

Count



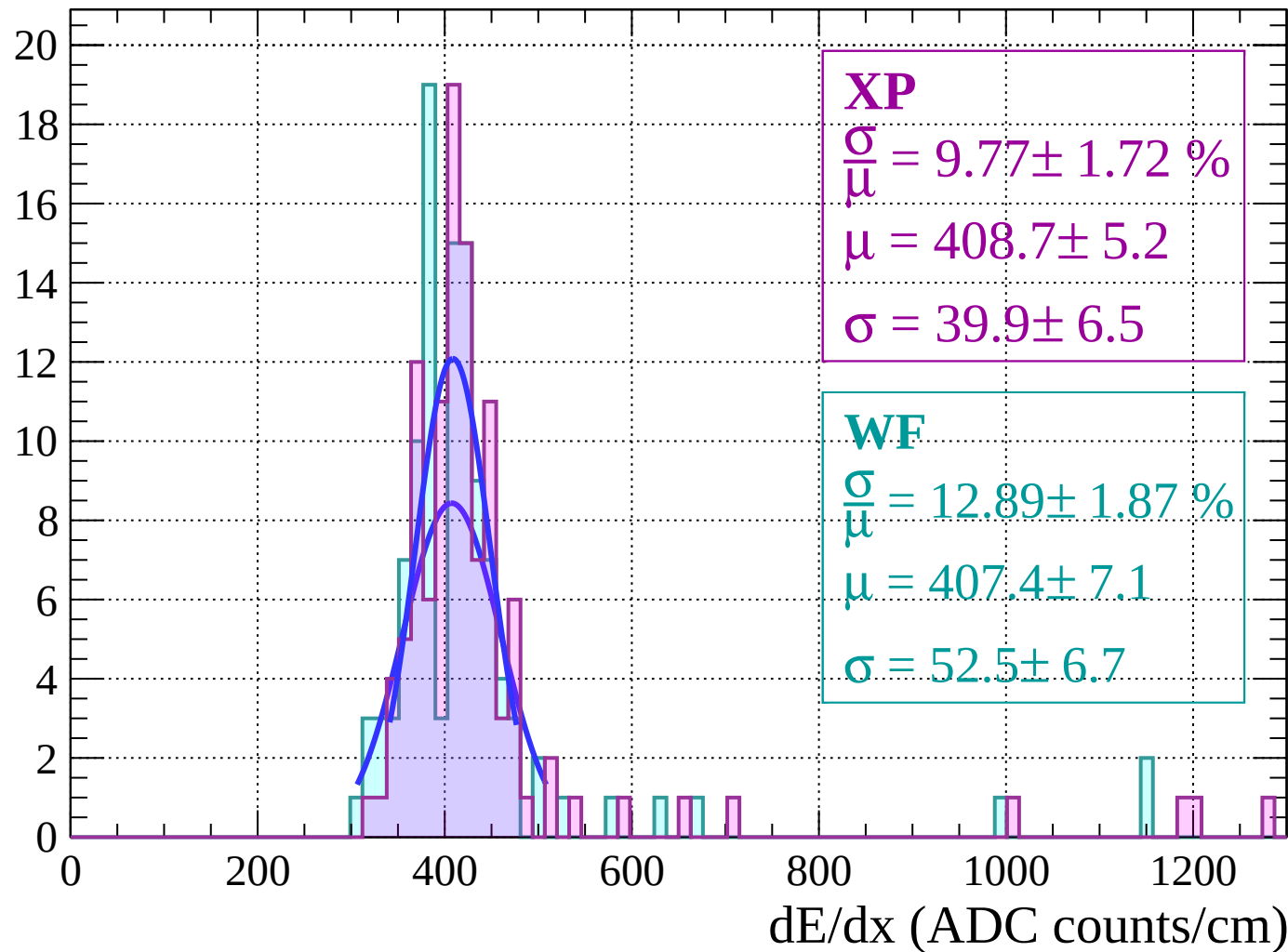
Energy loss | $30 < \theta < 32$

Count



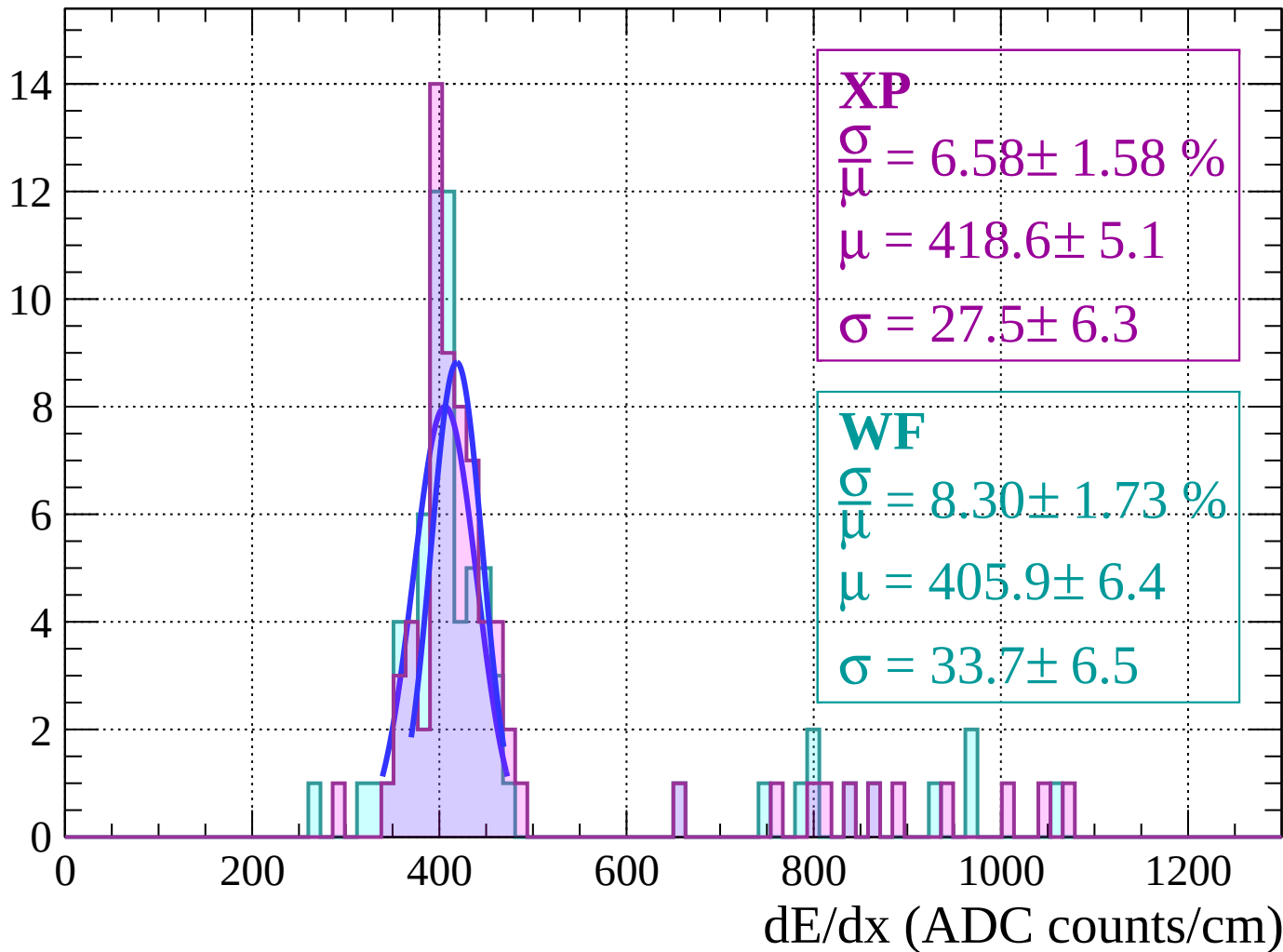
Energy loss | $32 < \theta < 34$

Count



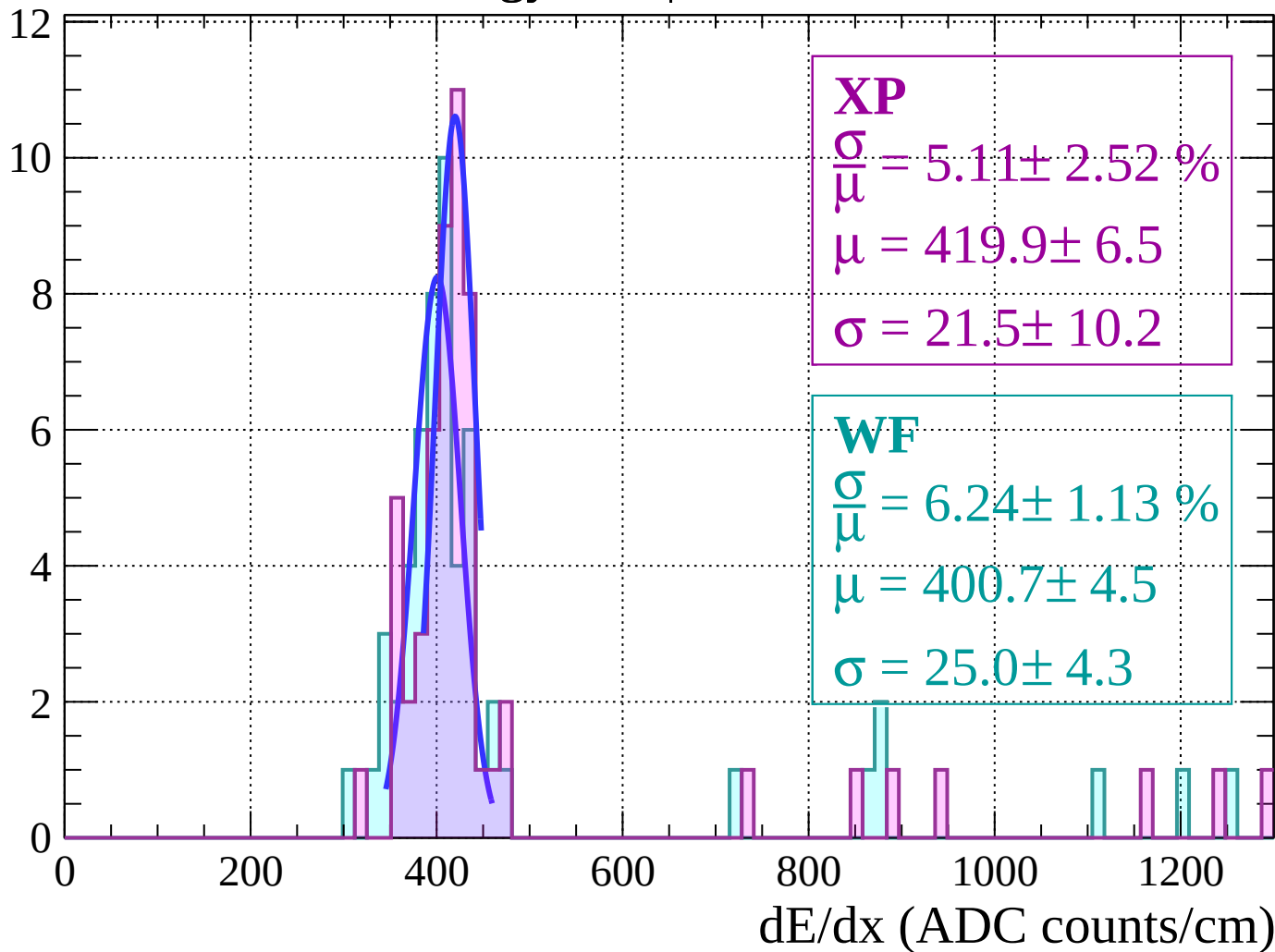
Energy loss | $34 < \theta < 36$

Count



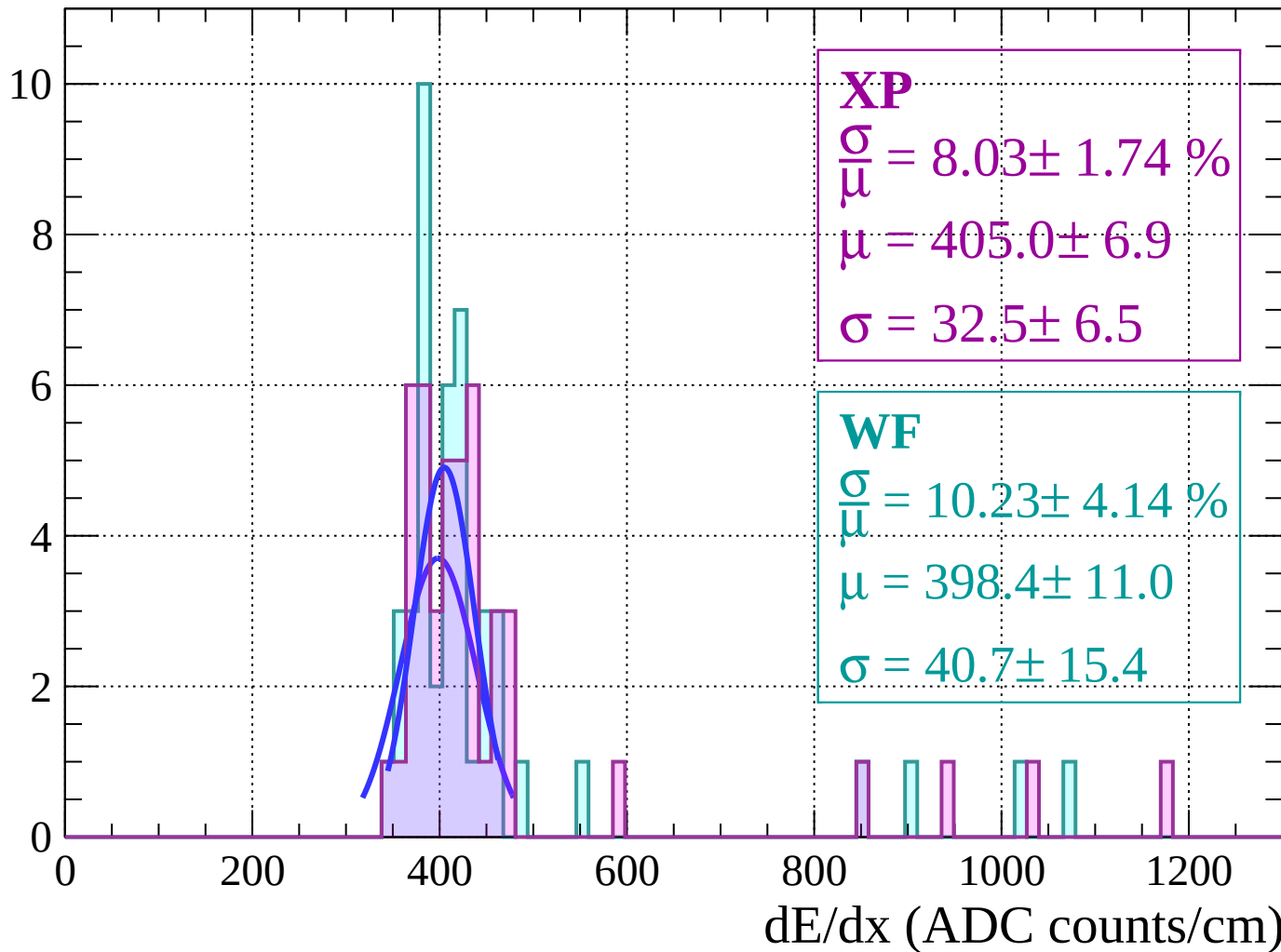
Energy loss | $36 < \theta < 38$

Count



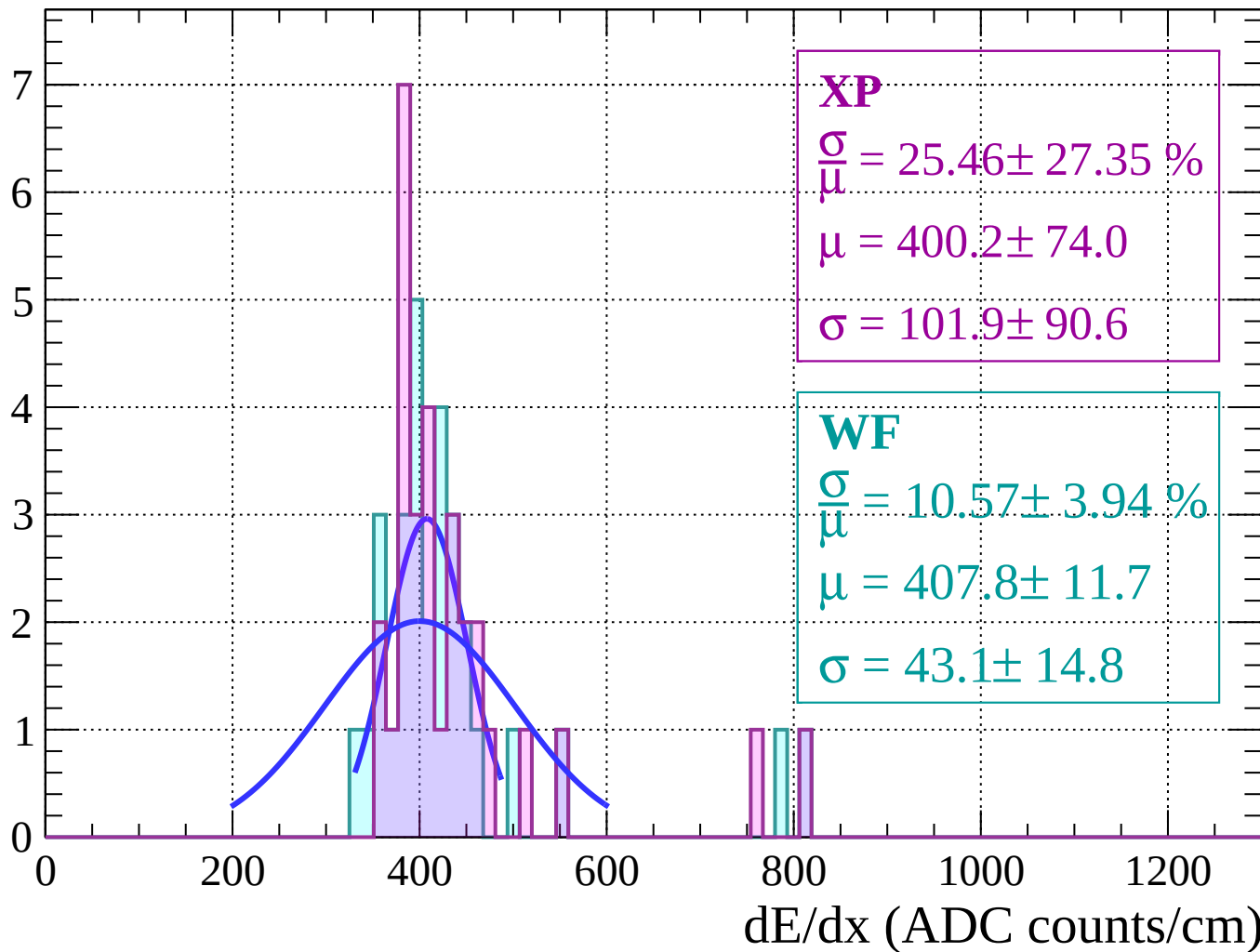
Energy loss | $38 < \theta < 40$

Count



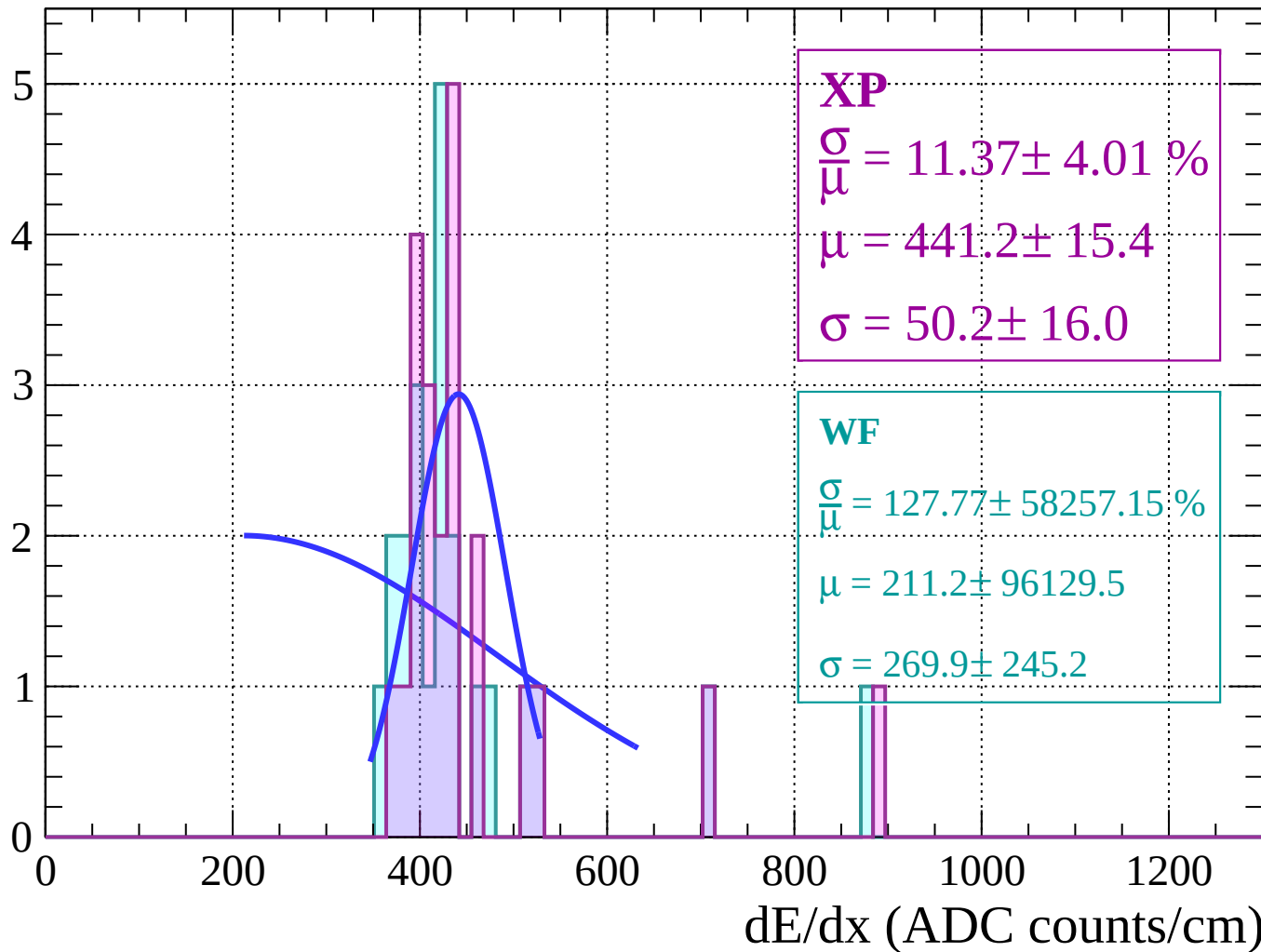
Energy loss | $40 < \theta < 42$

Count



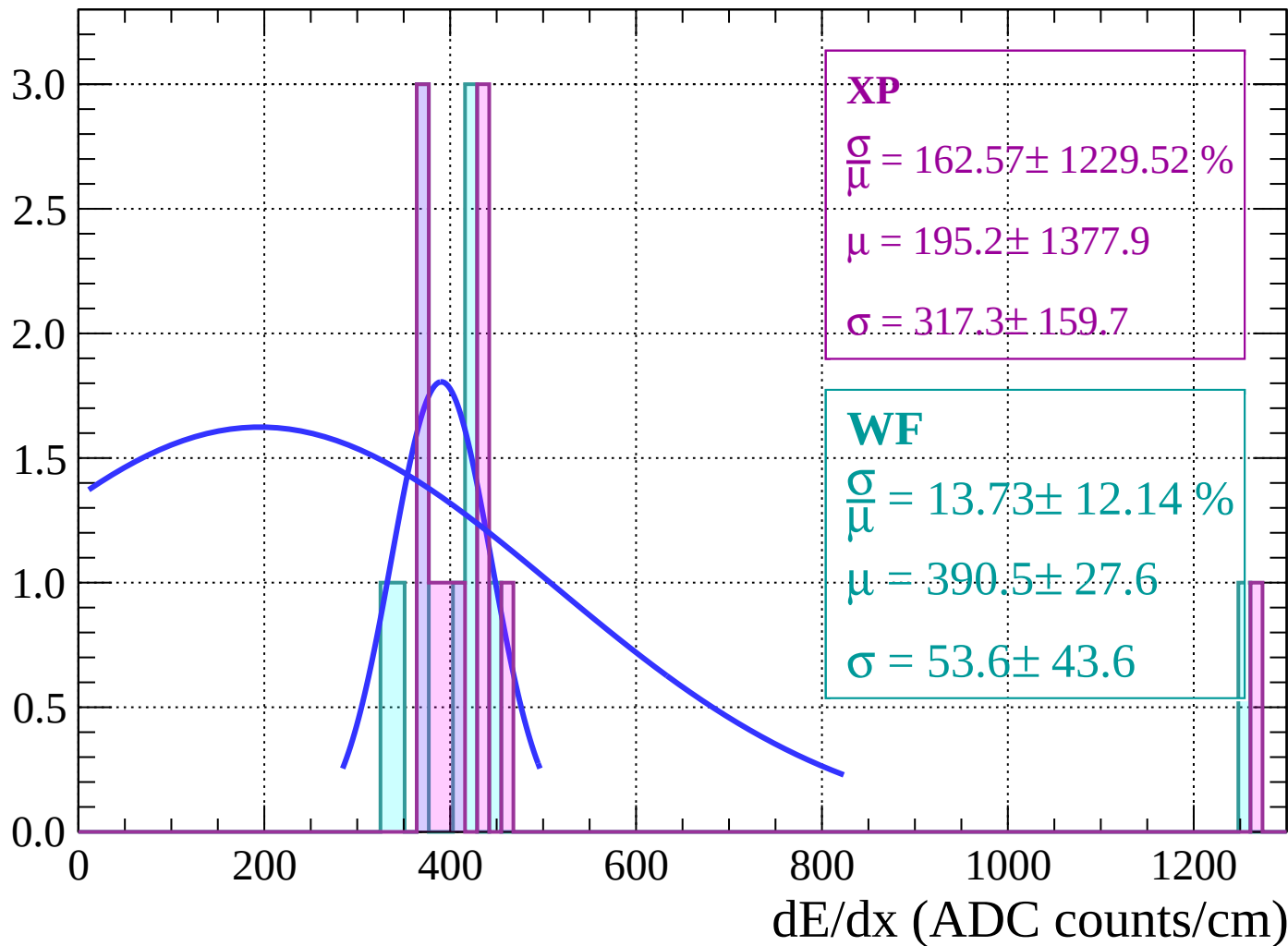
Energy loss | $42 < \theta < 44$

Count



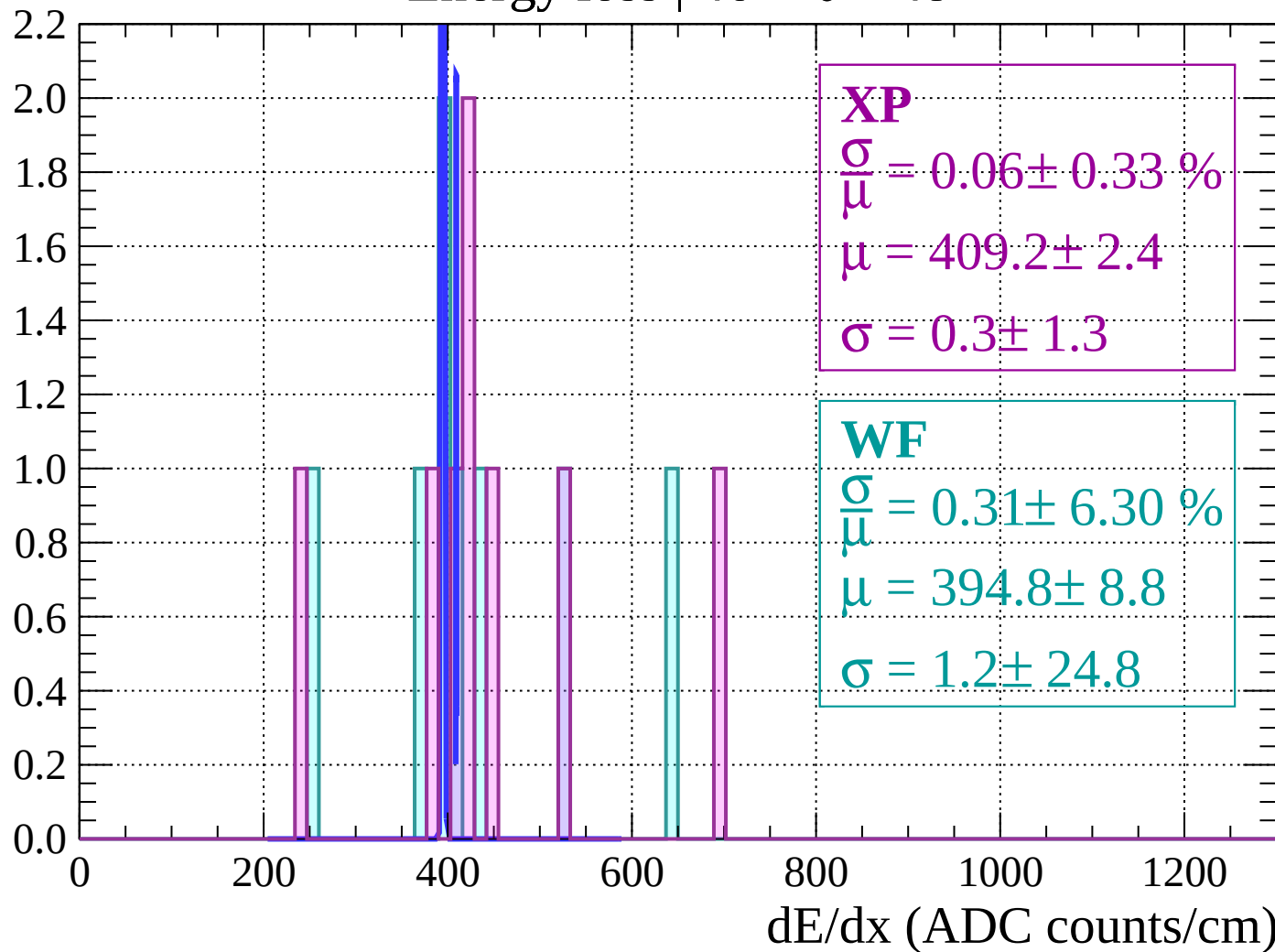
Energy loss | $44 < \theta < 46$

Count



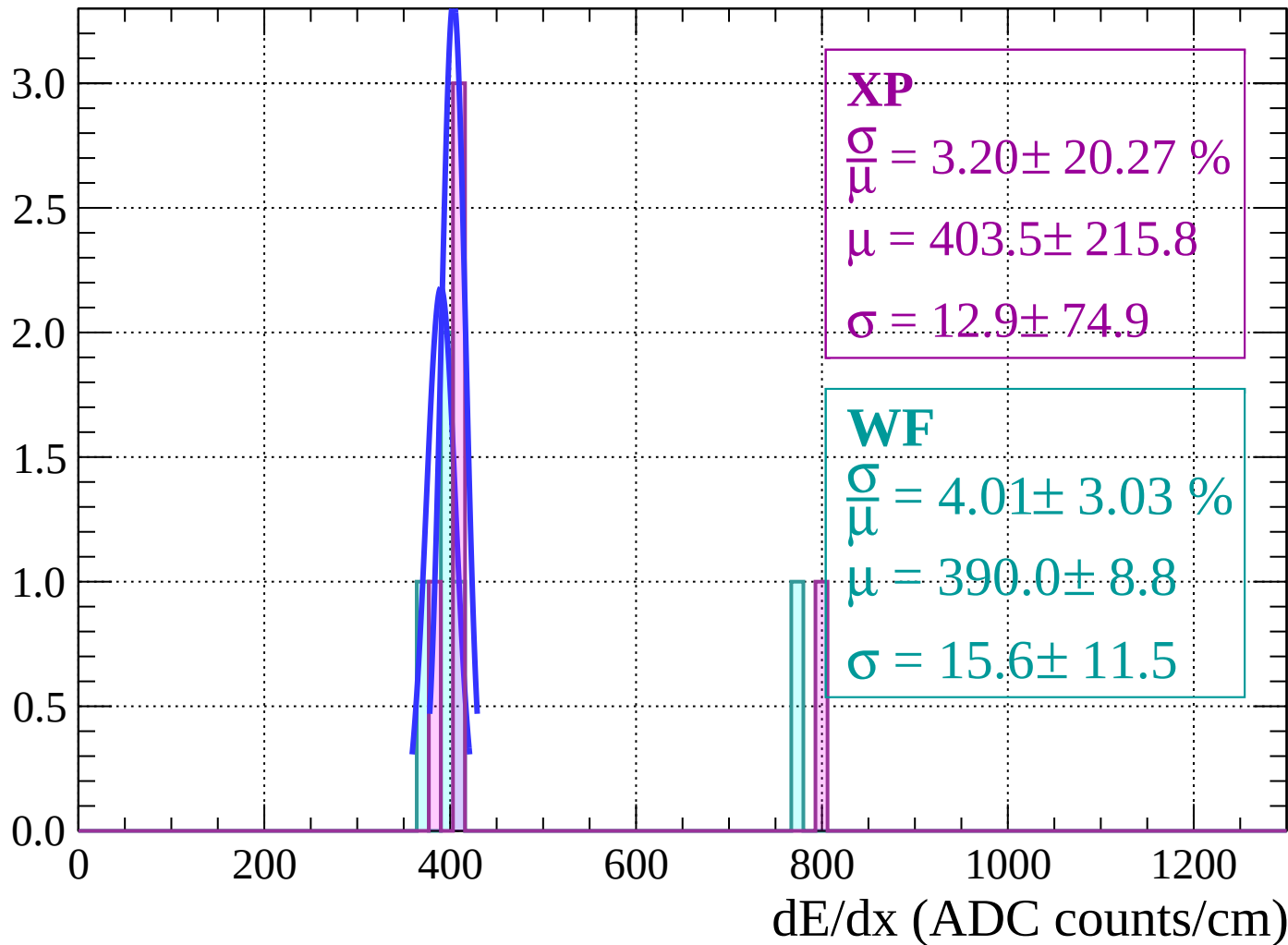
Energy loss | $46 < \theta < 48$

Count



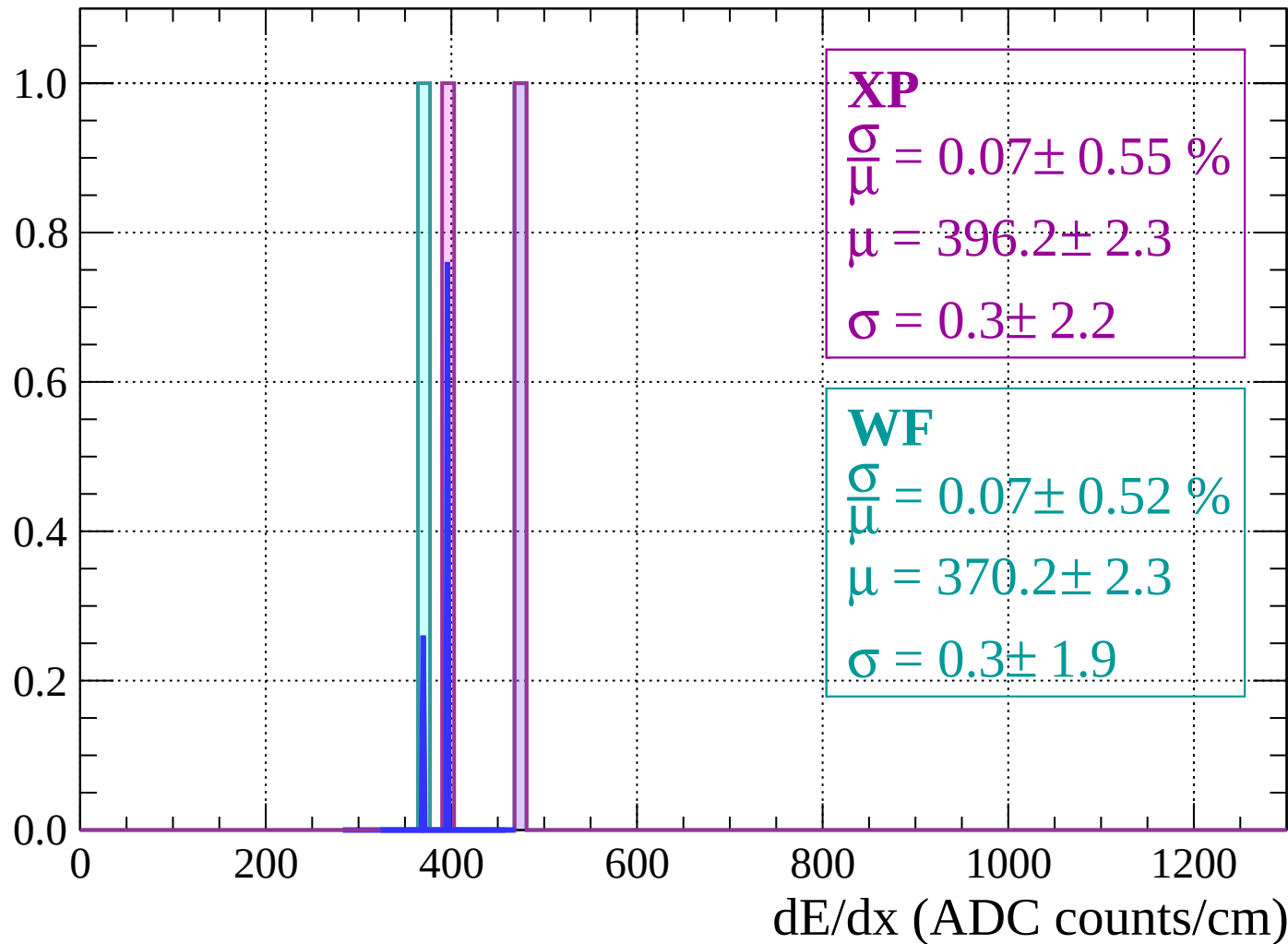
Energy loss | $48 < \theta < 50$

Count



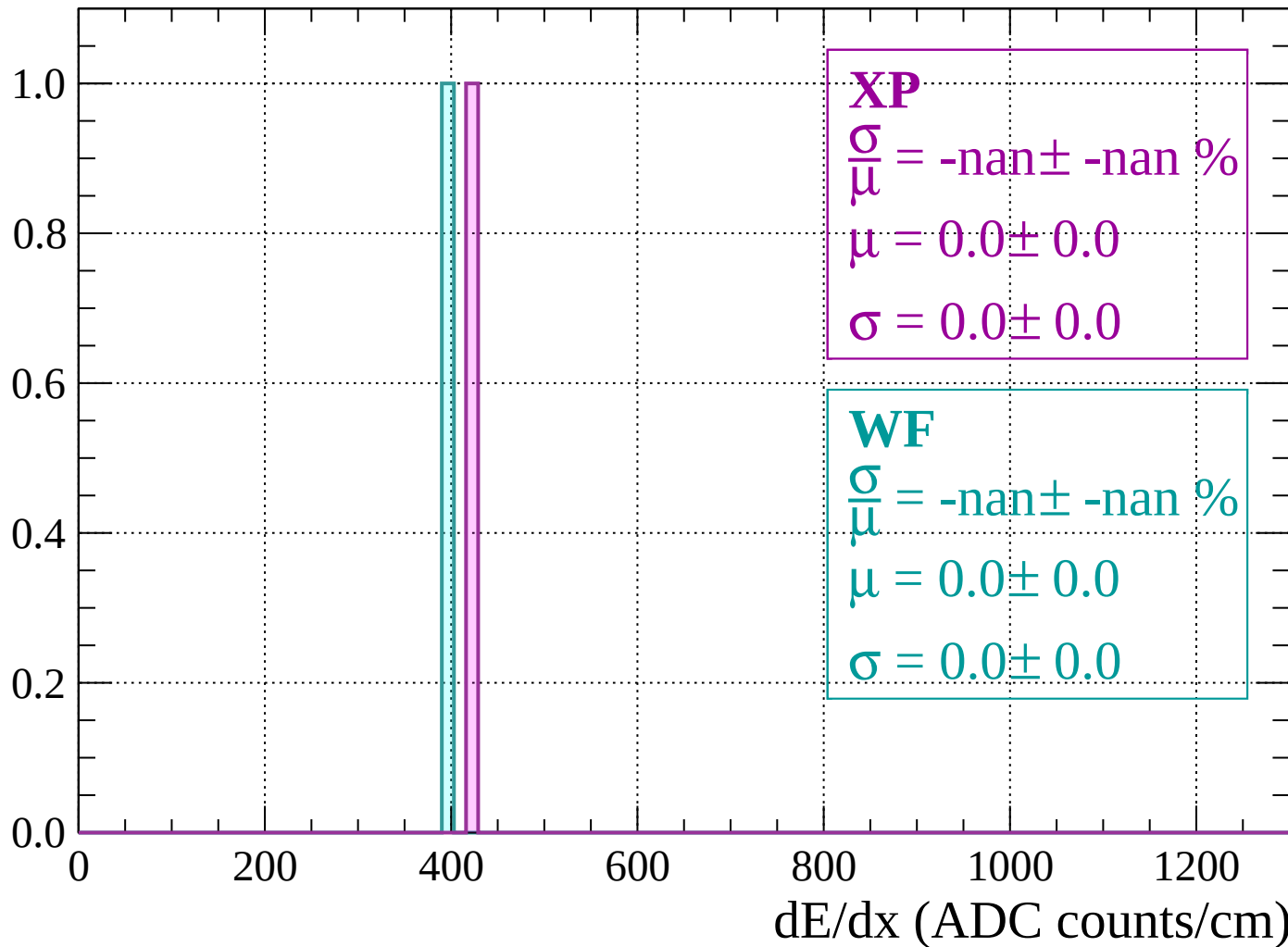
Energy loss | $50 < \theta < 52$

Count



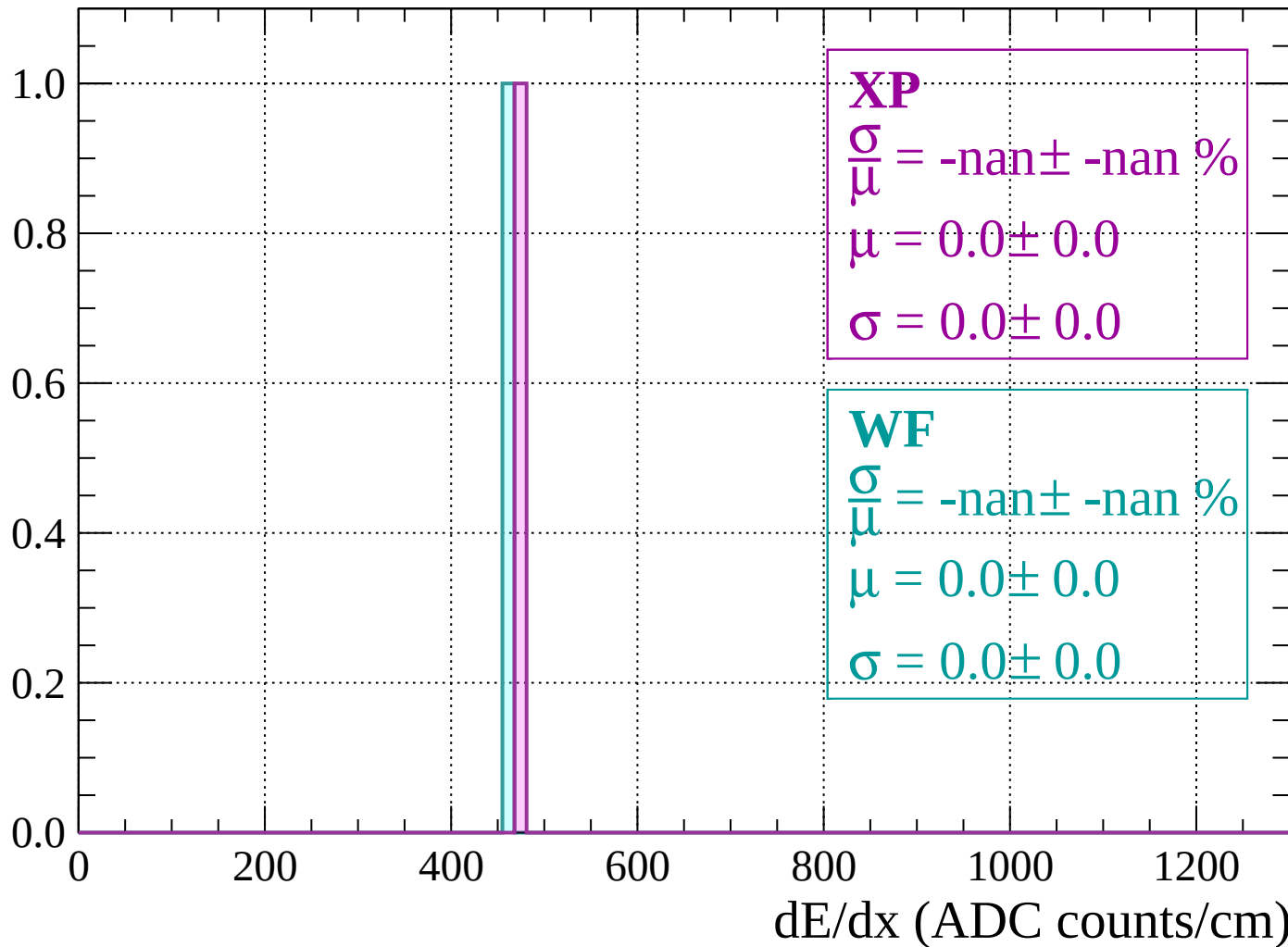
Energy loss | $52 < \theta < 54$

Count



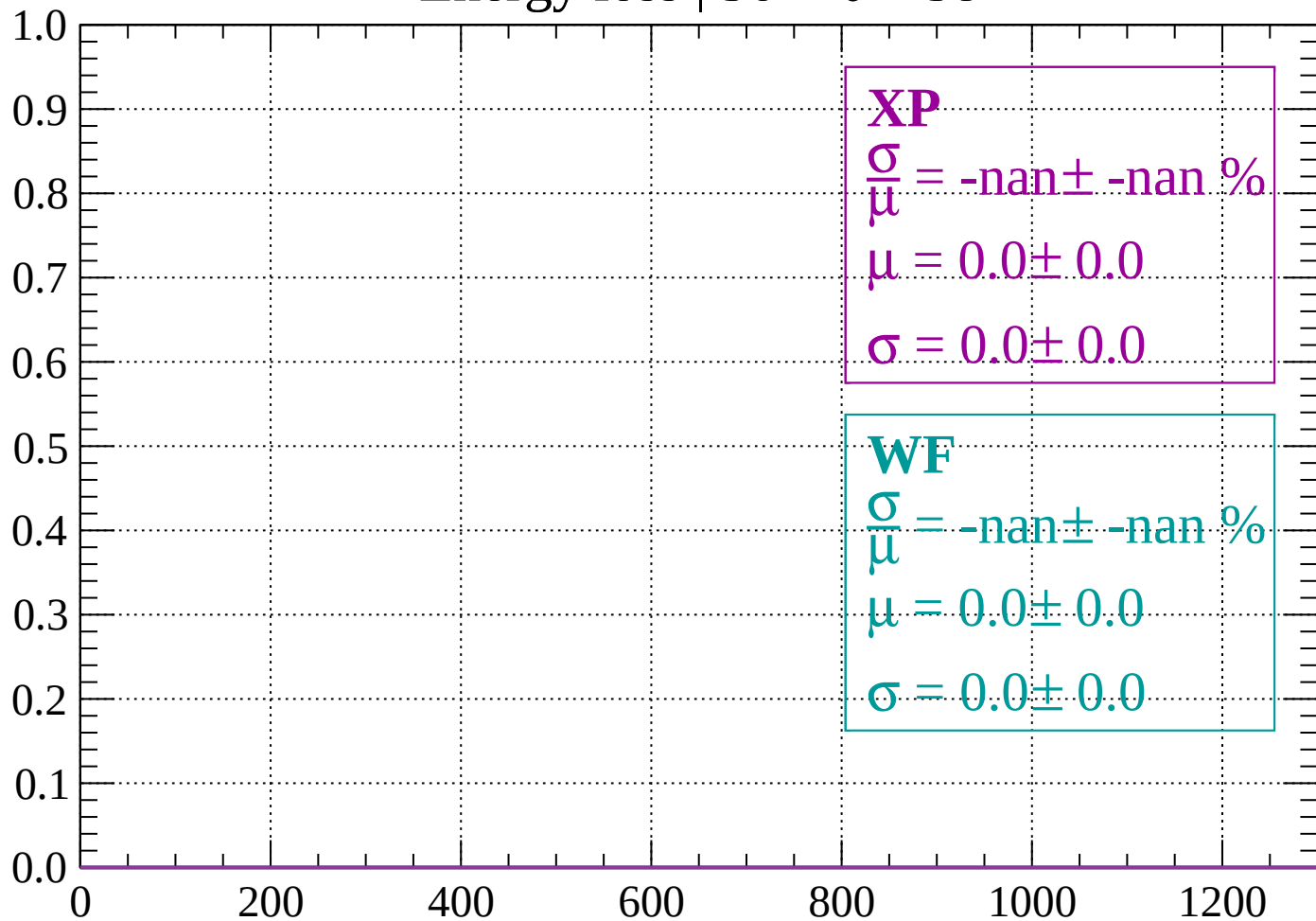
Energy loss | $54 < \theta < 56$

Count



Energy loss | $56 < \theta < 58$

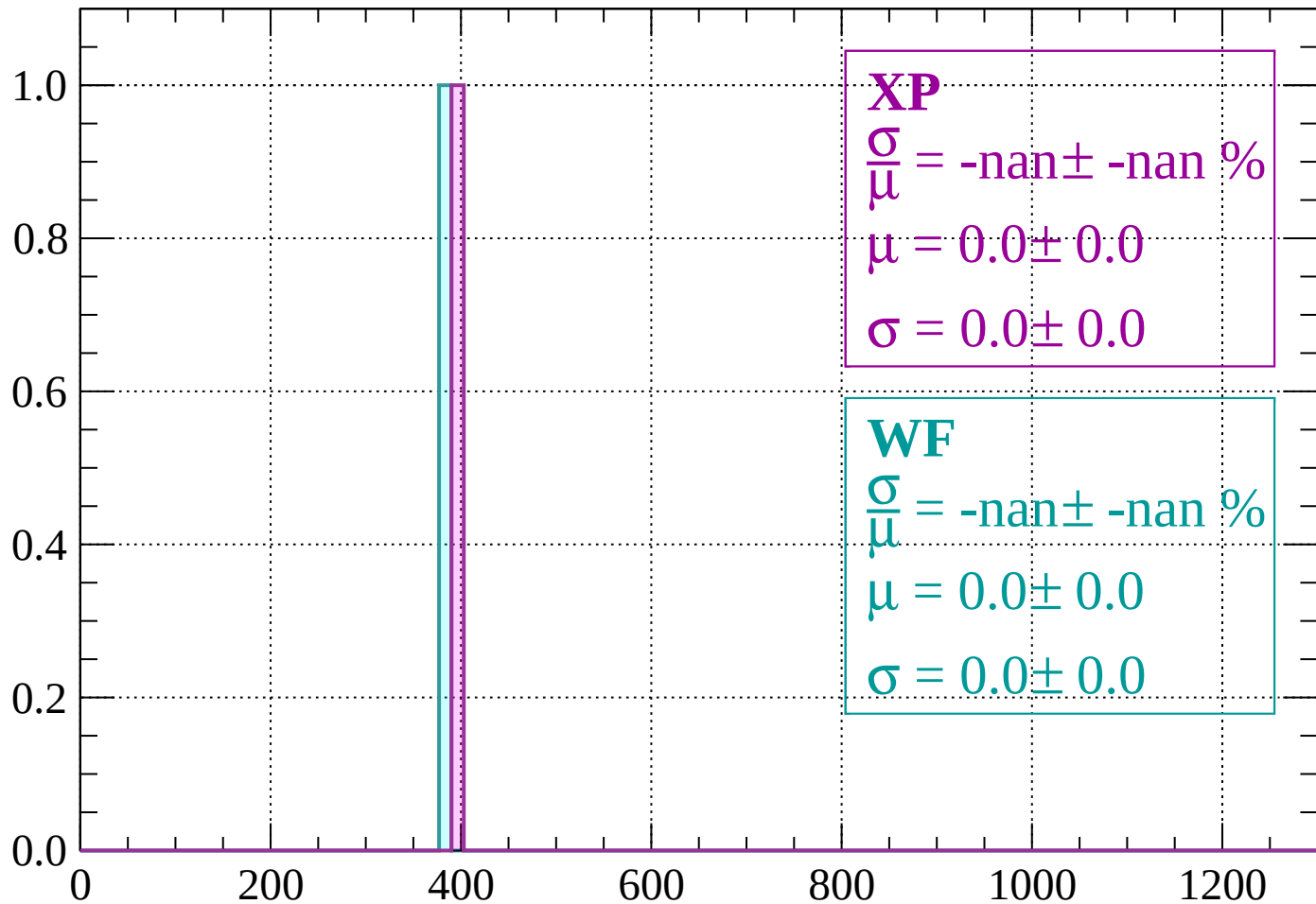
Count



dE/dx (ADC counts/cm)

Energy loss | $58 < \theta < 60$

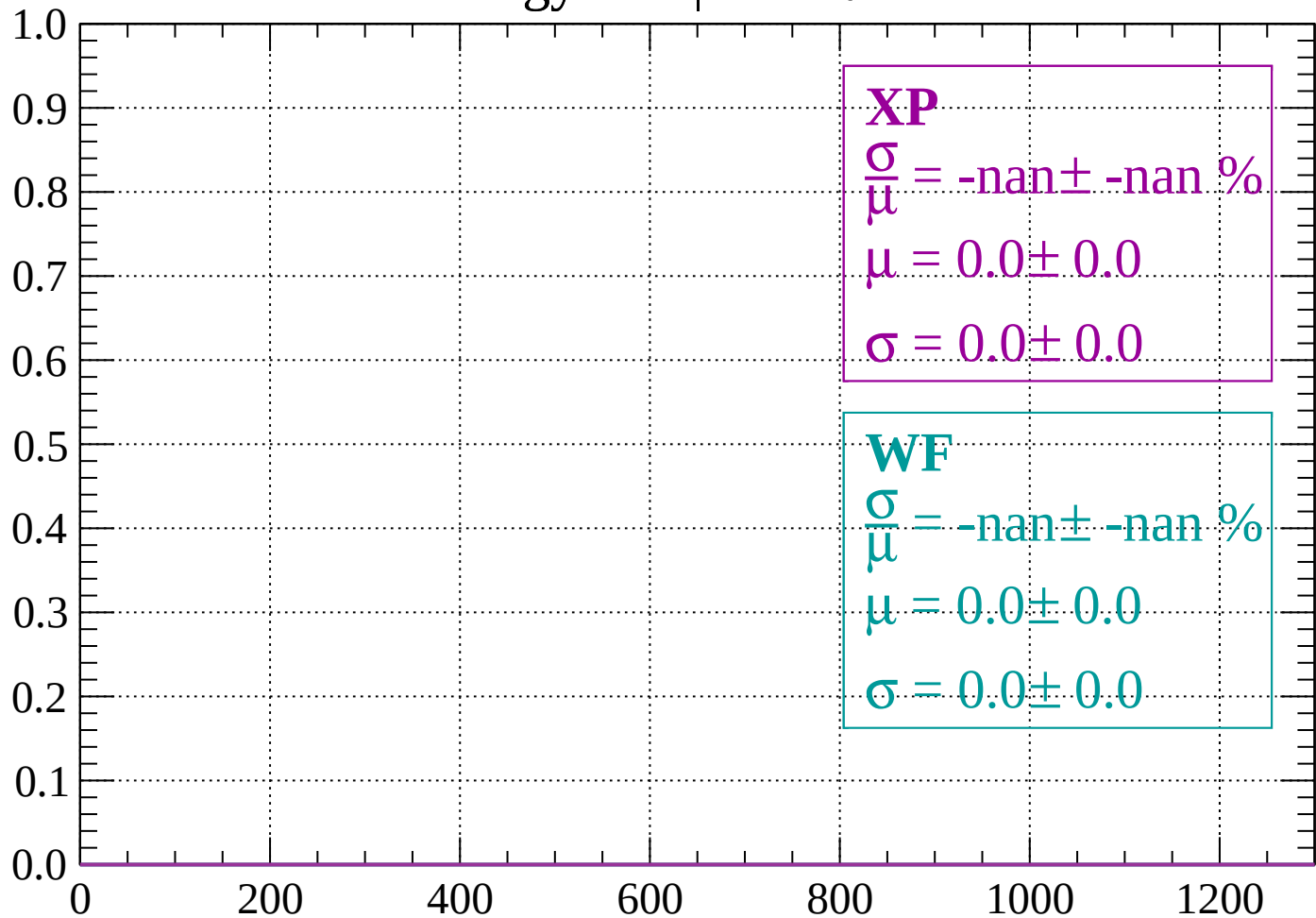
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 62$

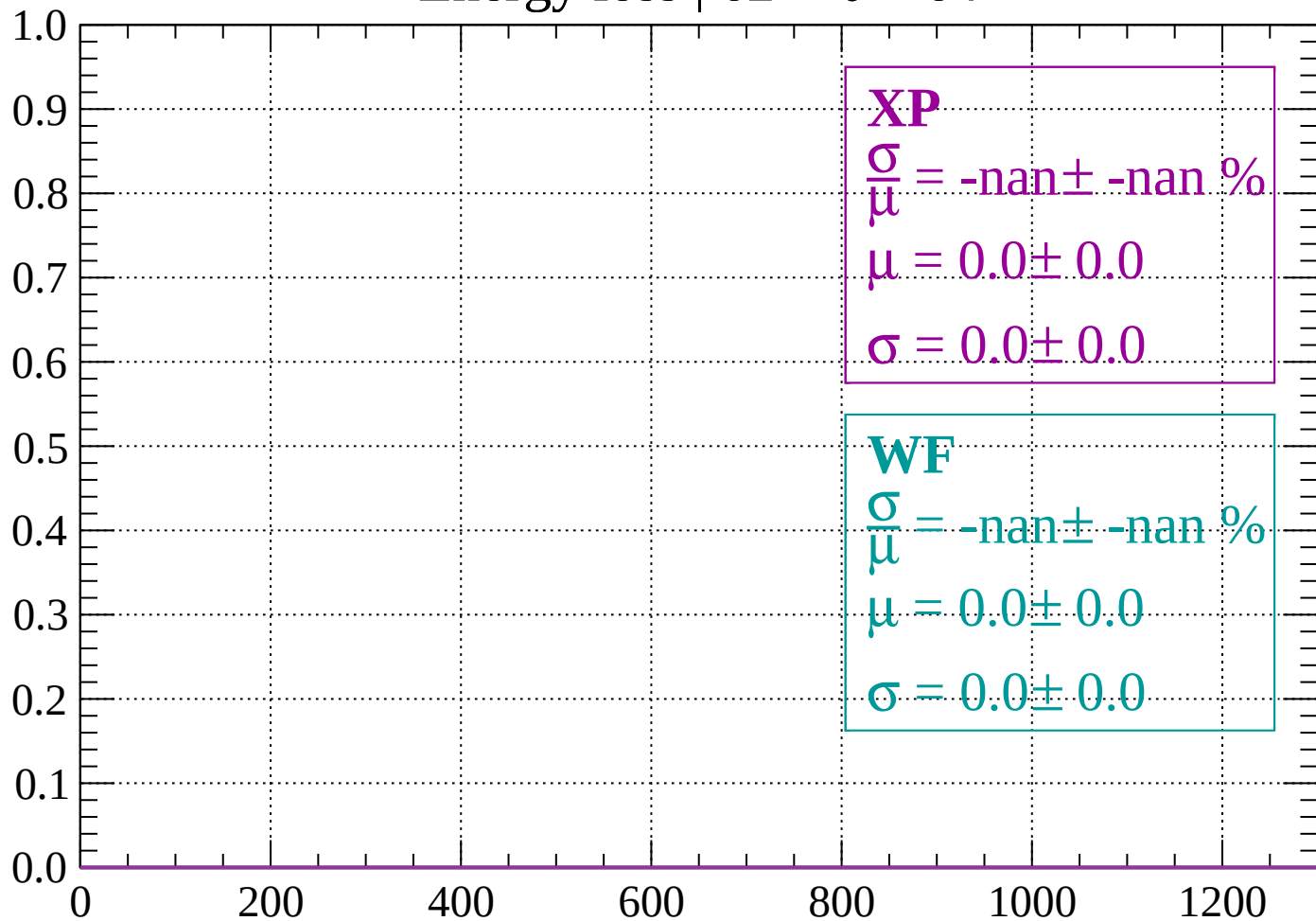
Count



dE/dx (ADC counts/cm)

Energy loss | $62 < \theta < 64$

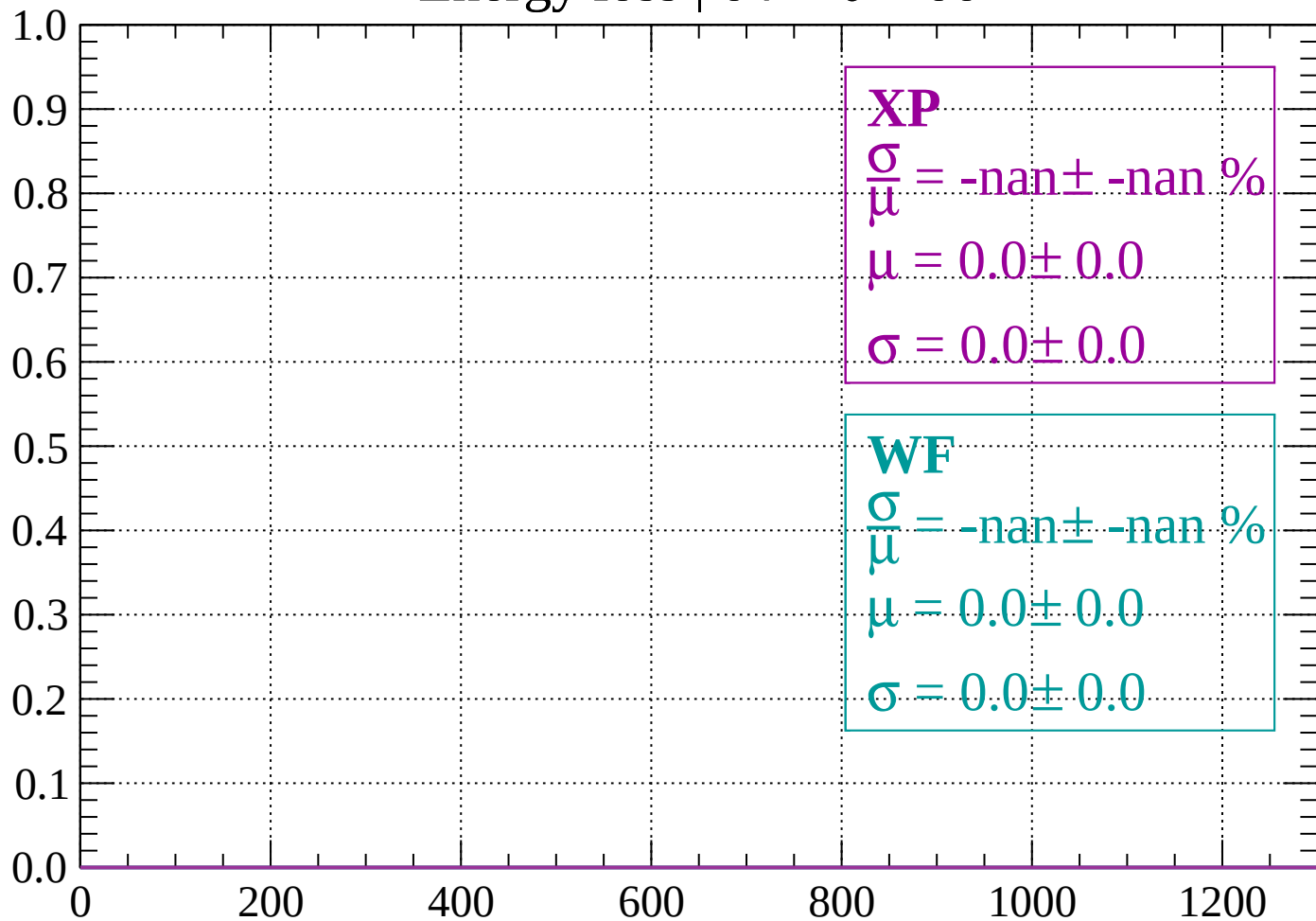
Count



dE/dx (ADC counts/cm)

Energy loss | $64 < \theta < 66$

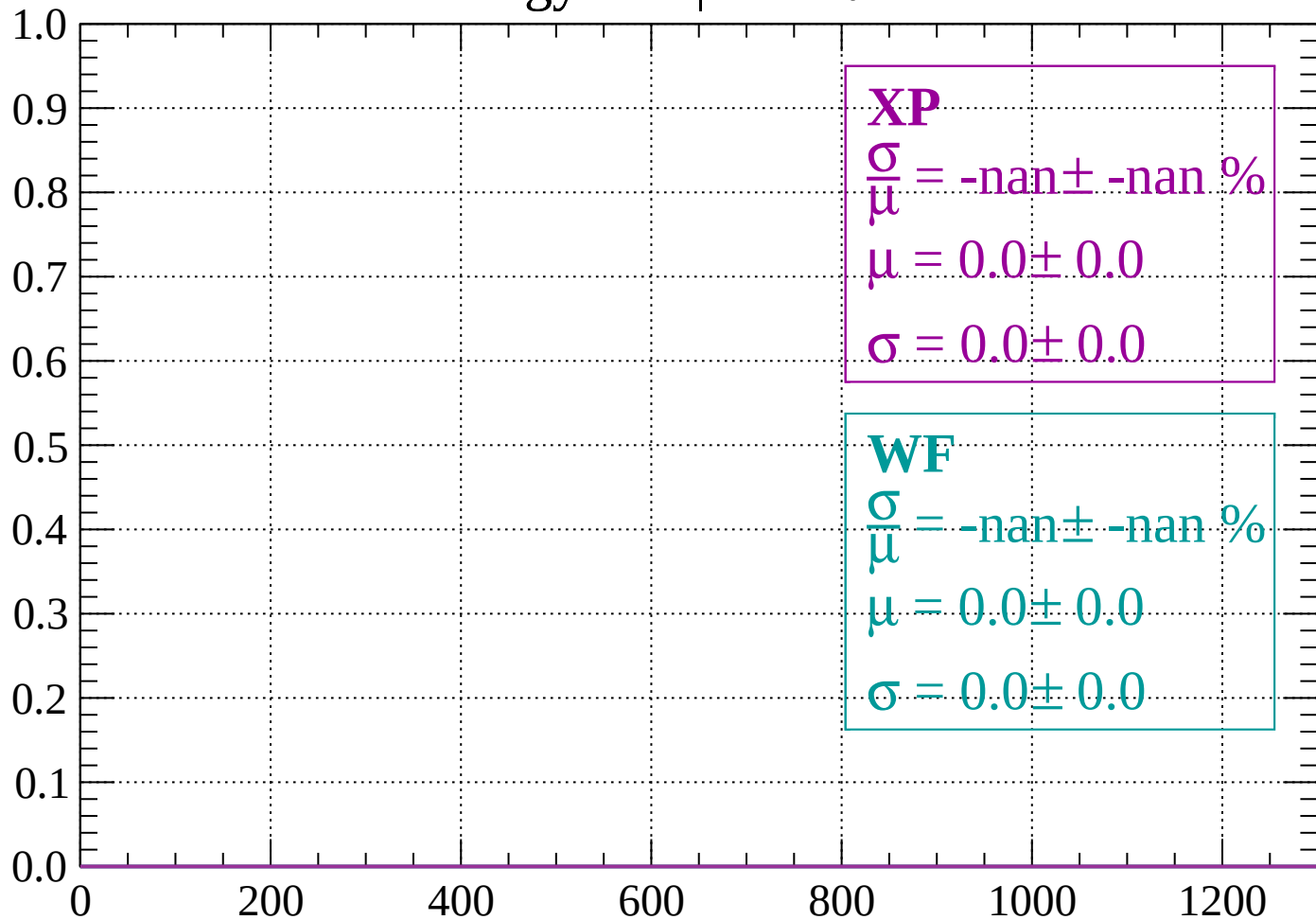
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

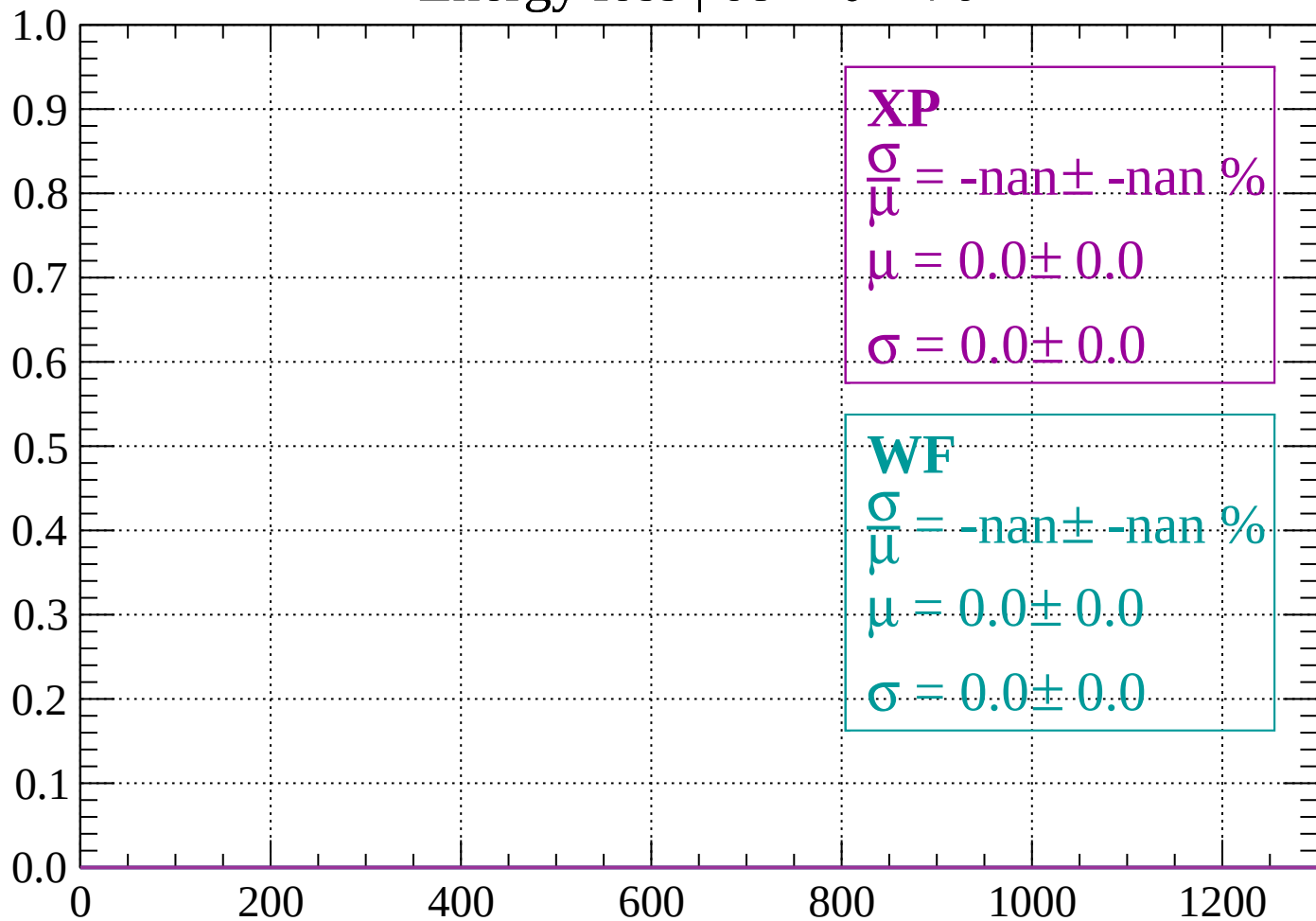
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

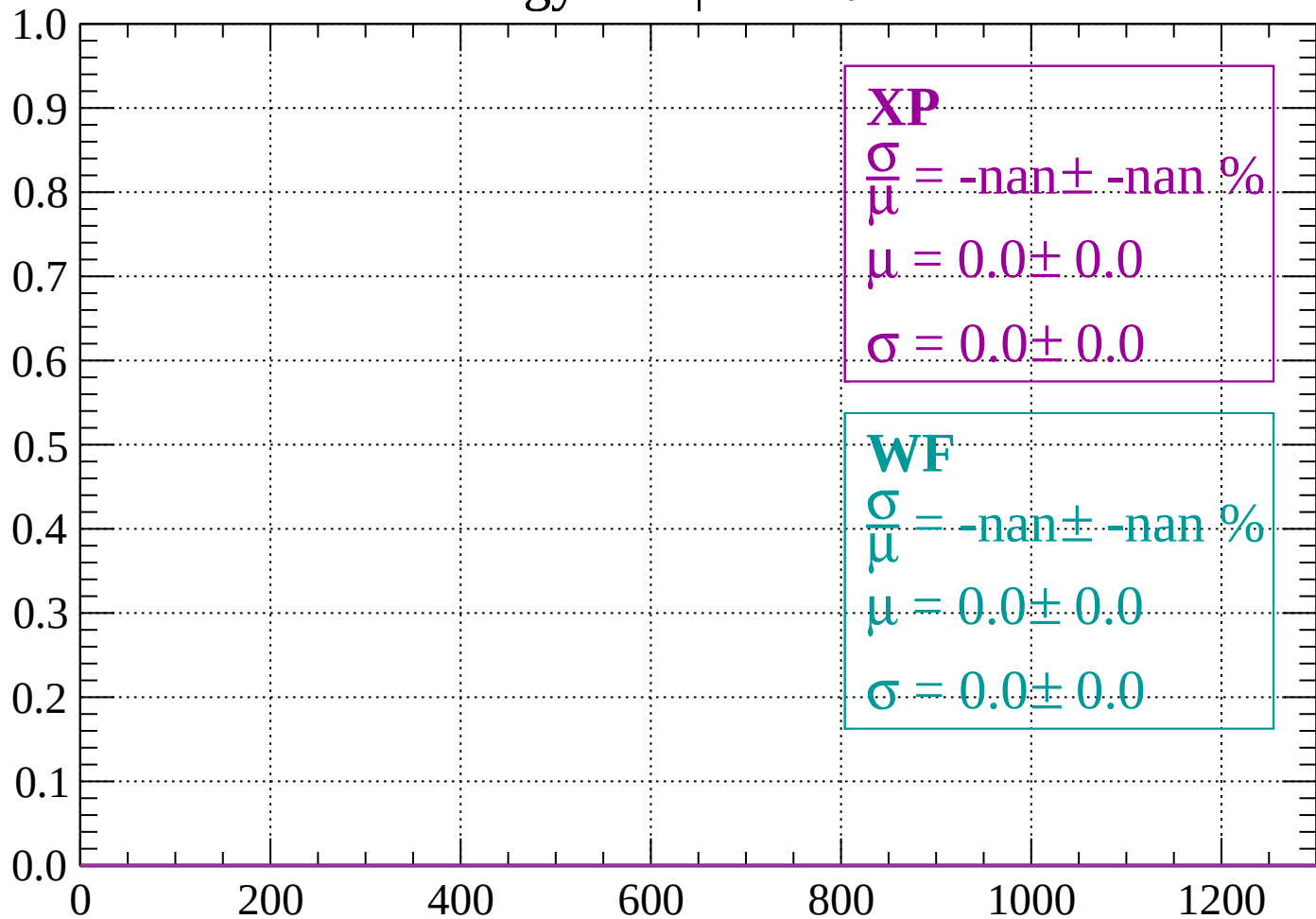
Count



dE/dx (ADC counts/cm)

Energy loss | $70 < \theta < 72$

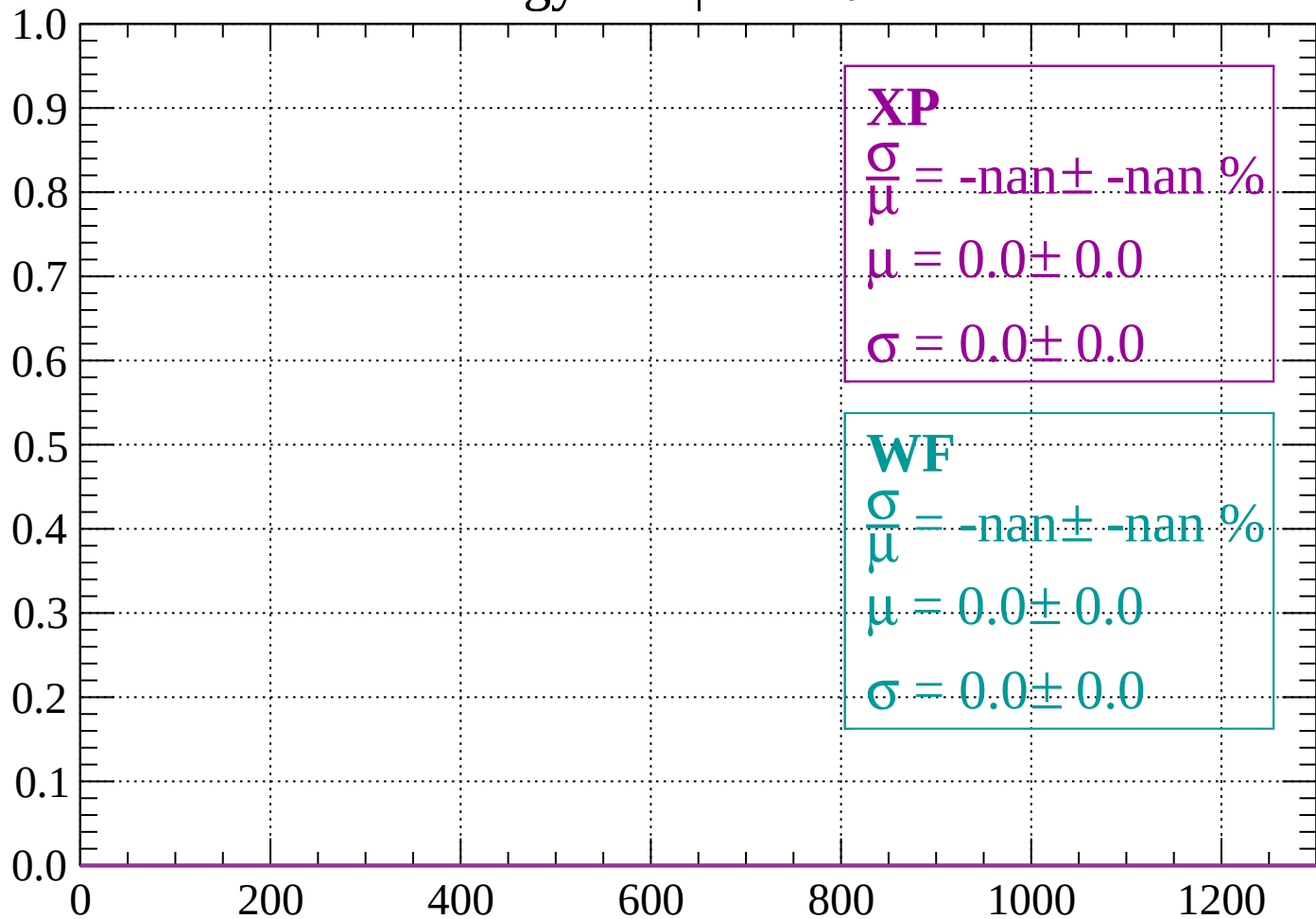
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

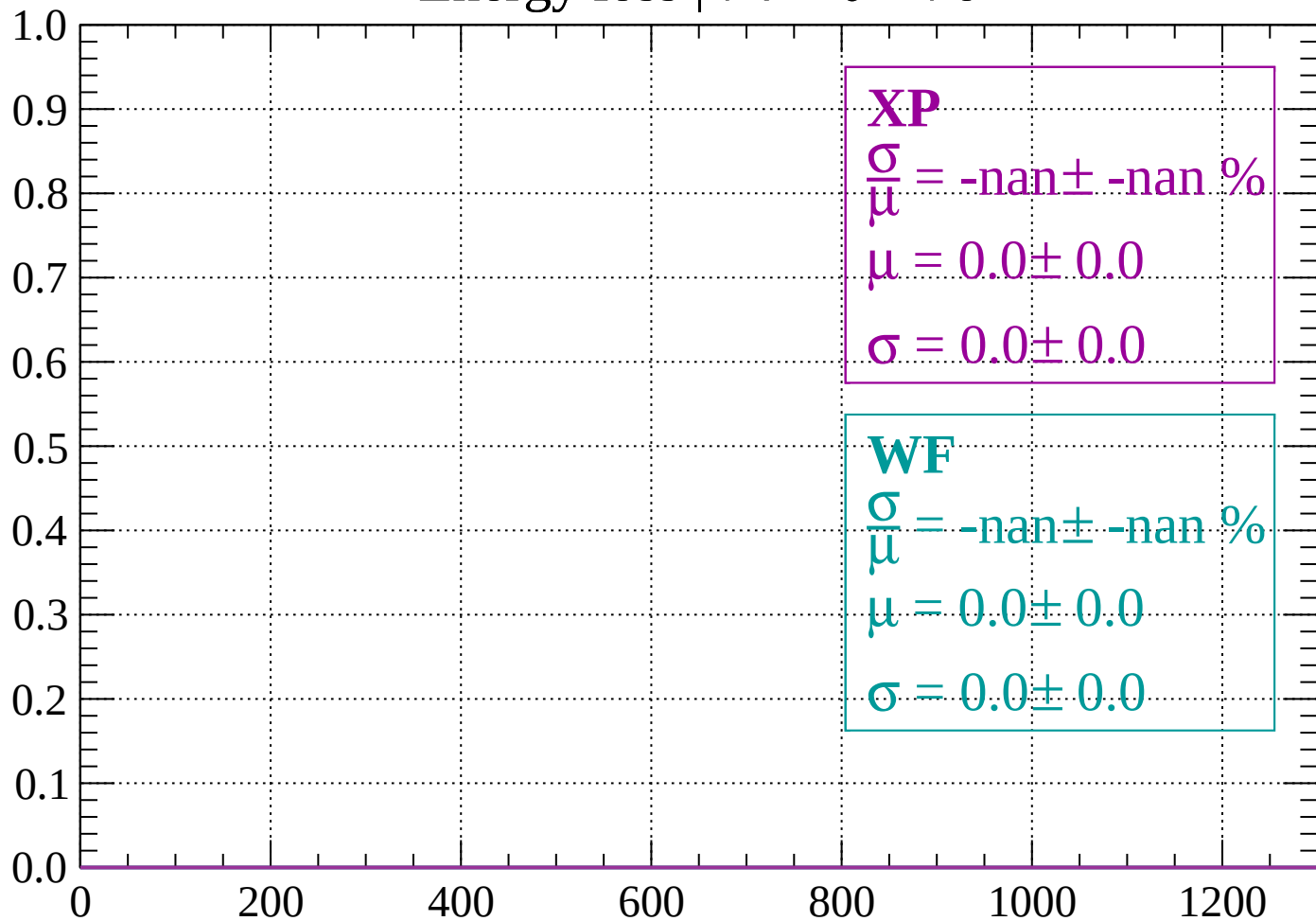
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

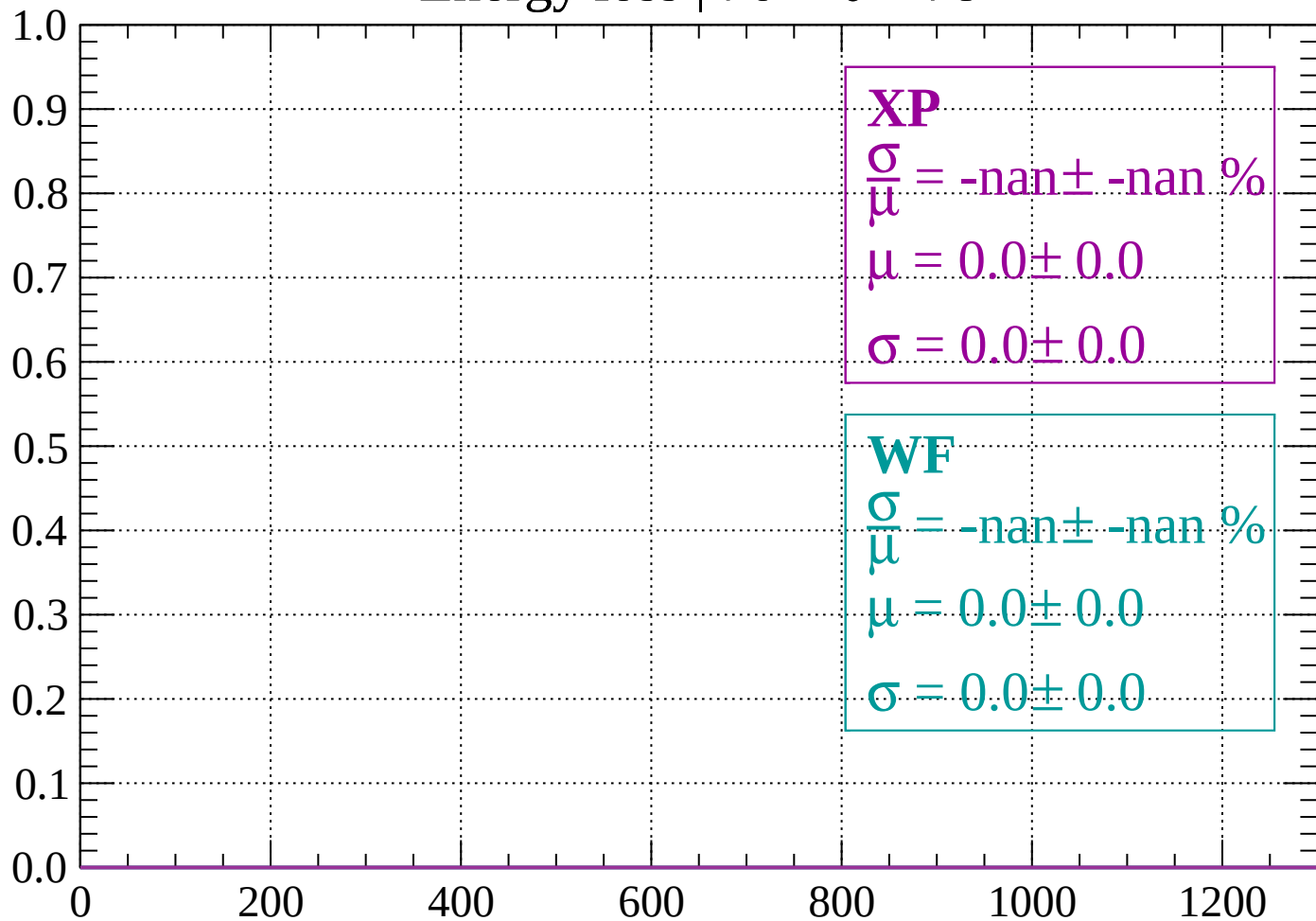
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

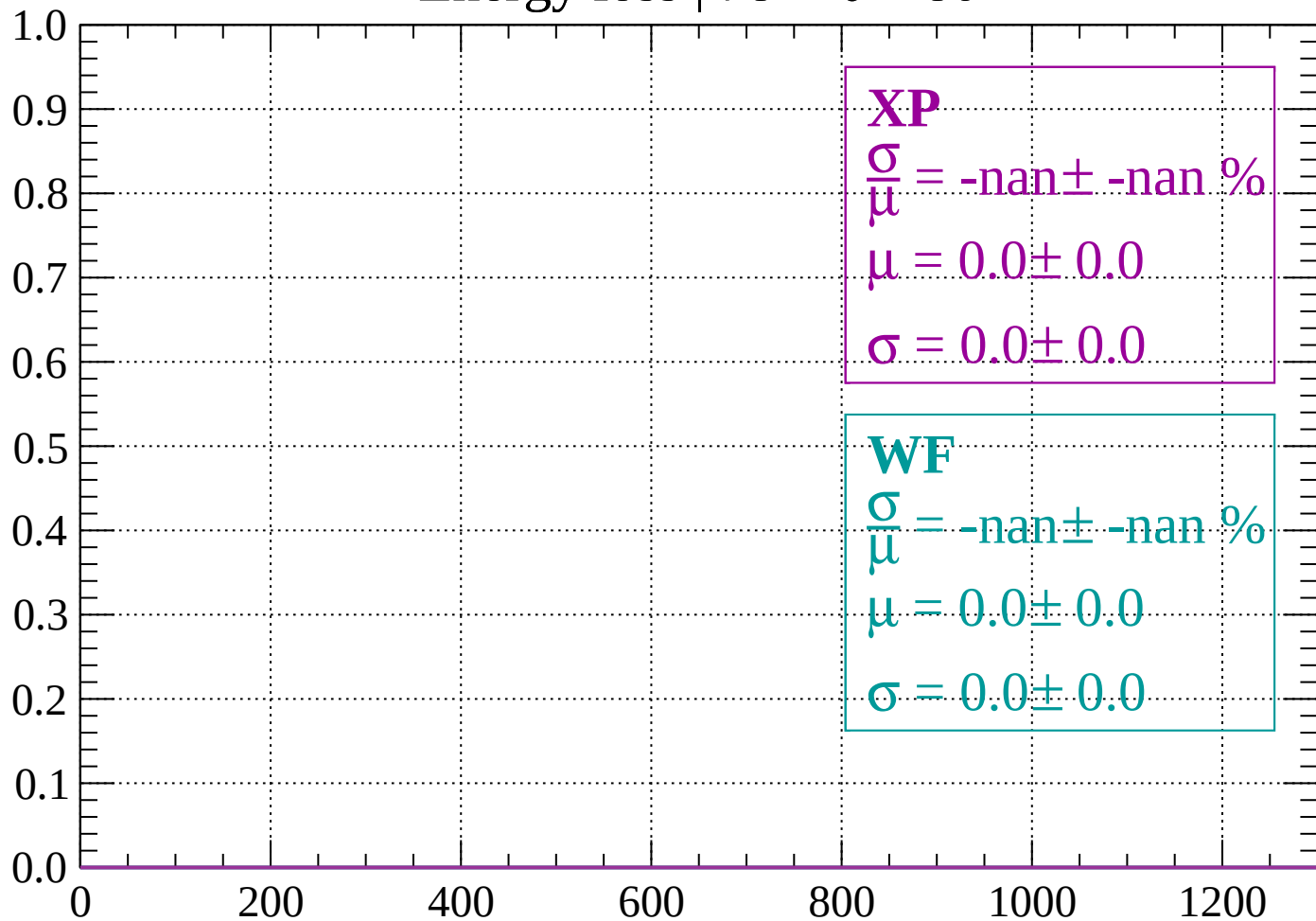
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

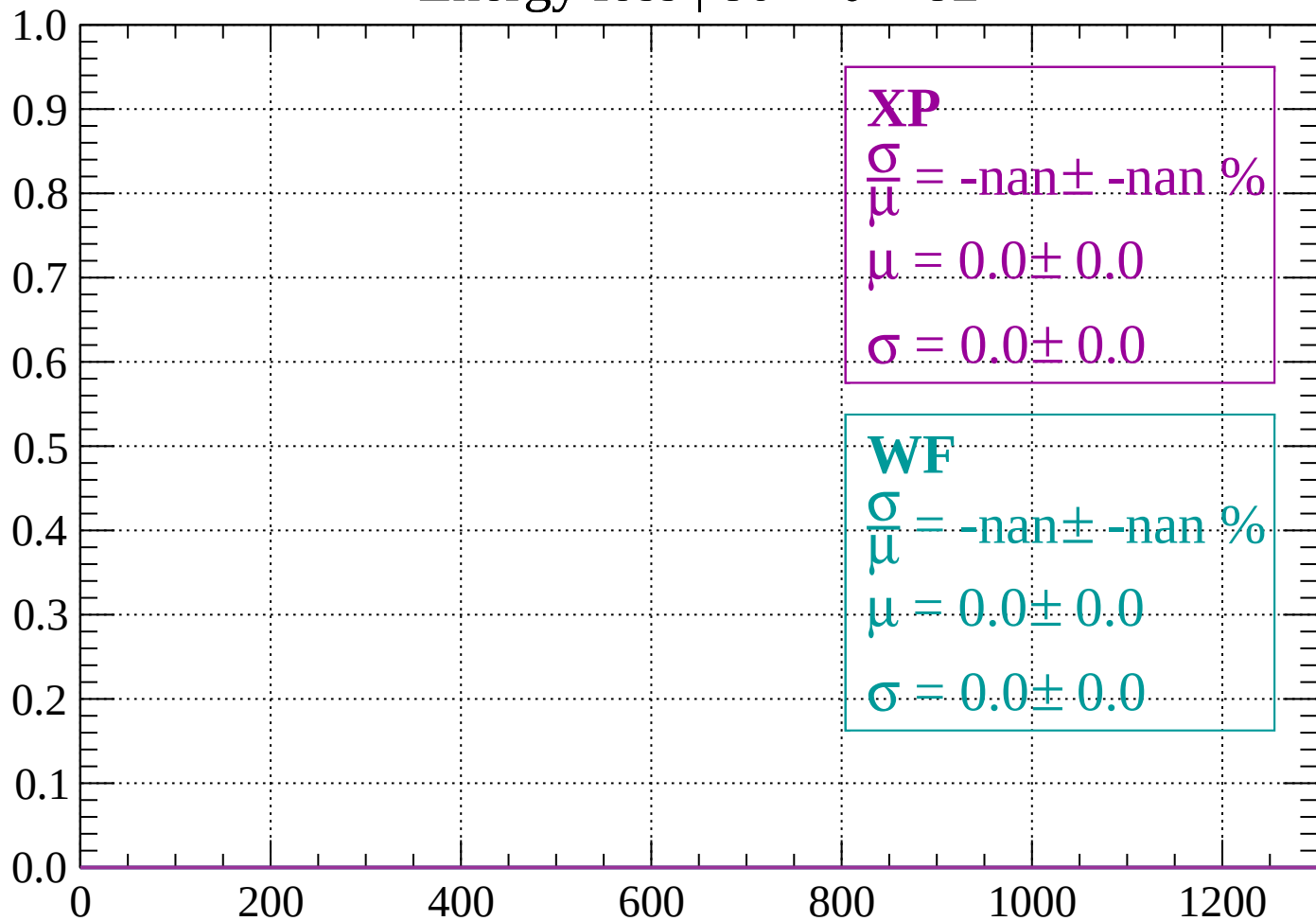
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

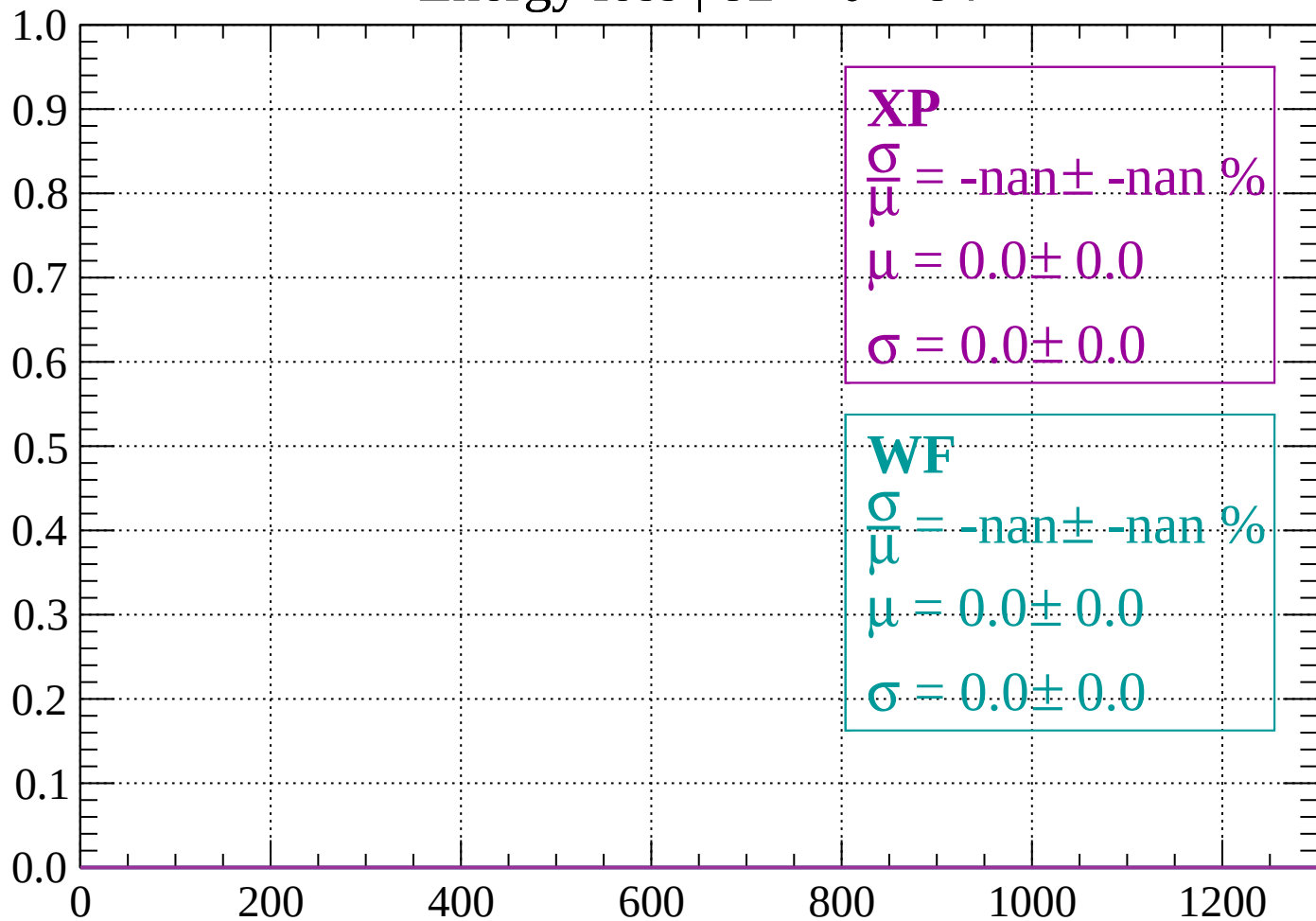
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

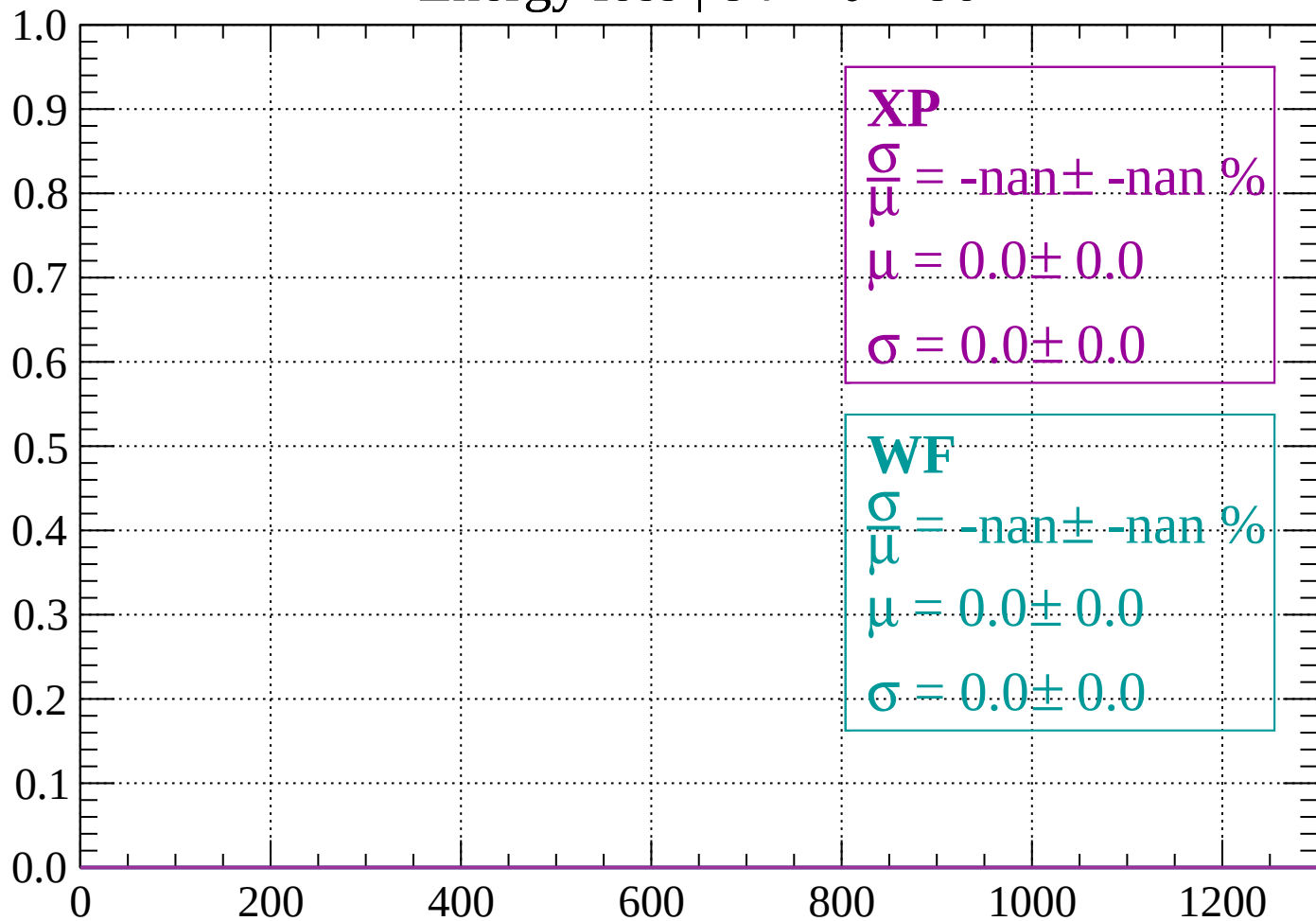
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

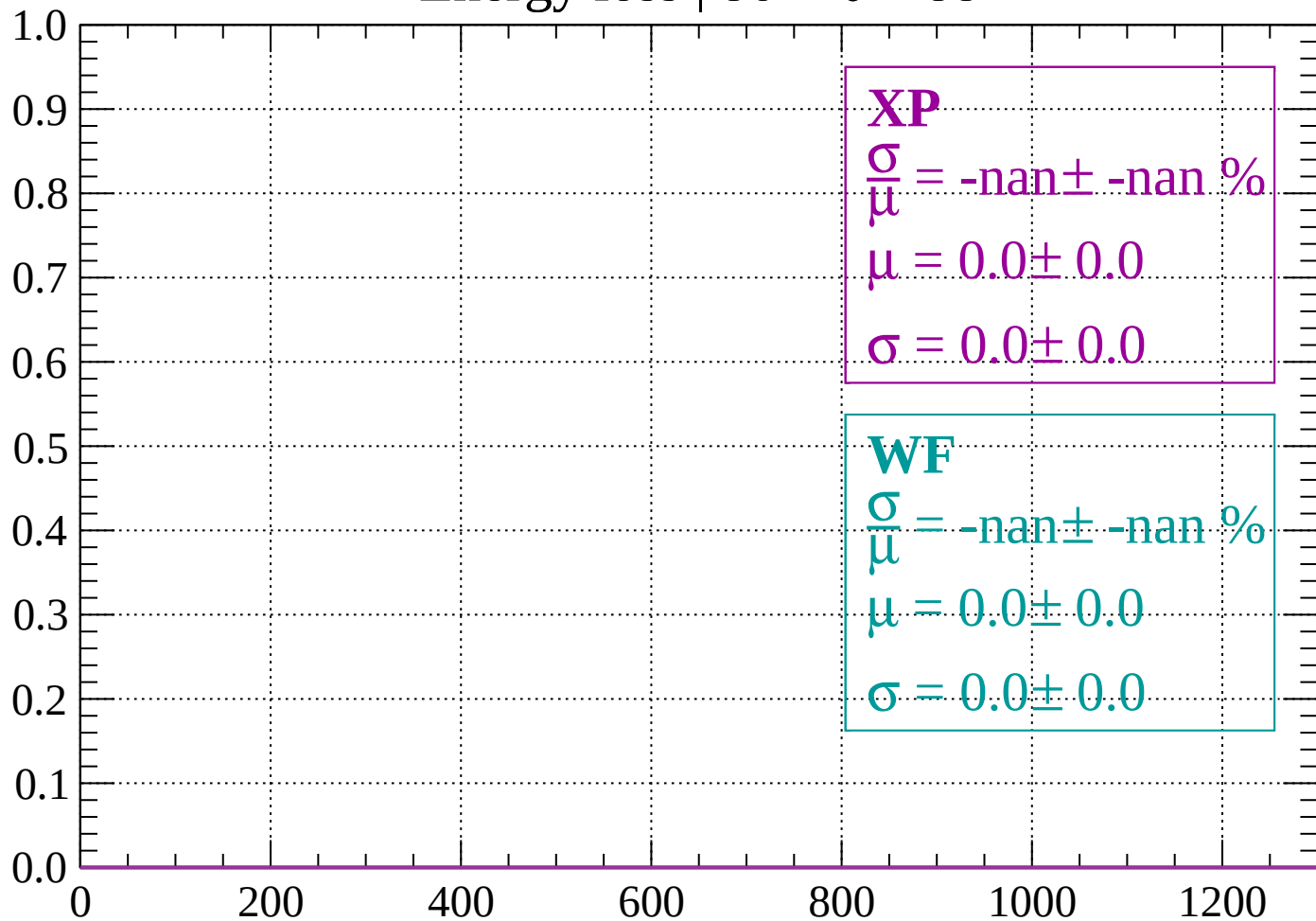
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

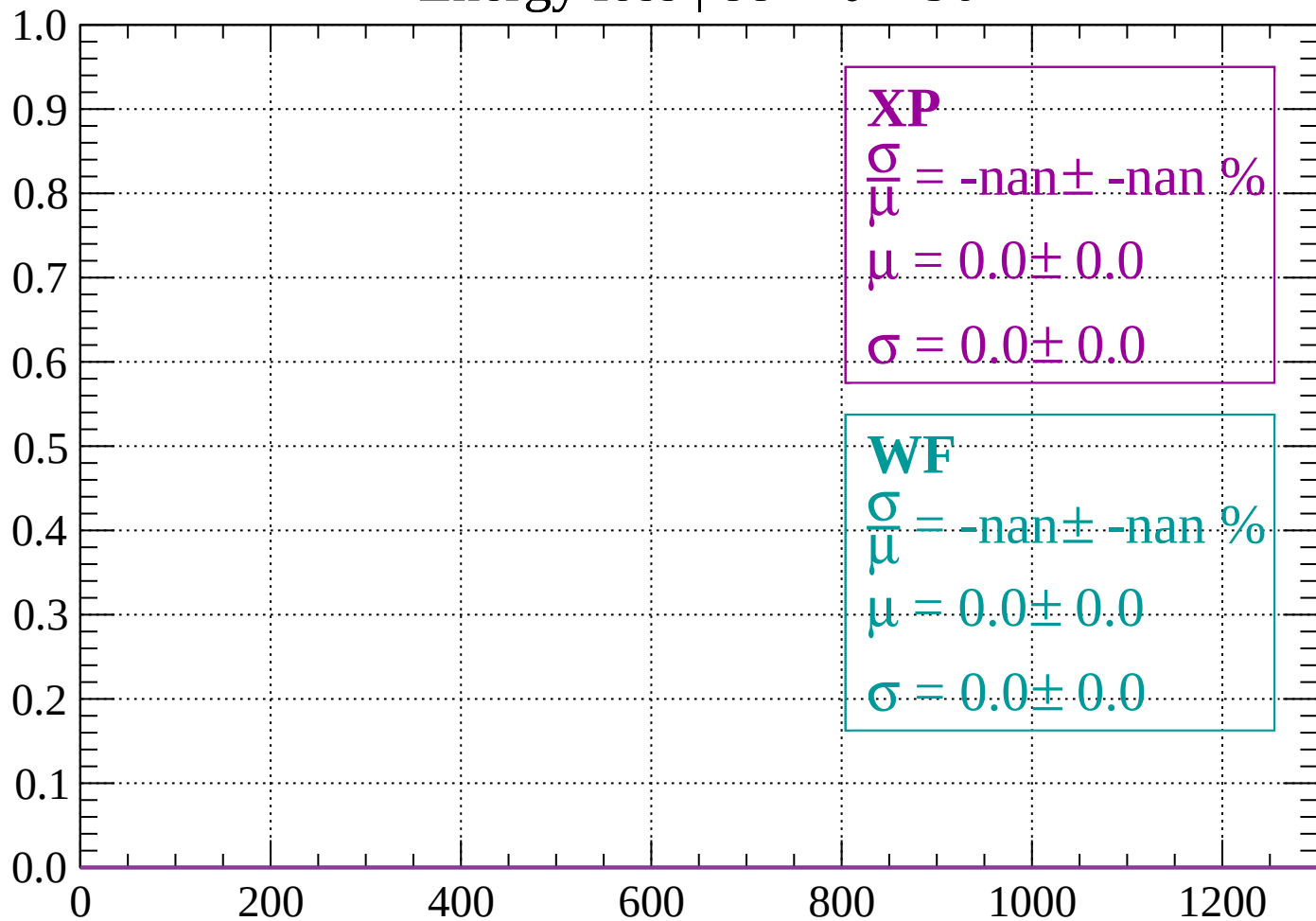
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

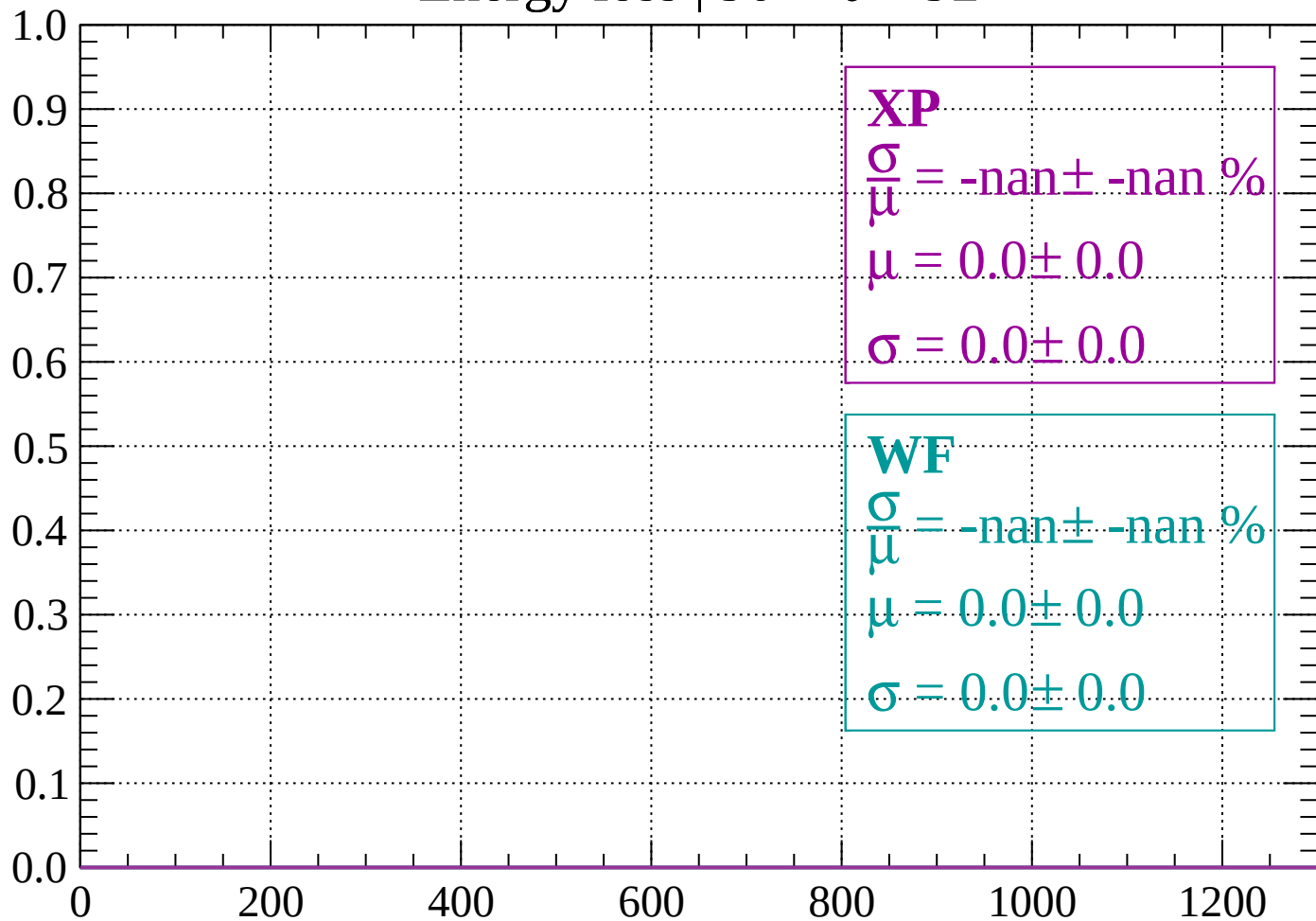
Count



dE/dx (ADC counts/cm)

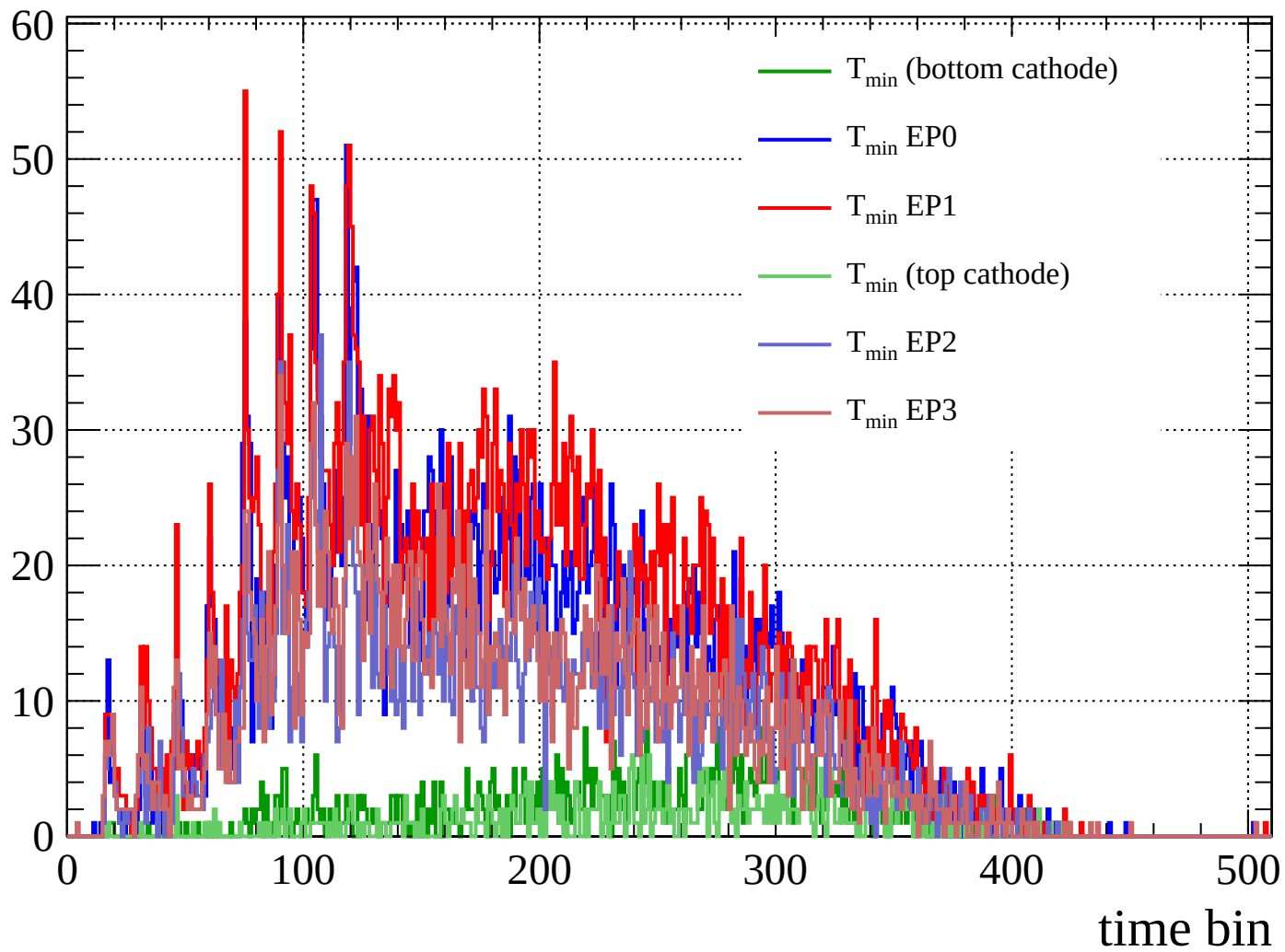
Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

Count



Count

